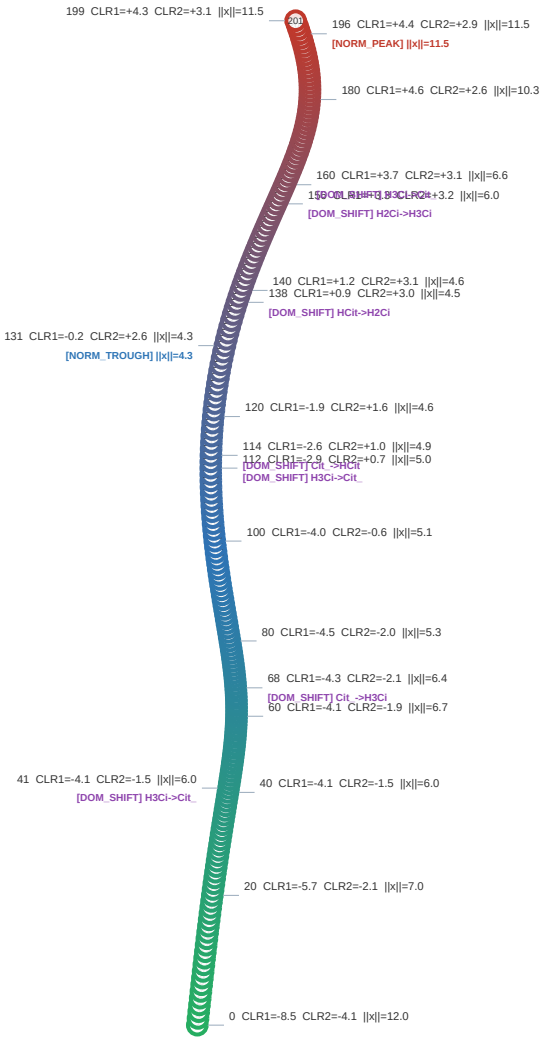
MANIFOLD HELIX — POLAR SLICE MAP

CITRIC_ACID_TITRATIO | Citric Acid Titration | 200 intervals | 4 carriers



Each numbered circle marks a polar slice. Coordinates: CLR1 (x), CLR2 (y), interval (z).

Intervals: 200 | Carriers: H3Cit, H2Cit_minus, HCit_2minus, Cit_3minus | Events detected: 9

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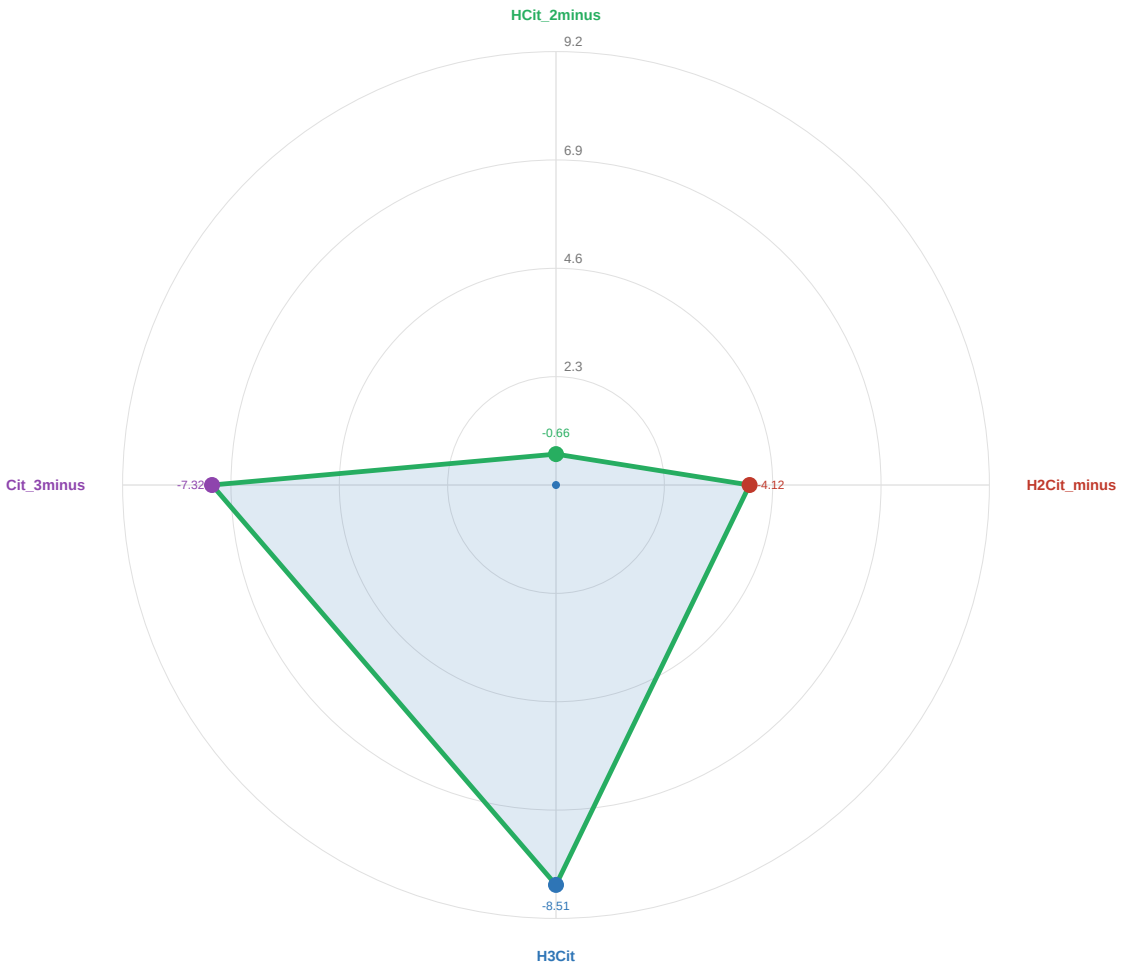
POLAR SLICE — t=0

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 1/200 | ||x||_A = 11.974



Phase = -154.2 deg

H3Cit=-8.51 H2Cit_=-4.12 HCit_2=-0.66 Cit_3m=-7.32

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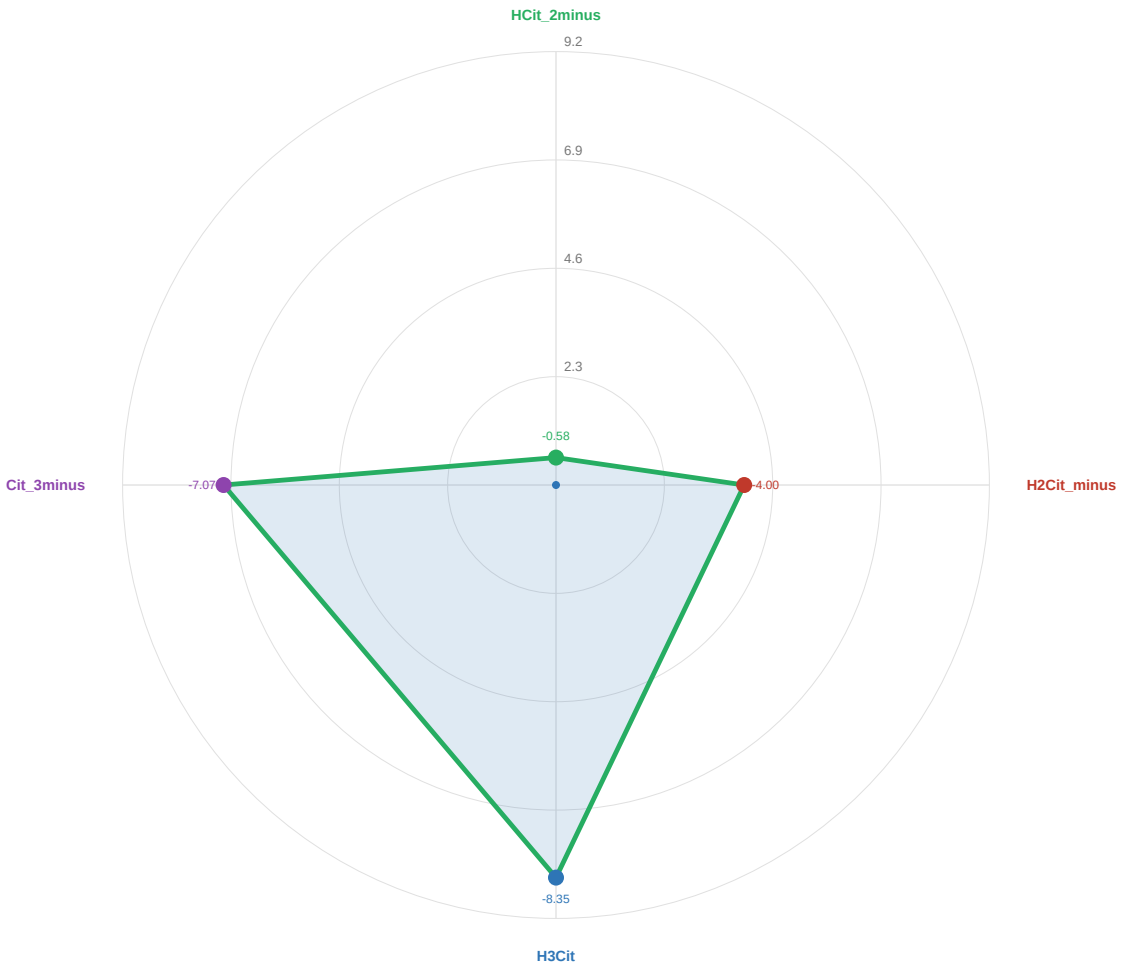
POLAR SLICE — t=1

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 2/200 | ||x||_A = 11.670



From 0: delta ||x||_A = -0.304

Phase = -154.4 deg

H3Cit=-8.35 H2Cit_=-4.00 HCit_2=-0.58 Cit_3m=-7.07

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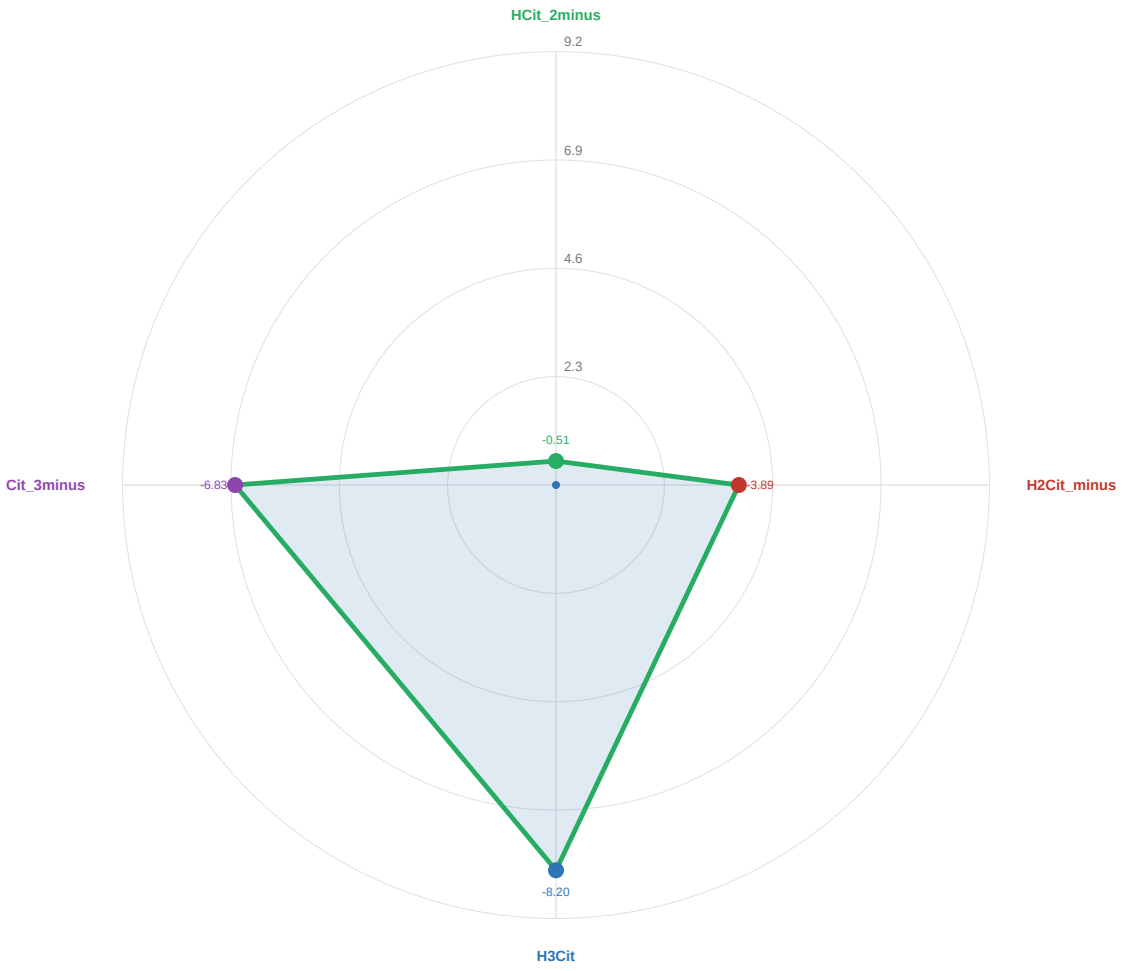
POLAR SLICE — t=2

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 3/200 | ||x||_A = 11.368



From 1: delta ||x||_A = -0.302

Phase = -154.6 deg

H3Cit=-8.20 H2Cit_=-3.89 HCit_2=-0.51 Cit_3m=-6.83

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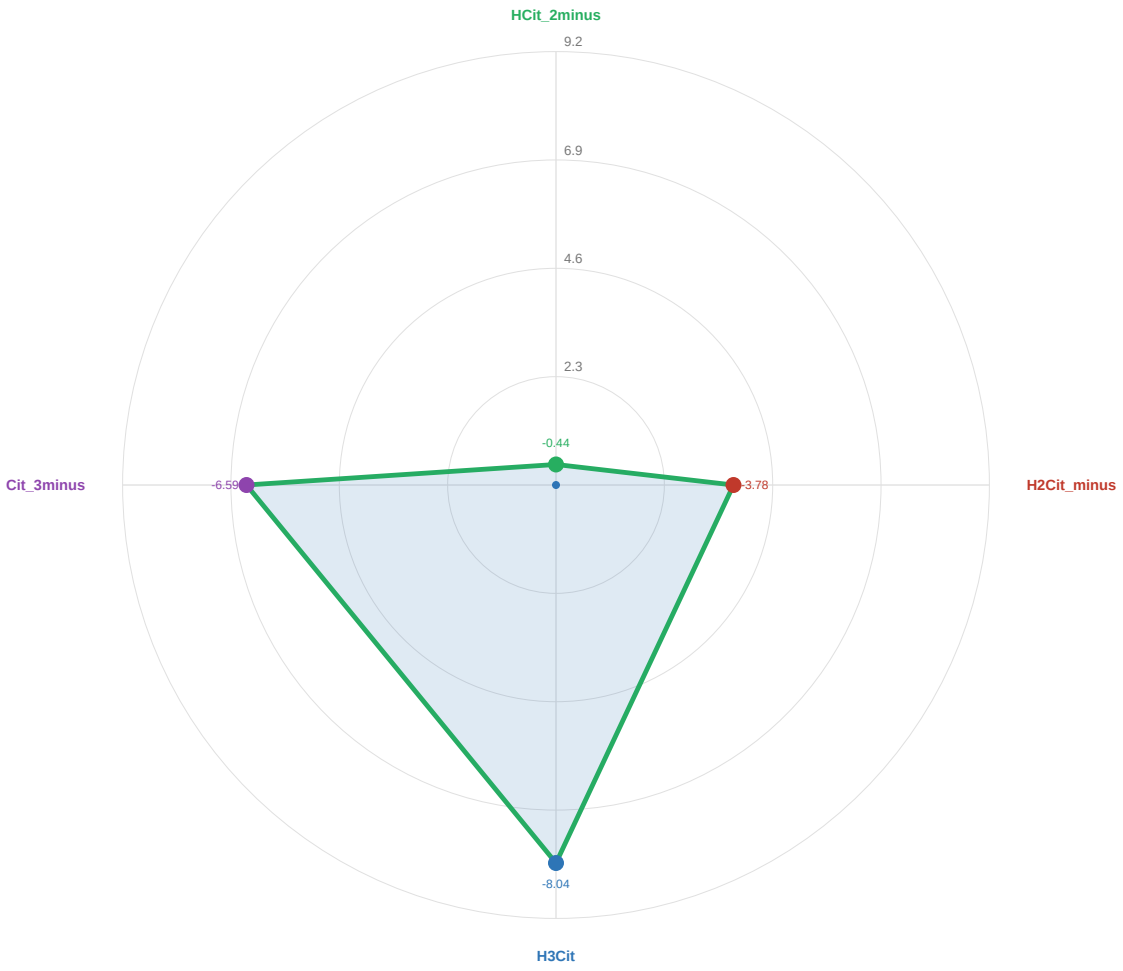
POLAR SLICE — t=3

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 4/200 | ||x||_A = 11.069



From 2: delta ||x||_A = -0.299

Phase = -154.8 deg

H3Cit=-8.04 H2Cit_=-3.78 HCit_2=-0.44 Cit_3m=-6.59

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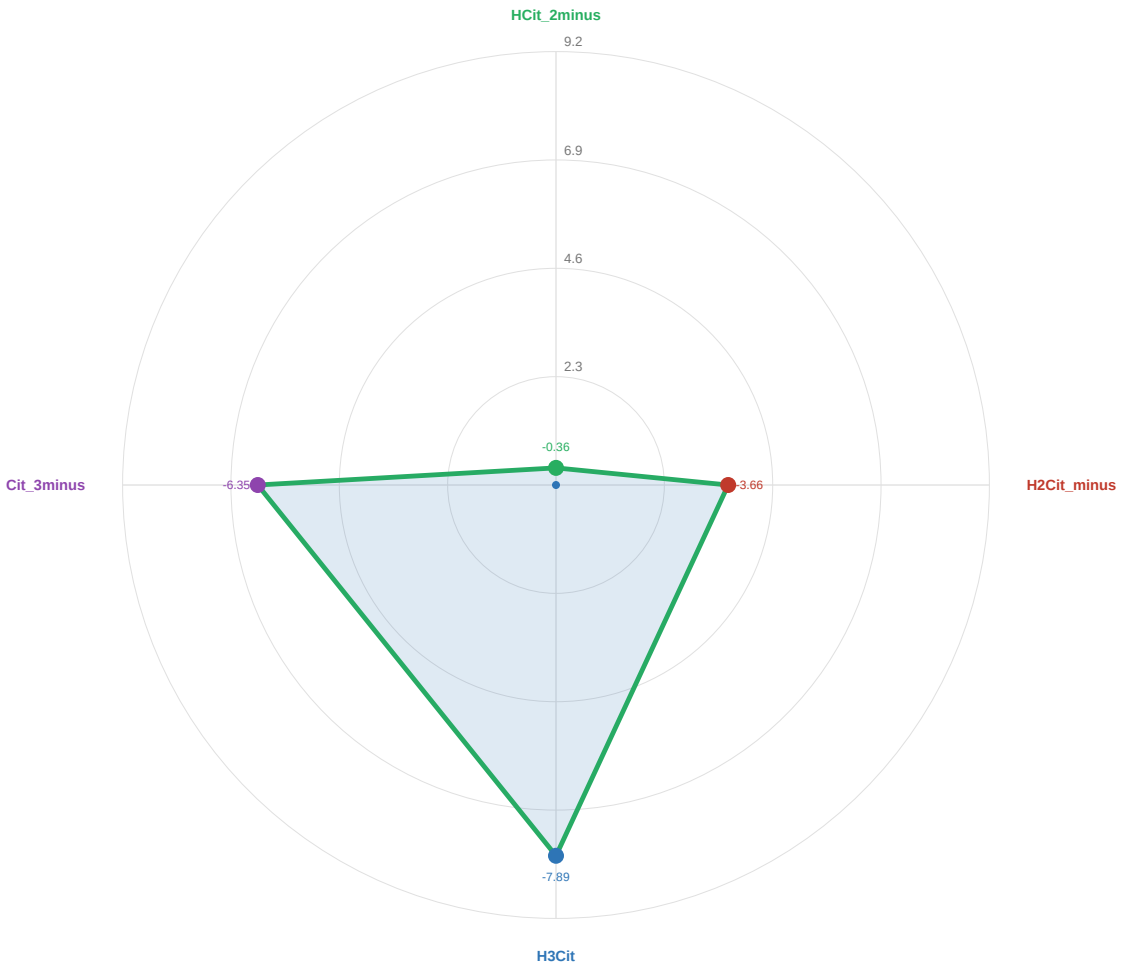
POLAR SLICE — t=4

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 5/200 | ||x||_A = 10.773



From 3: delta ||x||_A = -0.296

Phase = -155.1 deg

H3Cit=-7.89 H2Cit_=-3.66 HCit_2=-0.36 Cit_3m=-6.35

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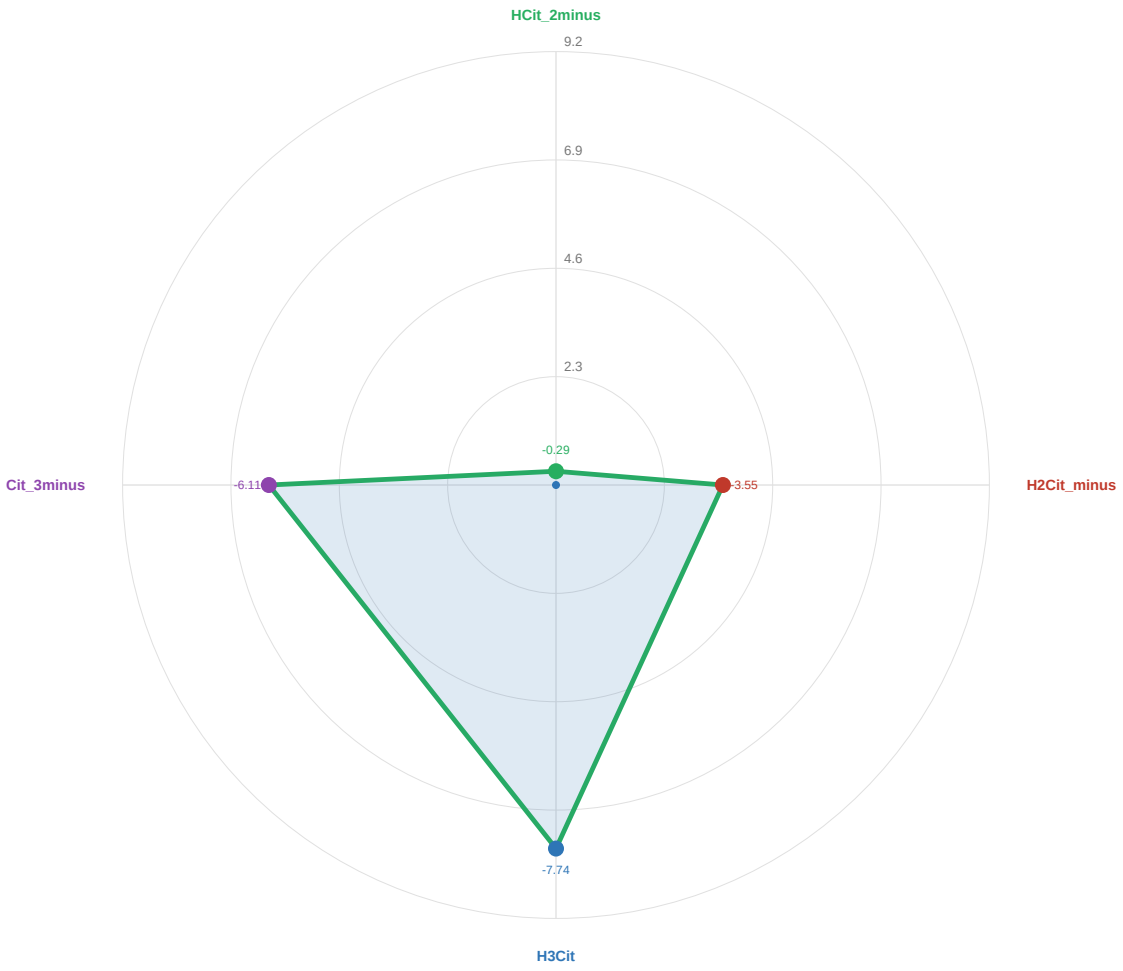
POLAR SLICE — t=5

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 6/200 | ||x||_A = 10.482



From 4: delta ||x||_A = -0.291

Phase = -155.3 deg

H3Cit=-7.74 H2Cit_=-3.55 HCit_2=-0.29 Cit_3m=-6.11

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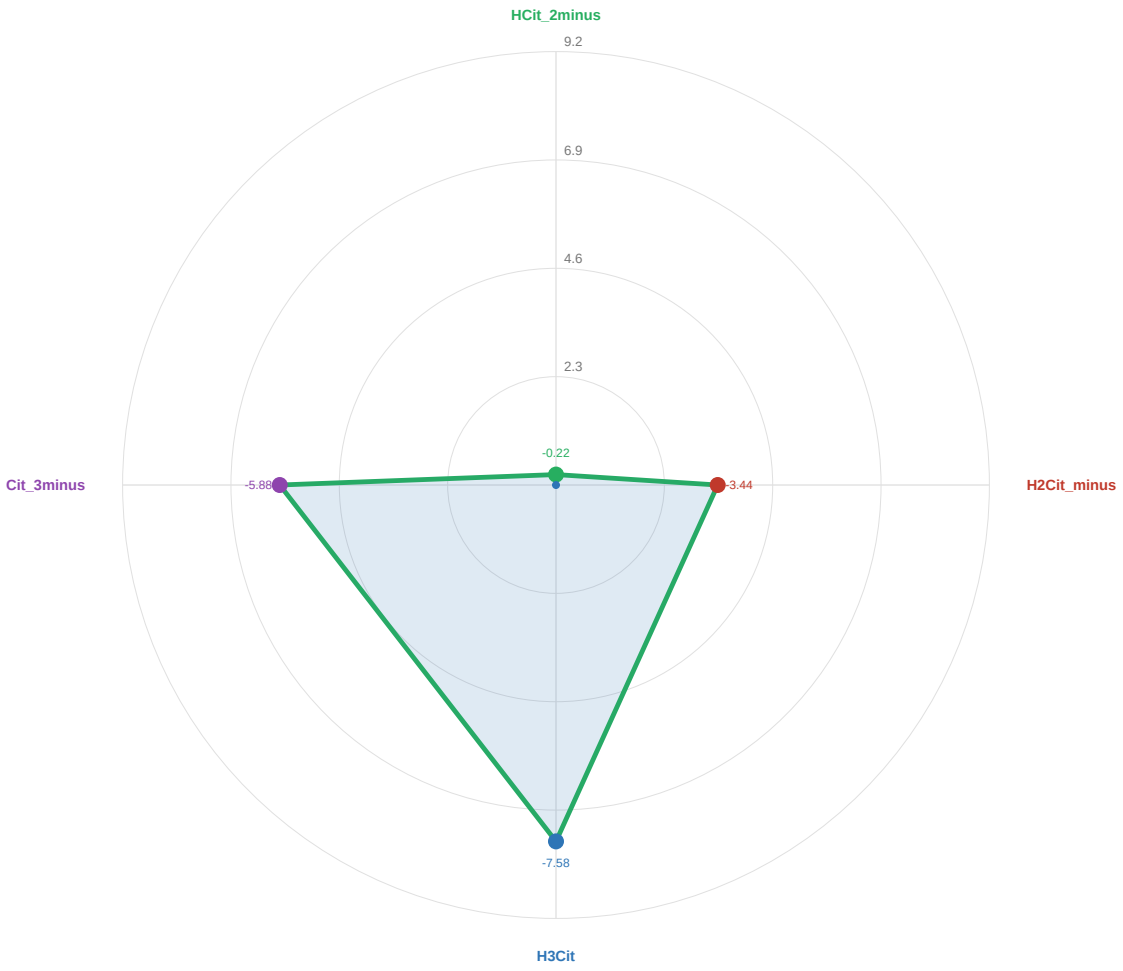
POLAR SLICE — t=6

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 7/200 | ||x||_A = 10.196



From 5: delta ||x||_A = -0.286

Phase = -155.6 deg

H3Cit=-7.58 H2Cit_=-3.44 HCit_2=-0.22 Cit_3m=-5.88

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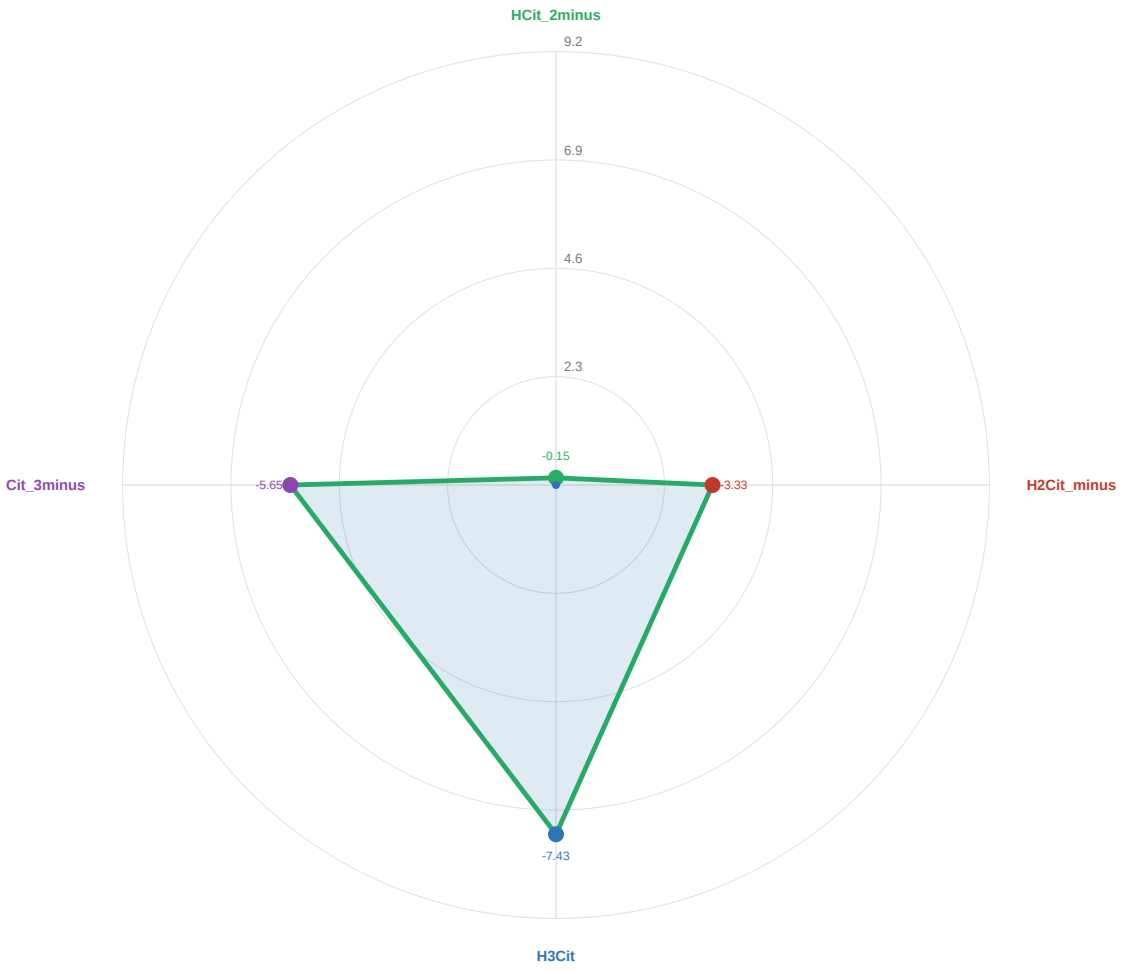
POLAR SLICE — t=7

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 8/200 | ||x||_A = 9.915



From 6: delta ||x||_A = -0.281

Phase = -155.9 deg

H3Cit=-7.43 H2Cit_=-3.33 HCit_2=-0.15 Cit_3m=-5.65

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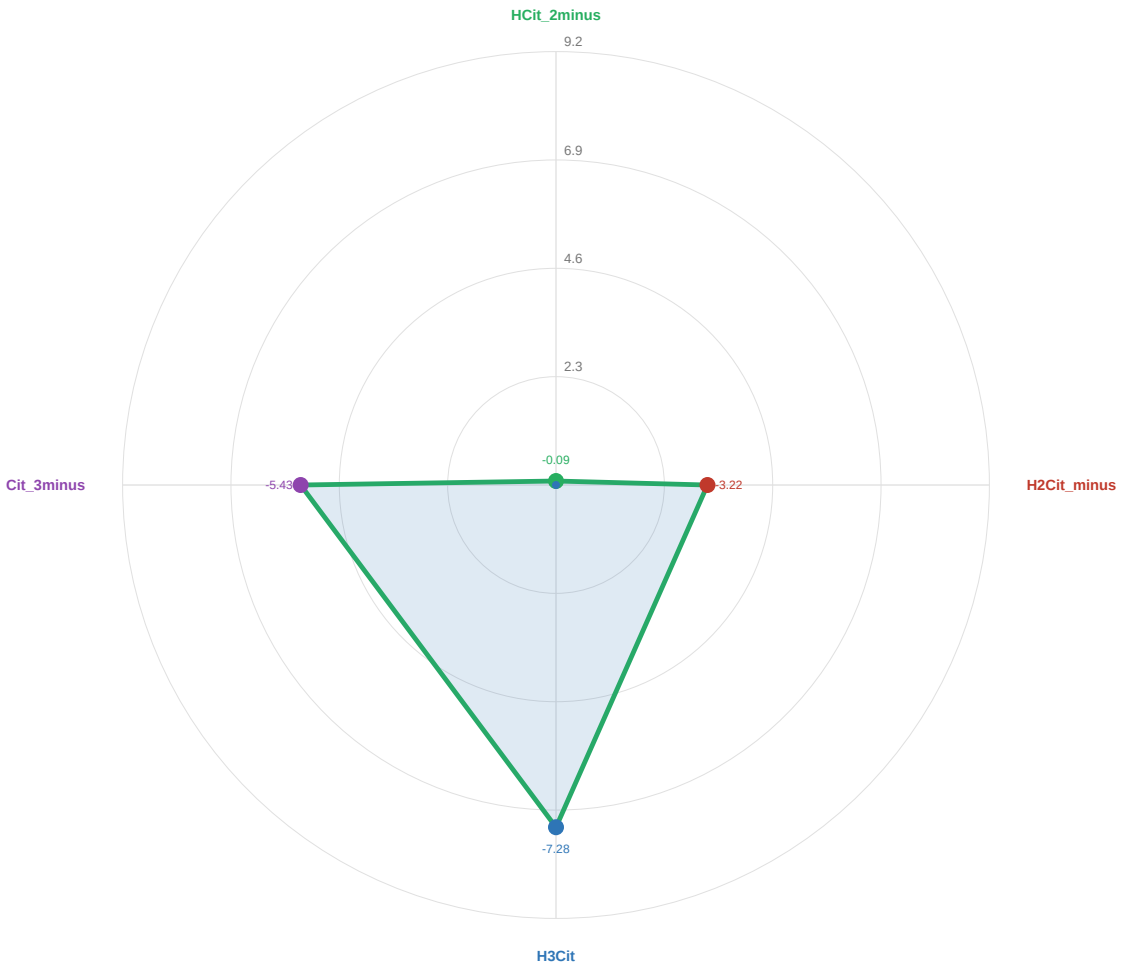
POLAR SLICE — t=8

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 9/200 | ||x||_A = 9.641



From 7: delta ||x||_A = -0.274

Phase = -156.1 deg

H3Cit=-7.28 H2Cit_=-3.22 HCit_2=-0.09 Cit_3m=-5.43

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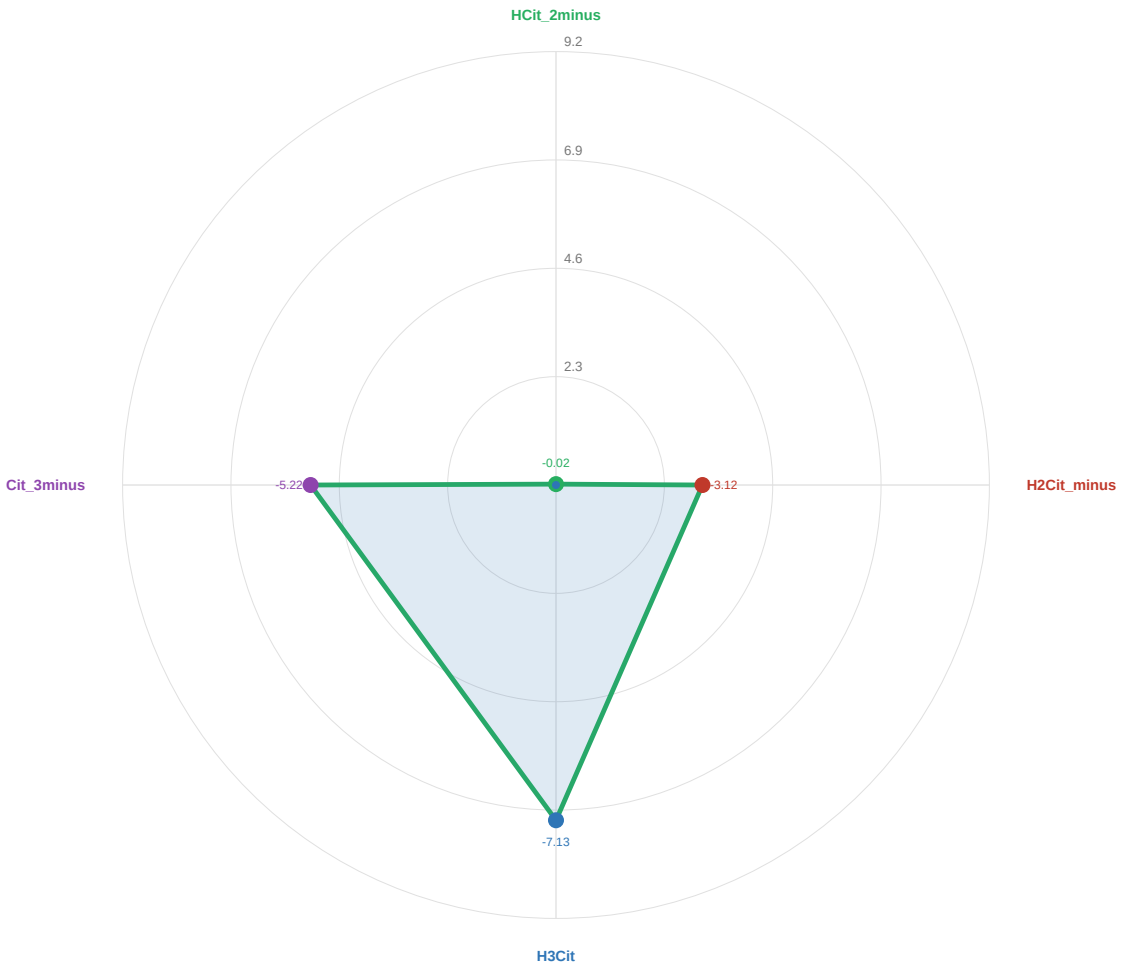
POLAR SLICE — t=9

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 10/200 | ||x||_A = 9.374



From 8: delta ||x||_A = -0.267

Phase = -156.4 deg

H3Cit=-7.13 H2Cit_=-3.12 HCit_2=-0.02 Cit_3m=-5.22

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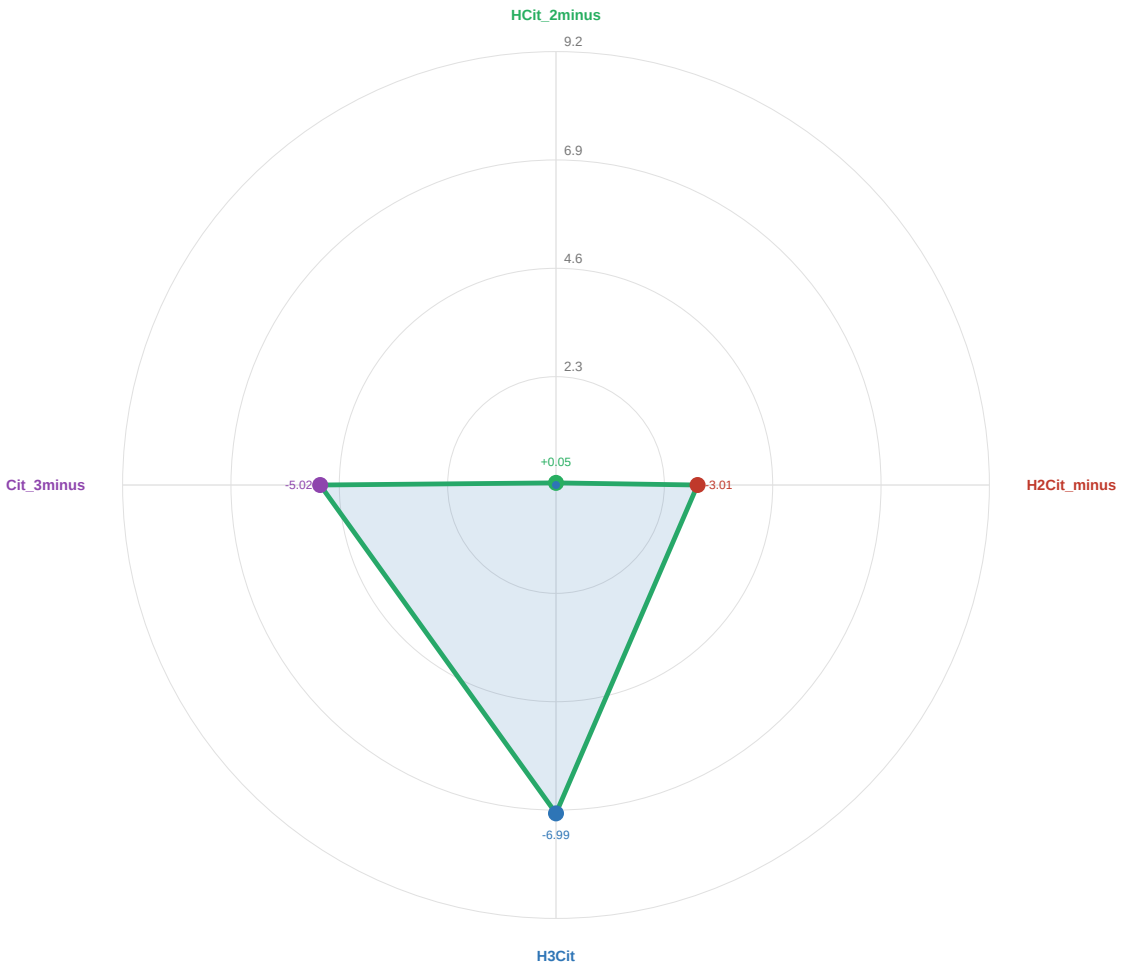
POLAR SLICE — t=10

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 11/200 | ||x||_A = 9.115



From 9: delta ||x||_A = -0.259

Phase = -156.7 deg

H3Cit=-6.99 H2Cit_=-3.01 HCit_2=+0.05 Cit_3m=-5.02

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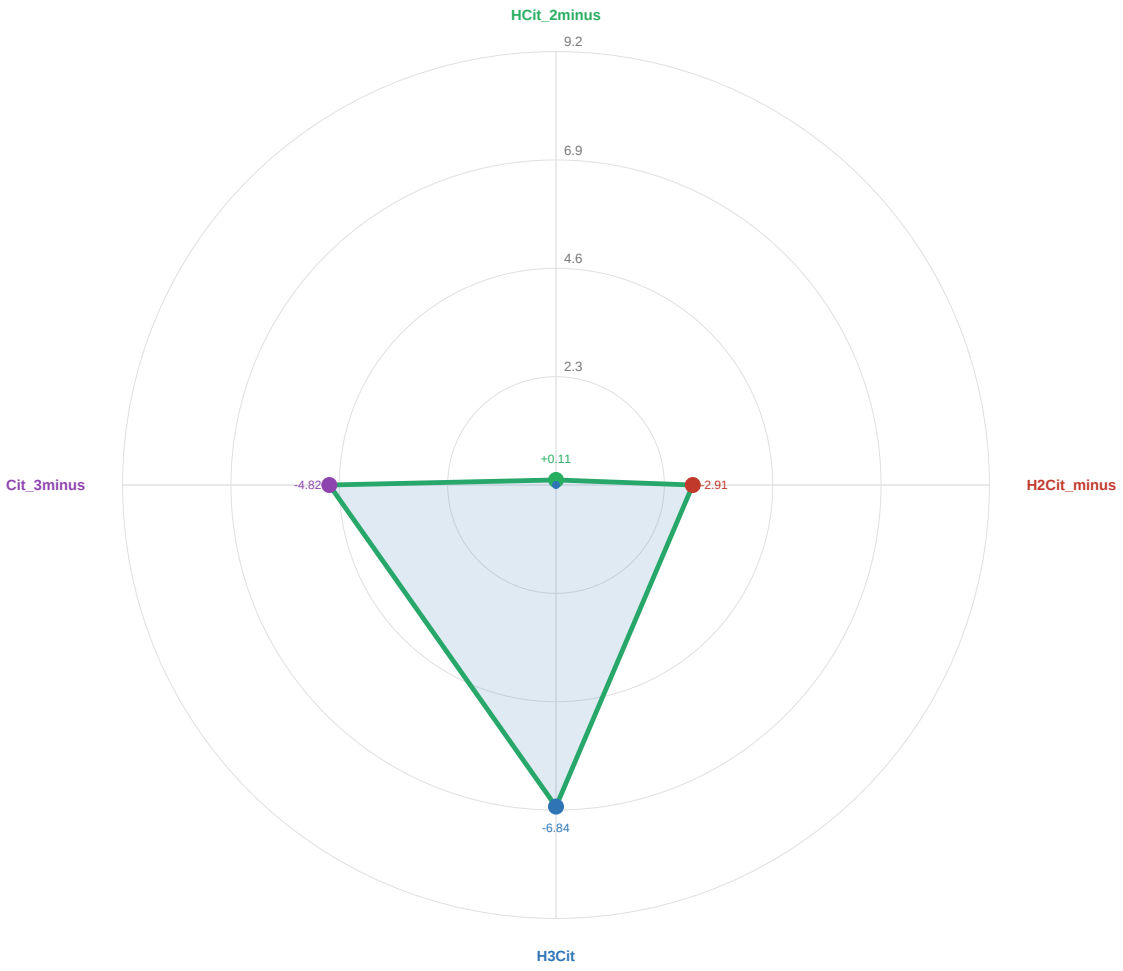
POLAR SLICE — t=11

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 12/200 | ||x||_A = 8.864



From 10: delta ||x||_A = -0.251

Phase = -157.0 deg

H3Cit=-6.84 H2Cit_=-2.91 HCit_2=+0.11 Cit_3m=-4.82

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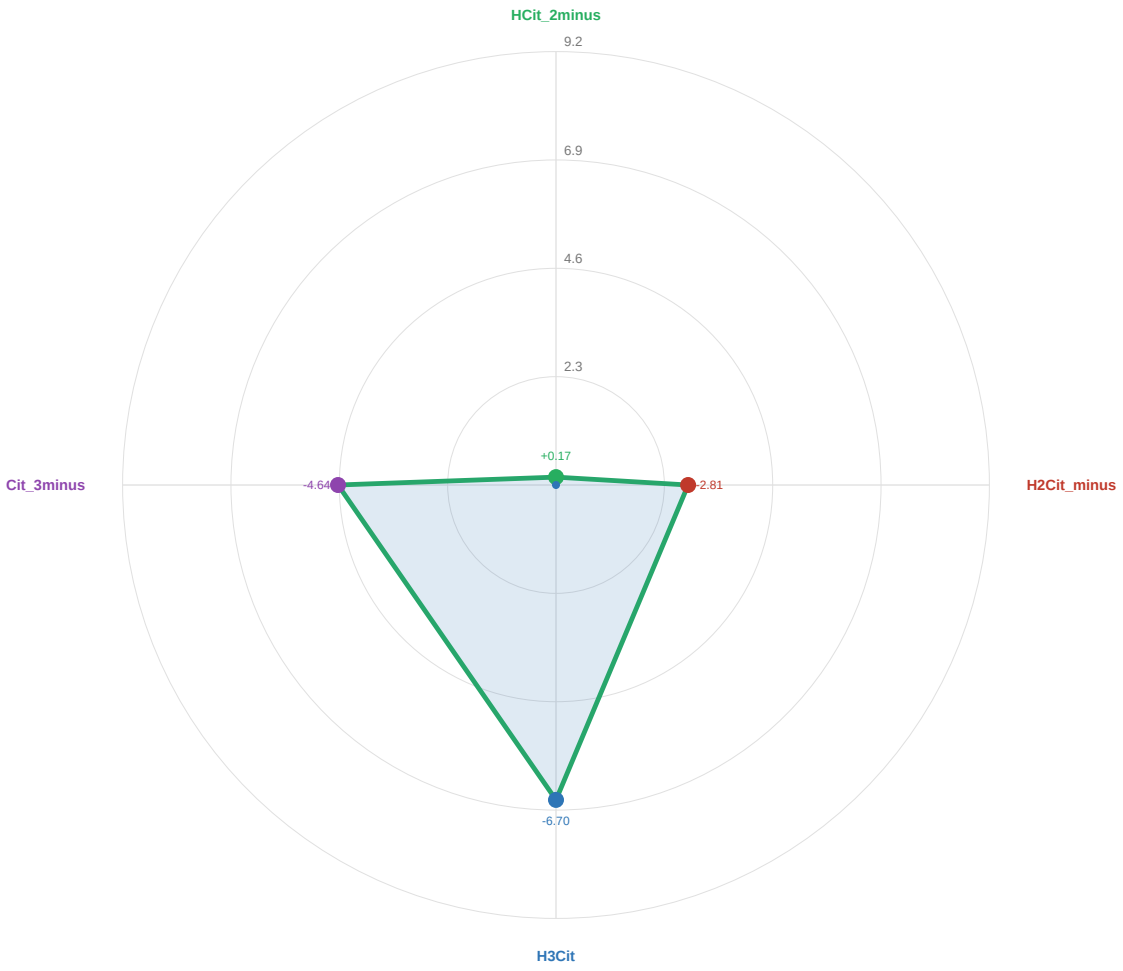
POLAR SLICE — t=12

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 13/200 | ||x||_A = 8.622



From 11: delta ||x||_A = -0.242

Phase = -157.2 deg

H3Cit=-6.70 H2Cit_=-2.81 HCit_2=+0.17 Cit_3m=-4.64

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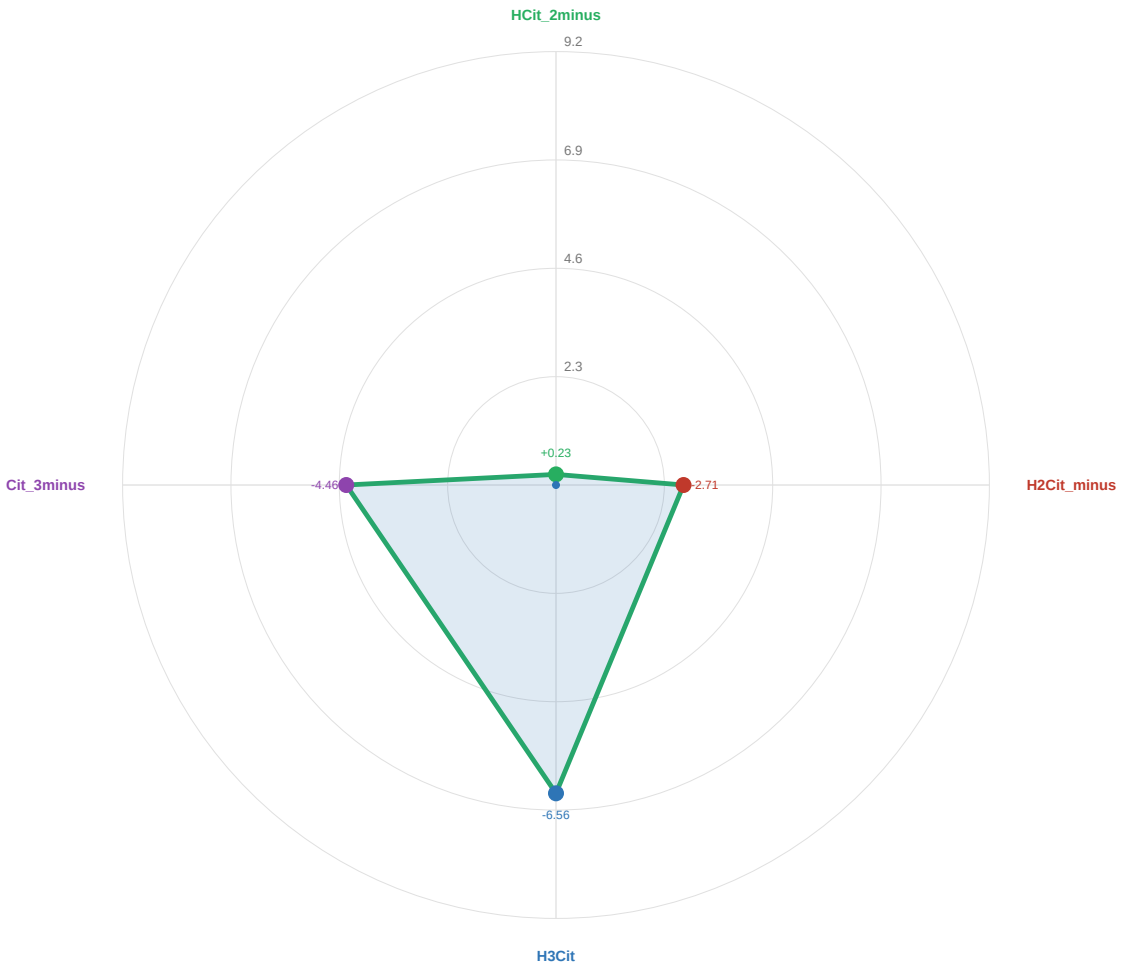
POLAR SLICE — t=13

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 14/200 | ||x||_A = 8.389



From 12: delta ||x||_A = -0.233

Phase = -157.5 deg

H3Cit=-6.56 H2Cit_=-2.71 HCit_2=+0.23 Cit_3m=-4.46

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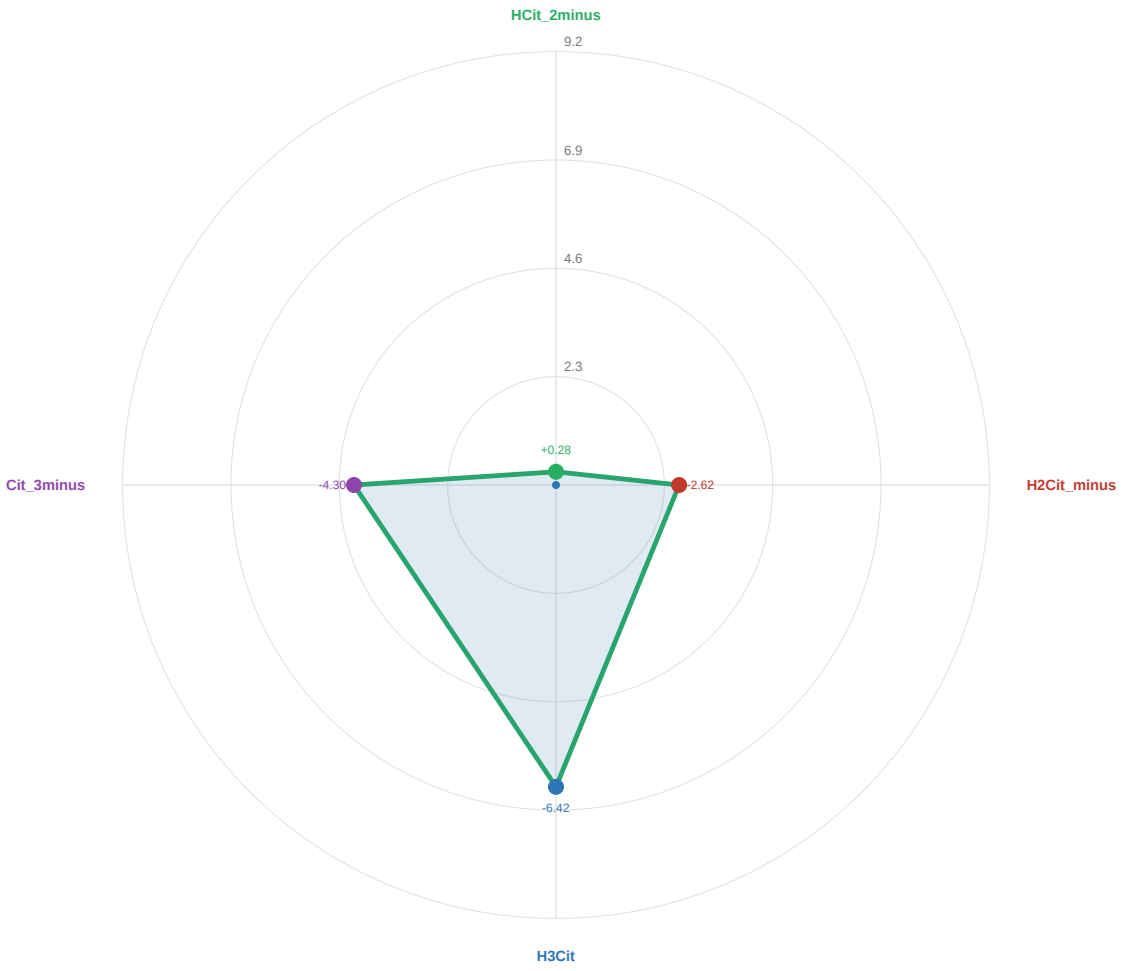
POLAR SLICE — t=14

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 15/200 | ||x||_A = 8.166



From 13: delta ||x||_A = -0.223

Phase = -157.8 deg

H3Cit=-6.42 H2Cit_=-2.62 HCit_2=+0.28 Cit_3m=-4.30

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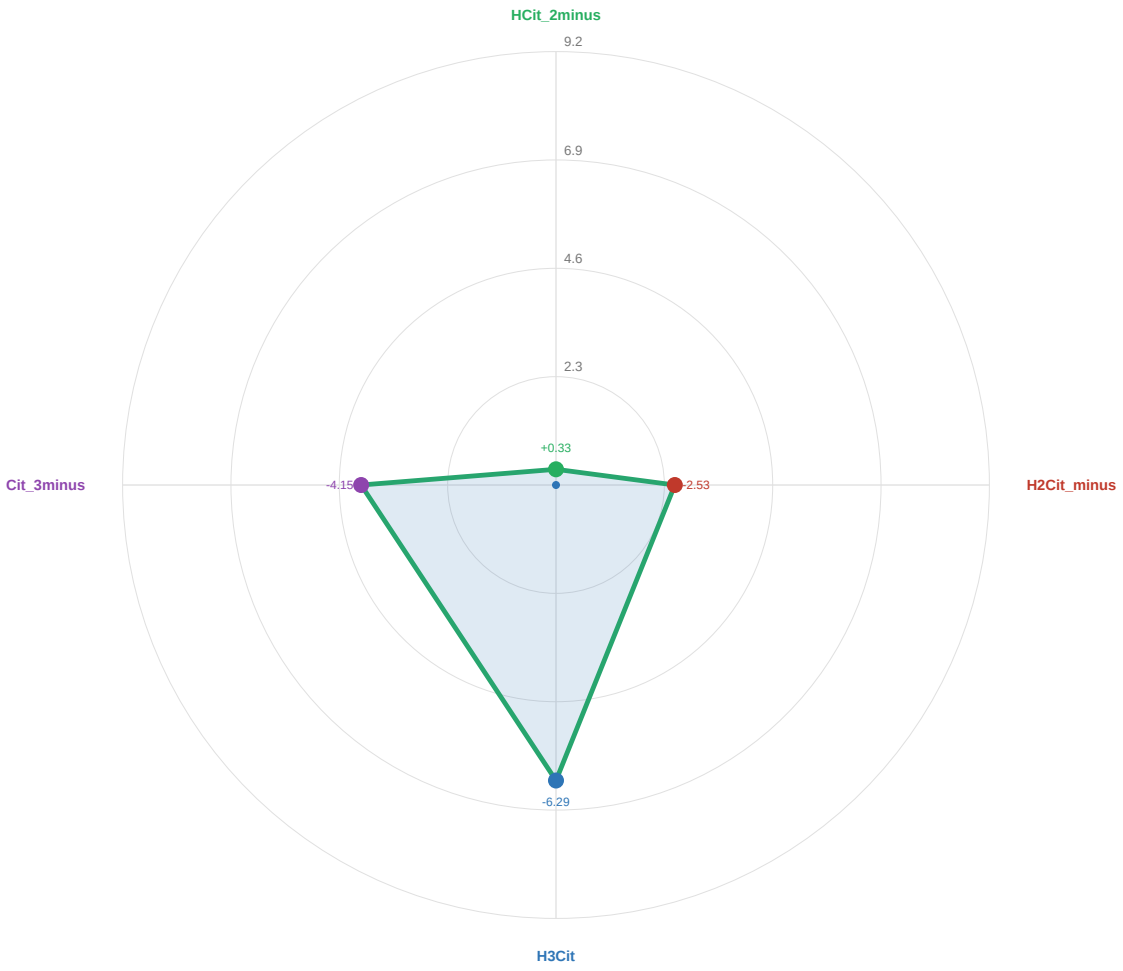
POLAR SLICE — t=15

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 16/200 | ||x||_A = 7.952



From 14: delta ||x||_A = -0.214

Phase = -158.1 deg

H3Cit=-6.29 H2Cit_=-2.53 HCit_2=+0.33 Cit_3m=-4.15

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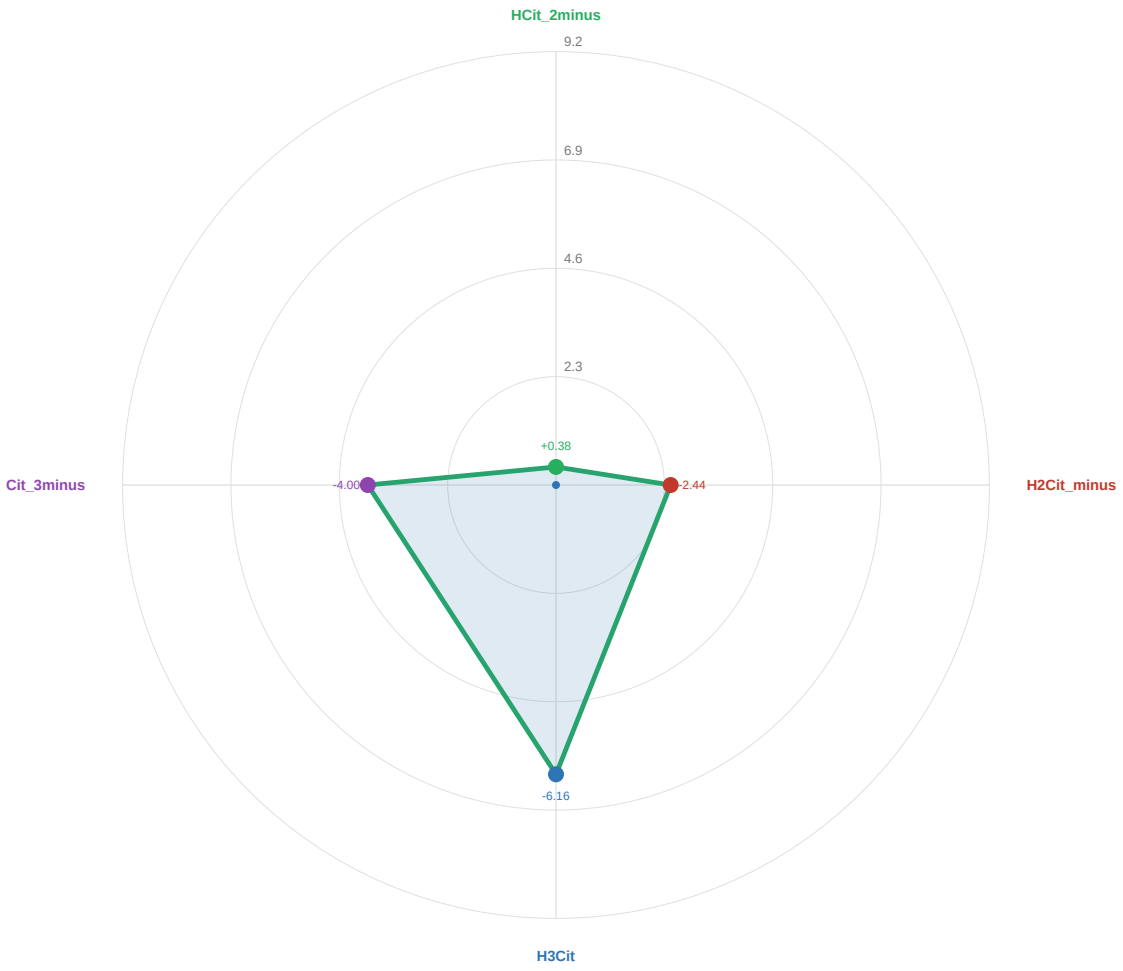
POLAR SLICE — t=16

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 17/200 | ||x||_A = 7.748



From 15: delta ||x||_A = -0.204

Phase = -158.4 deg

H3Cit=-6.16 H2Cit_=-2.44 HCit_2=+0.38 Cit_3m=-4.00

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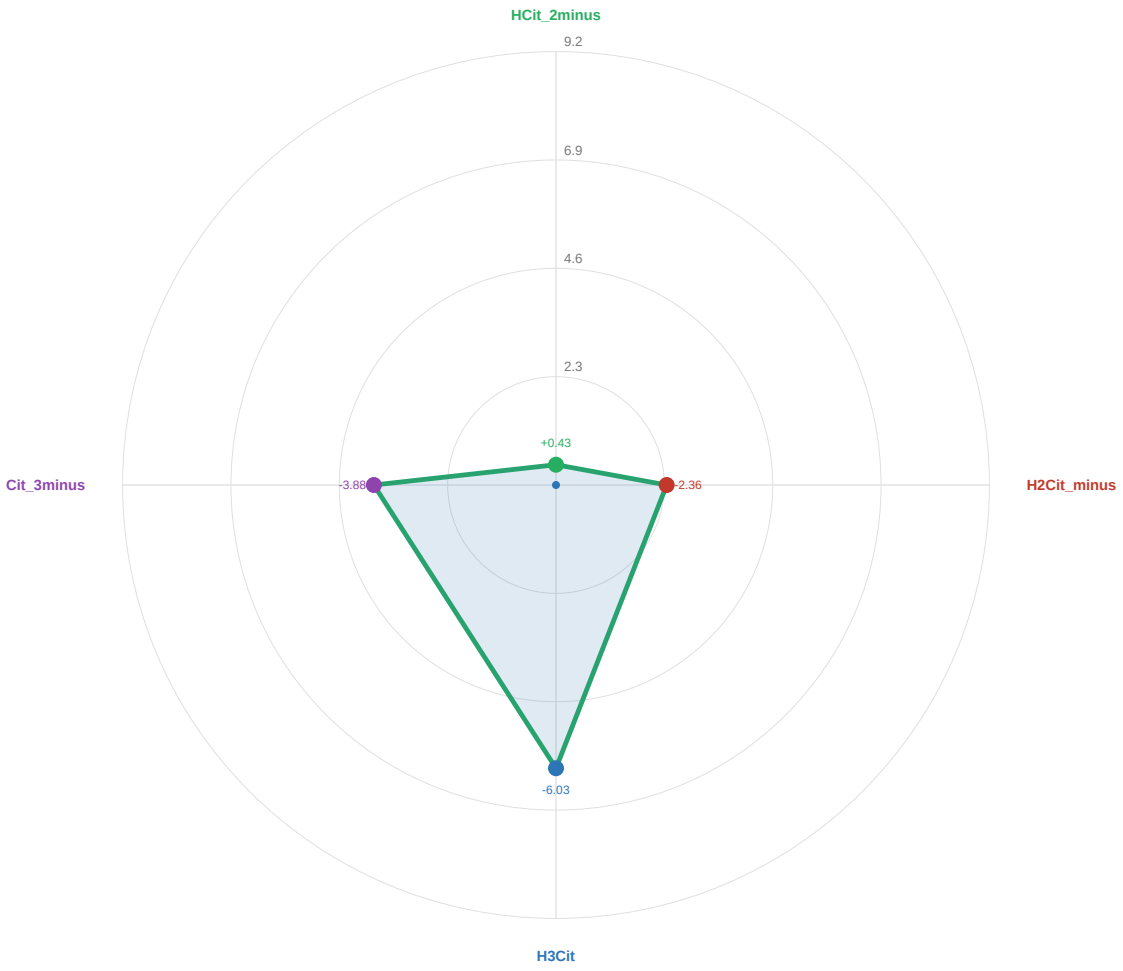
POLAR SLICE — t=17

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 18/200 | ||x||_A = 7.555



From 16: delta ||x||_A = -0.193

Phase = -158.7 deg

H3Cit=-6.03 H2Cit_=-2.36 HCit_2=+0.43 Cit_3m=-3.88

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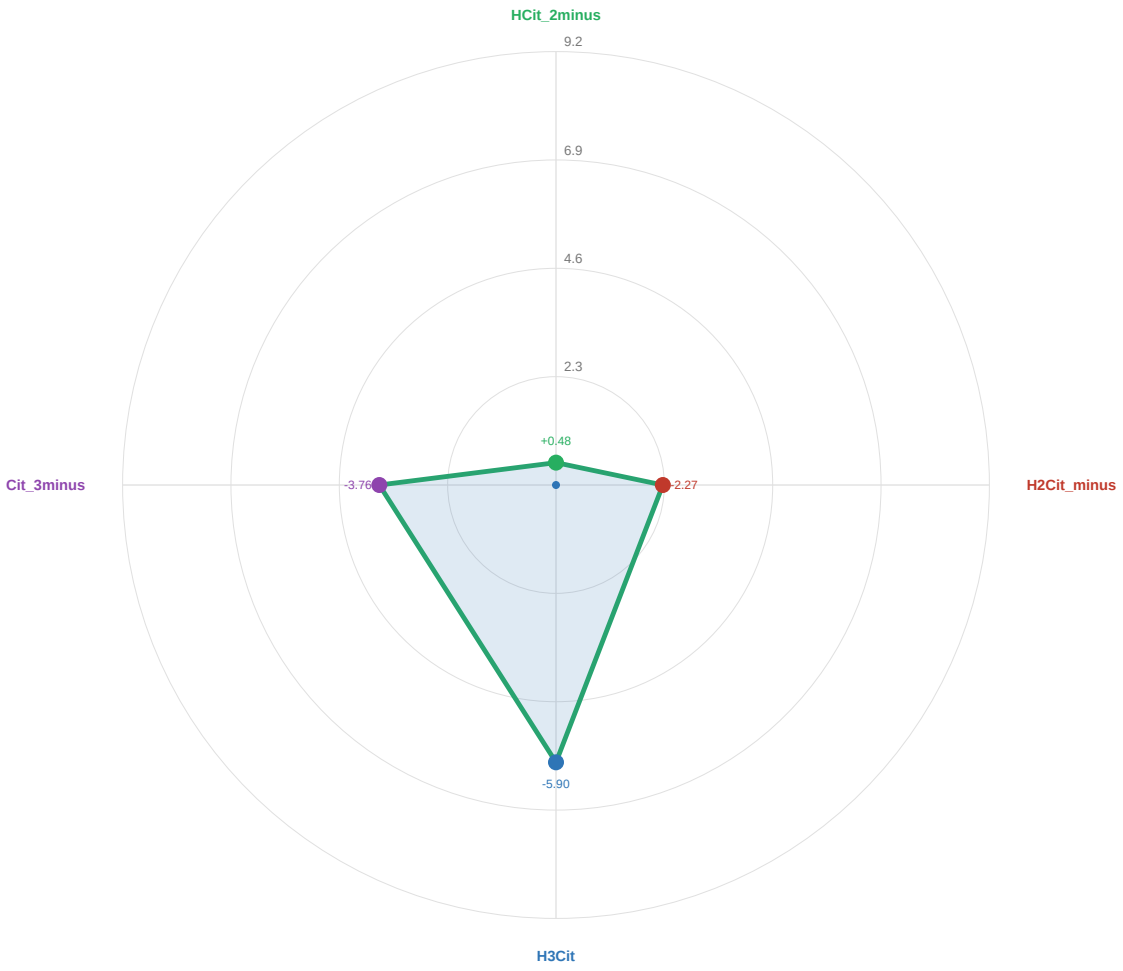
POLAR SLICE — t=18

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 19/200 | ||x||_A = 7.372



From 17: delta ||x||_A = -0.183

Phase = -158.9 deg

H3Cit=-5.90 H2Cit_=-2.27 HCit_2=+0.48 Cit_3m=-3.76

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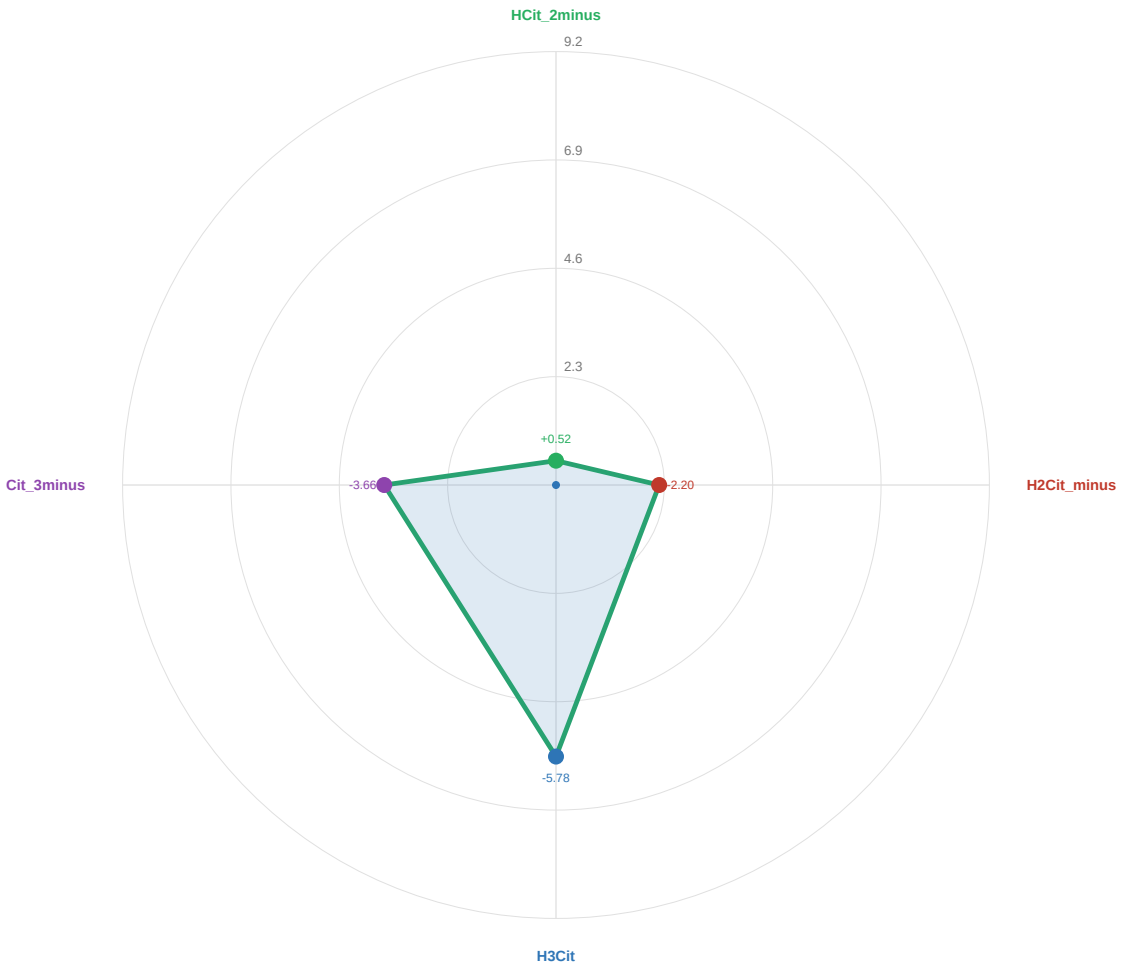
POLAR SLICE — t=19

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 20/200 | ||x||_A = 7.199



From 18: delta ||x||_A = -0.173

Phase = -159.2 deg

H3Cit=-5.78 H2Cit_=-2.20 HCit_2=+0.52 Cit_3m=-3.66

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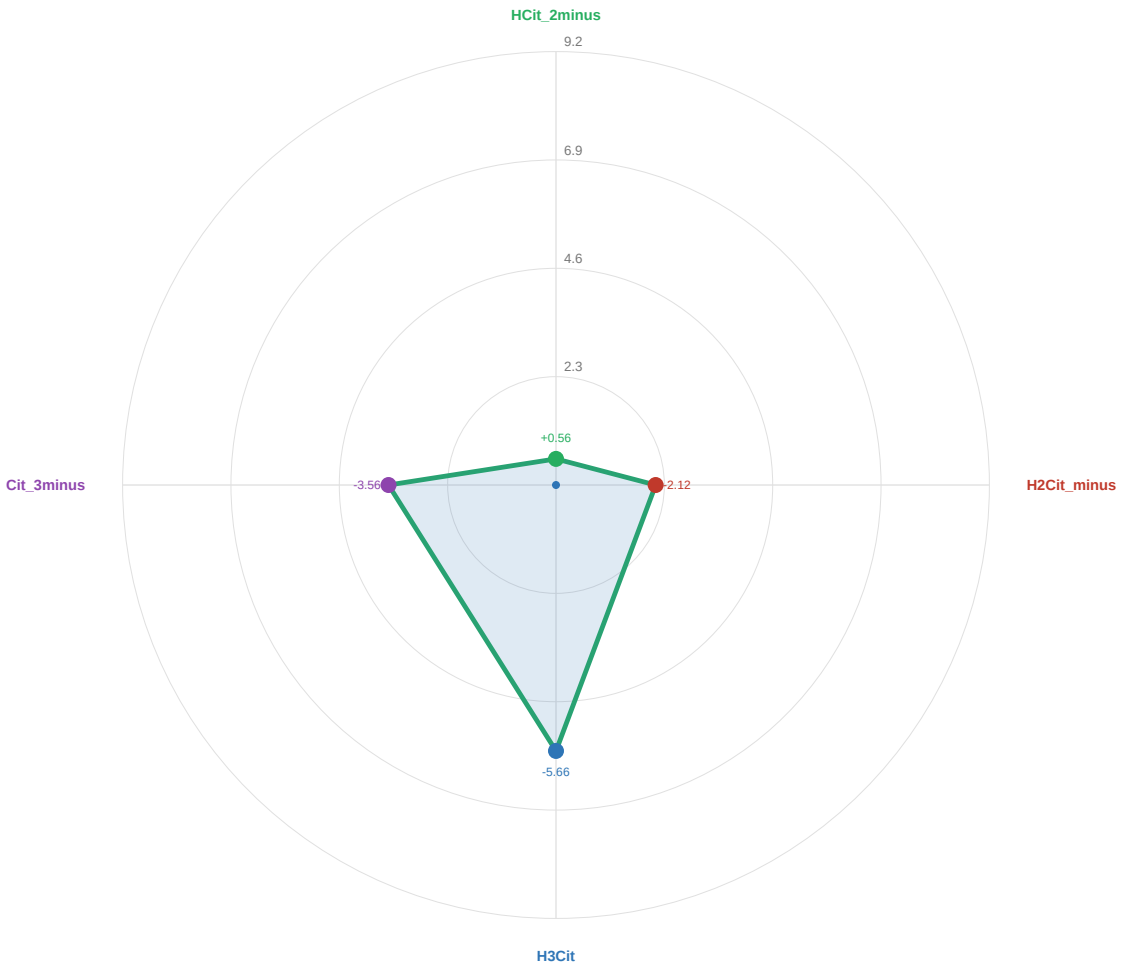
POLAR SLICE — t=20

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 21/200 | ||x||_A = 7.037



From 19: delta ||x||_A = -0.162

Phase = -159.5 deg

H3Cit=-5.66 H2Cit_=-2.12 HCit_2=+0.56 Cit_3m=-3.56

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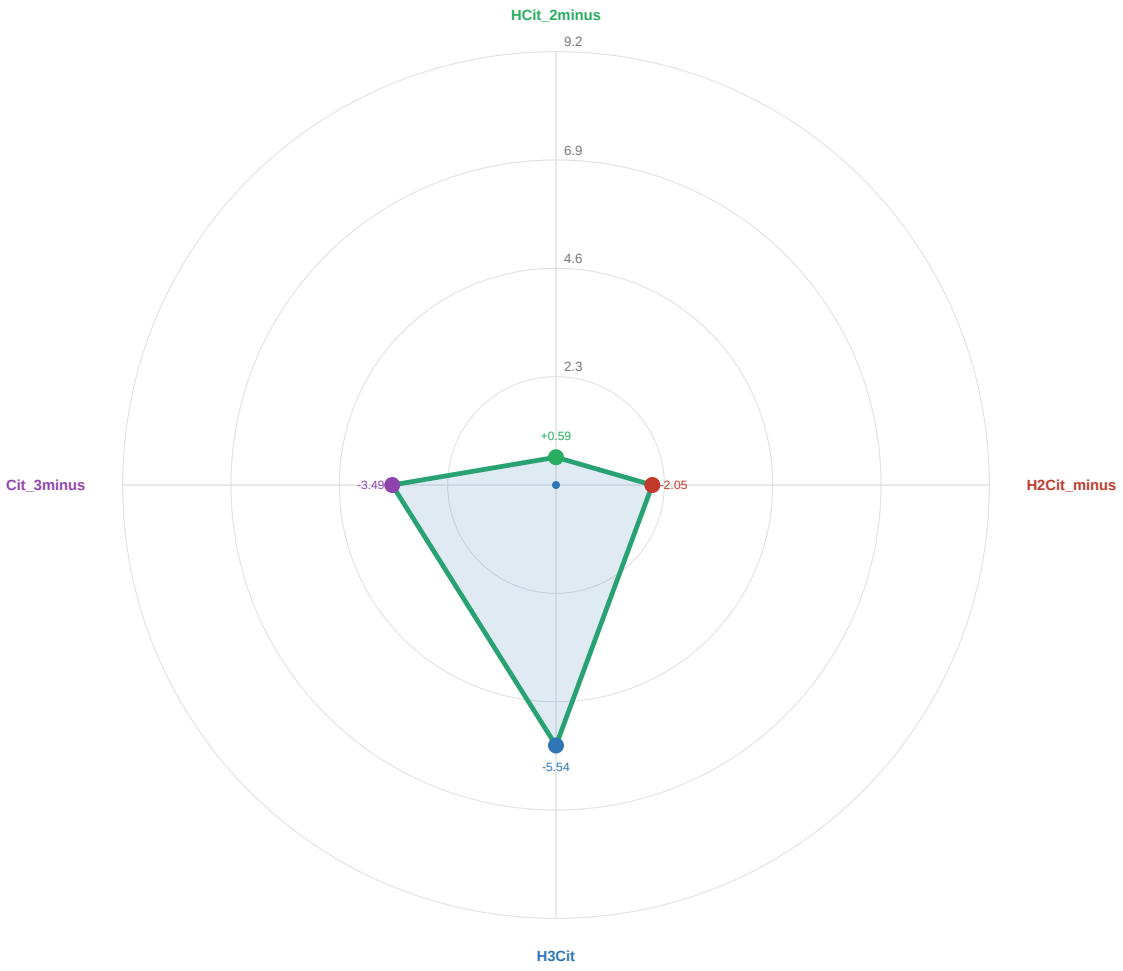
POLAR SLICE — t=21

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 22/200 | ||x||_A = 6.886



From 20: delta ||x||_A = -0.151

Phase = -159.7 deg

H3Cit=-5.54 H2Cit_=-2.05 HCit_2=+0.59 Cit_3m=-3.49

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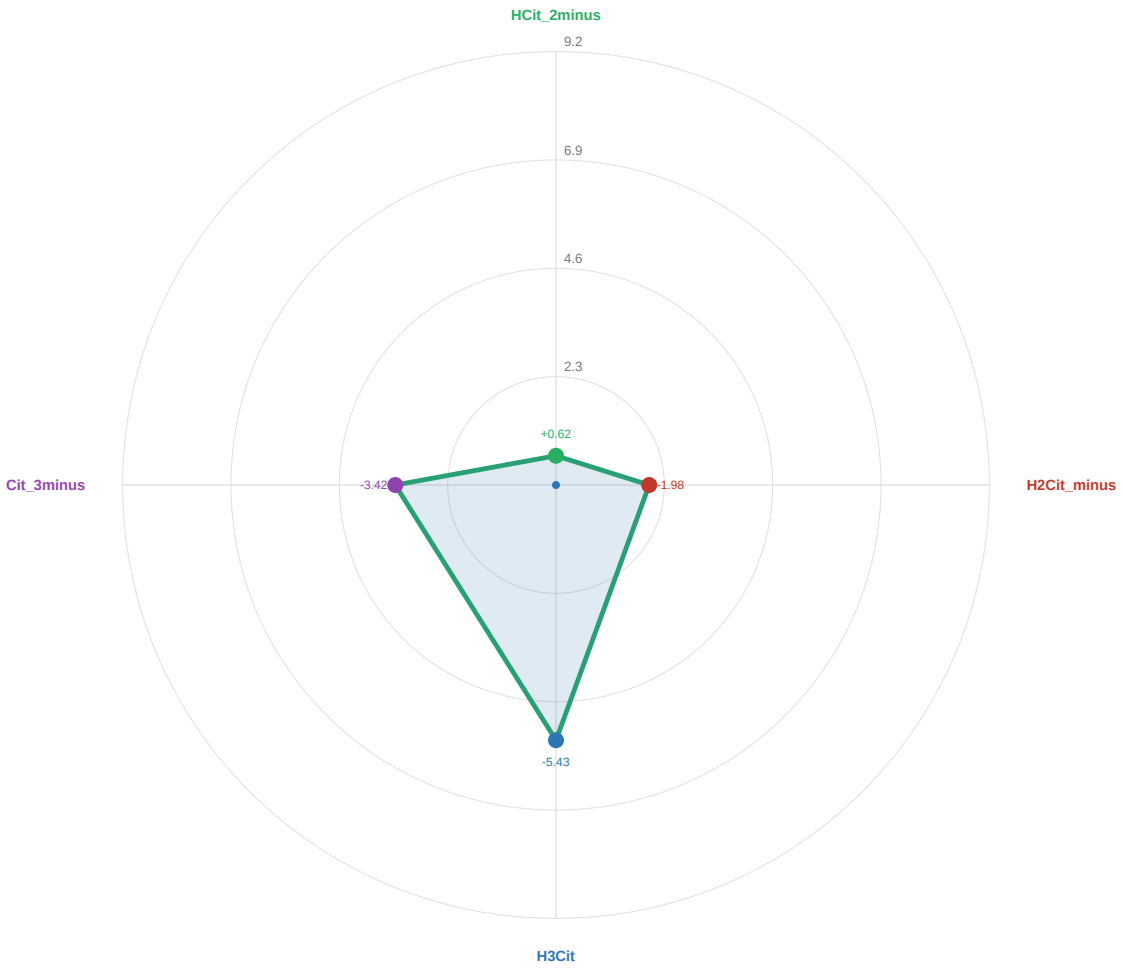
POLAR SLICE — t=22

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 23/200 | ||x||_A = 6.746



From 21: delta ||x||_A = -0.141

Phase = -159.9 deg

H3Cit=-5.43 H2Cit_=-1.98 HCit_2=+0.62 Cit_3m=-3.42

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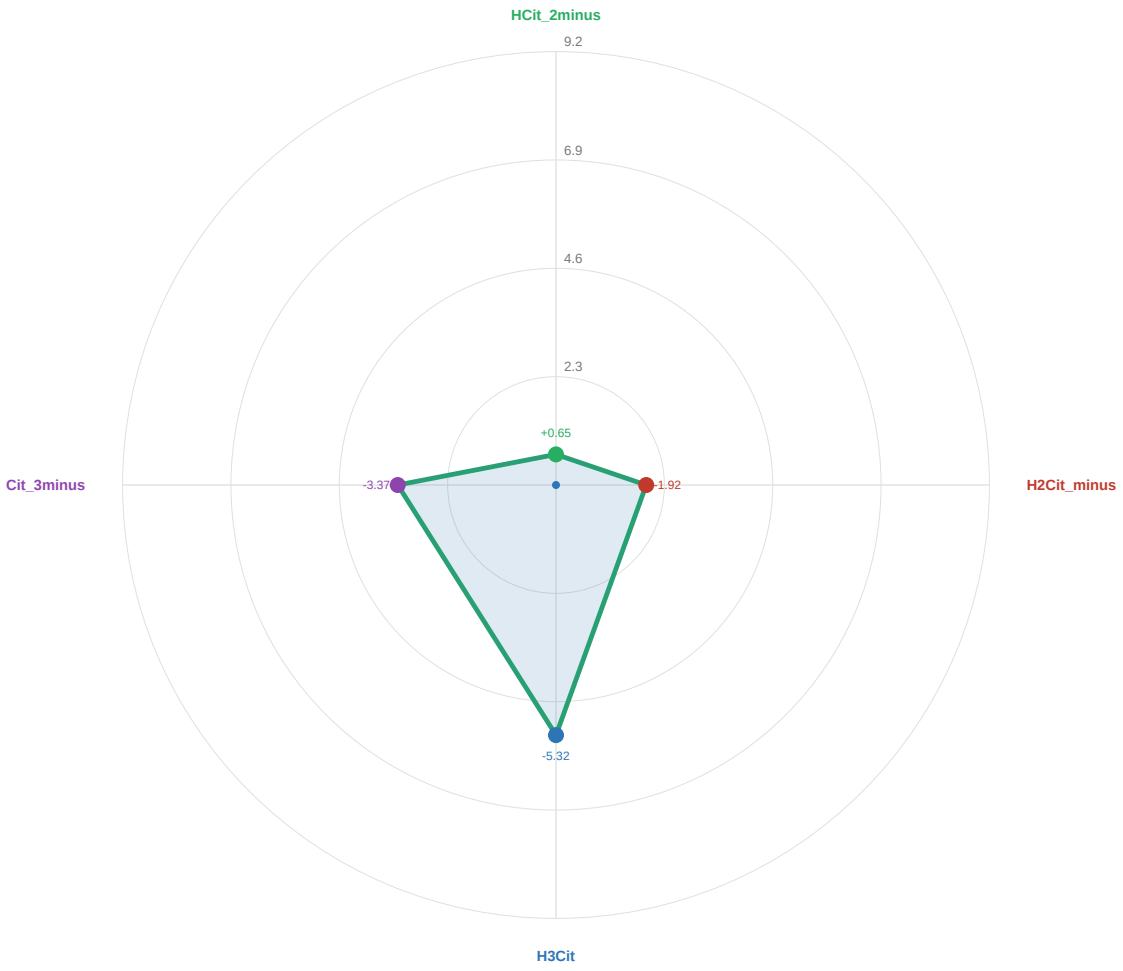
POLAR SLICE — t=23

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 24/200 | ||x||_A = 6.616



From 22: delta ||x||_A = -0.130

Phase = -160.2 deg

H3Cit=-5.32 H2Cit_=-1.92 HCit_2=+0.65 Cit_3m=-3.37

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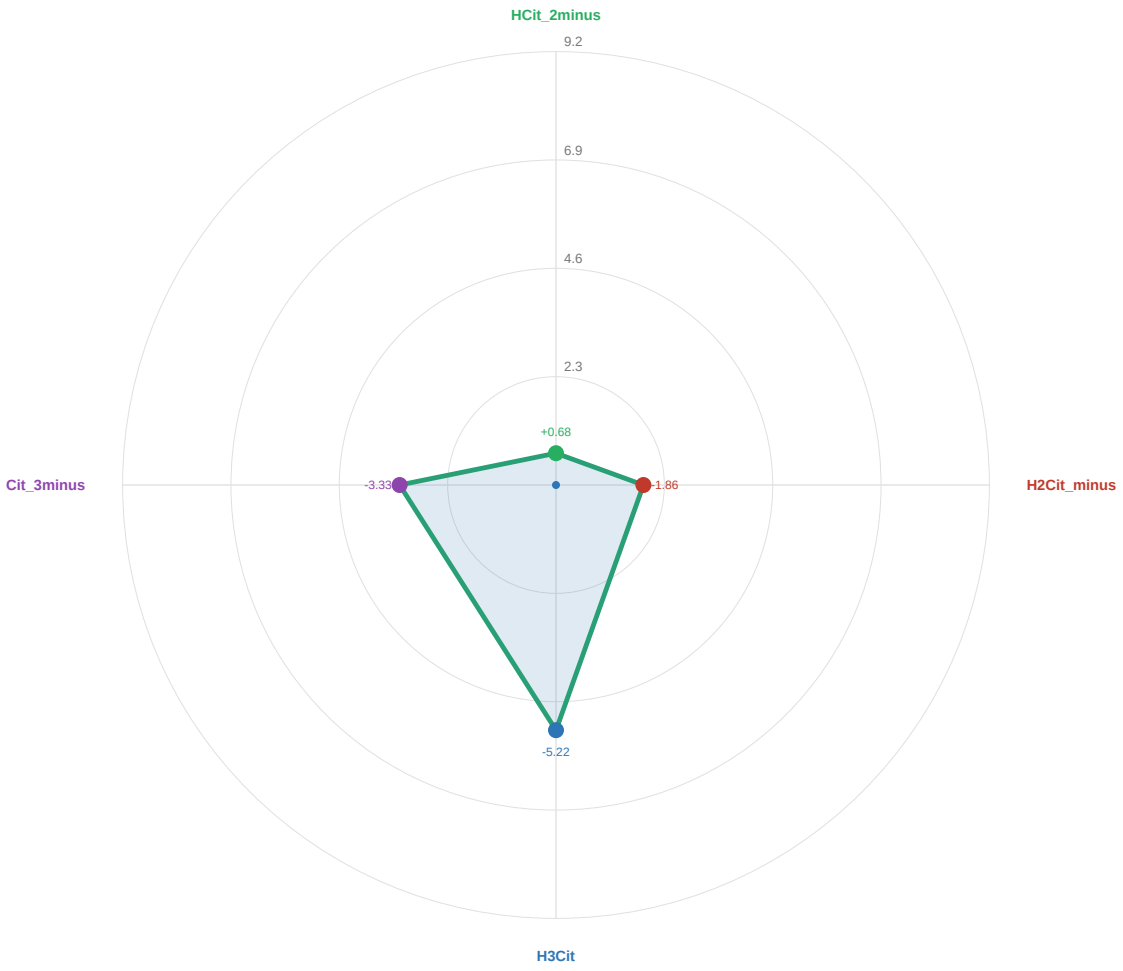
POLAR SLICE — t=24

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 25/200 | ||x||_A = 6.497



From 23: delta ||x||_A = -0.119

Phase = -160.4 deg

H3Cit=-5.22 H2Cit_=-1.86 HCit_2=+0.68 Cit_3m=-3.33

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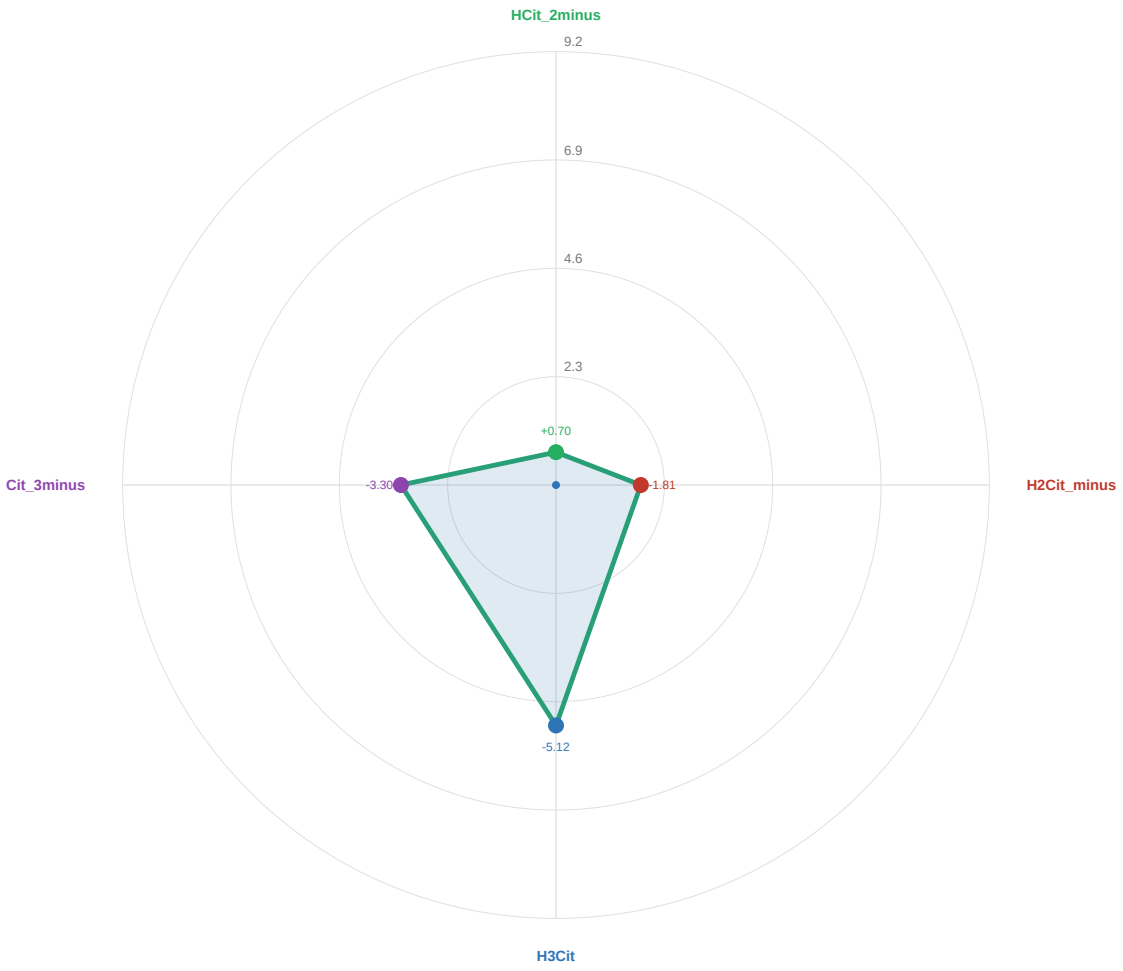
POLAR SLICE — t=25

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 26/200 | ||x||_A = 6.389



From 24: delta ||x||_A = -0.108

Phase = -160.6 deg

H3Cit=-5.12 H2Cit_=-1.81 HCit_2=+0.70 Cit_3m=-3.30

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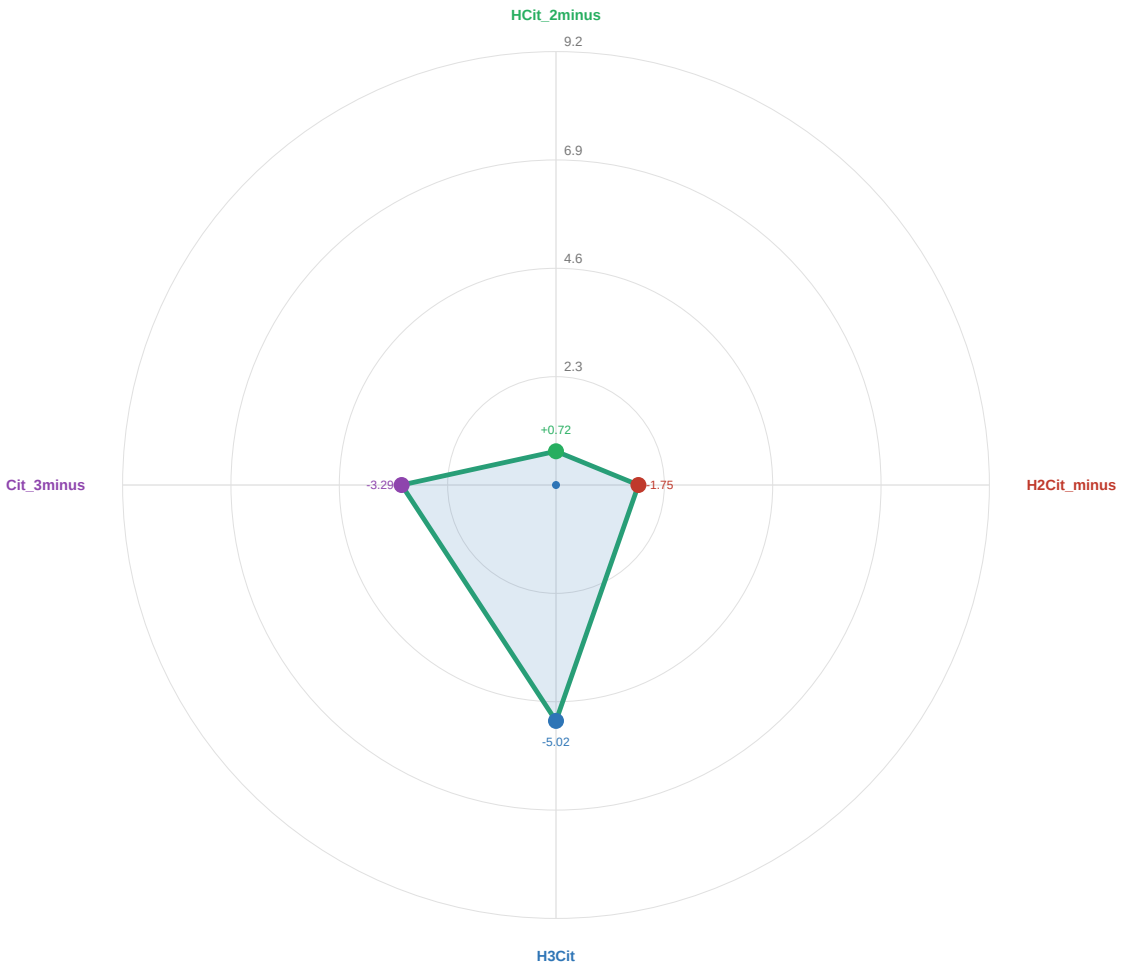
POLAR SLICE — t=26

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 27/200 | ||x||_A = 6.291



From 25: delta ||x||_A = -0.097

Phase = -160.7 deg

H3Cit=-5.02 H2Cit_=-1.75 HCit_2=+0.72 Cit_3m=-3.29

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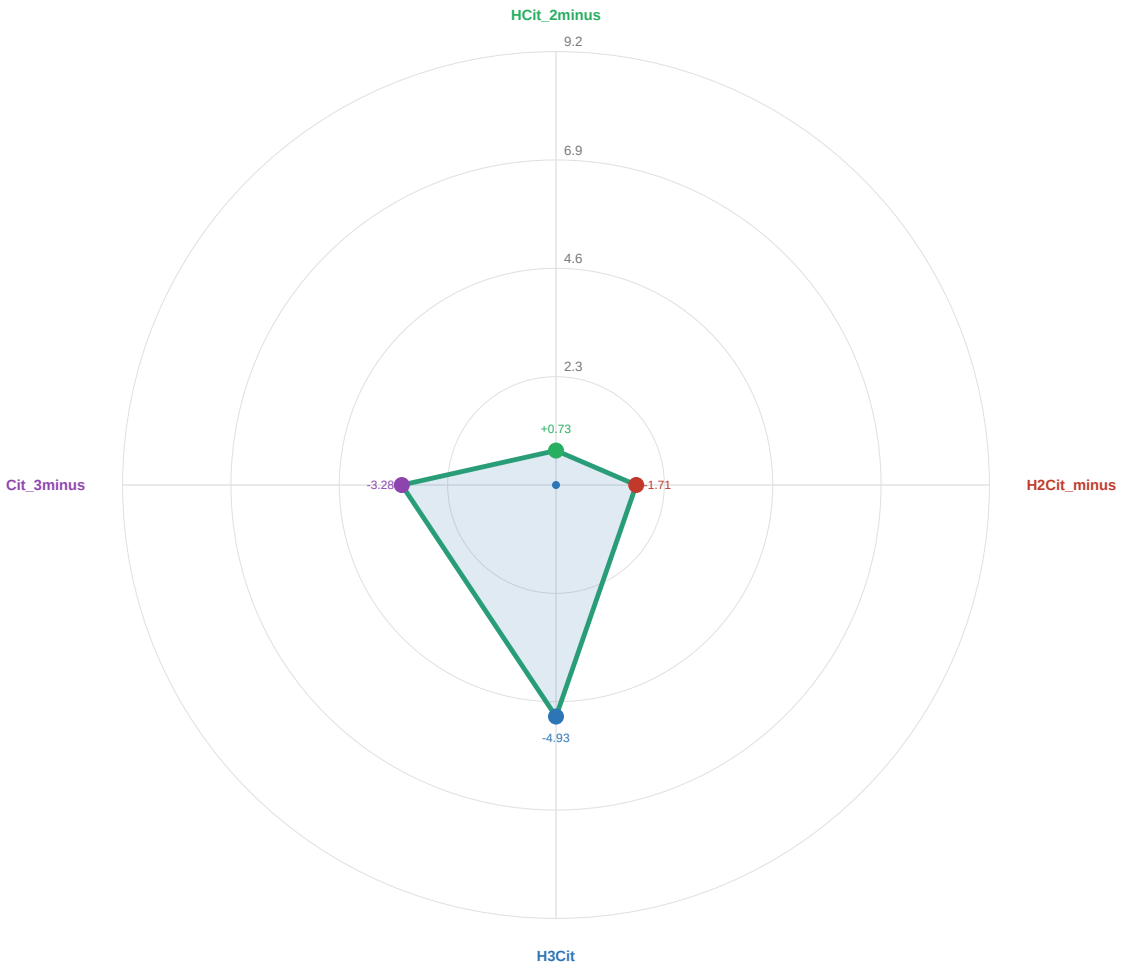
POLAR SLICE — t=27

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 28/200 | ||x||_A = 6.204



From 26: delta ||x||_A = -0.087

Phase = -160.9 deg

H3Cit=-4.93 H2Cit_=-1.71 HCit_2=+0.73 Cit_3m=-3.28

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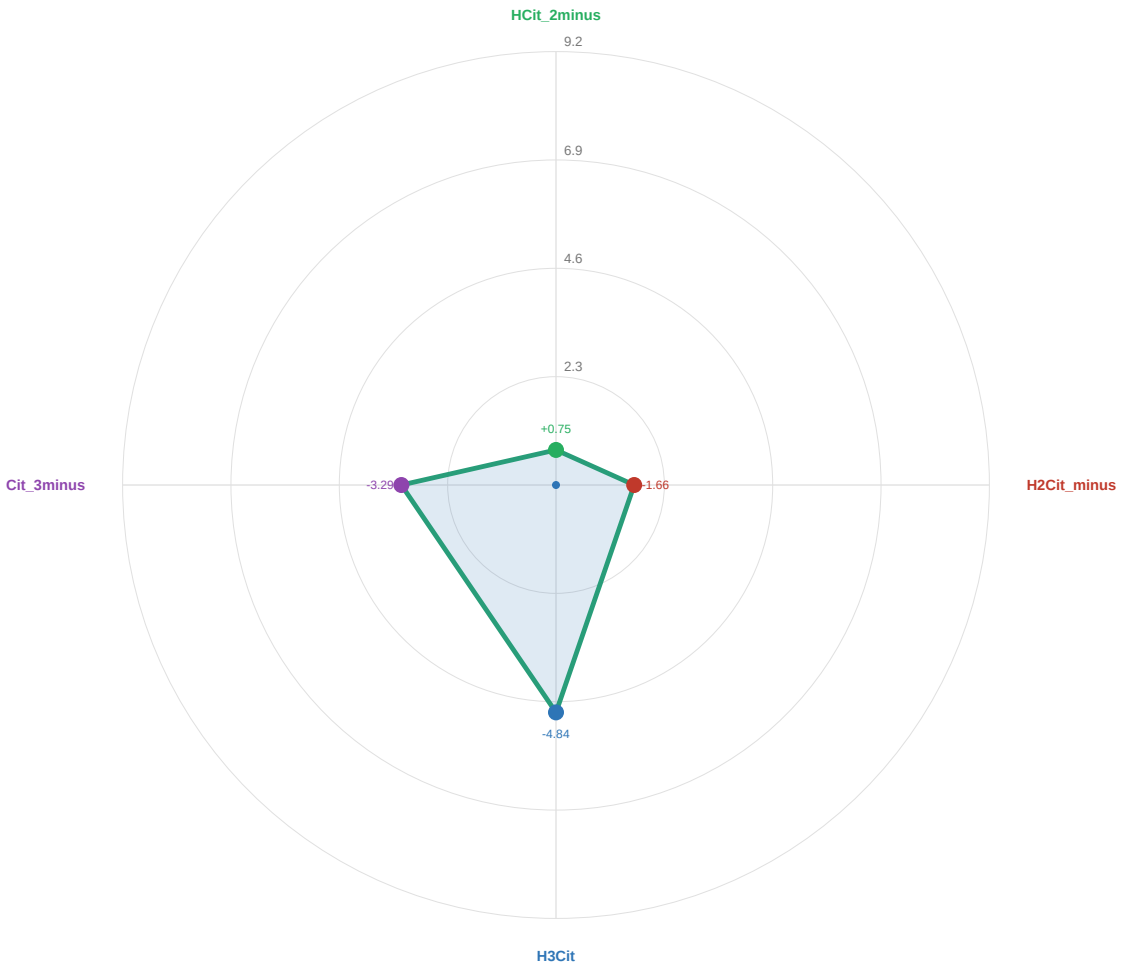
POLAR SLICE — t=28

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 29/200 | ||x||_A = 6.128



From 27: delta ||x||_A = -0.076

Phase = -161.0 deg

H3Cit=-4.84 H2Cit_=-1.66 HCit_2=+0.75 Cit_3m=-3.29

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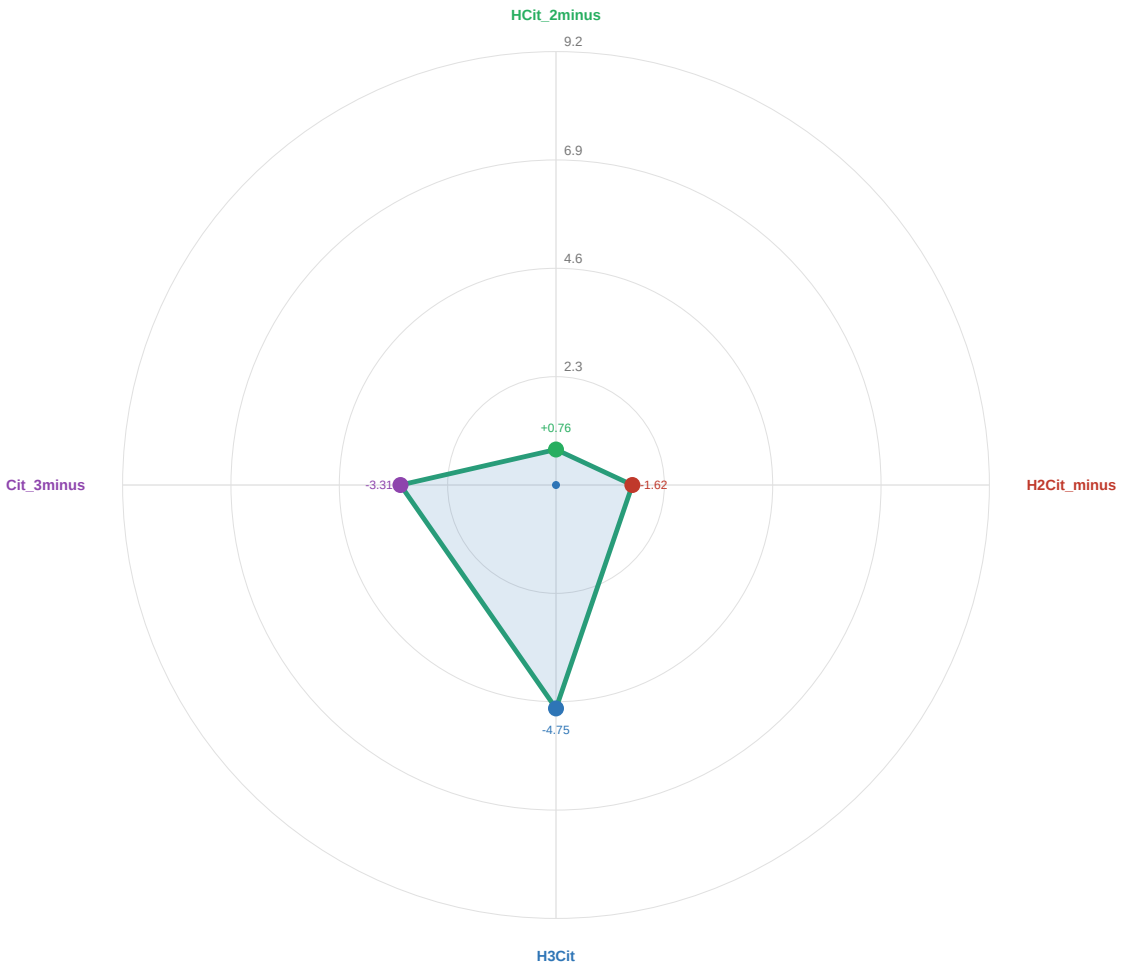
POLAR SLICE — t=29

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 30/200 | ||x||_A = 6.063



From 28: delta ||x||_A = -0.065

Phase = -161.1 deg

H3Cit=-4.75 H2Cit_=-1.62 HCit_2=+0.76 Cit_3m=-3.31

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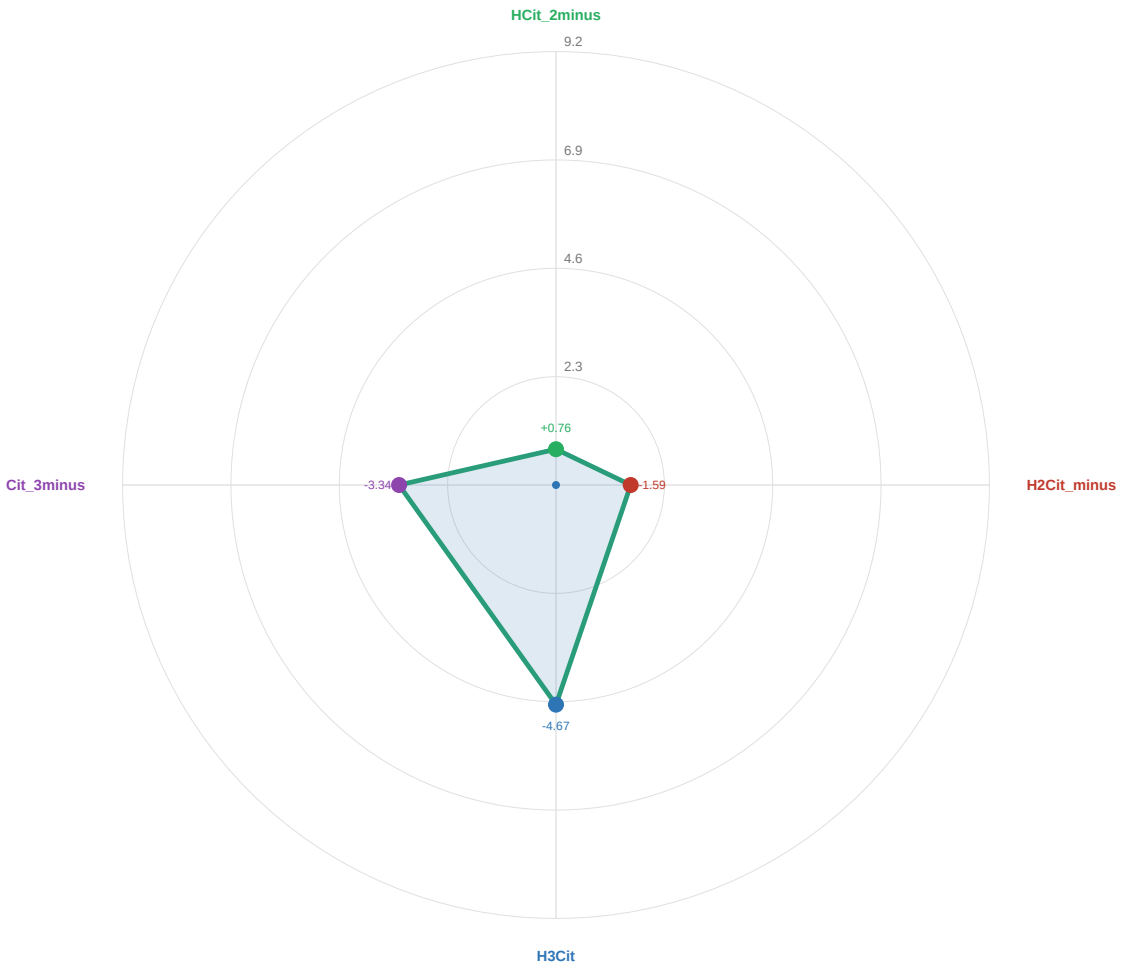
POLAR SLICE — t=30

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 31/200 | ||x||_A = 6.008



From 29: delta ||x||_A = -0.055

Phase = -161.2 deg

H3Cit=-4.67 H2Cit_=-1.59 HCit_2=+0.76 Cit_3m=-3.34

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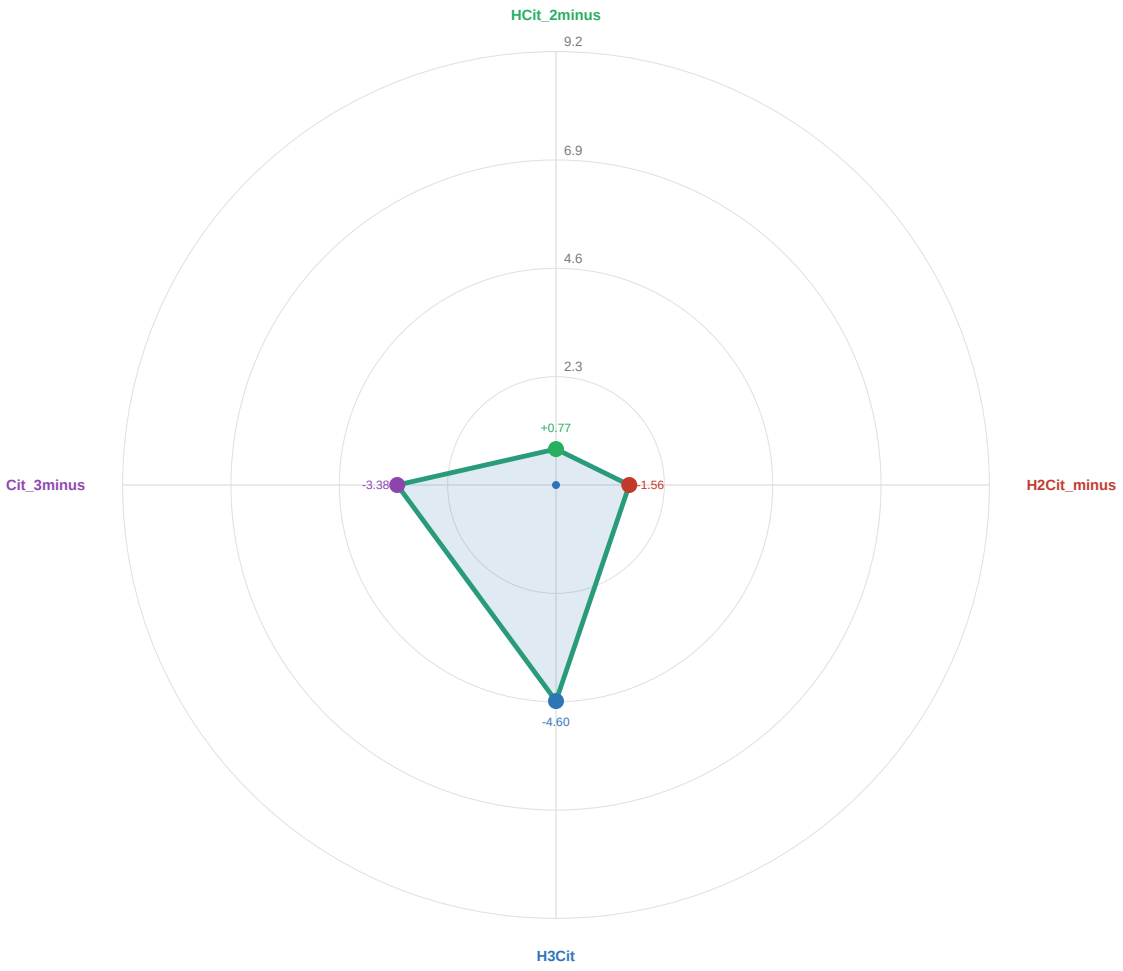
POLAR SLICE — t=31

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 32/200 | ||x||_A = 5.964



From 30: delta ||x||_A = -0.044

Phase = -161.3 deg

H3Cit=-4.60 H2Cit_=-1.56 HCit_2=+0.77 Cit_3m=-3.38

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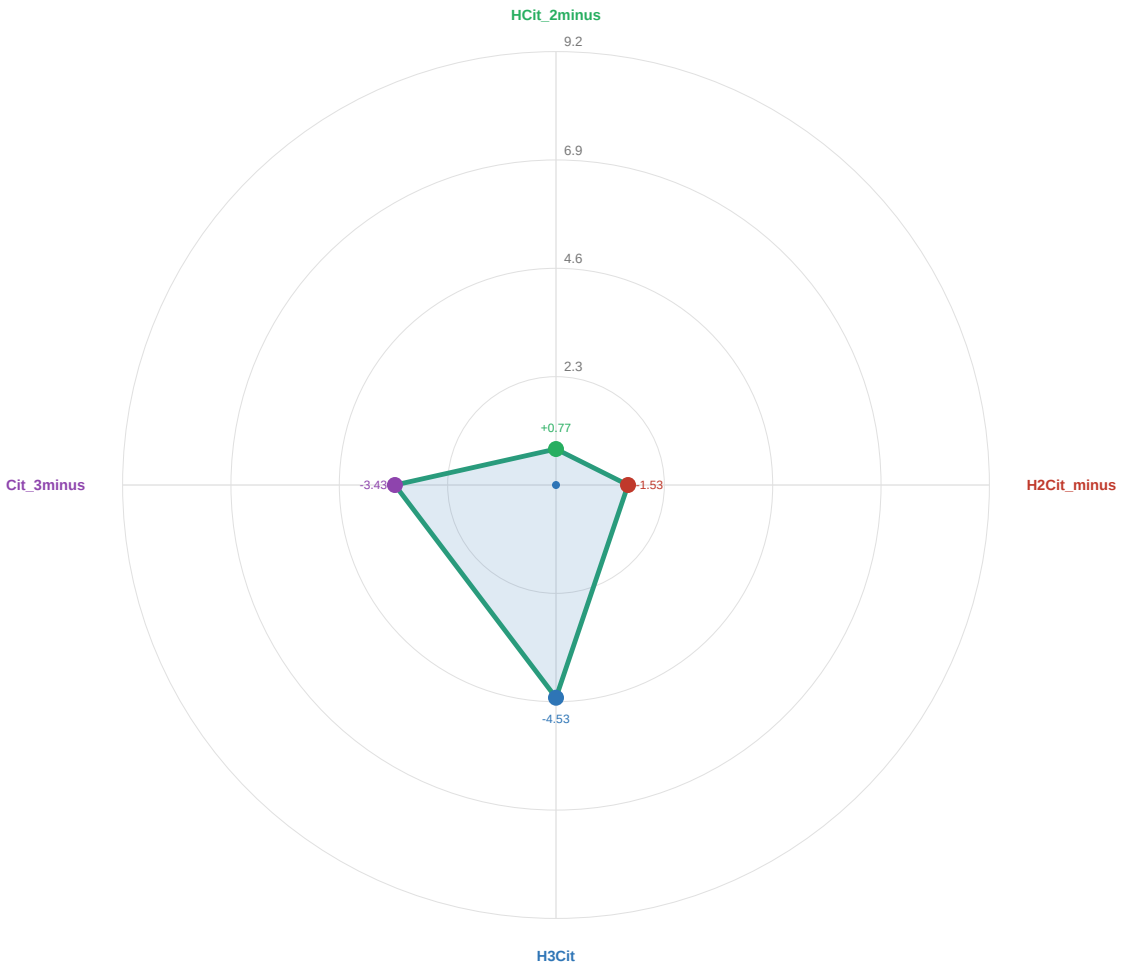
POLAR SLICE — t=32

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 33/200 | ||x||_A = 5.930



From 31: delta ||x||_A = -0.034

Phase = -161.3 deg

H3Cit=-4.53 H2Cit_=-1.53 HCit_2=+0.77 Cit_3m=-3.43

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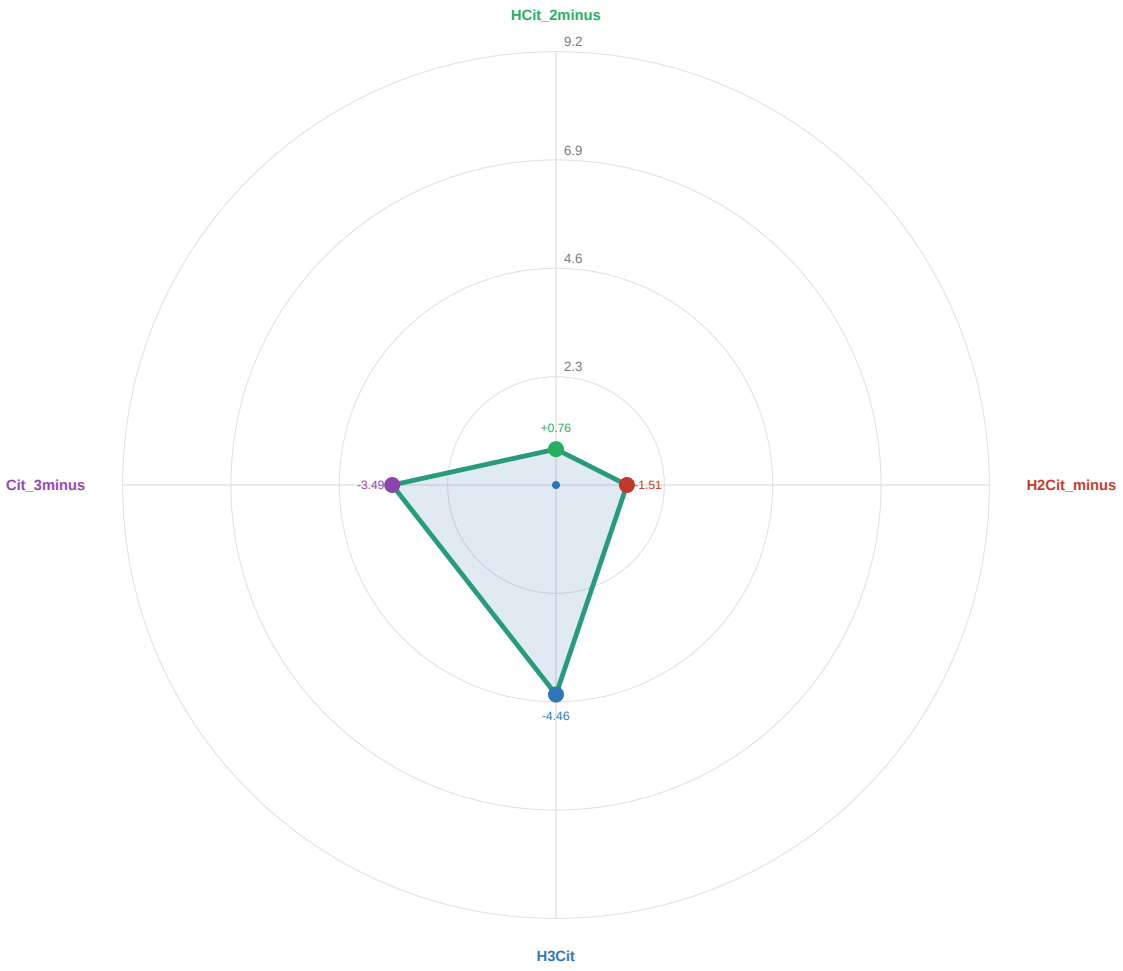
POLAR SLICE — t=33

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 34/200 | ||x||_A = 5.907



From 32: delta ||x||_A = -0.024

Phase = -161.3 deg

H3Cit=-4.46 H2Cit_=-1.51 HCit_2=+0.76 Cit_3m=-3.49

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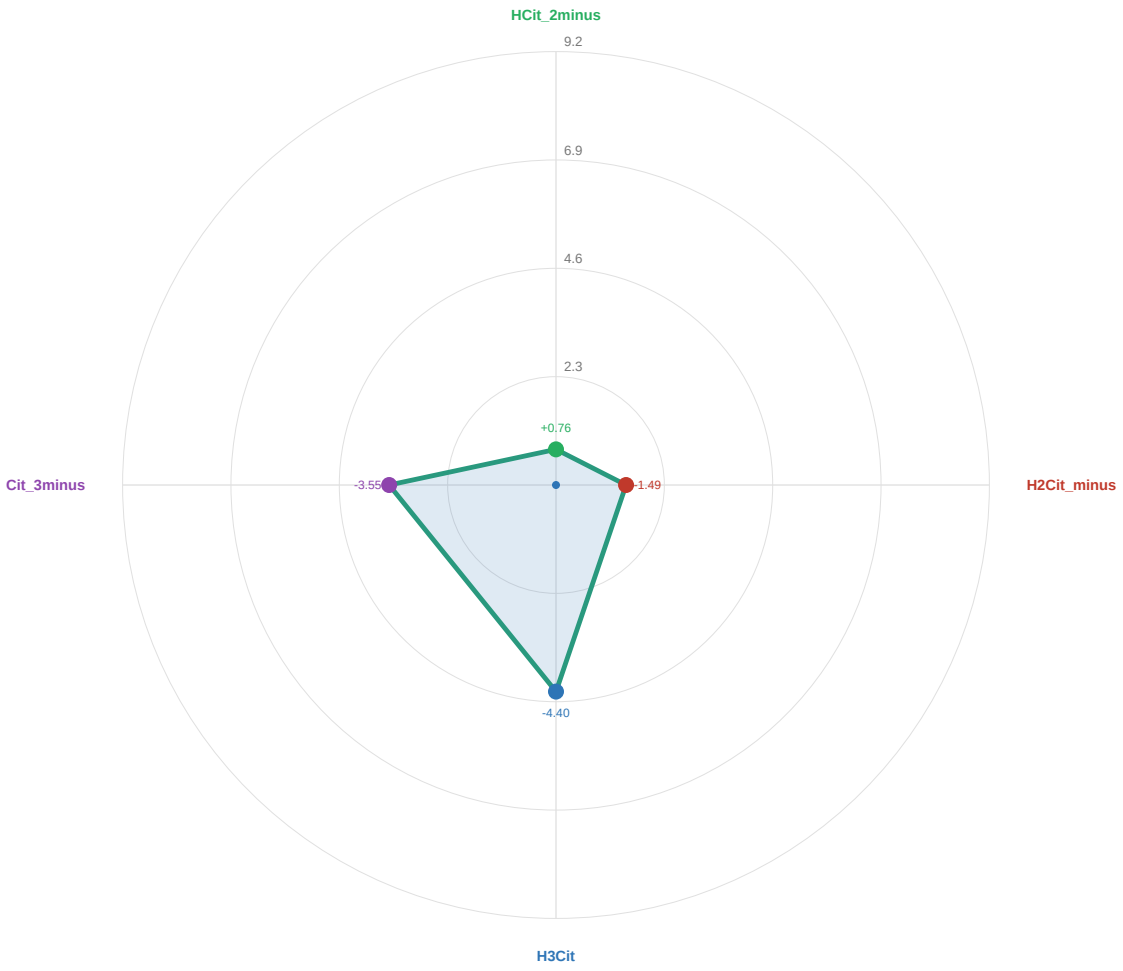
POLAR SLICE — t=34

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 35/200 | ||x||_A = 5.893



From 33: delta ||x||_A = -0.014

Phase = -161.3 deg

H3Cit=-4.40 H2Cit_=-1.49 HCit_2=+0.76 Cit_3m=-3.55

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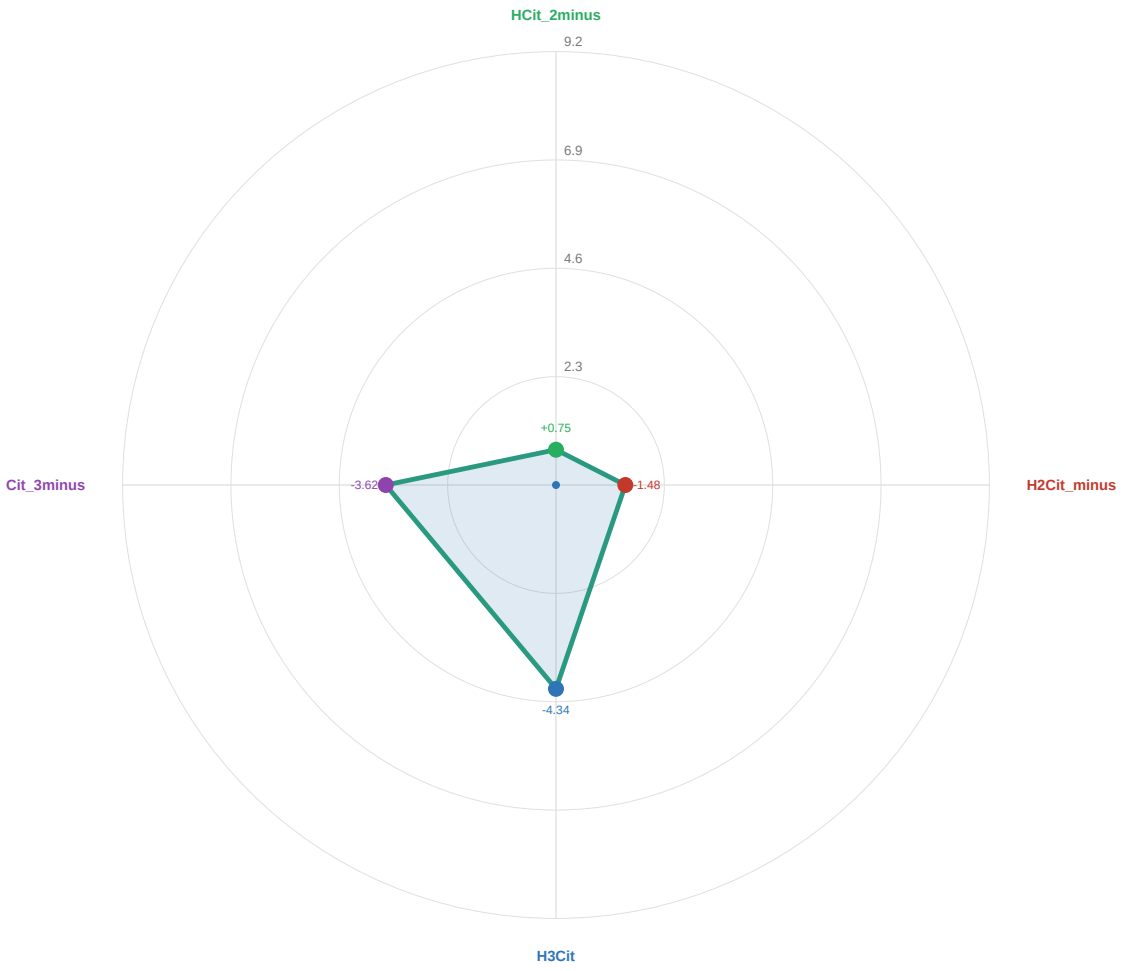
POLAR SLICE — t=35

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 36/200 | ||x||_A = 5.888



From 34: delta ||x||_A = -0.004

Phase = -161.2 deg

H3Cit=-4.34 H2Cit_=-1.48 HCit_2=+0.75 Cit_3m=-3.62

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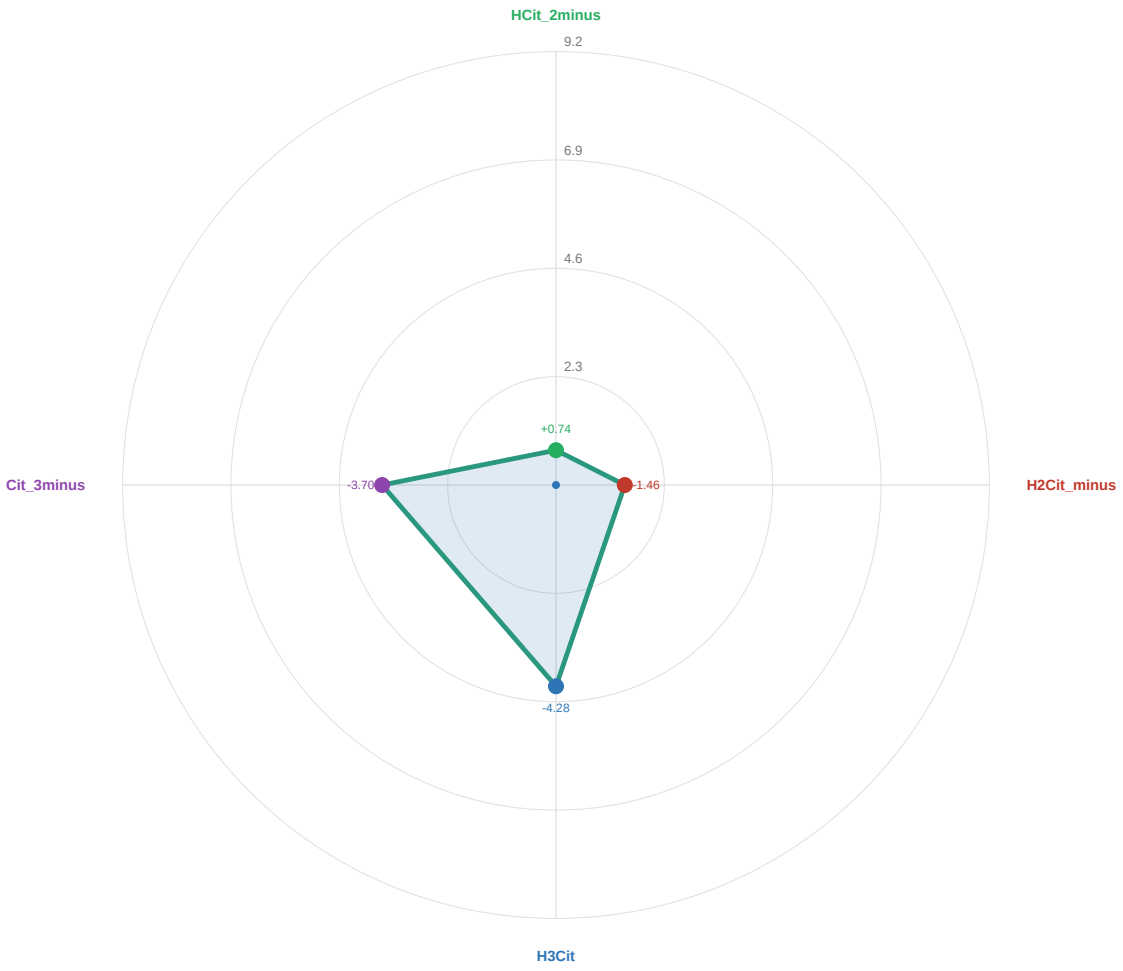
POLAR SLICE — t=36

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 37/200 | ||x||_A = 5.893



From 35: delta ||x||_A = +0.005

Phase = -161.1 deg

H3Cit=-4.28 H2Cit_=-1.46 HCit_2=+0.74 Cit_3m=-3.70

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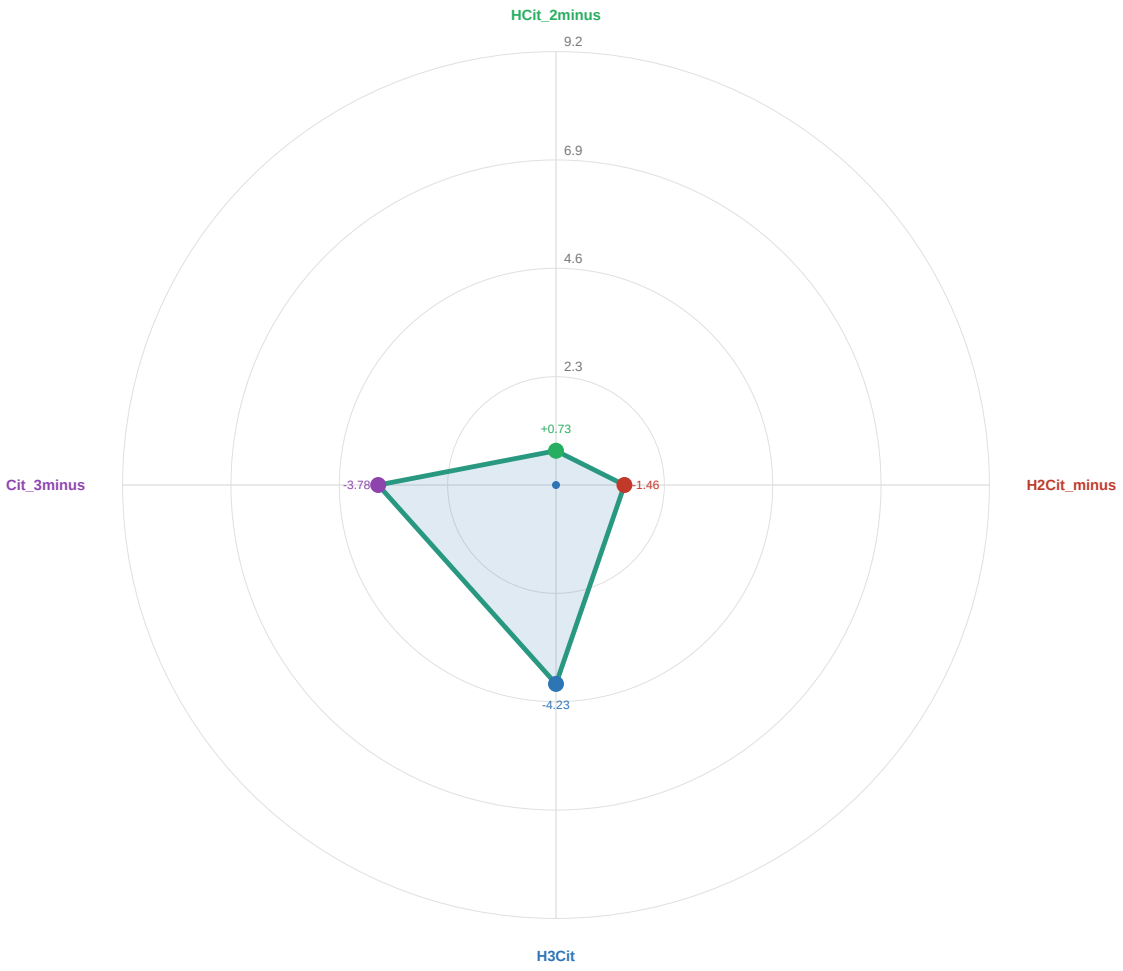
POLAR SLICE — t=37

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 38/200 | ||x||_A = 5.906



From 36: delta ||x||_A = +0.013

Phase = -161.0 deg

H3Cit=-4.23 H2Cit_=-1.46 HCit_2=+0.73 Cit_3m=-3.78

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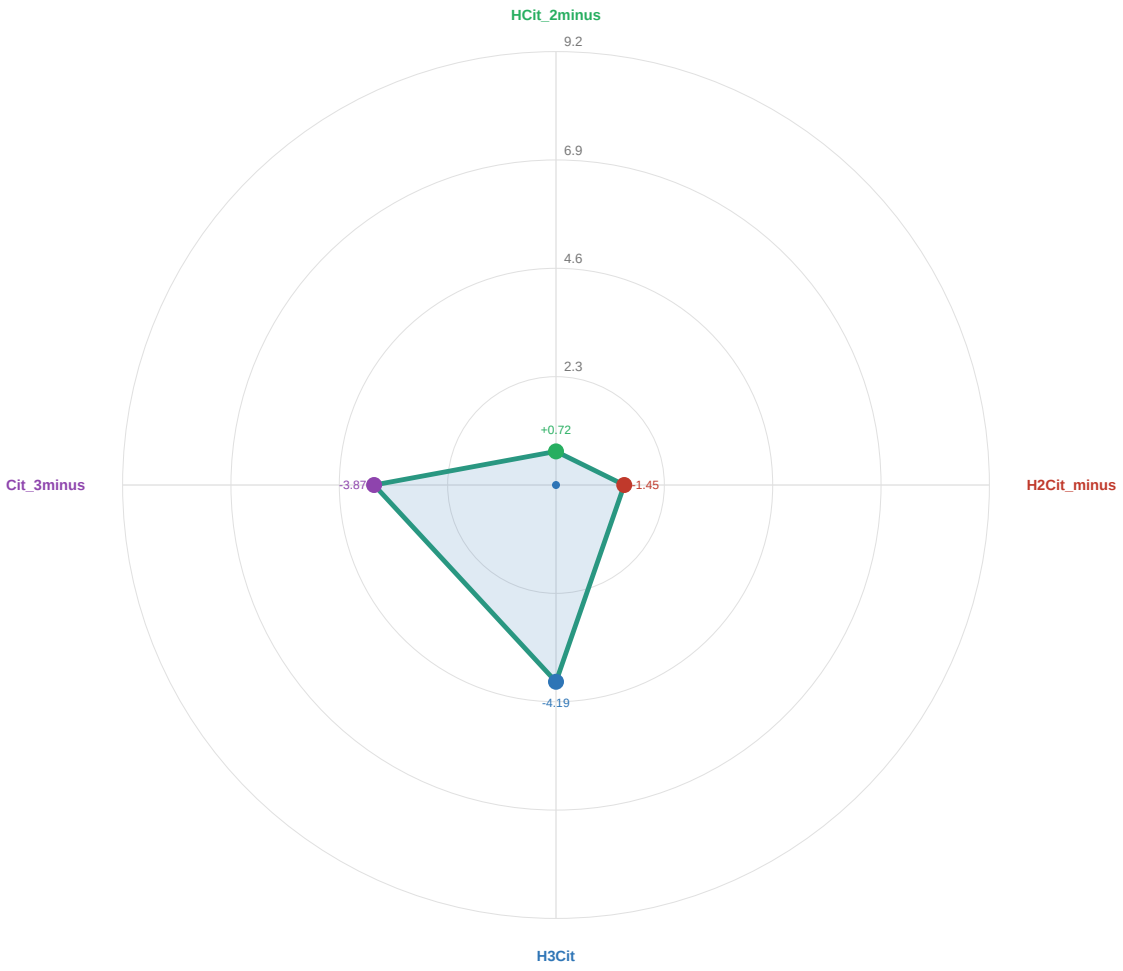
POLAR SLICE — t=38

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 39/200 | ||x||_A = 5.927



From 37: delta ||x||_A = +0.021

Phase = -160.9 deg

H3Cit=-4.19 H2Cit_=-1.45 HCit_2=+0.72 Cit_3m=-3.87

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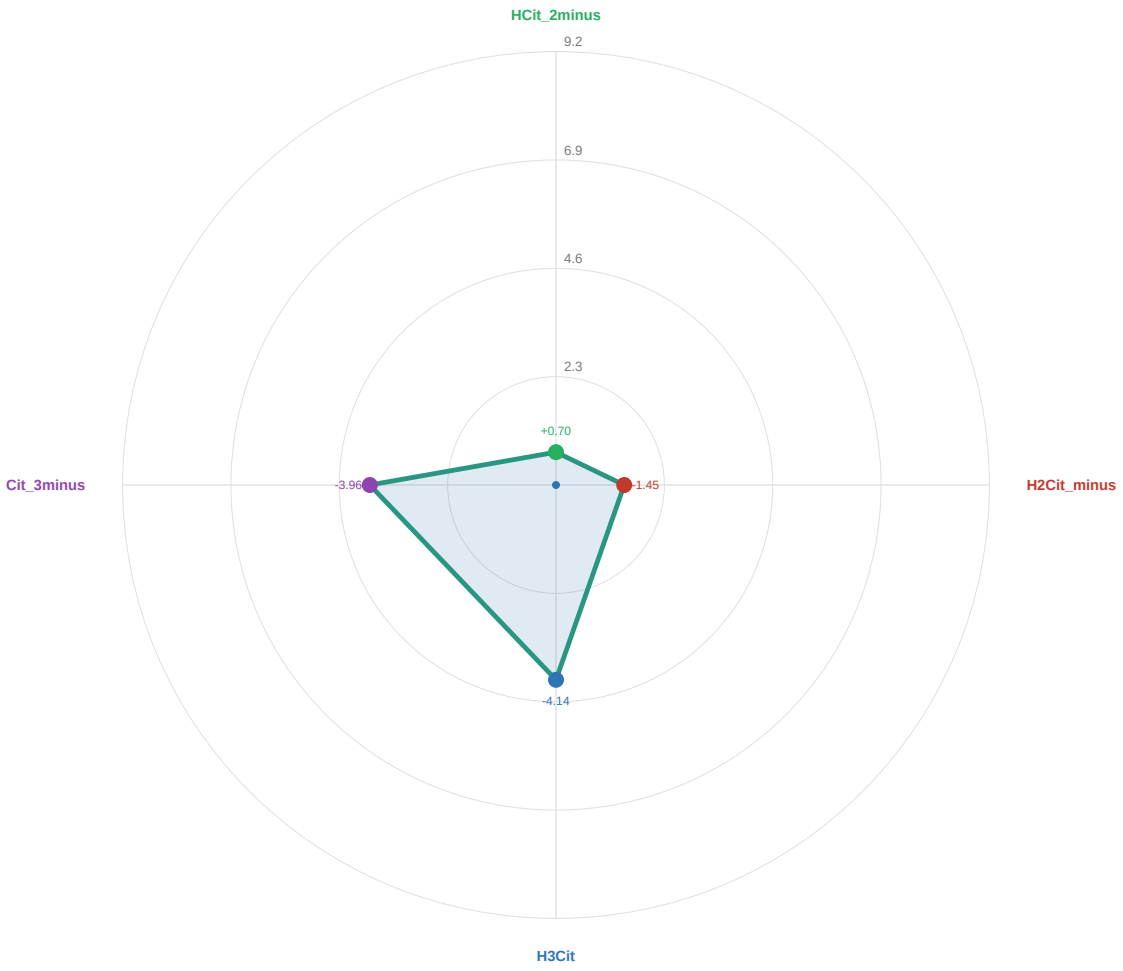
POLAR SLICE — t=39

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 40/200 | ||x||_A = 5.956



From 38: delta ||x||_A = +0.029

Phase = -160.7 deg

H3Cit=-4.14 H2Cit_=-1.45 HCit_2=+0.70 Cit_3m=-3.96

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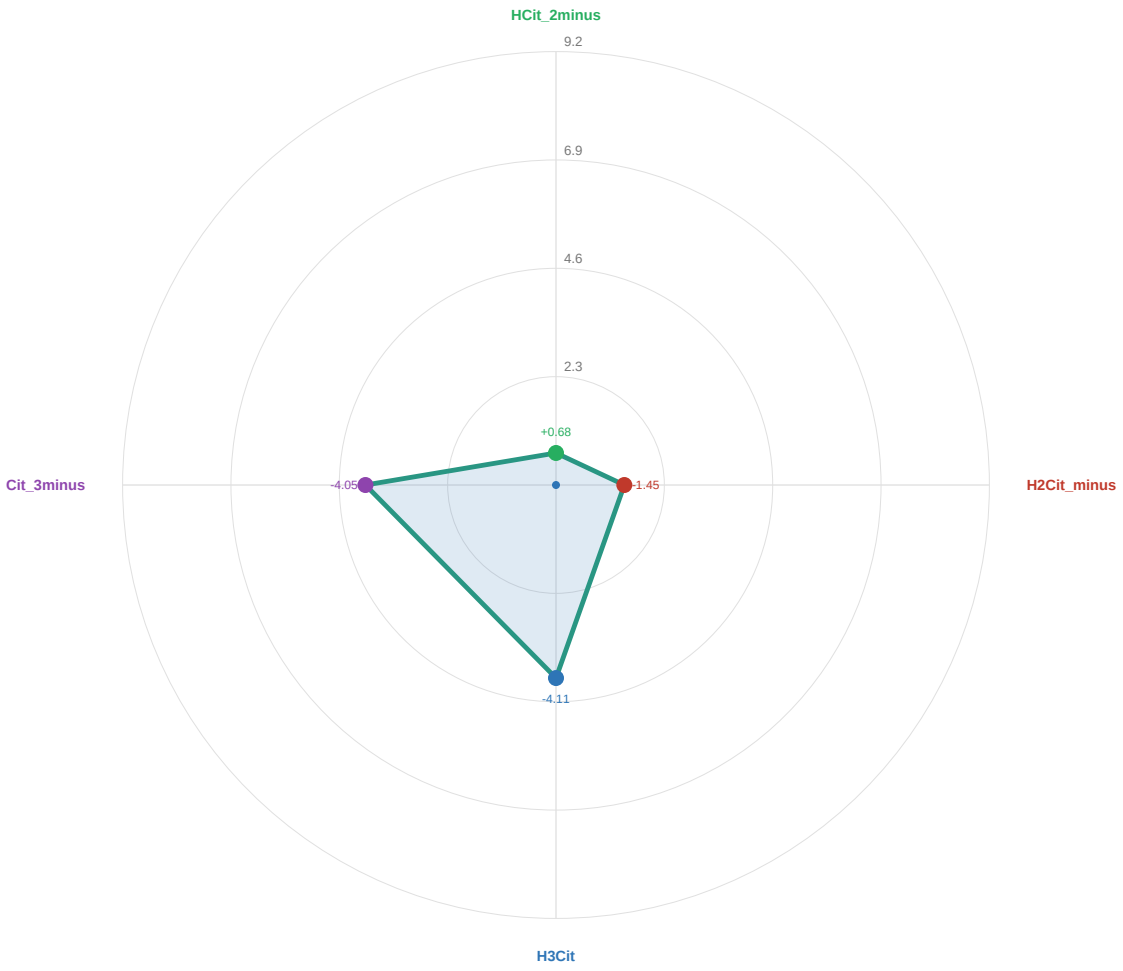
POLAR SLICE — t=40

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 41/200 | ||x||_A = 5.991



From 39: delta ||x||_A = +0.035

Phase = -160.5 deg

H3Cit=-4.11 H2Cit_=-1.45 HCit_2=+0.68 Cit_3m=-4.05

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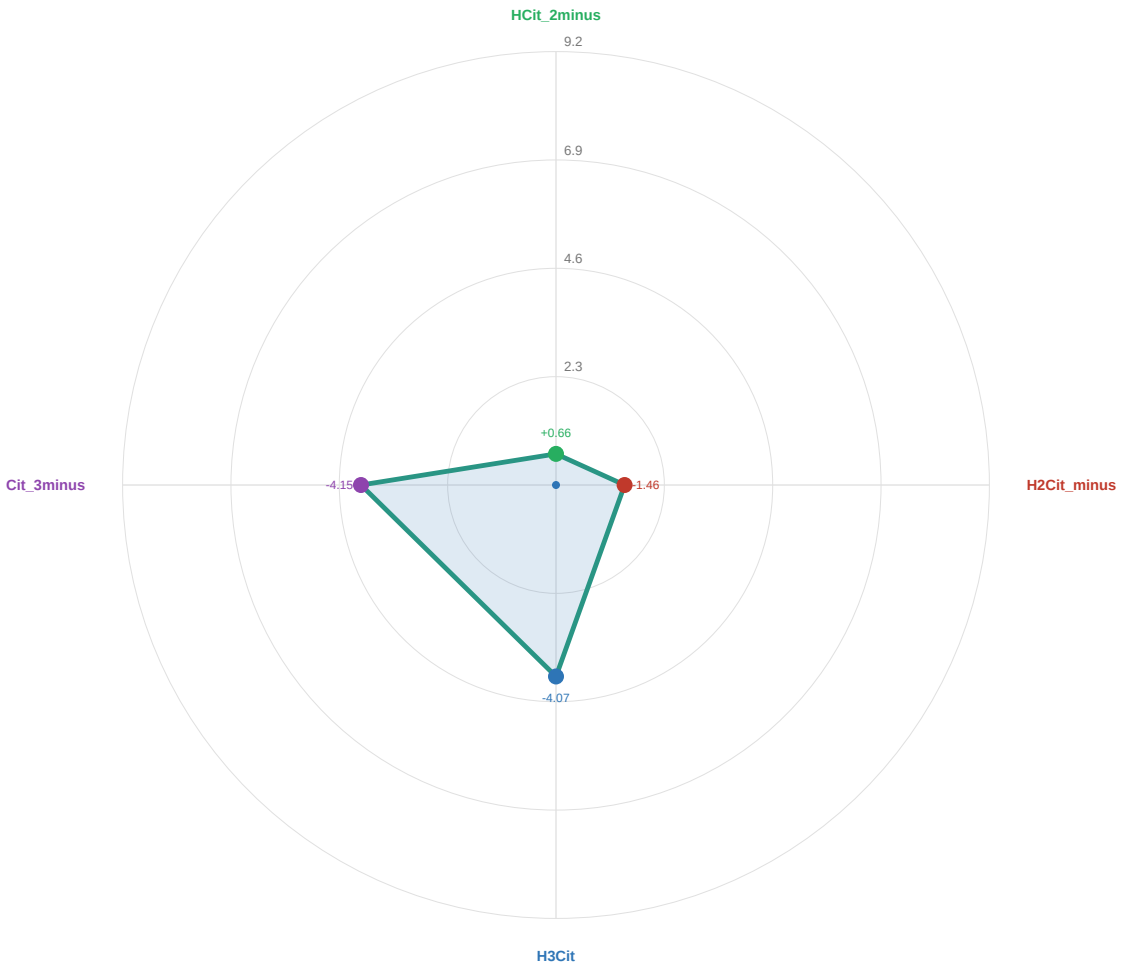
POLAR SLICE — t=41

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 42/200 | ||x||_A = 6.032



From 40: delta ||x||_A = +0.041

Phase = -160.3 deg

H3Cit=-4.07 H2Cit_=-1.46 HCit_2=+0.66 Cit_3m=-4.15

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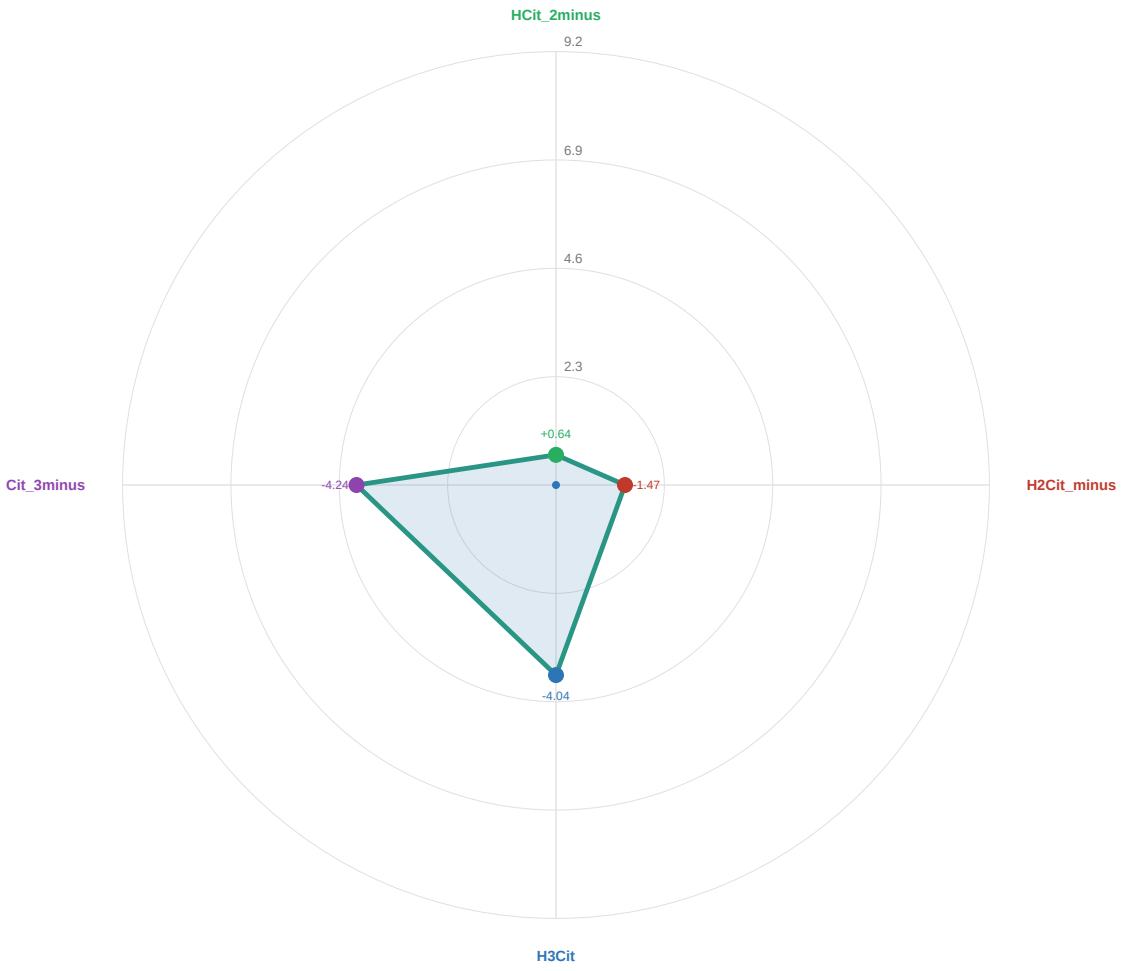
POLAR SLICE — t=42

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 43/200 | ||x||_A = 6.078



From 41: delta ||x||_A = +0.046

Phase = -160.0 deg

H3Cit=-4.04 H2Cit_=-1.47 HCit_2=+0.64 Cit_3m=-4.24

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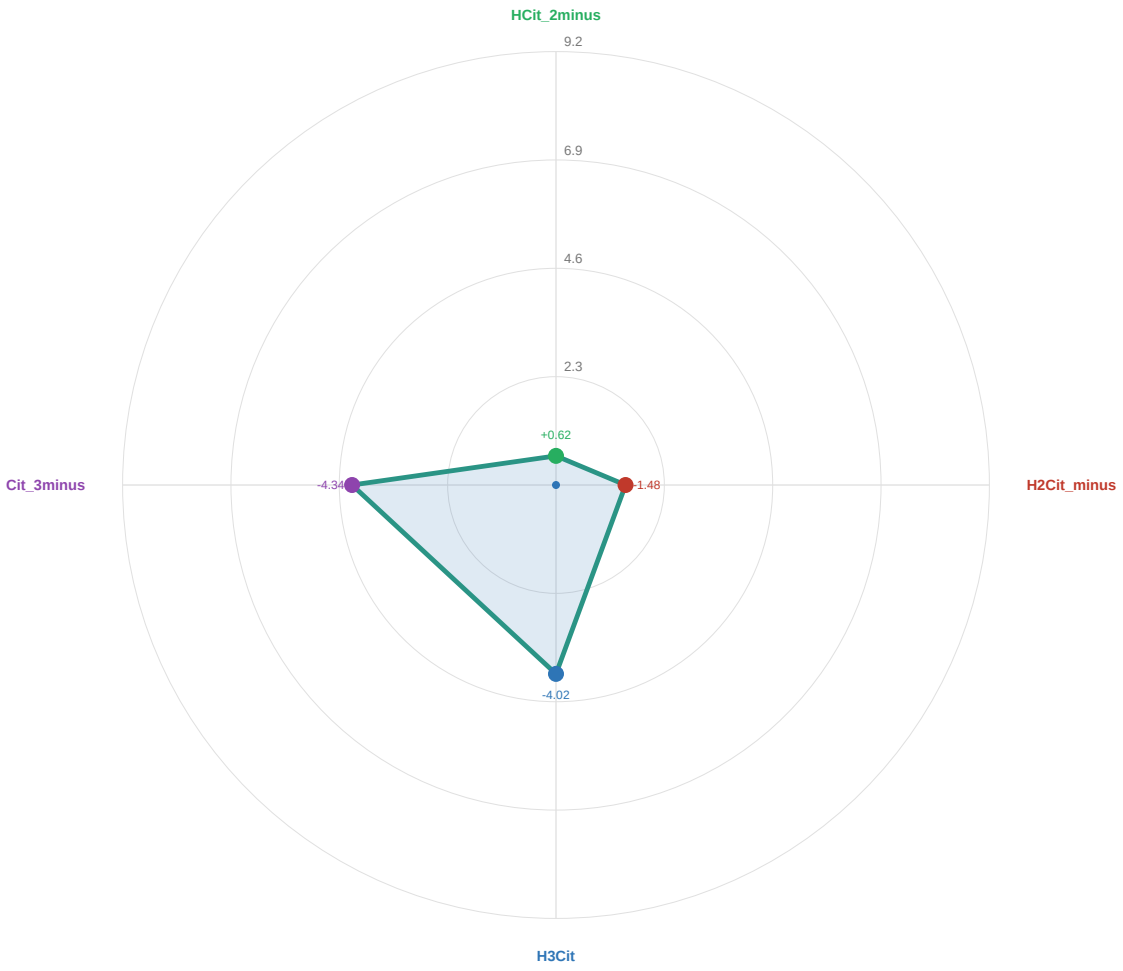
POLAR SLICE — t=43

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 44/200 | ||x||_A = 6.128



From 42: delta ||x||_A = +0.050

Phase = -159.8 deg

H3Cit=-4.02 H2Cit_=-1.48 HCit_2=+0.62 Cit_3m=-4.34

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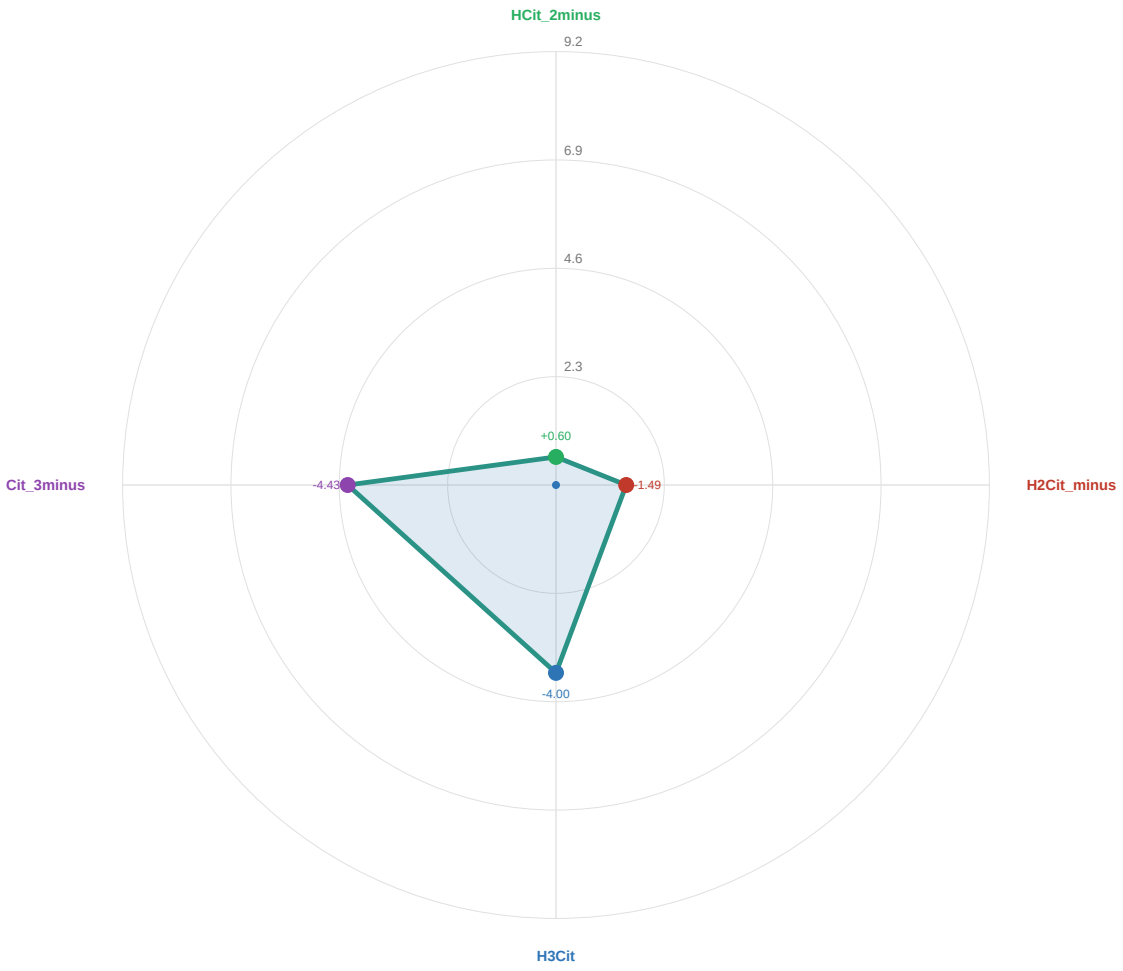
POLAR SLICE — t=44

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 45/200 | ||x||_A = 6.182



From 43: delta ||x||_A = +0.053

Phase = -159.5 deg

H3Cit=-4.00 H2Cit_=-1.49 HCit_2=+0.60 Cit_3m=-4.43

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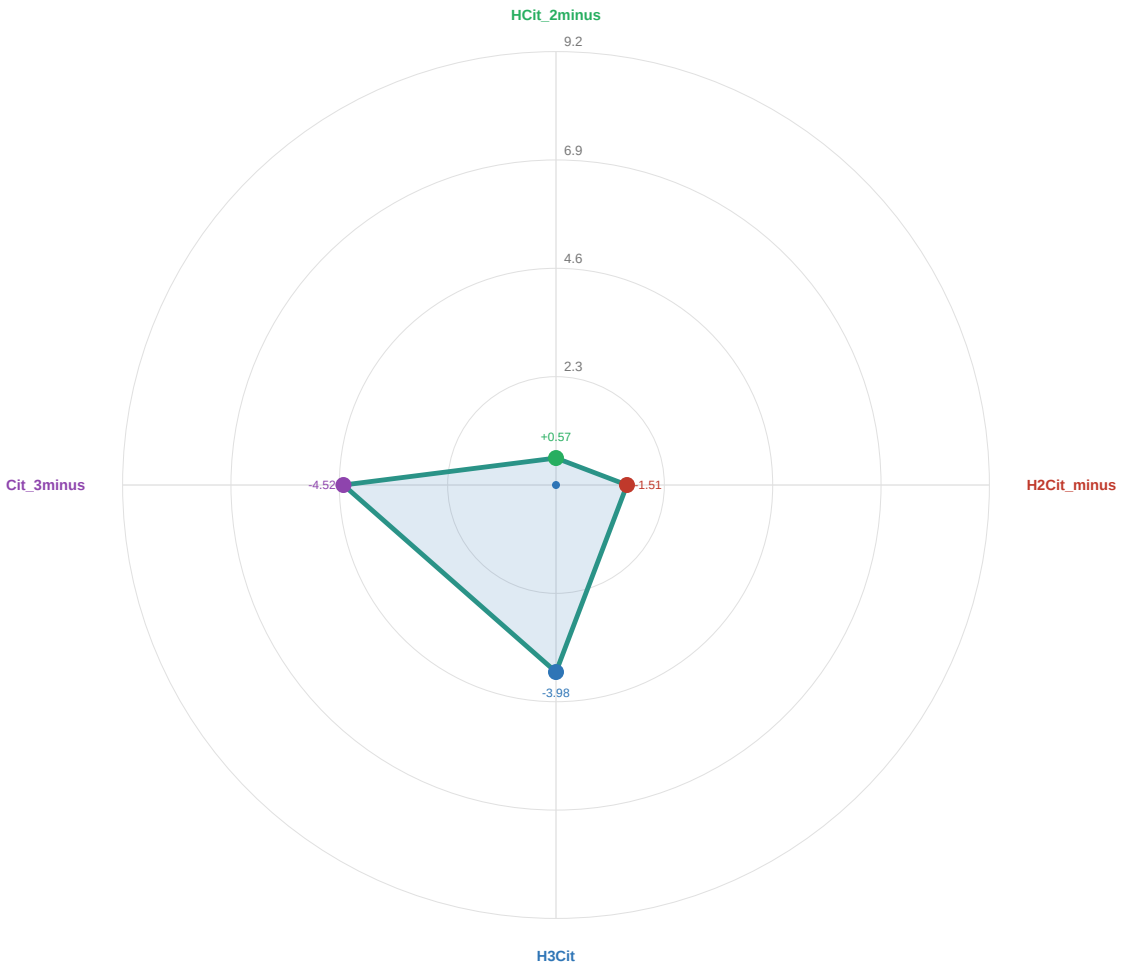
POLAR SLICE — t=45

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 46/200 | ||x||_A = 6.237



From 44: delta ||x||_A = +0.056

Phase = -159.2 deg

H3Cit=-3.98 H2Cit_=-1.51 HCit_2=+0.57 Cit_3m=-4.52

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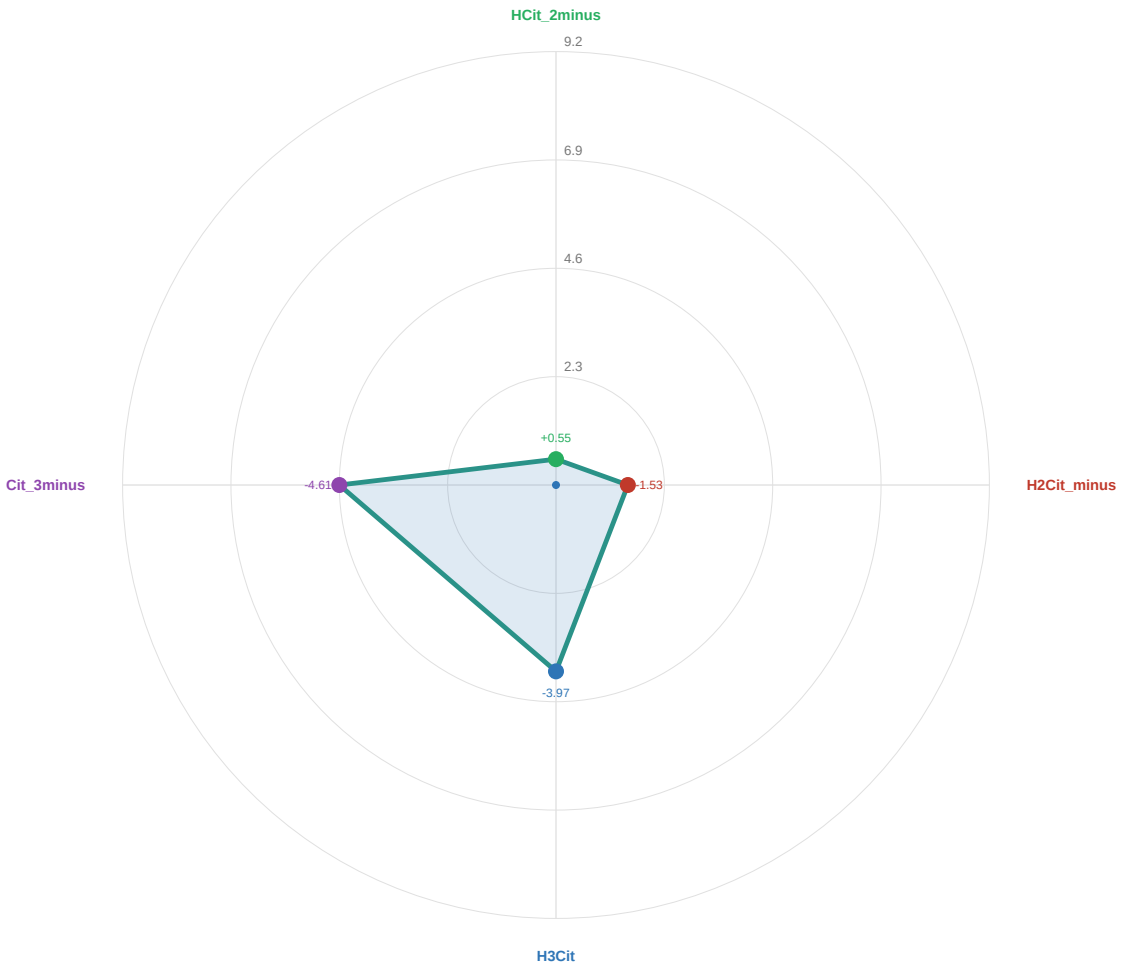
POLAR SLICE — t=46

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 47/200 | ||x||_A = 6.294



From 45: delta ||x||_A = +0.057

Phase = -158.9 deg

H3Cit=-3.97 H2Cit_=-1.53 HCit_2=+0.55 Cit_3m=-4.61

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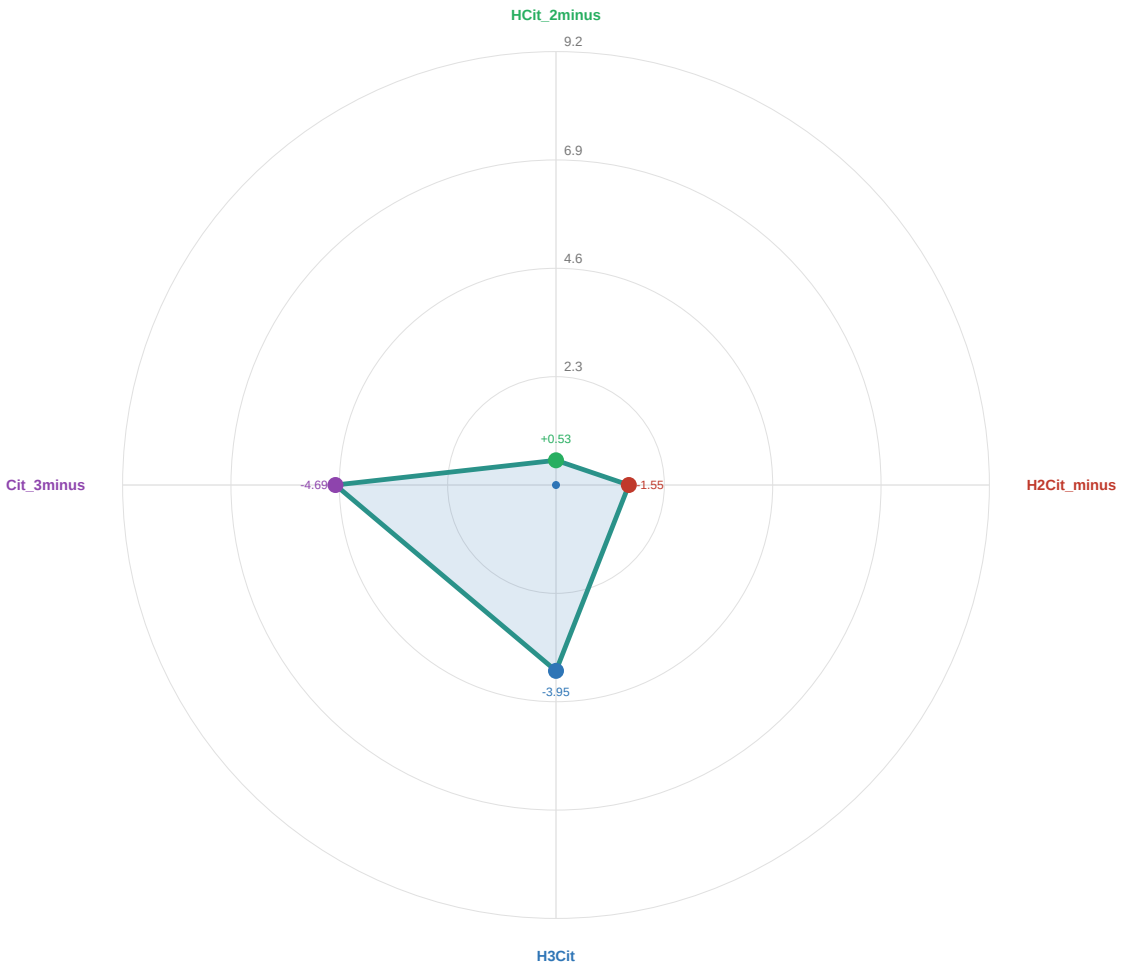
POLAR SLICE — t=47

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 48/200 | ||x||_A = 6.351



From 46: delta ||x||_A = +0.057

Phase = -158.6 deg

H3Cit=-3.95 H2Cit_=-1.55 HCit_2=+0.53 Cit_3m=-4.69

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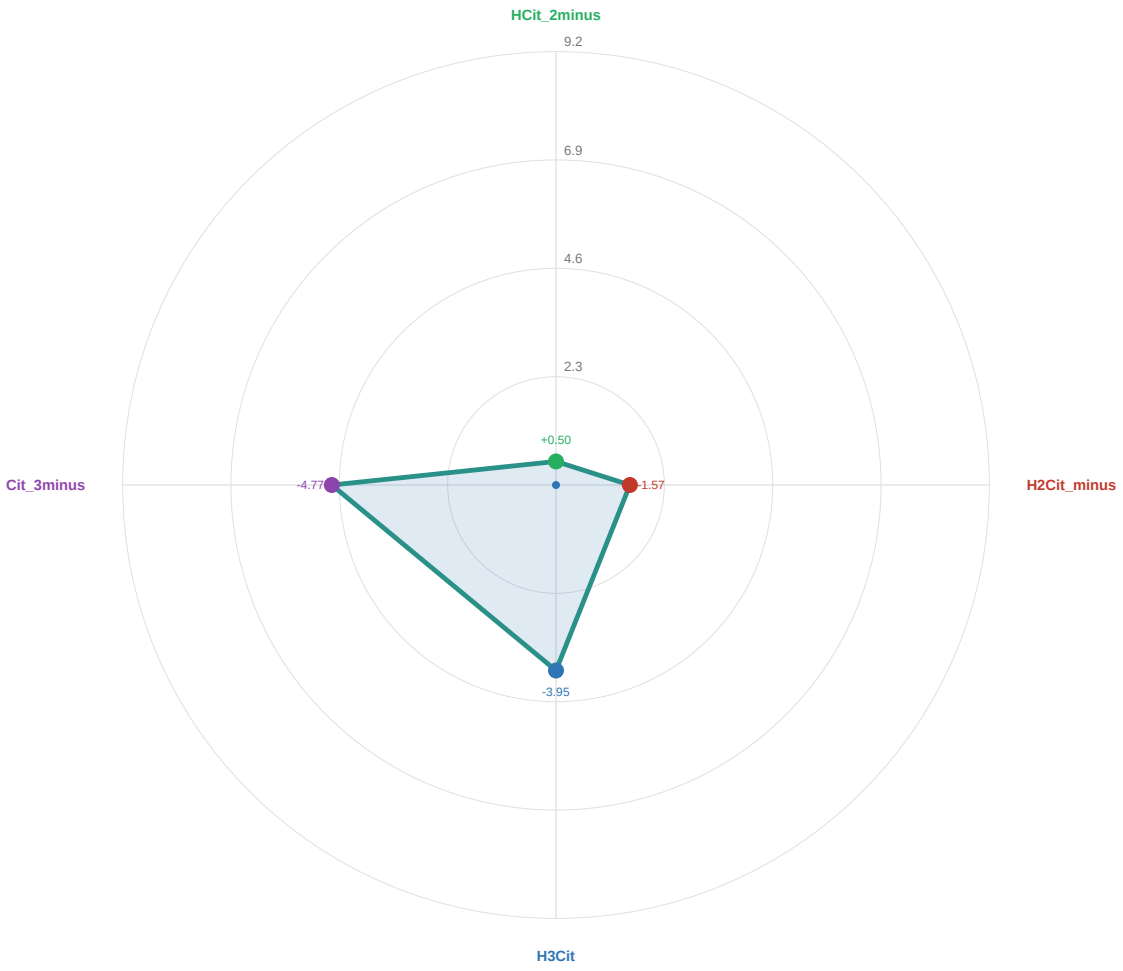
POLAR SLICE — t=48

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 49/200 | ||x||_A = 6.408



From 47: delta ||x||_A = +0.057

Phase = -158.3 deg

H3Cit=-3.95 H2Cit_=-1.57 HCit_2=+0.50 Cit_3m=-4.77

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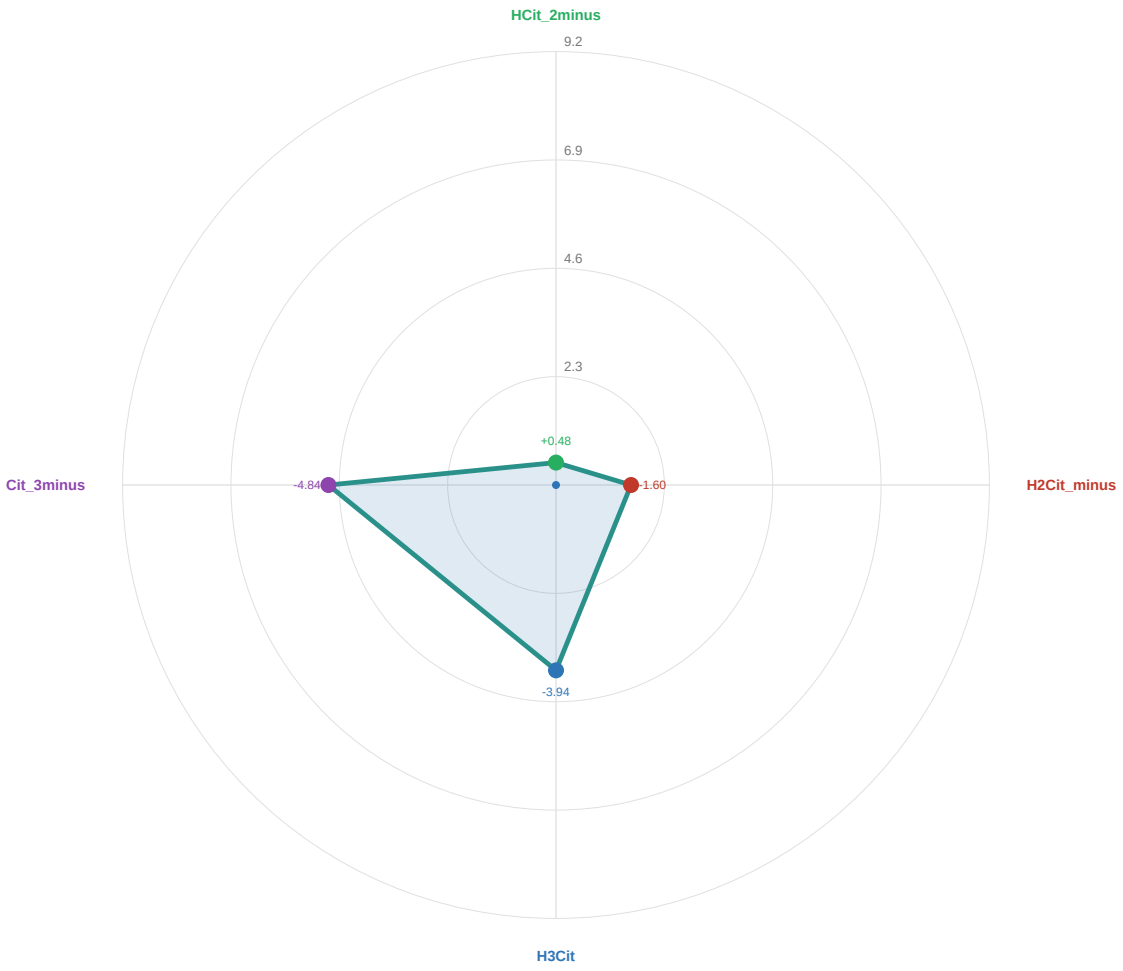
POLAR SLICE — t=49

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 50/200 | ||x||_A = 6.463



From 48: delta ||x||_A = +0.055

Phase = -158.0 deg

H3Cit=-3.94 H2Cit_=-1.60 HCit_2=+0.48 Cit_3m=-4.84

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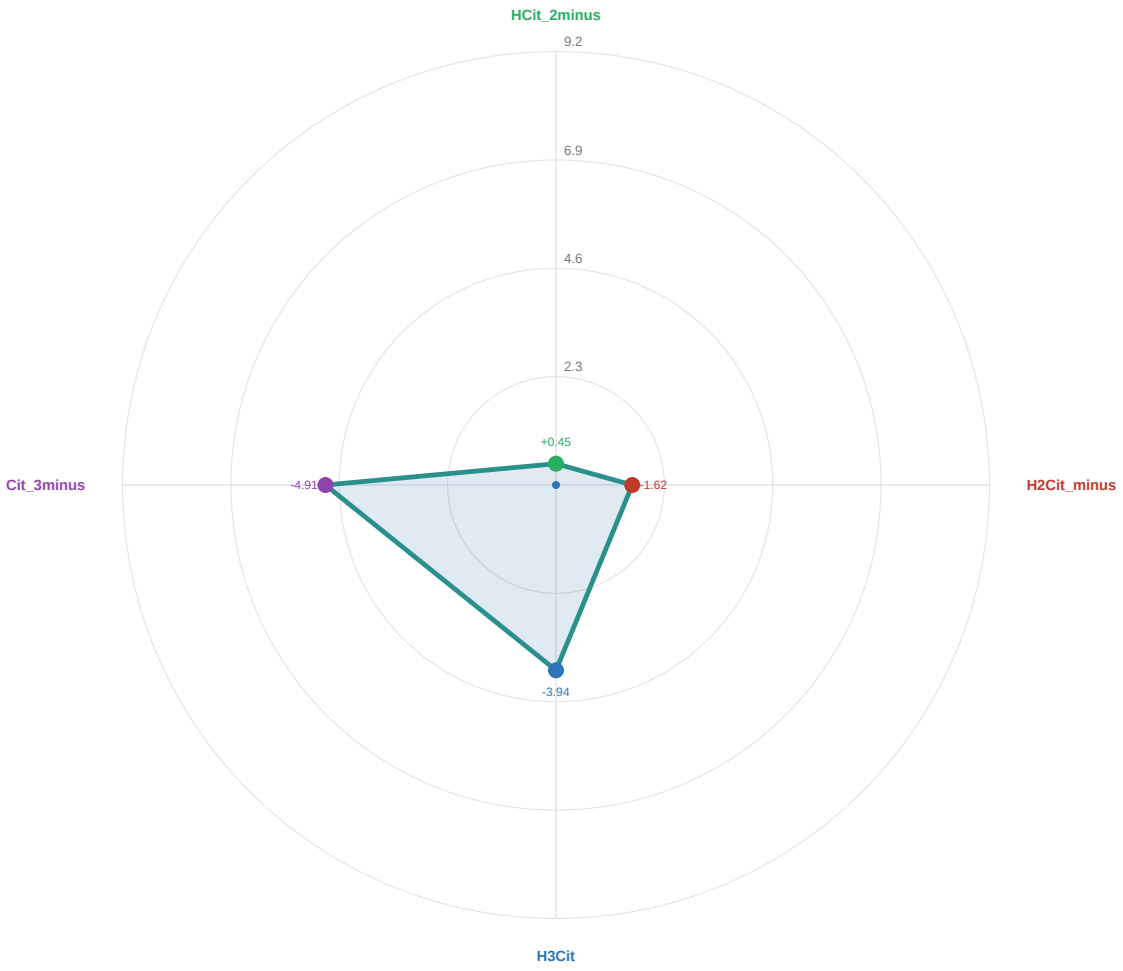
POLAR SLICE — t=50

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 51/200 | ||x||_A = 6.516



From 49: delta ||x||_A = +0.053

Phase = -157.6 deg

H3Cit=-3.94 H2Cit_=-1.62 HCit_2=+0.45 Cit_3m=-4.91

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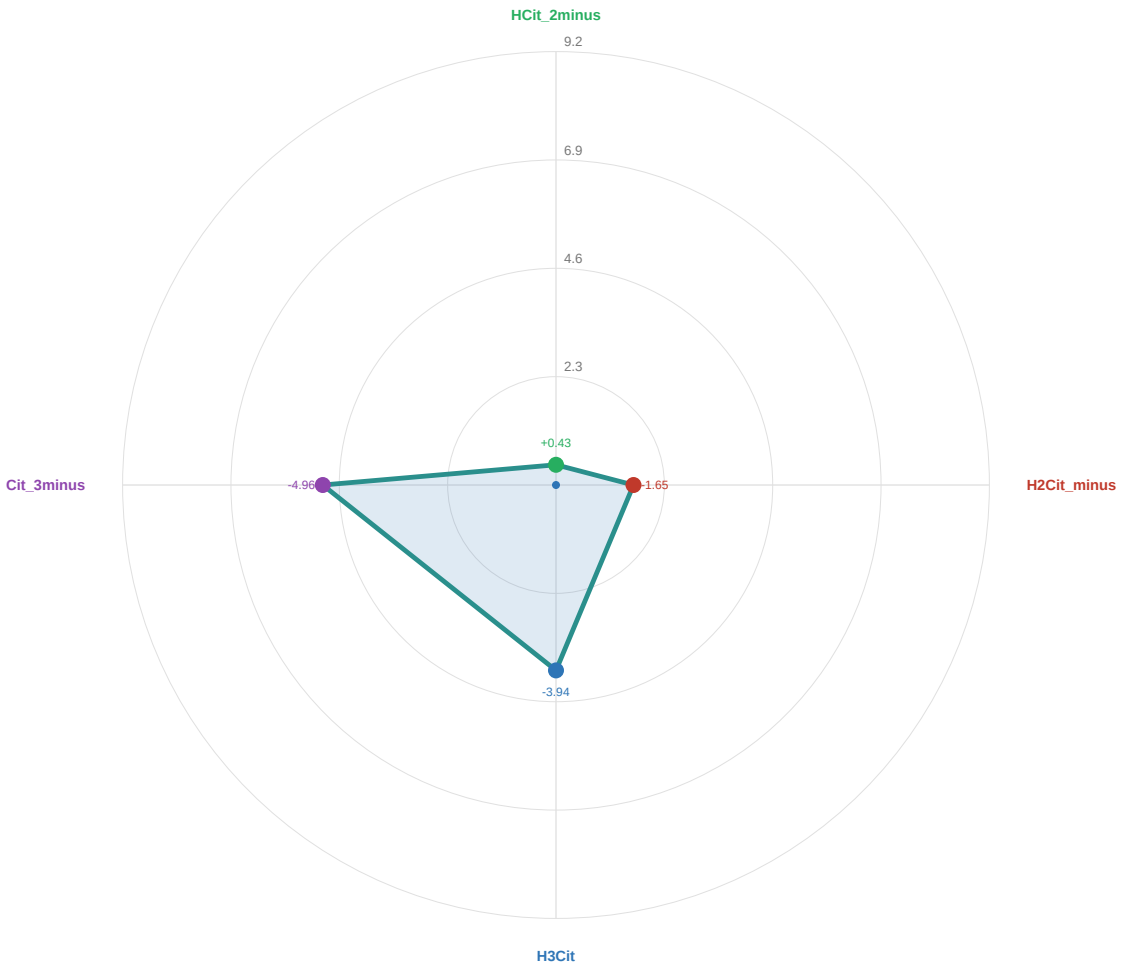
POLAR SLICE — t=51

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 52/200 | ||x||_A = 6.565



From 50: delta ||x||_A = +0.049

Phase = -157.3 deg

H3Cit=-3.94 H2Cit_=-1.65 HCit_2=+0.43 Cit_3m=-4.96

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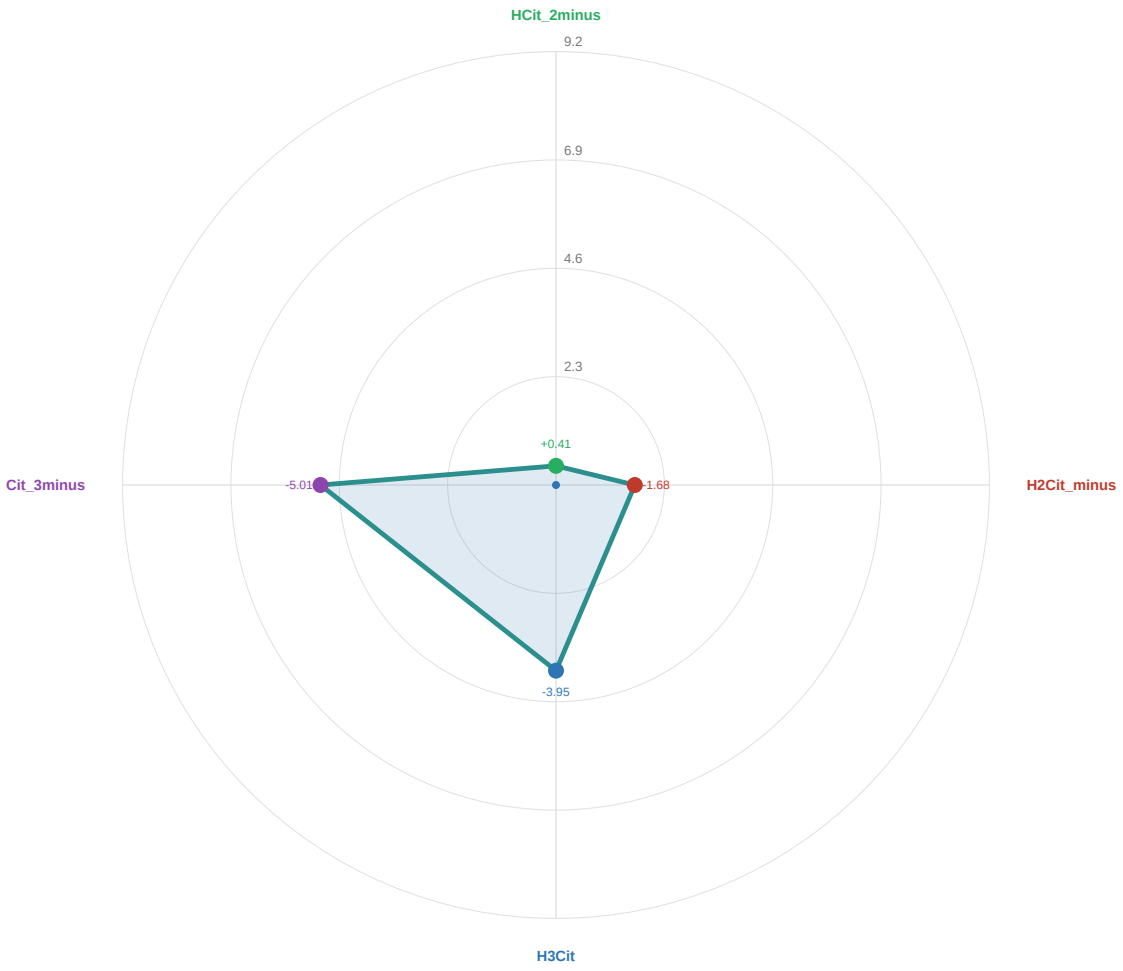
POLAR SLICE — t=52

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 53/200 | ||x||_A = 6.610



From 51: delta ||x||_A = +0.045

Phase = -157.0 deg

H3Cit=-3.95 H2Cit_=-1.68 HCit_2=+0.41 Cit_3m=-5.01

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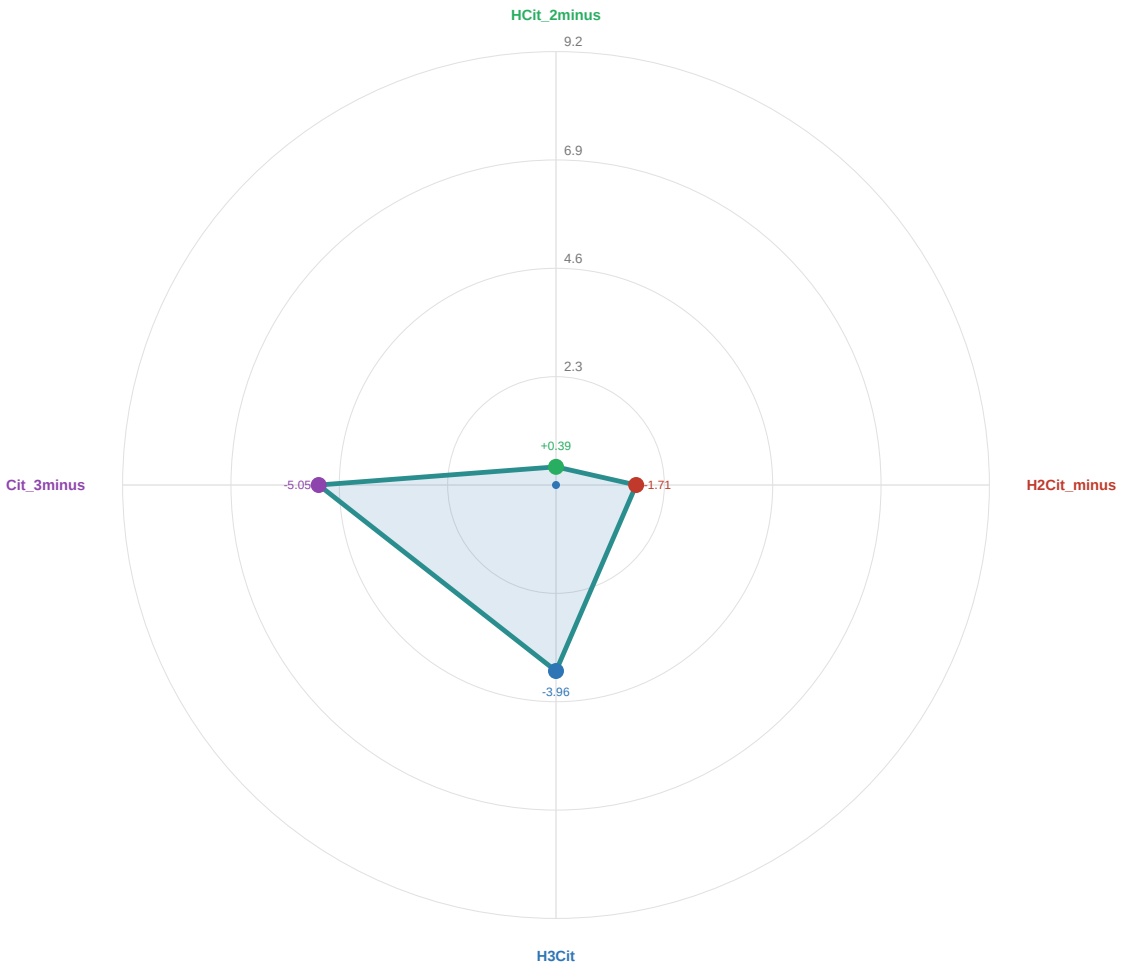
POLAR SLICE — t=53

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 54/200 | ||x||_A = 6.650



From 52: delta ||x||_A = +0.040

Phase = -156.7 deg

H3Cit=-3.96 H2Cit_=-1.71 HCit_2=+0.39 Cit_3m=-5.05

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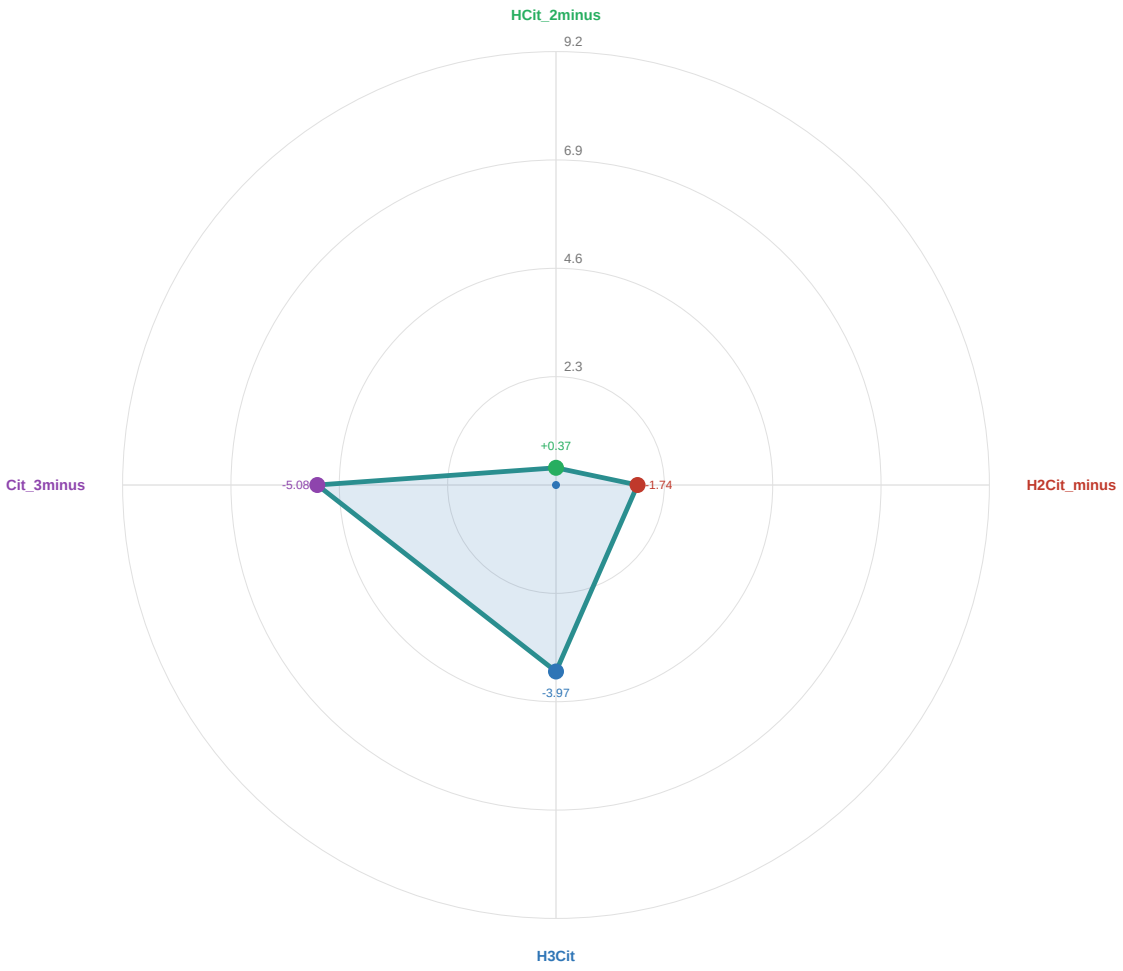
POLAR SLICE — t=54

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 55/200 | ||x||_A = 6.685



From 53: delta ||x||_A = +0.034

Phase = -156.4 deg

H3Cit=-3.97 H2Cit_=-1.74 HCit_2=+0.37 Cit_3m=-5.08

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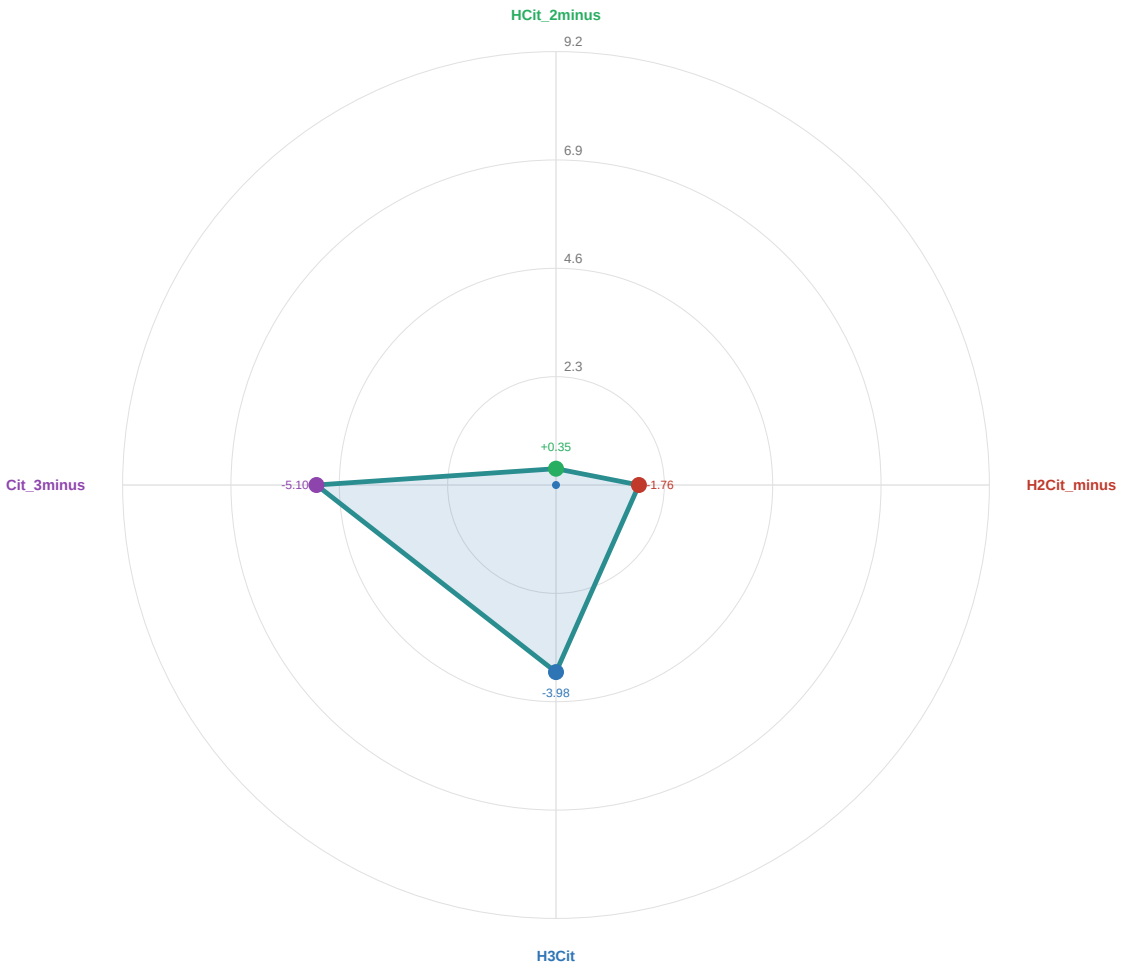
POLAR SLICE — t=55

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 56/200 | ||x||_A = 6.713



From 54: delta ||x||_A = +0.028

Phase = -156.1 deg

H3Cit=-3.98 H2Cit_=-1.76 HCit_2=+0.35 Cit_3m=-5.10

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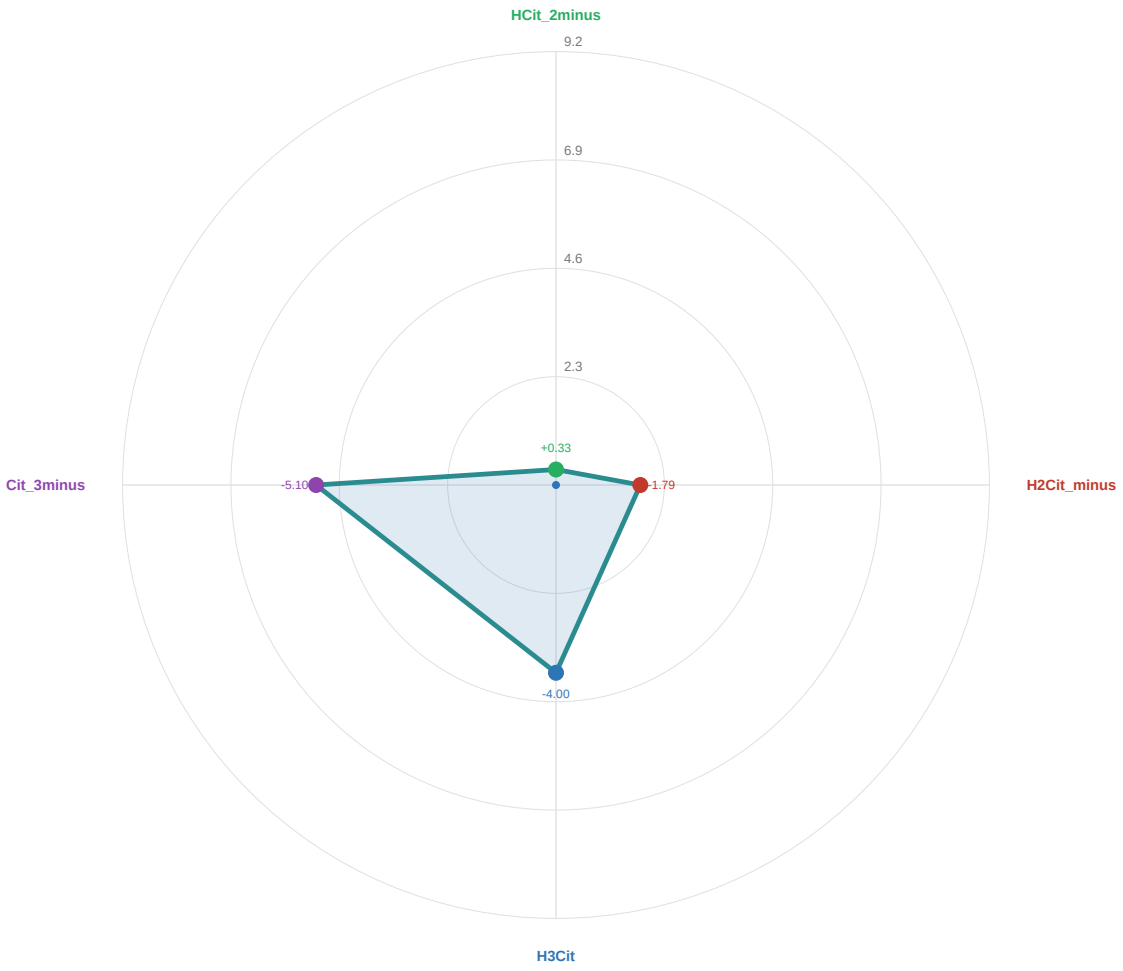
POLAR SLICE — t=56

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 57/200 | ||x||_A = 6.734



From 55: delta ||x||_A = +0.021

Phase = -155.8 deg

H3Cit=-4.00 H2Cit_=-1.79 HCit_2=+0.33 Cit_3m=-5.10

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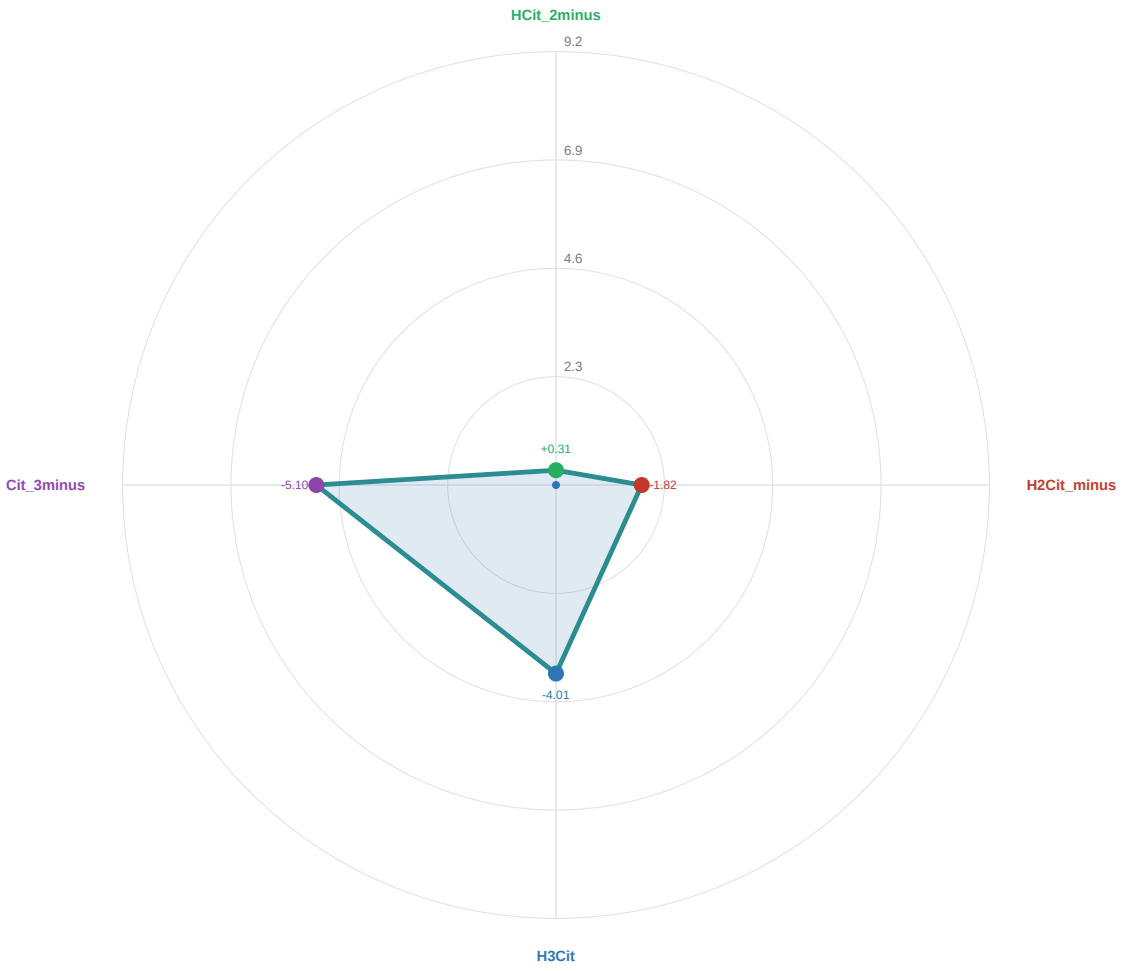
POLAR SLICE — t=57

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 58/200 | ||x||_A = 6.748



From 56: delta ||x||_A = +0.014

Phase = -155.6 deg

H3Cit=-4.01 H2Cit_=-1.82 HCit_2=+0.31 Cit_3m=-5.10

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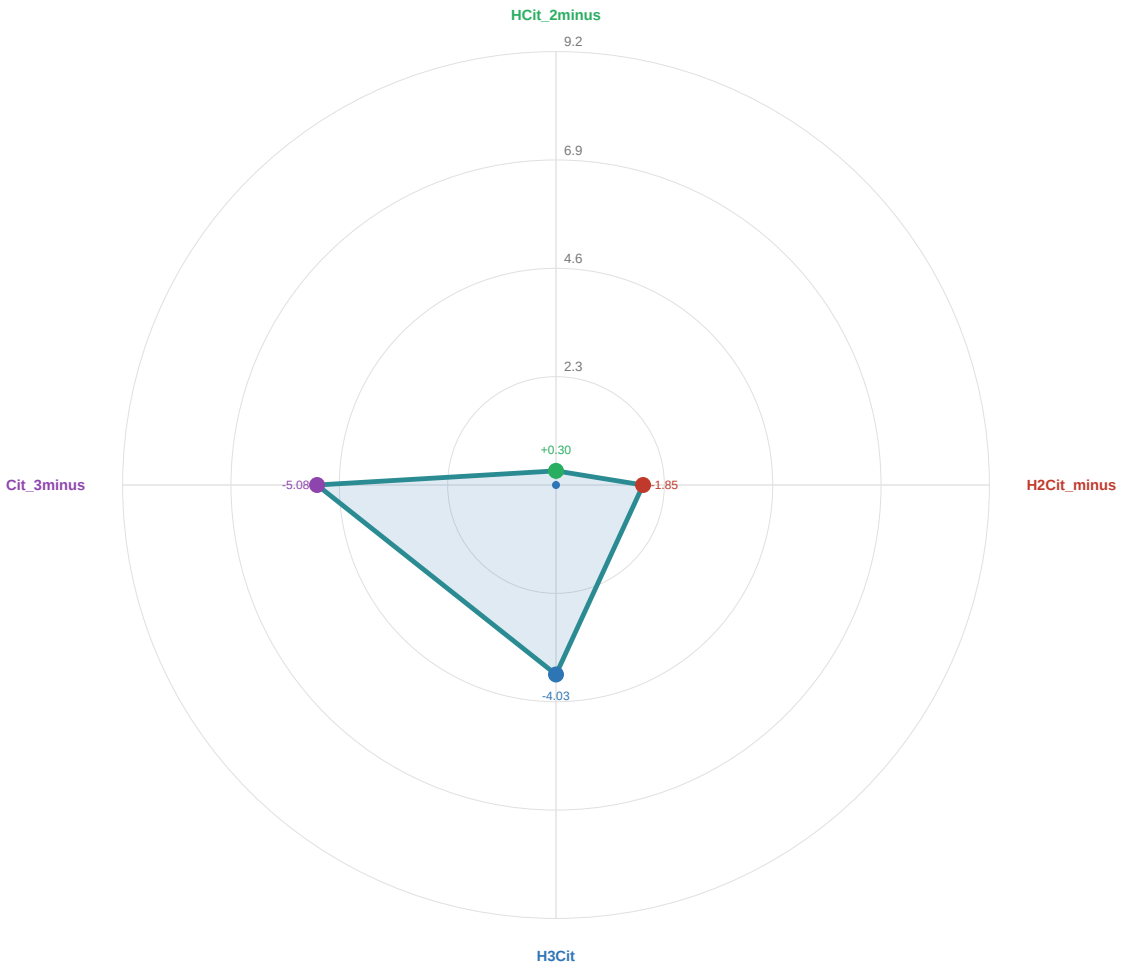
POLAR SLICE — t=58

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 59/200 | ||x||_A = 6.755



From 57: delta ||x||_A = +0.006

Phase = -155.3 deg

H3Cit=-4.03 H2Cit_=-1.85 HCit_2=+0.30 Cit_3m=-5.08

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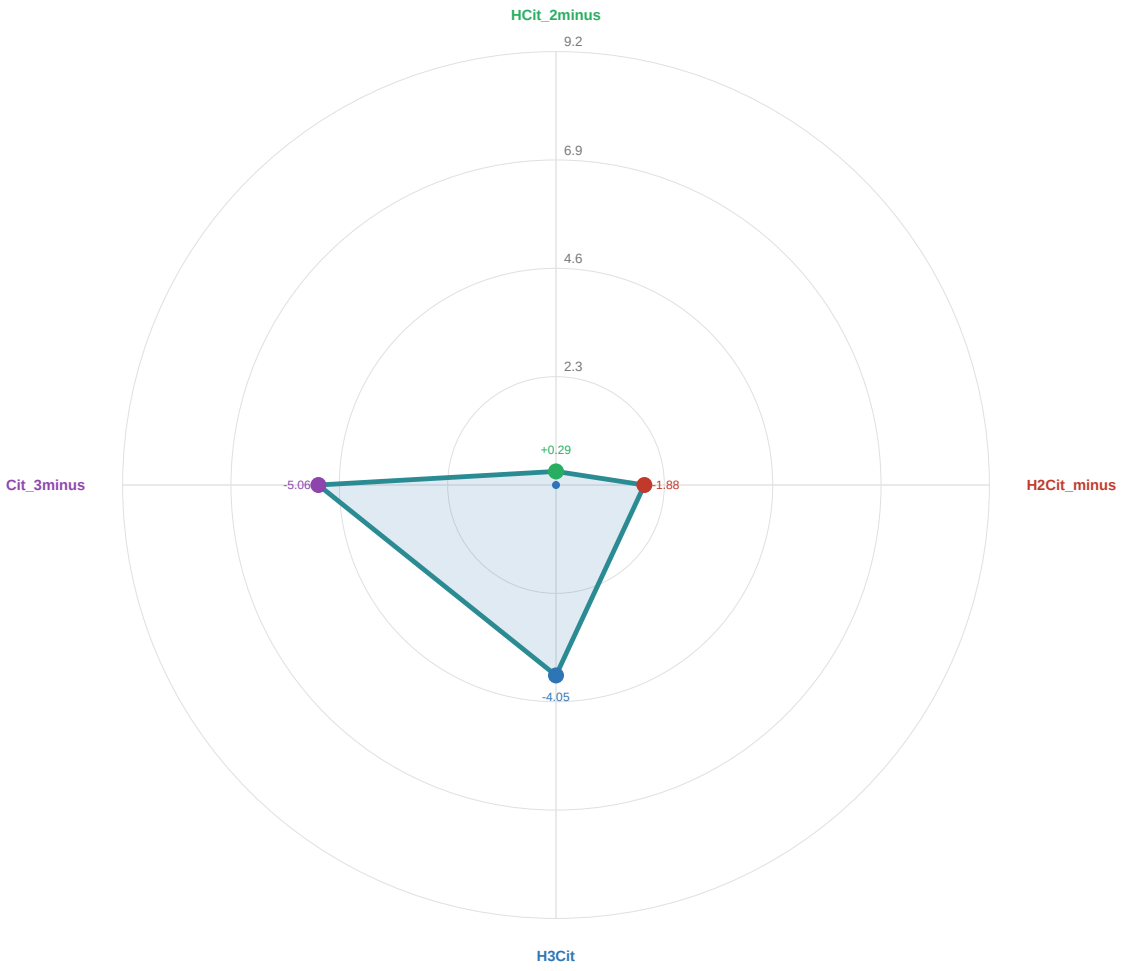
POLAR SLICE — t=59

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 60/200 | ||x||_A = 6.753



From 58: delta ||x||_A = -0.002

Phase = -155.1 deg

H3Cit=-4.05 H2Cit_=-1.88 HCit_2=+0.29 Cit_3m=-5.06

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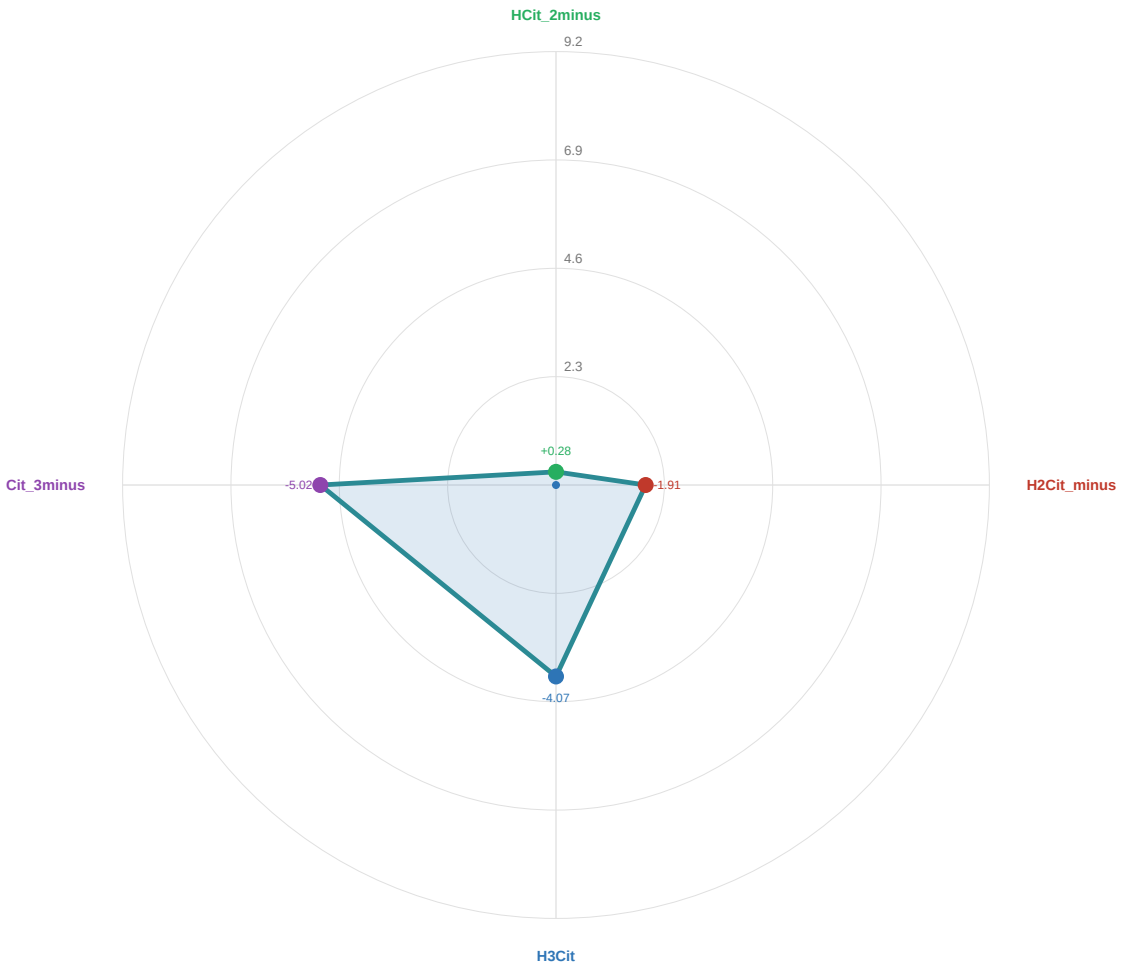
POLAR SLICE — t=60

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 61/200 | ||x||_A = 6.743



From 59: delta ||x||_A = -0.010

Phase = -154.9 deg

H3Cit=-4.07 H2Cit_=-1.91 HCit_2=+0.28 Cit_3m=-5.02

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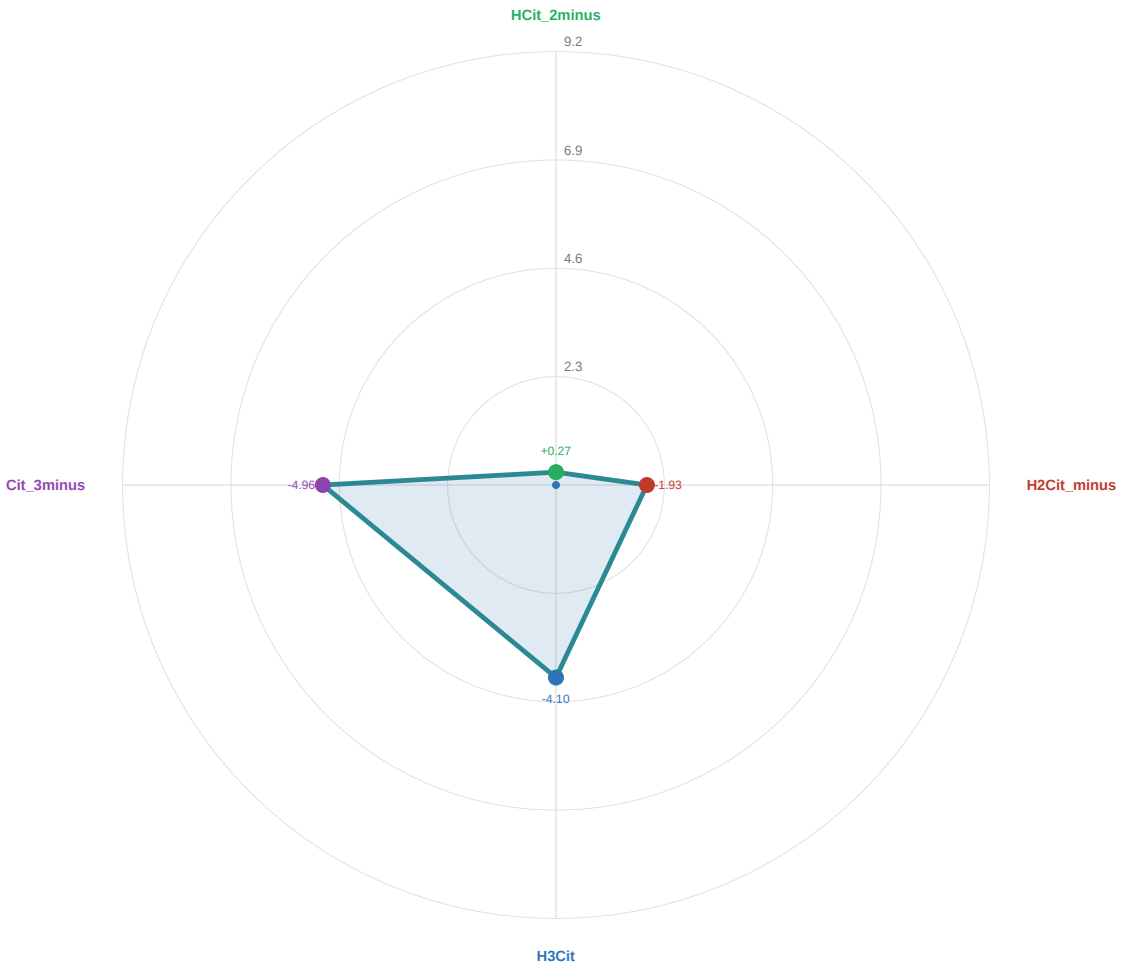
POLAR SLICE — t=61

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 62/200 | ||x||_A = 6.724



From 60: delta ||x||_A = -0.018

Phase = -154.7 deg

H3Cit=-4.10 H2Cit_=-1.93 HCit_2=+0.27 Cit_3m=-4.96

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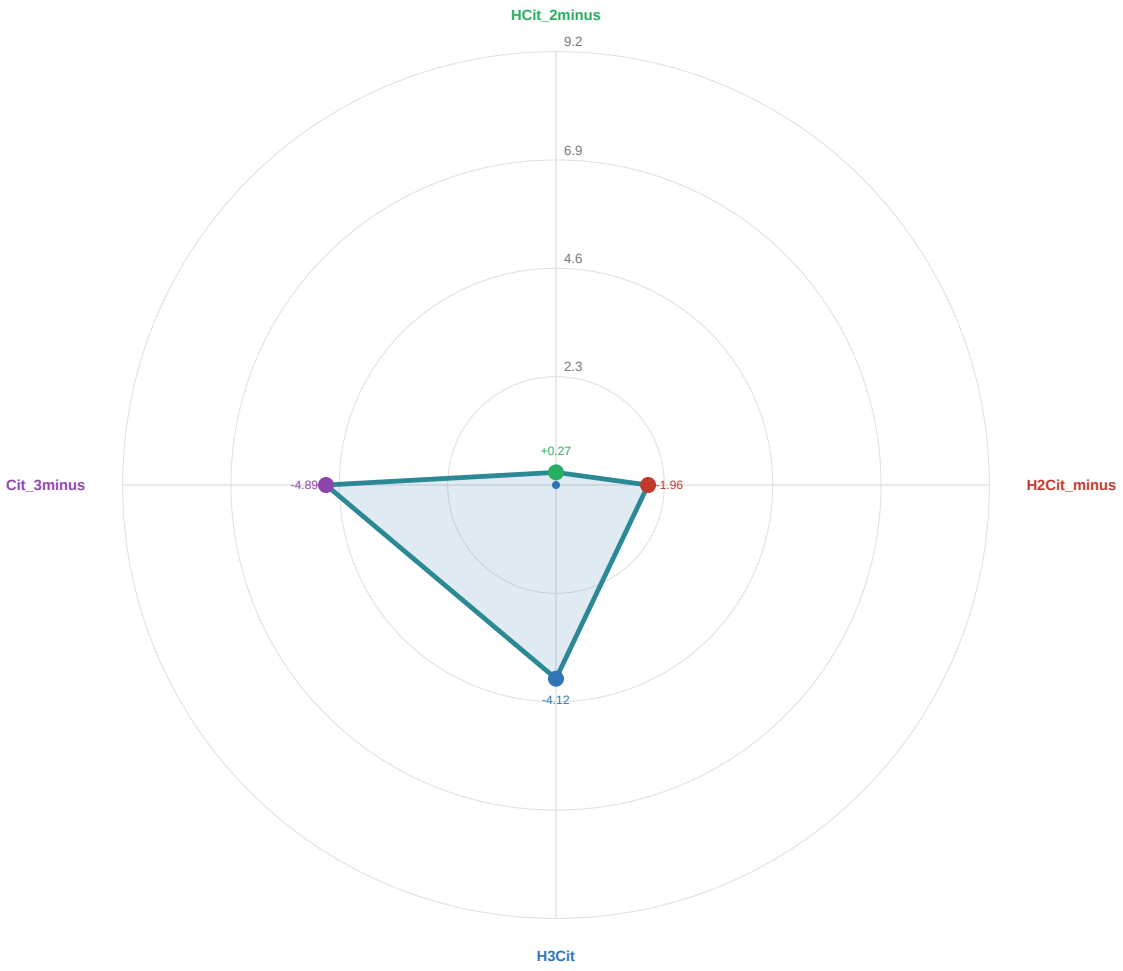
POLAR SLICE — t=62

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 63/200 | ||x||_A = 6.698



From 61: delta ||x||_A = -0.027

Phase = -154.6 deg

H3Cit=-4.12 H2Cit_=-1.96 HCit_2=+0.27 Cit_3m=-4.89

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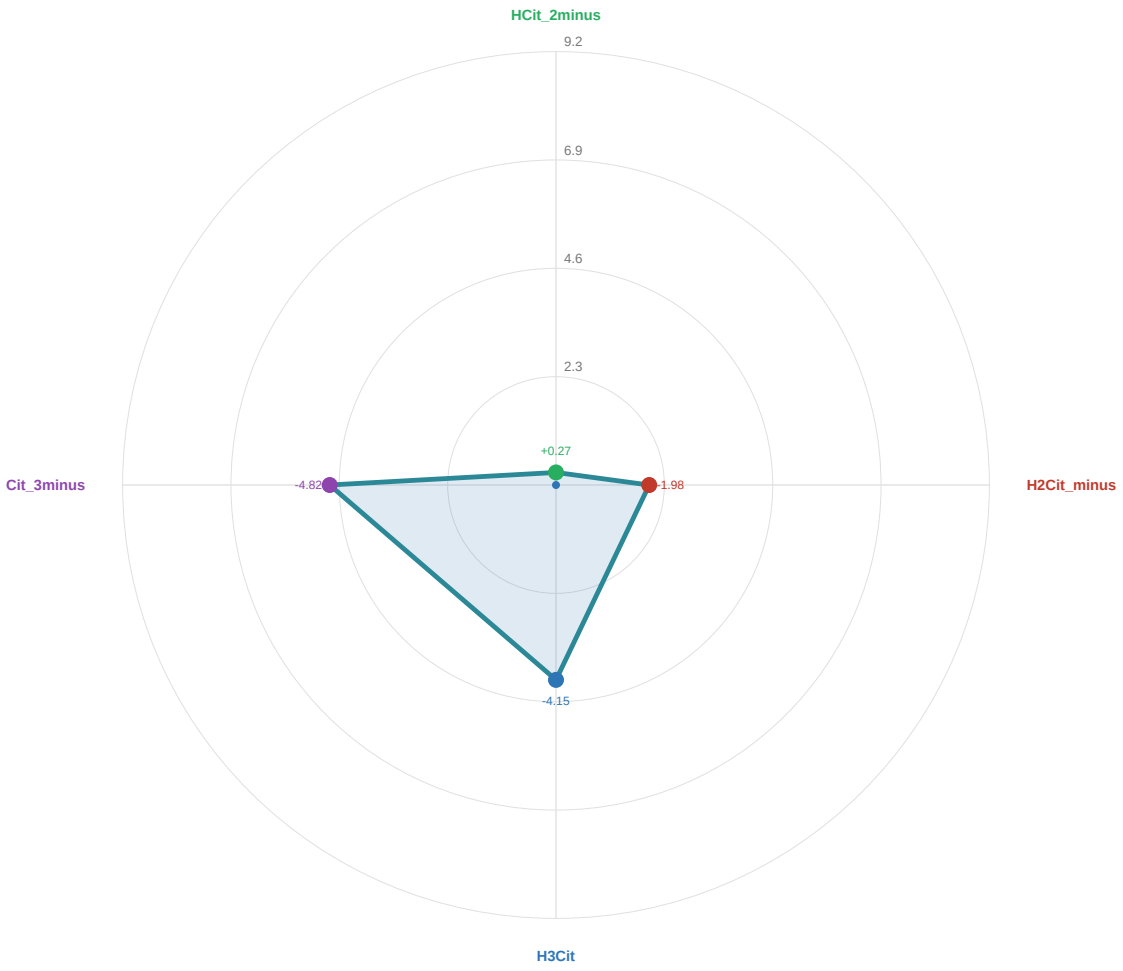
POLAR SLICE — t=63

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 64/200 | ||x||_A = 6.662



From 62: delta ||x||_A = -0.035

Phase = -154.4 deg

H3Cit=-4.15 H2Cit_=-1.98 HCit_2=+0.27 Cit_3m=-4.82

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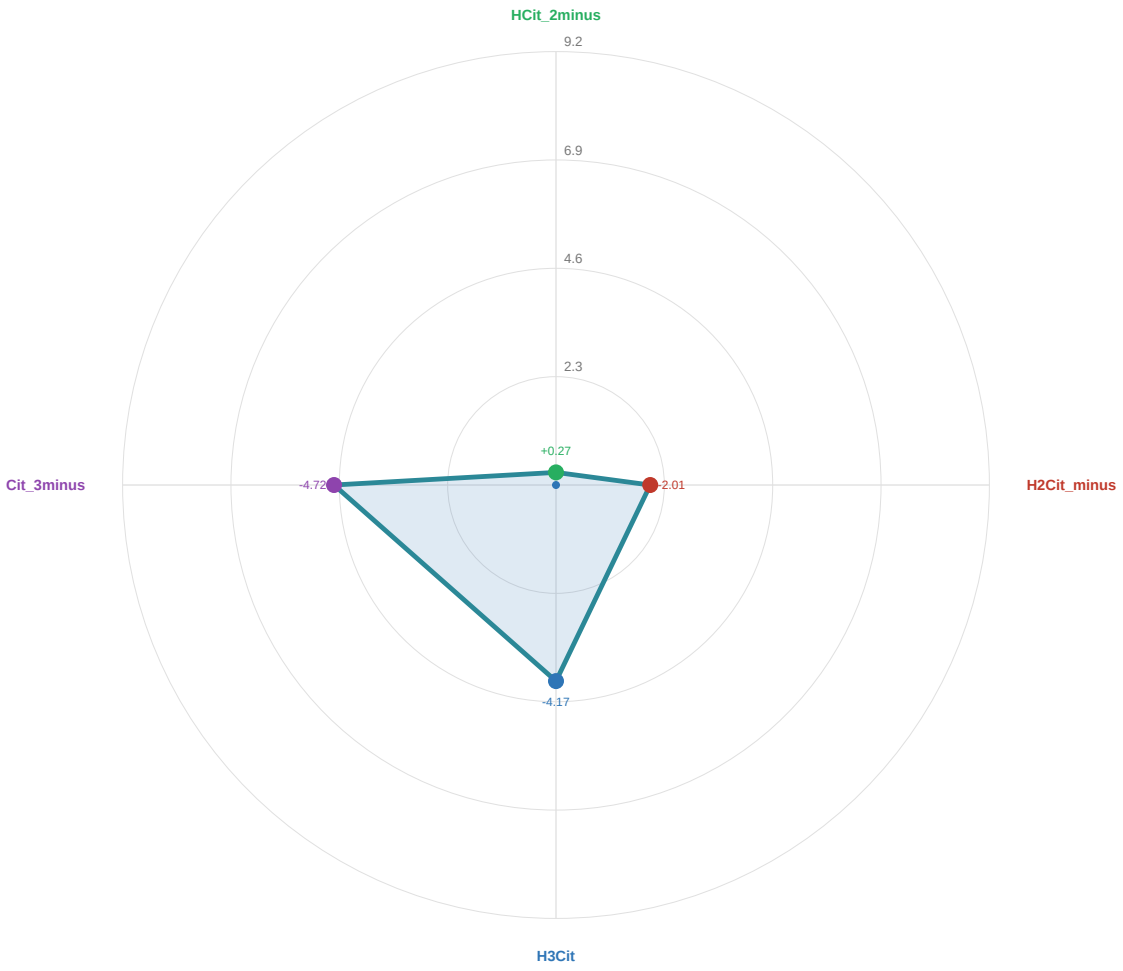
POLAR SLICE — t=64

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 65/200 | ||x||_A = 6.619



From 63: delta ||x||_A = -0.044

Phase = -154.3 deg

H3Cit=-4.17 H2Cit_=-2.01 HCit_2=+0.27 Cit_3m=-4.72

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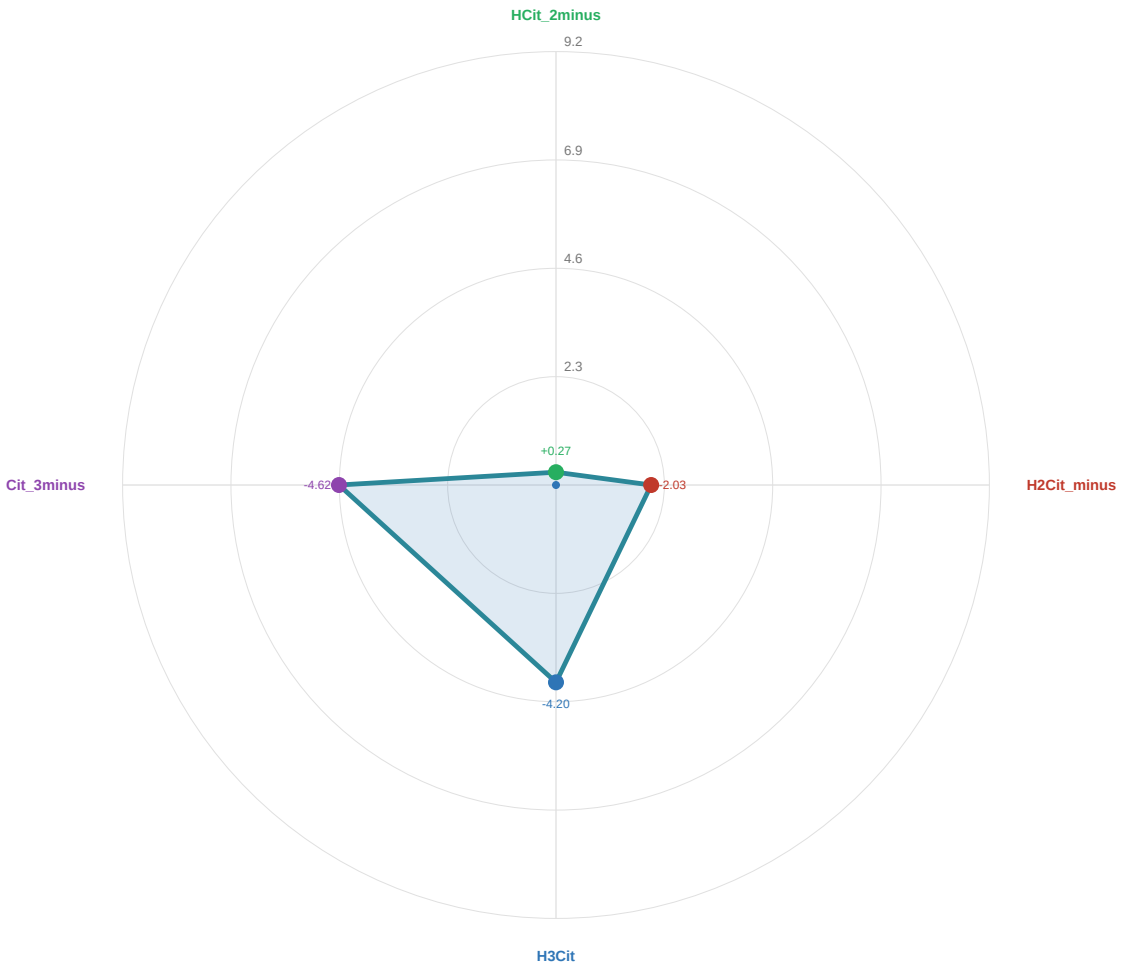
POLAR SLICE — t=65

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 66/200 | ||x||_A = 6.567



From 64: delta ||x||_A = -0.052

Phase = -154.3 deg

H3Cit=-4.20 H2Cit_=-2.03 HCit_2=+0.27 Cit_3m=-4.62

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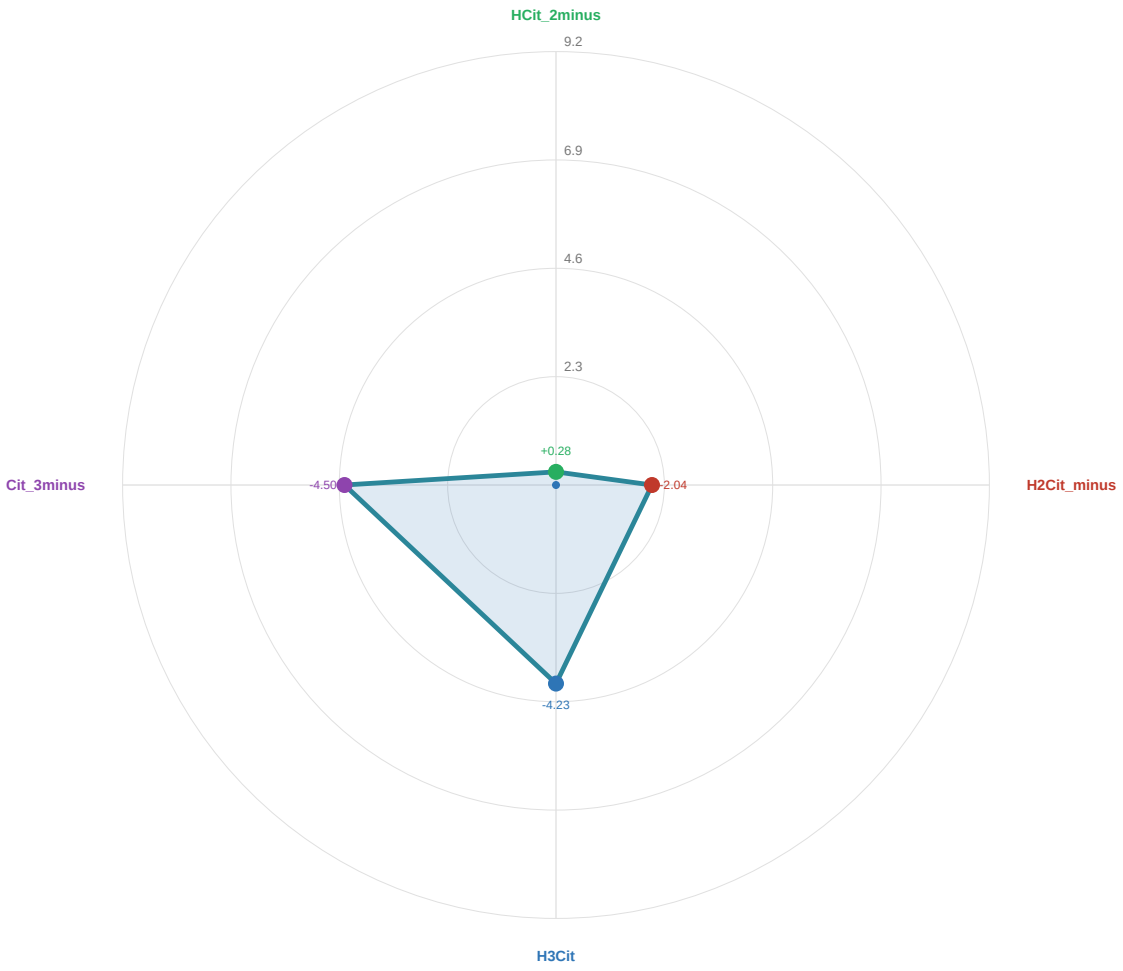
POLAR SLICE — t=66

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 67/200 | ||x||_A = 6.508



From 65: delta ||x||_A = -0.059

Phase = -154.2 deg

H3Cit=-4.23 H2Cit_=-2.04 HCit_2=+0.28 Cit_3m=-4.50

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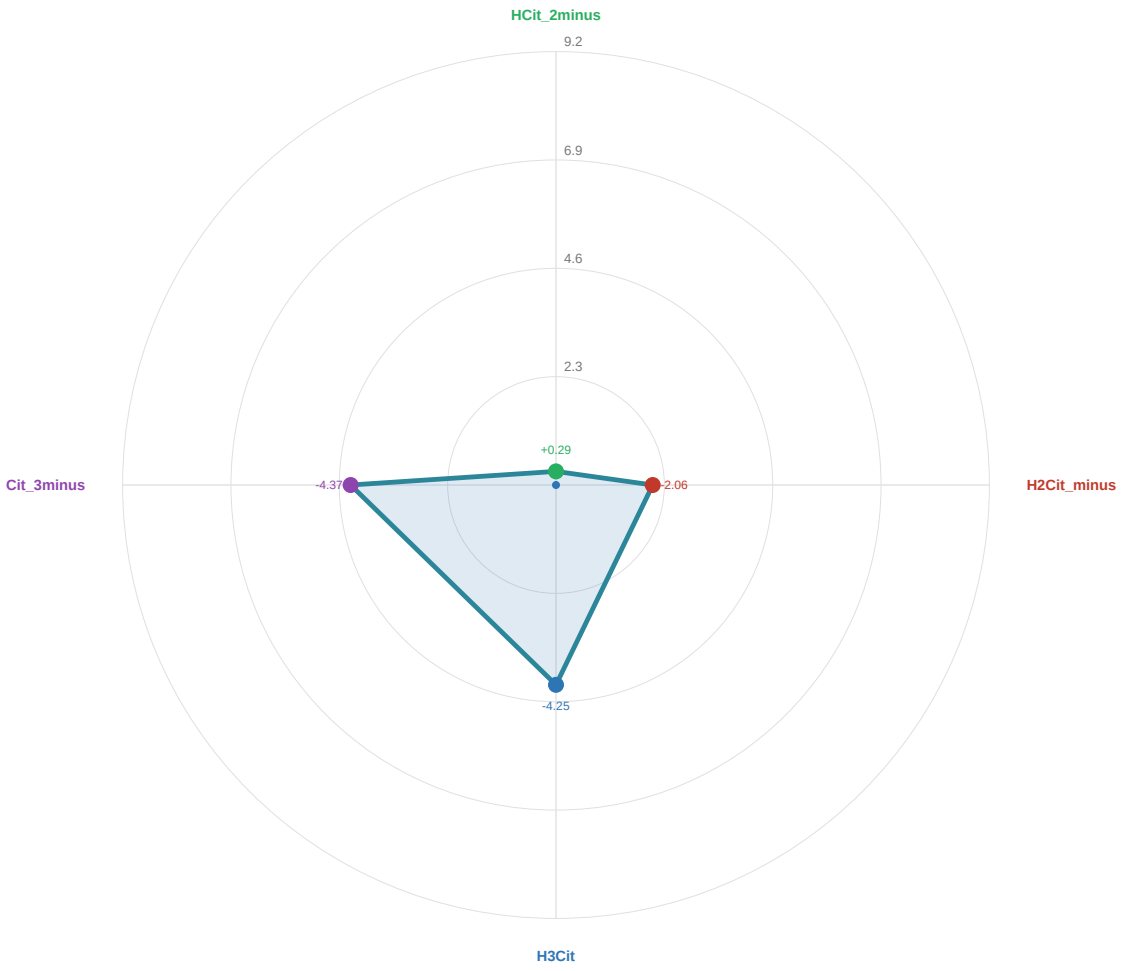
POLAR SLICE — t=67

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 68/200 | ||x||_A = 6.442



From 66: delta ||x||_A = -0.066

Phase = -154.2 deg

H3Cit=-4.25 H2Cit_=-2.06 HCit_2=+0.29 Cit_3m=-4.37

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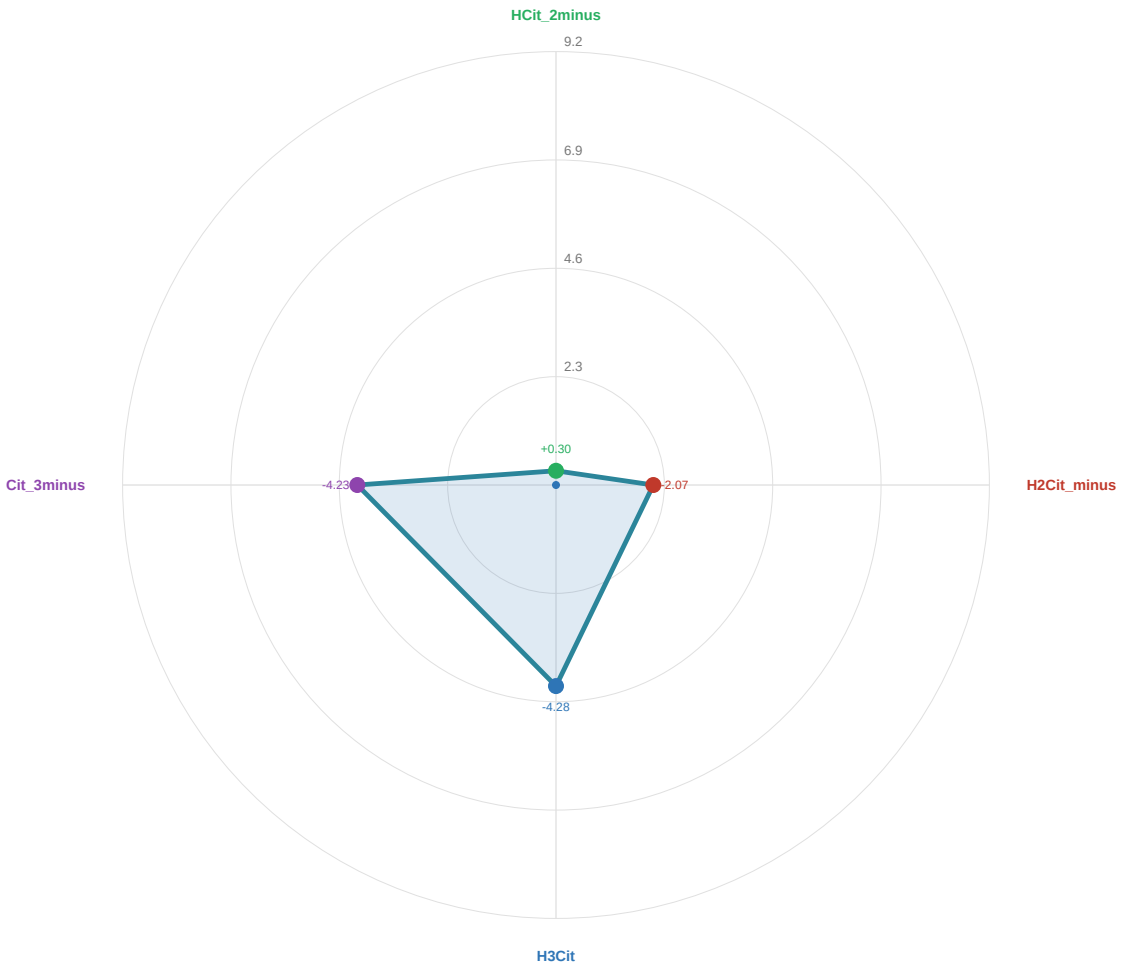
POLAR SLICE — t=68

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 69/200 | ||x||_A = 6.369



From 67: delta ||x||_A = -0.073

Phase = -154.2 deg

H3Cit=-4.28 H2Cit_=-2.07 HCit_2=+0.30 Cit_3m=-4.23

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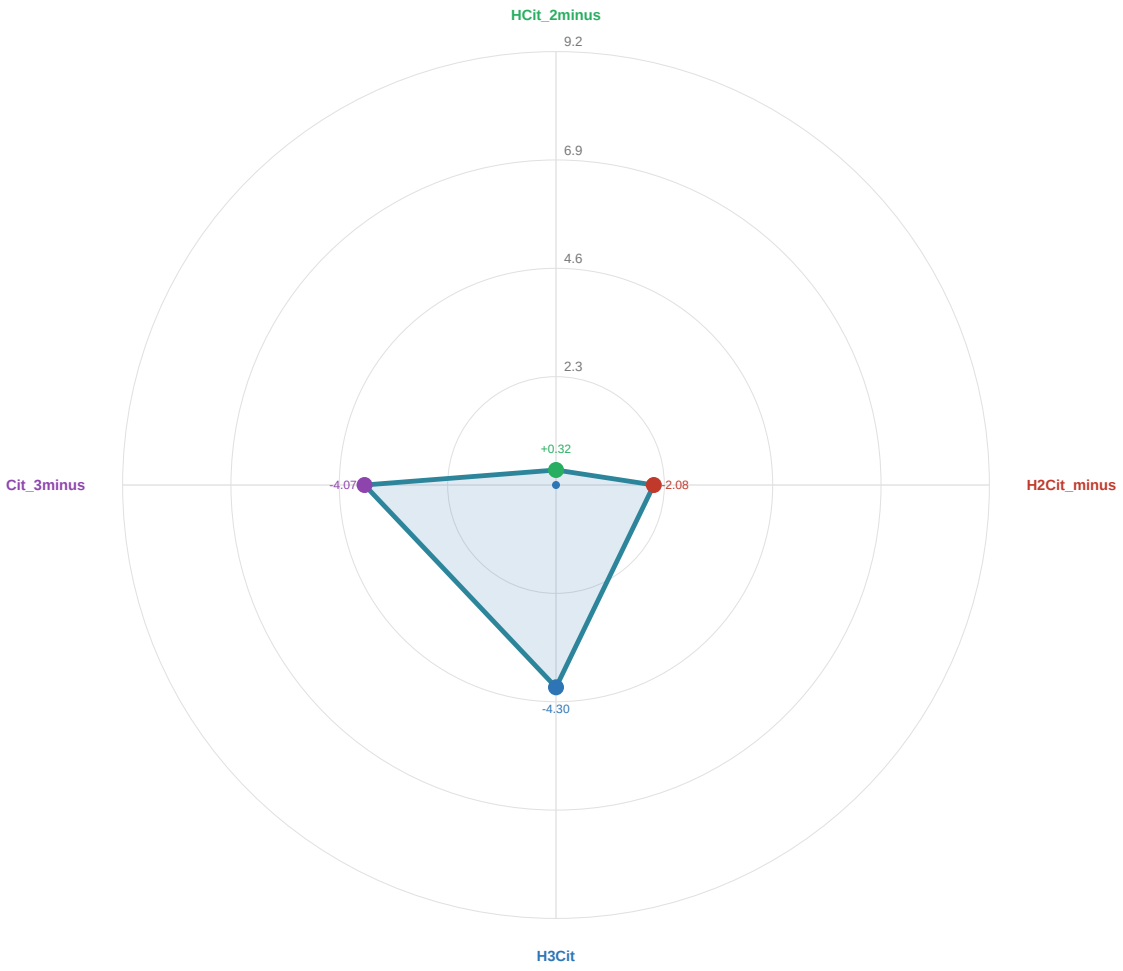
POLAR SLICE — t=69

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 70/200 | ||x||_A = 6.290



From 68: delta ||x||_A = -0.079

Phase = -154.2 deg

H3Cit=-4.30 H2Cit_=-2.08 HCit_2=+0.32 Cit_3m=-4.07

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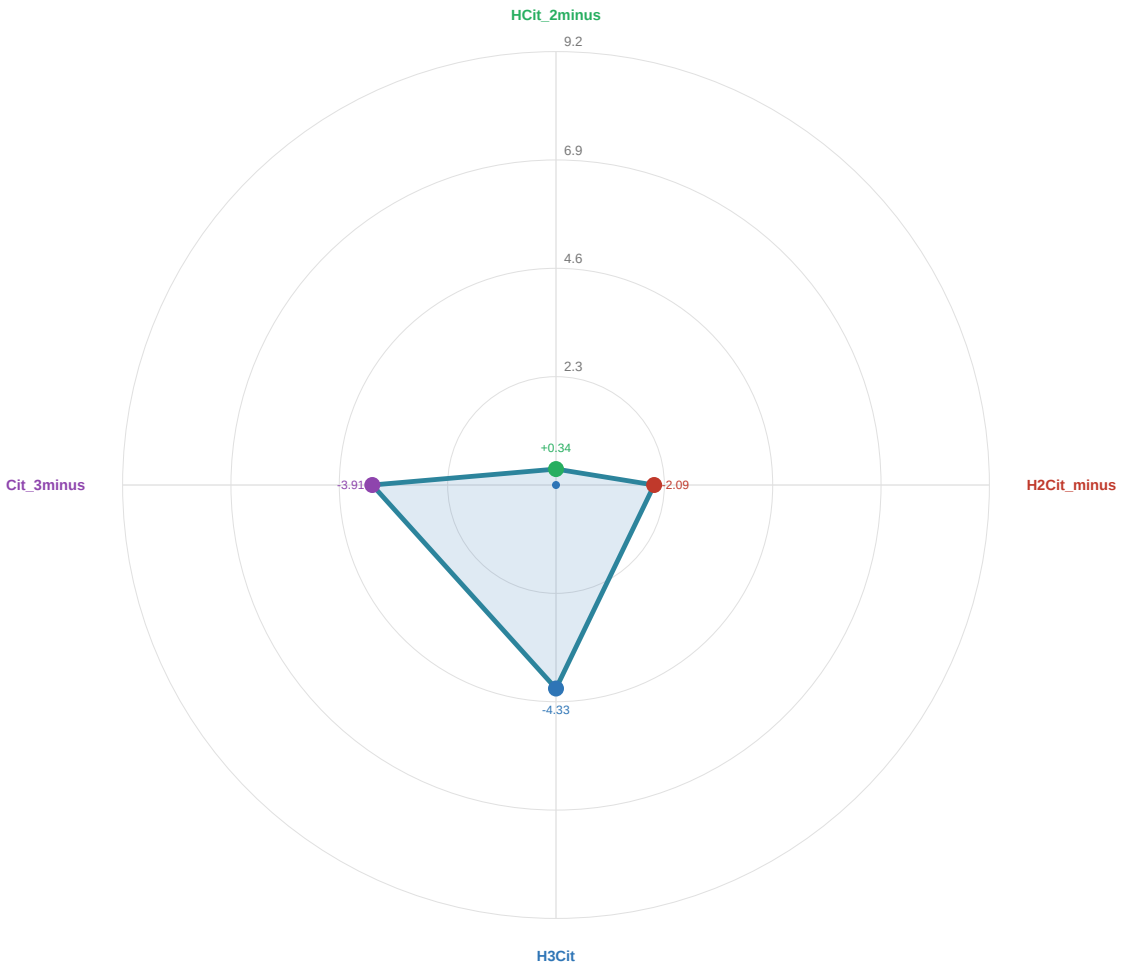
POLAR SLICE — t=70

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 71/200 | ||x||_A = 6.205



From 69: delta ||x||_A = -0.084

Phase = -154.3 deg

H3Cit=-4.33 H2Cit_=-2.09 HCit_2=+0.34 Cit_3m=-3.91

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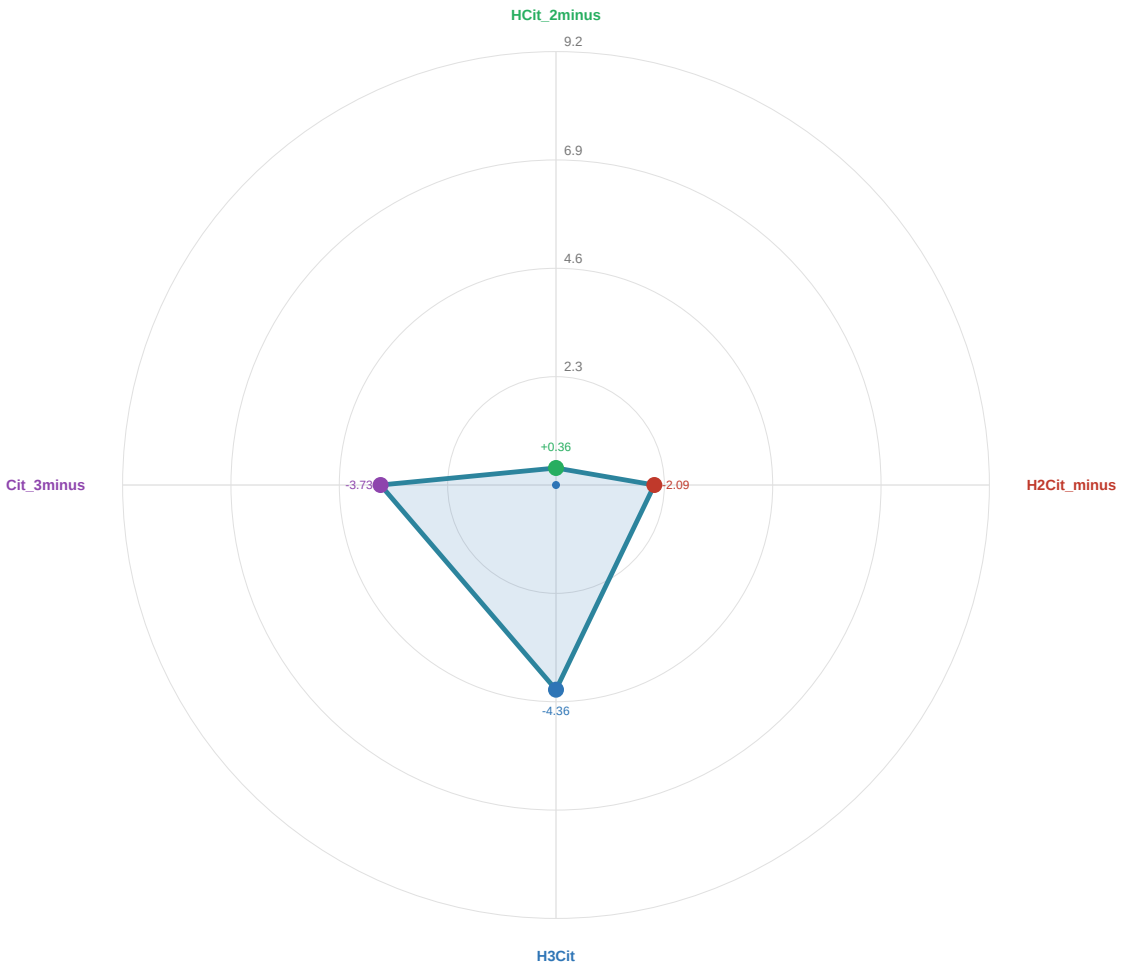
POLAR SLICE — t=71

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 72/200 | ||x||_A = 6.117



From 70: delta ||x||_A = -0.089

Phase = -154.3 deg

H3Cit=-4.36 H2Cit_=-2.09 HCit_2=+0.36 Cit_3m=-3.73

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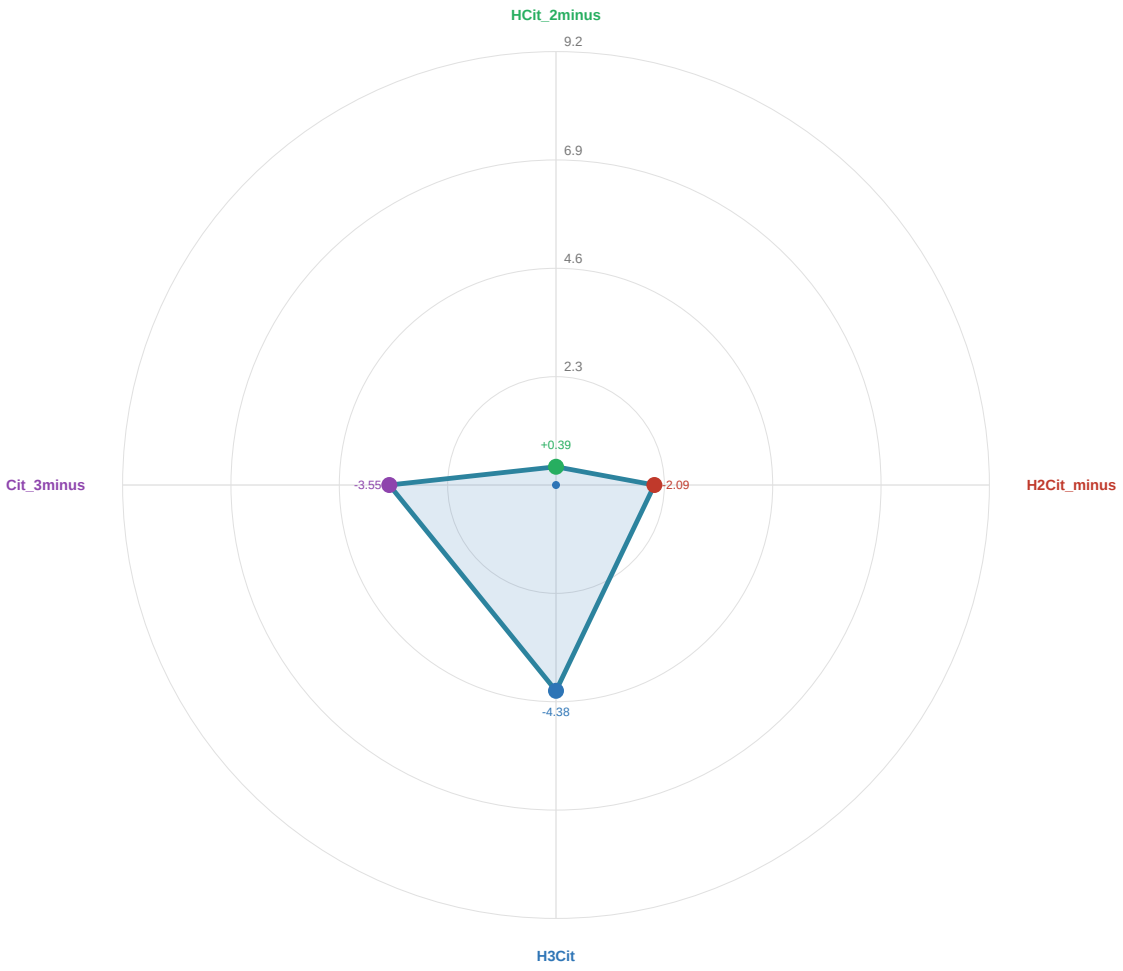
POLAR SLICE — t=72

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 73/200 | ||x||_A = 6.024



From 71: delta ||x||_A = -0.092

Phase = -154.4 deg

H3Cit=-4.38 H2Cit_=-2.09 HCit_2=+0.39 Cit_3m=-3.55

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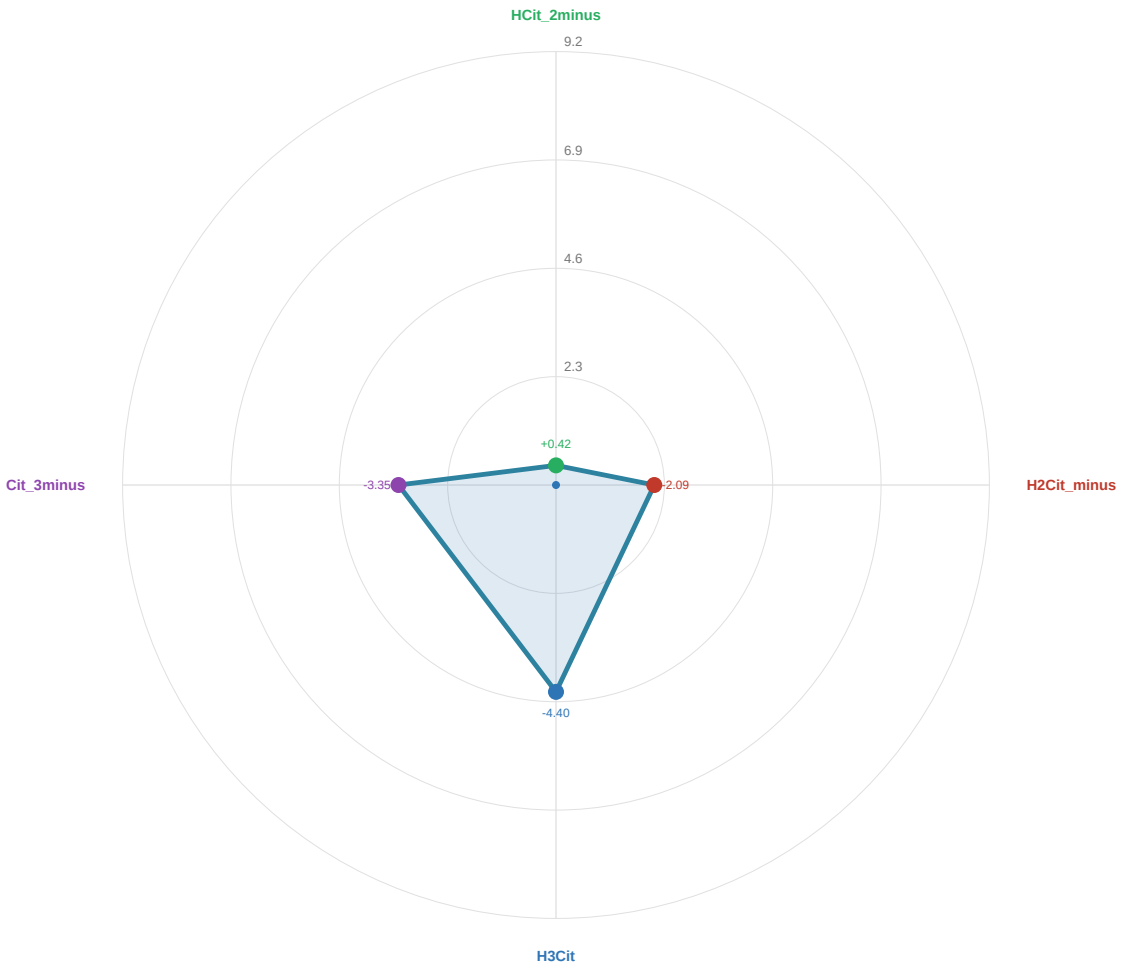
POLAR SLICE — t=73

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 74/200 | ||x||_A = 5.929



From 72: delta ||x||_A = -0.095

Phase = -154.6 deg

H3Cit=-4.40 H2Cit_=-2.09 HCit_2=+0.42 Cit_3m=-3.35

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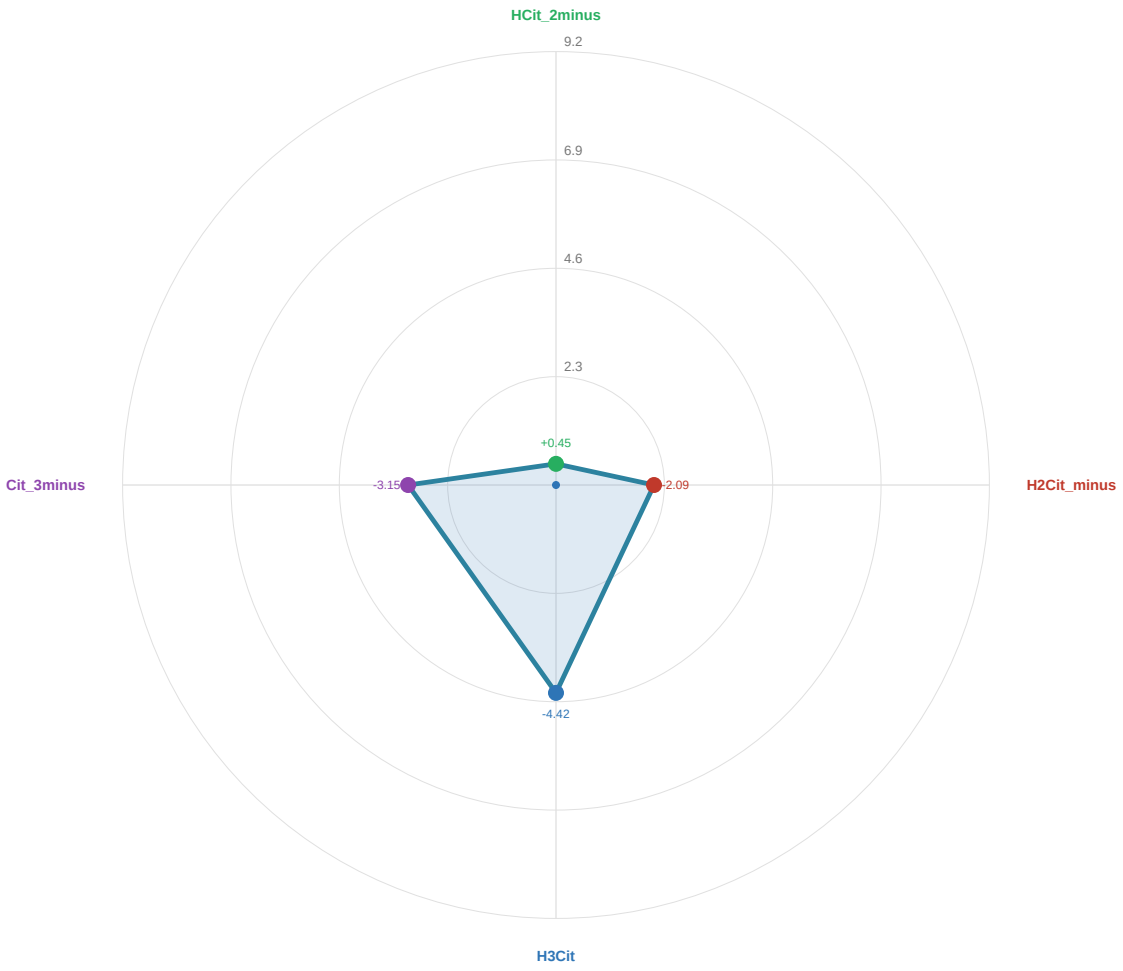
POLAR SLICE — t=74

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 75/200 | ||x||_A = 5.833



From 73: delta ||x||_A = -0.096

Phase = -154.8 deg

H3Cit=-4.42 H2Cit_=-2.09 HCit_2=+0.45 Cit_3m=-3.15

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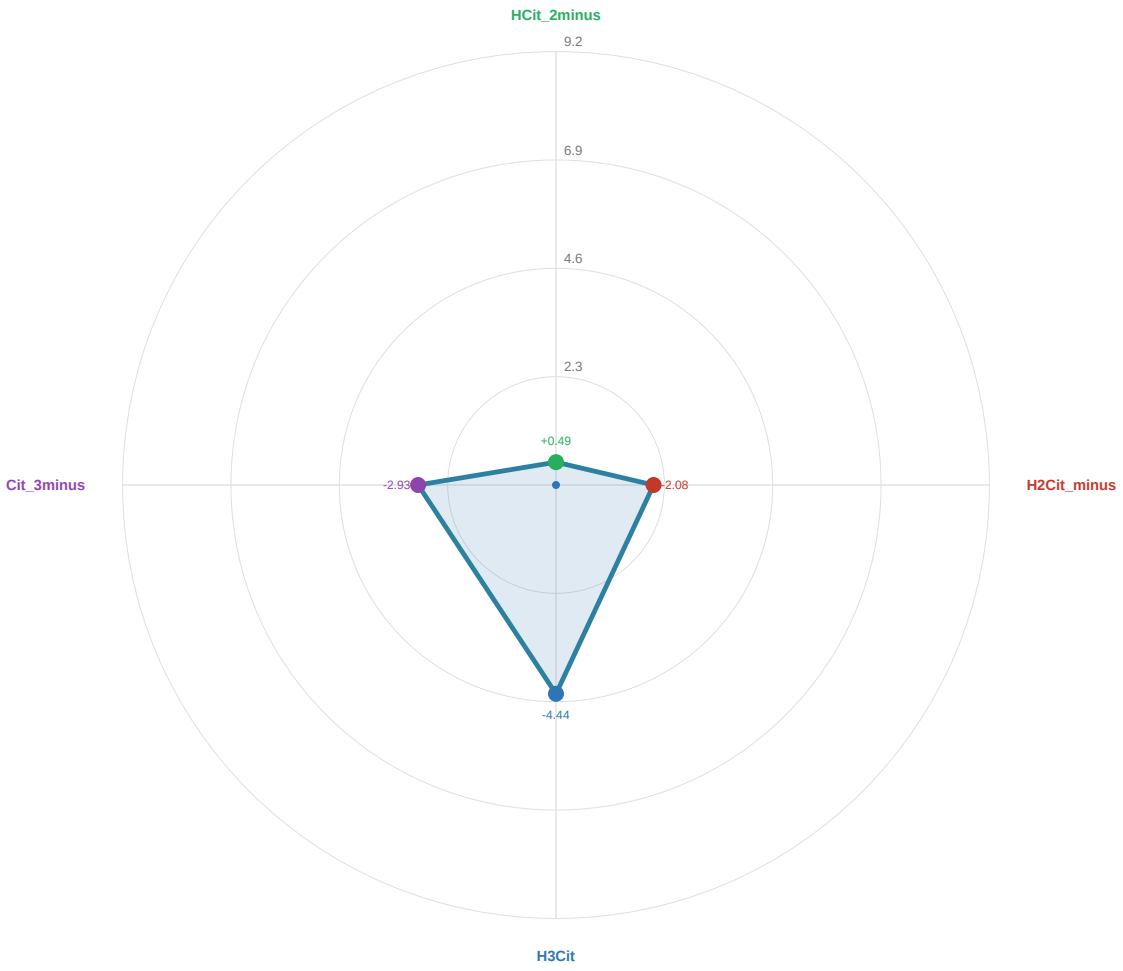
POLAR SLICE — t=75

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 76/200 | ||x||_A = 5.736



From 74: delta ||x||_A = -0.097

Phase = -154.9 deg

H3Cit=-4.44 H2Cit_=-2.08 HCit_2=+0.49 Cit_3m=-2.93

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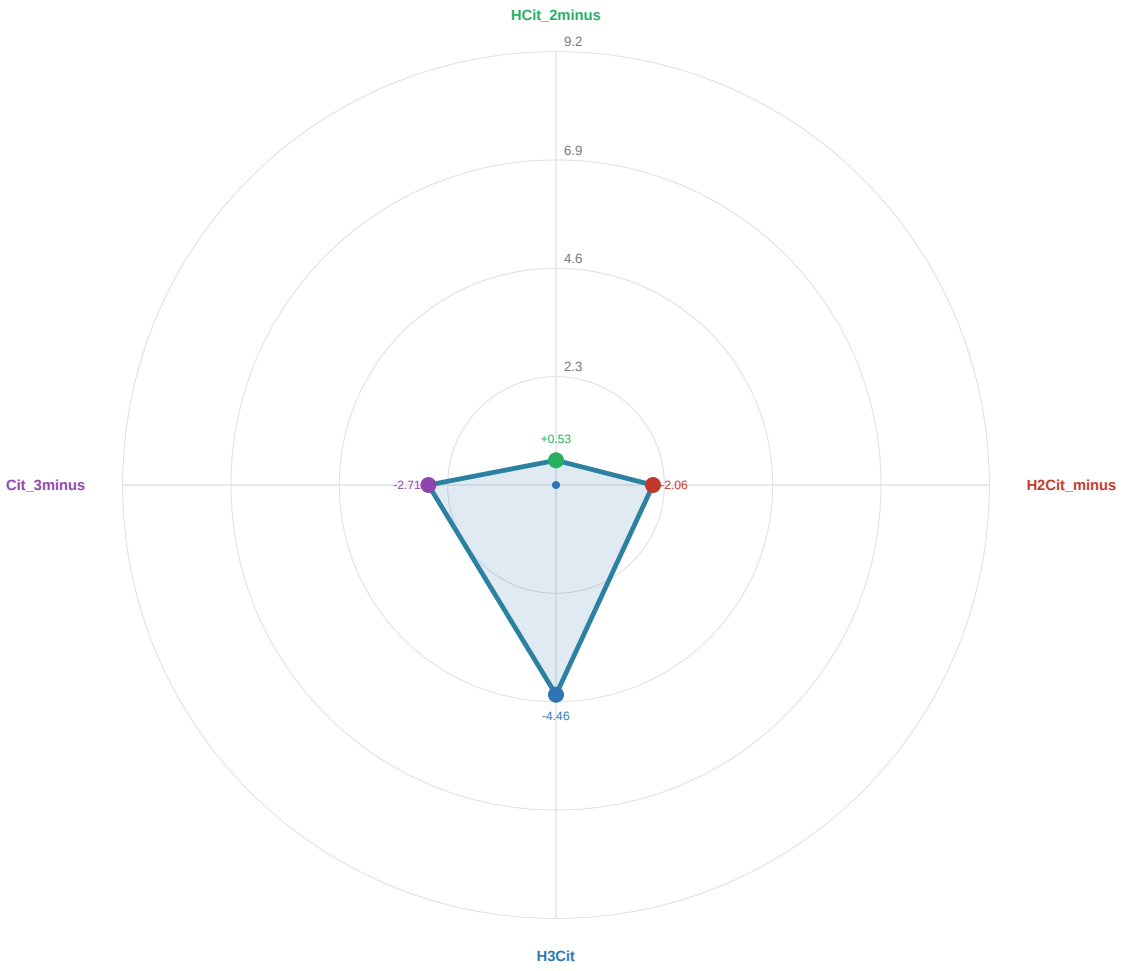
POLAR SLICE — t=76

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 77/200 | ||x||_A = 5.640



From 75: delta ||x||_A = -0.096

Phase = -155.2 deg

H3Cit=-4.46 H2Cit_=-2.06 HCit_2=+0.53 Cit_3m=-2.71

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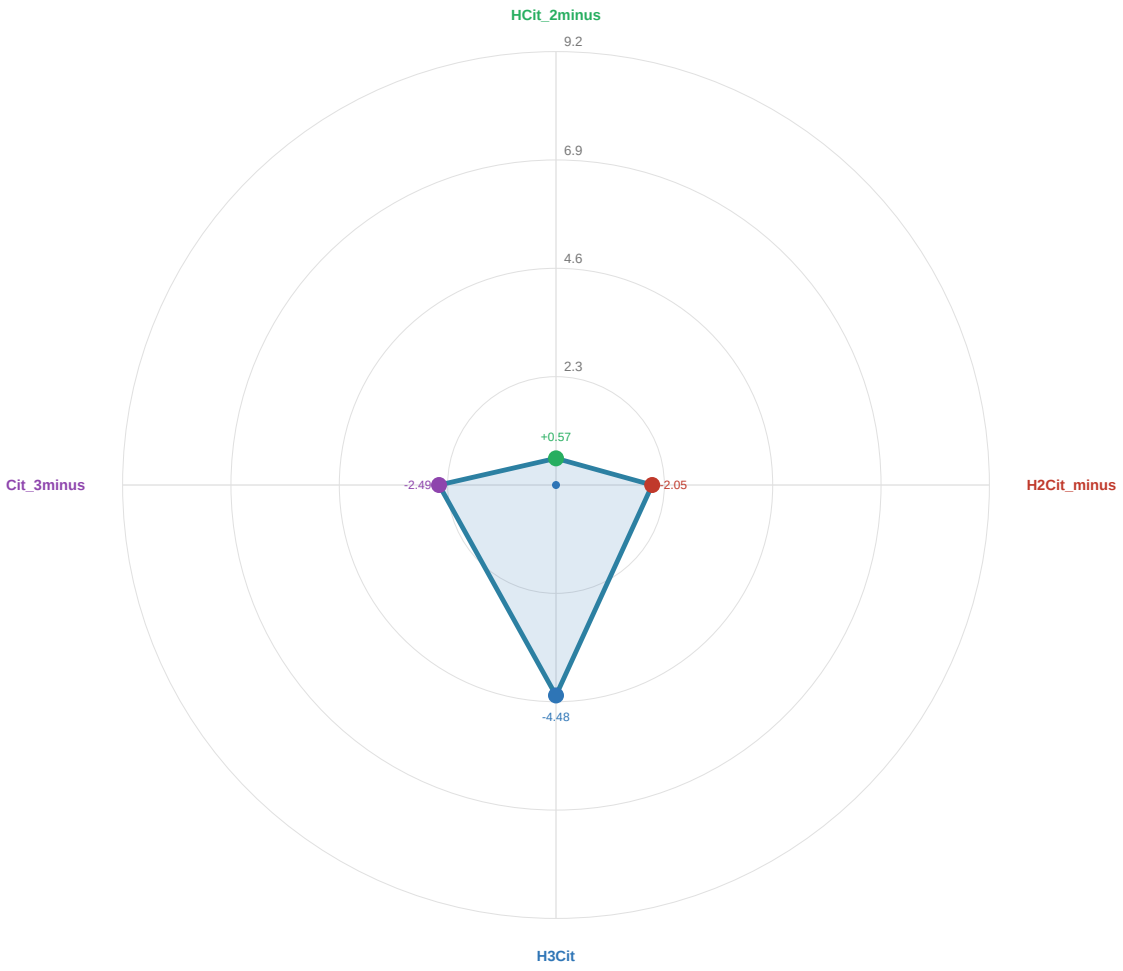
POLAR SLICE — t=77

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 78/200 | ||x||_A = 5.546



From 76: delta ||x||_A = -0.094

Phase = -155.4 deg

H3Cit=-4.48 H2Cit_=-2.05 HCit_2=+0.57 Cit_3m=-2.49

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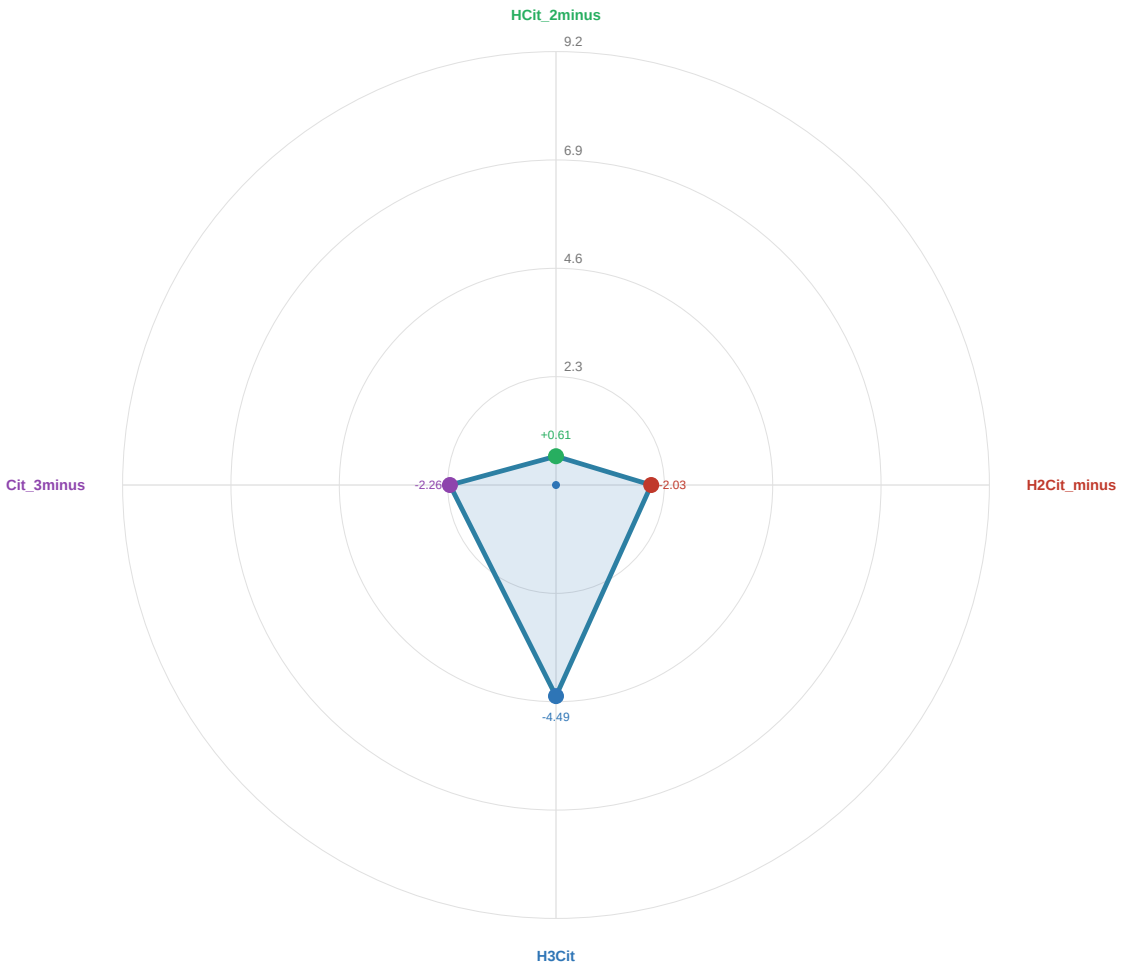
POLAR SLICE — t=78

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 79/200 | ||x||_A = 5.455



From 77: delta ||x||_A = -0.091

Phase = -155.7 deg

H3Cit=-4.49 H2Cit_=-2.03 HCit_2=+0.61 Cit_3m=-2.26

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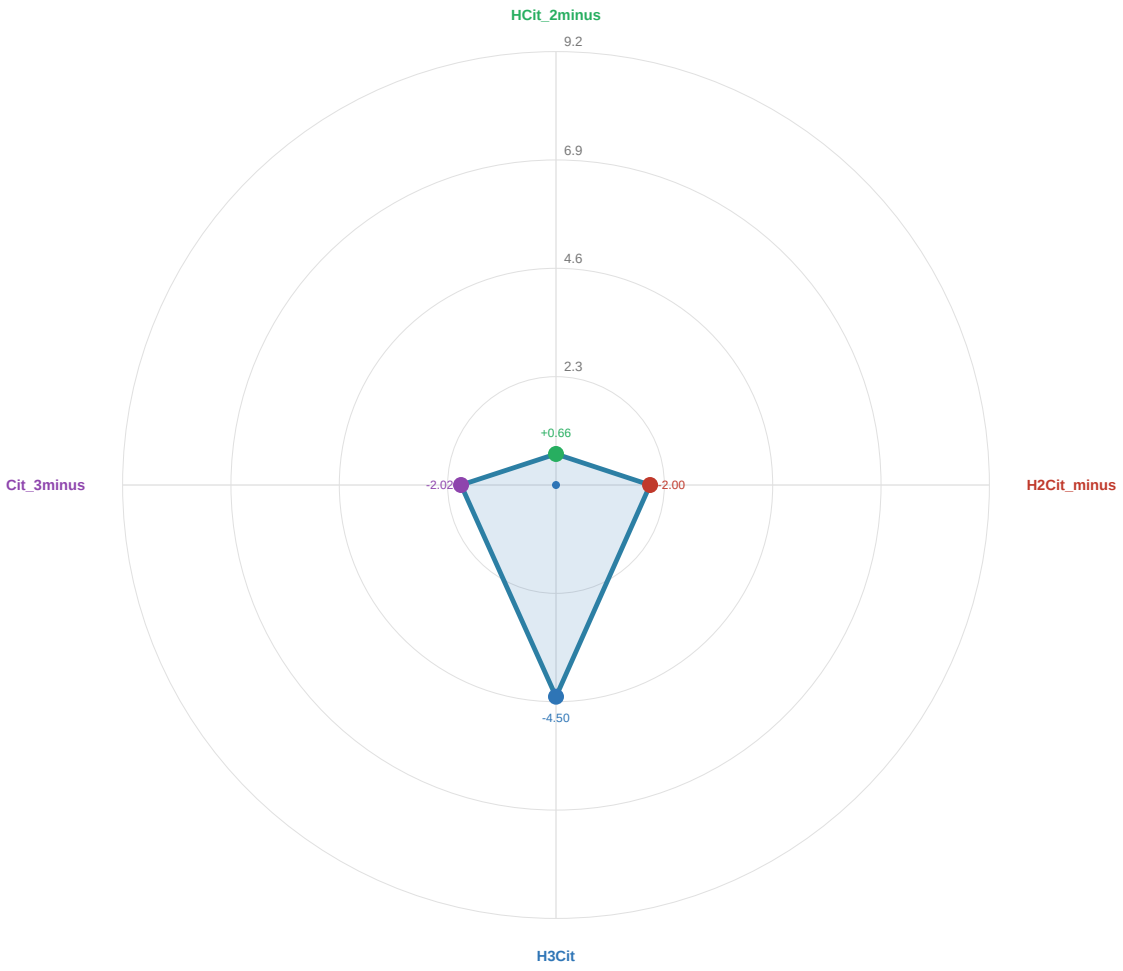
POLAR SLICE — t=79

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 80/200 | ||x||_A = 5.368



From 78: delta ||x||_A = -0.087

Phase = -156.0 deg

H3Cit=-4.50 H2Cit_=-2.00 HCit_2=+0.66 Cit_3m=-2.02

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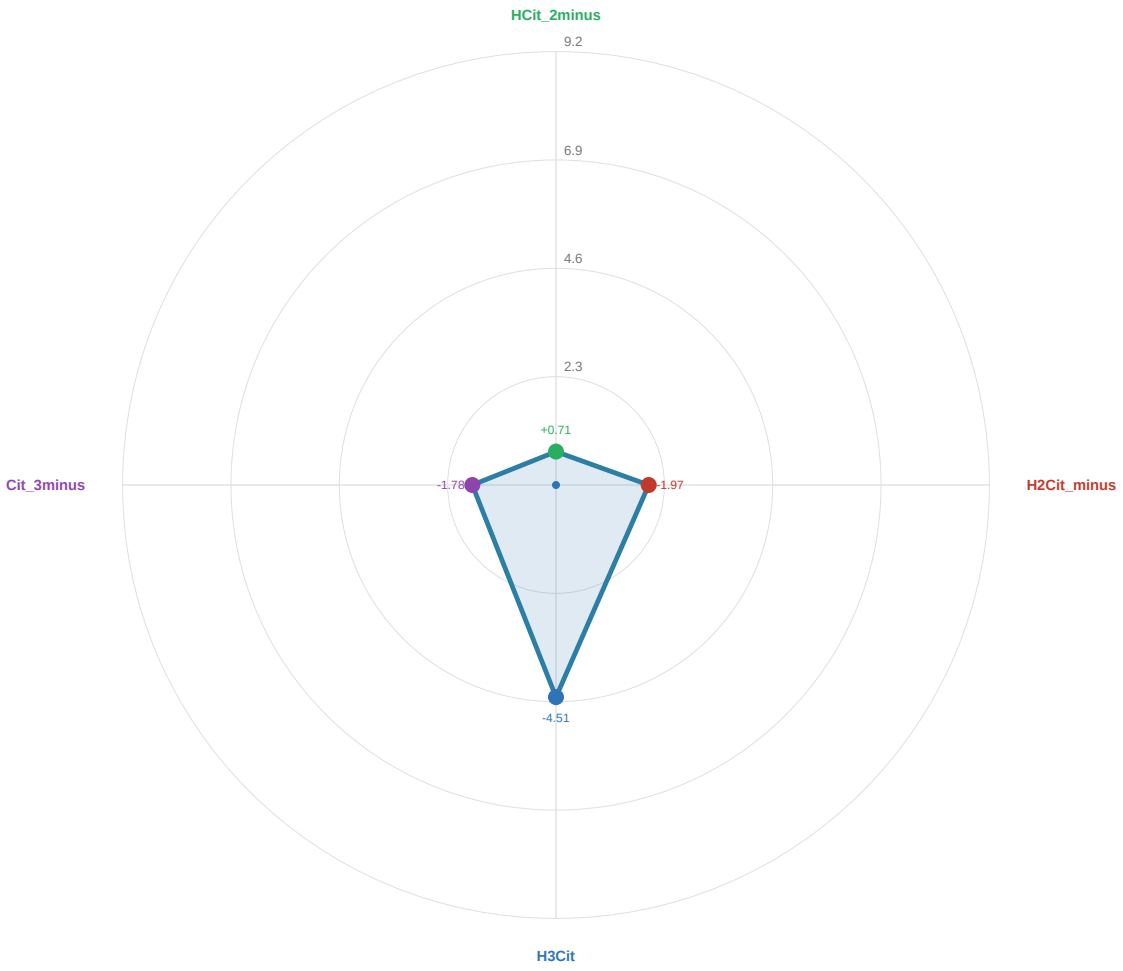
POLAR SLICE — t=80

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 81/200 | ||x||_A = 5.286



From 79: delta ||x||_A = -0.082

Phase = -156.4 deg

H3Cit=-4.51 H2Cit_=-1.97 HCit_2=+0.71 Cit_3m=-1.78

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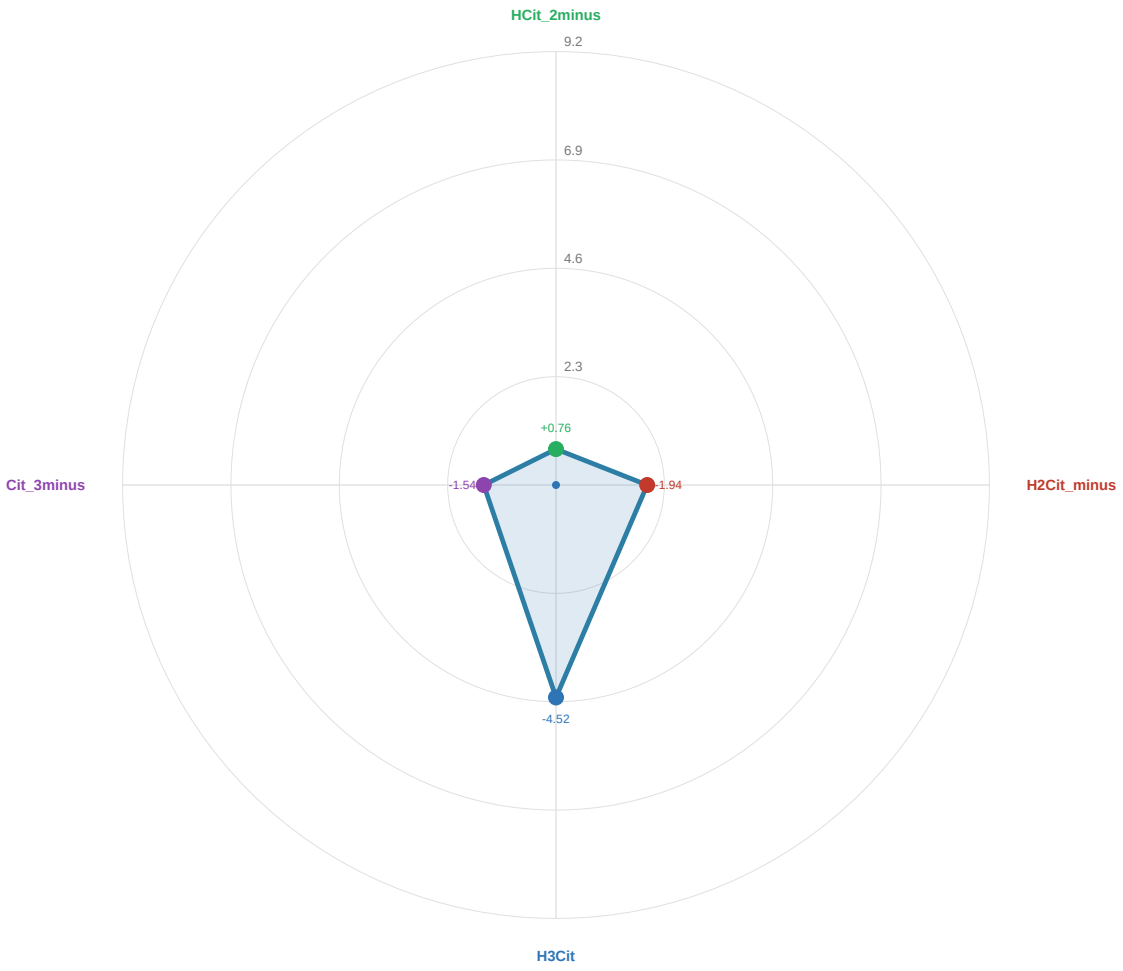
POLAR SLICE — t=81

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 82/200 | ||x||_A = 5.210



From 80: delta ||x||_A = -0.076

Phase = -156.8 deg

H3Cit=-4.52 H2Cit_=-1.94 HCit_2=+0.76 Cit_3m=-1.54

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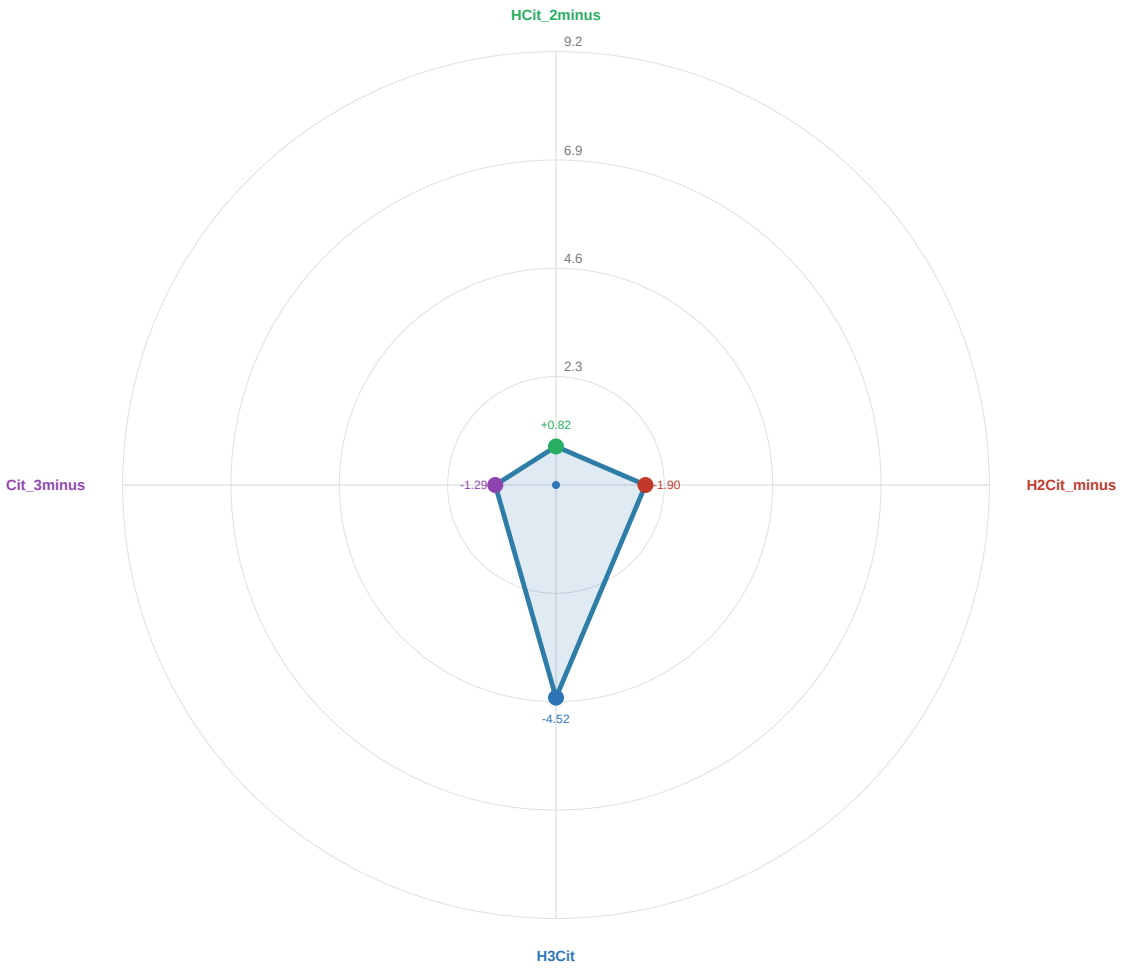
POLAR SLICE — t=82

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 83/200 | ||x||_A = 5.142



From 81: delta ||x||_A = -0.068

Phase = -157.2 deg

H3Cit=-4.52 H2Cit_=-1.90 HCit_2=+0.82 Cit_3m=-1.29

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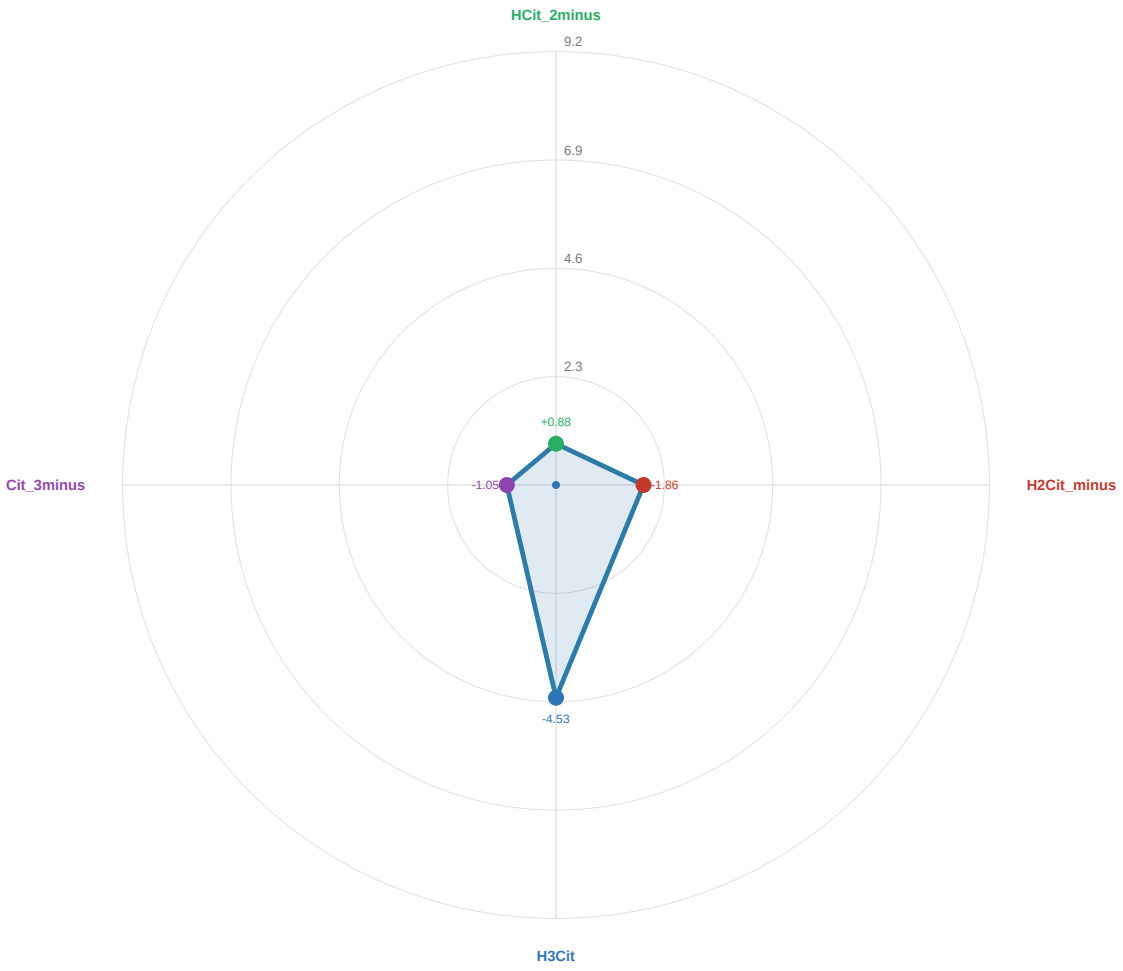
POLAR SLICE — t=83

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 84/200 | ||x||_A = 5.081



From 82: delta ||x||_A = -0.060

Phase = -157.6 deg

H3Cit=-4.53 H2Cit_=-1.86 HCit_2=+0.88 Cit_3m=-1.05

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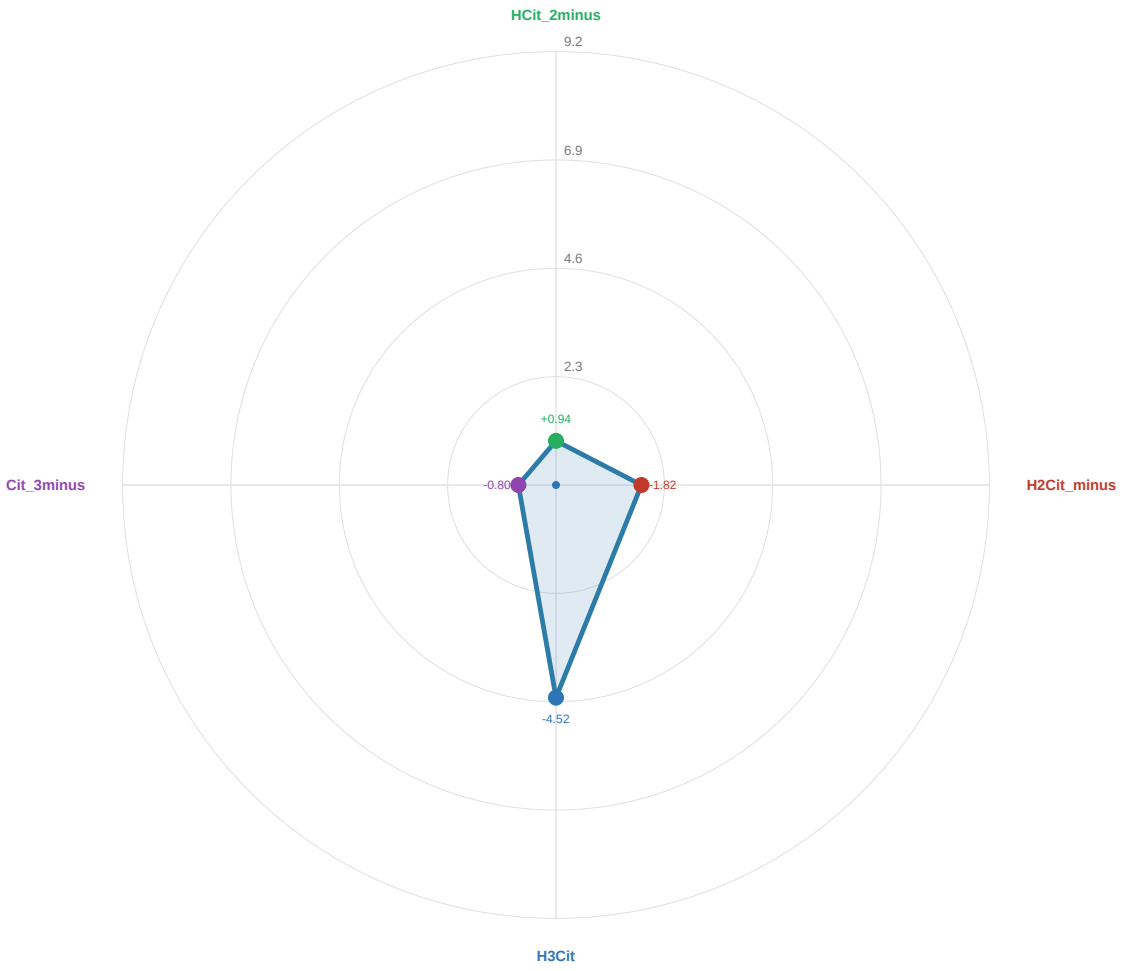
POLAR SLICE — t=84

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 85/200 | ||x||_A = 5.029



From 83: delta ||x||_A = -0.052

Phase = -158.1 deg

H3Cit=-4.52 H2Cit_=-1.82 HCit_2=+0.94 Cit_3m=-0.80

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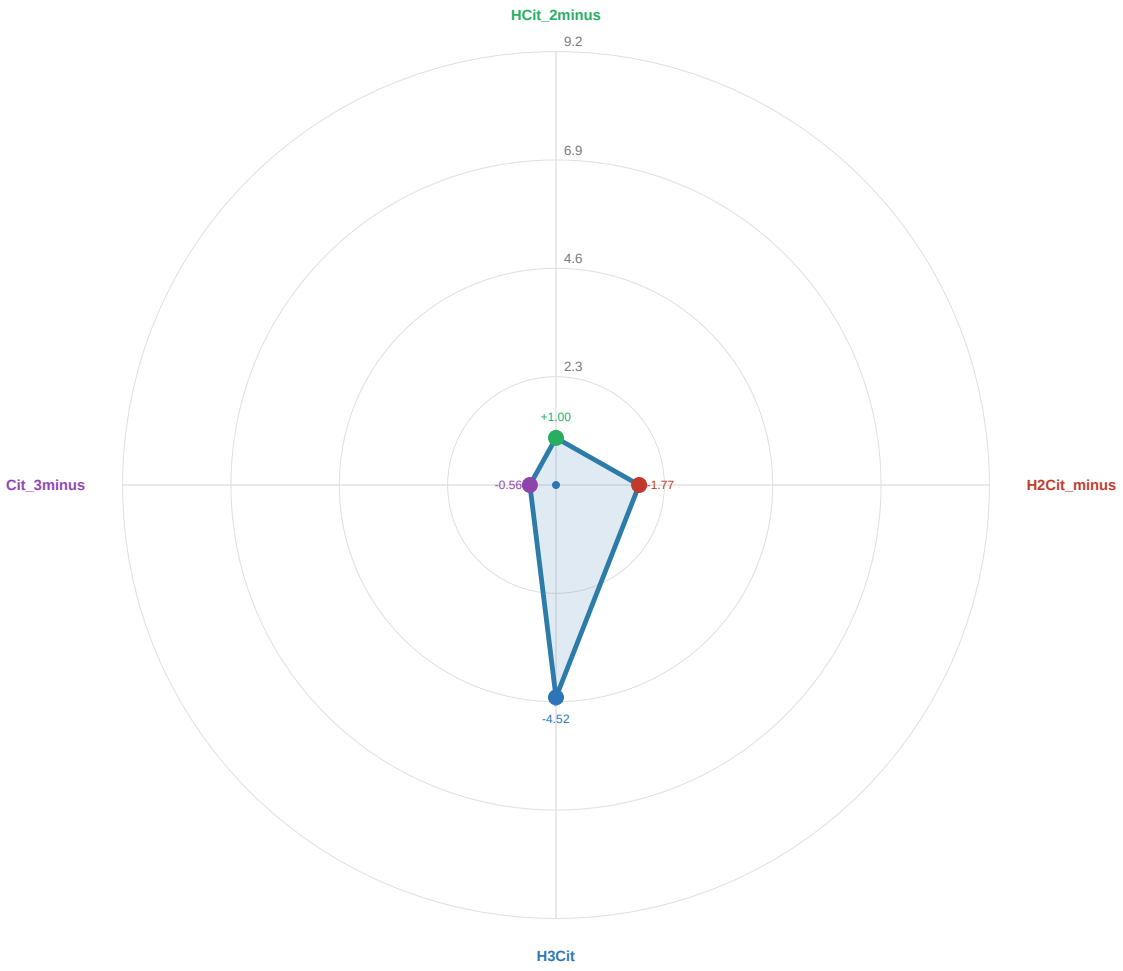
POLAR SLICE — t=85

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 86/200 | ||x||_A = 4.986



From 84: delta ||x||_A = -0.043

Phase = -158.6 deg

H3Cit=-4.52 H2Cit_=-1.77 HCit_2=+1.00 Cit_3m=-0.56

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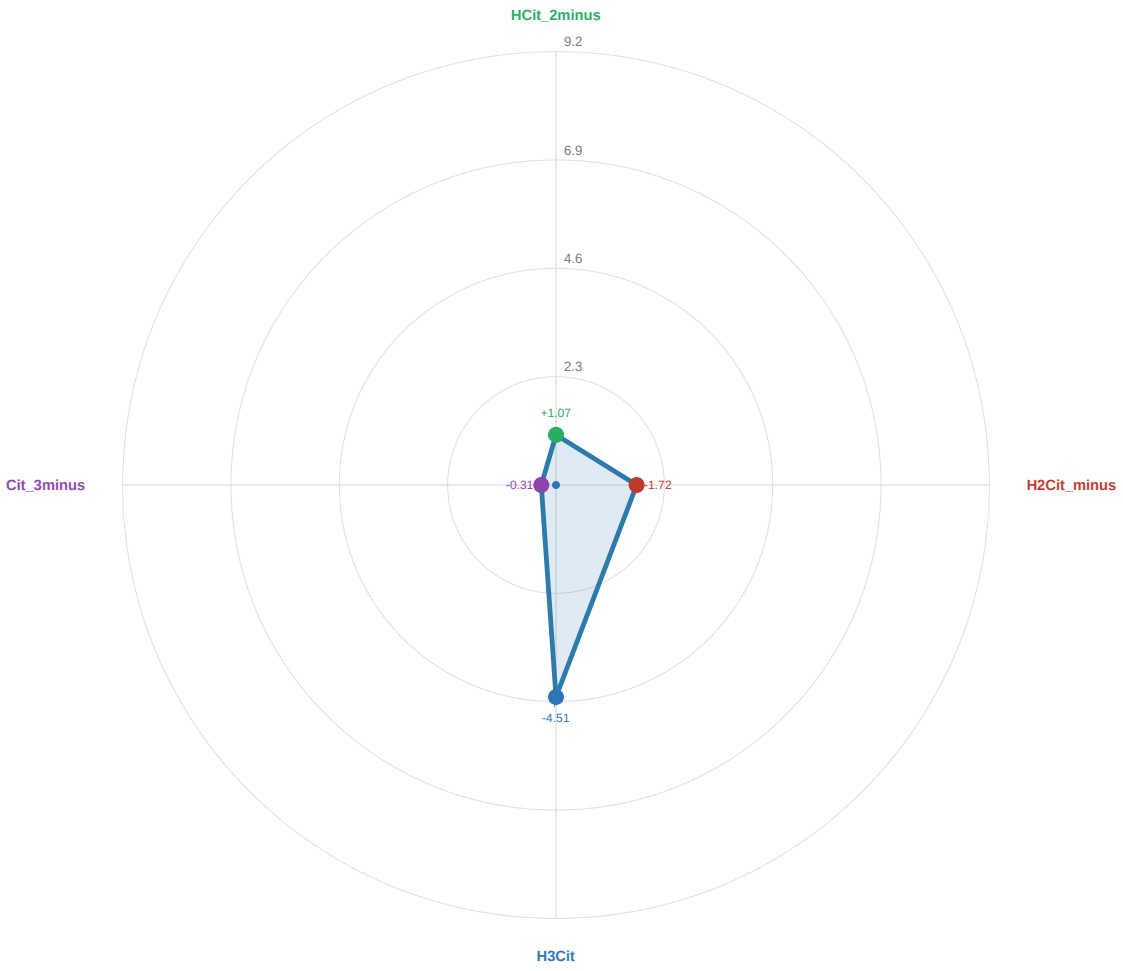
POLAR SLICE — t=86

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 87/200 | ||x||_A = 4.953



From 85: delta ||x||_A = -0.034

Phase = -159.2 deg

H3Cit=-4.51 H2Cit_=-1.72 HCit_2=+1.07 Cit_3m=-0.31

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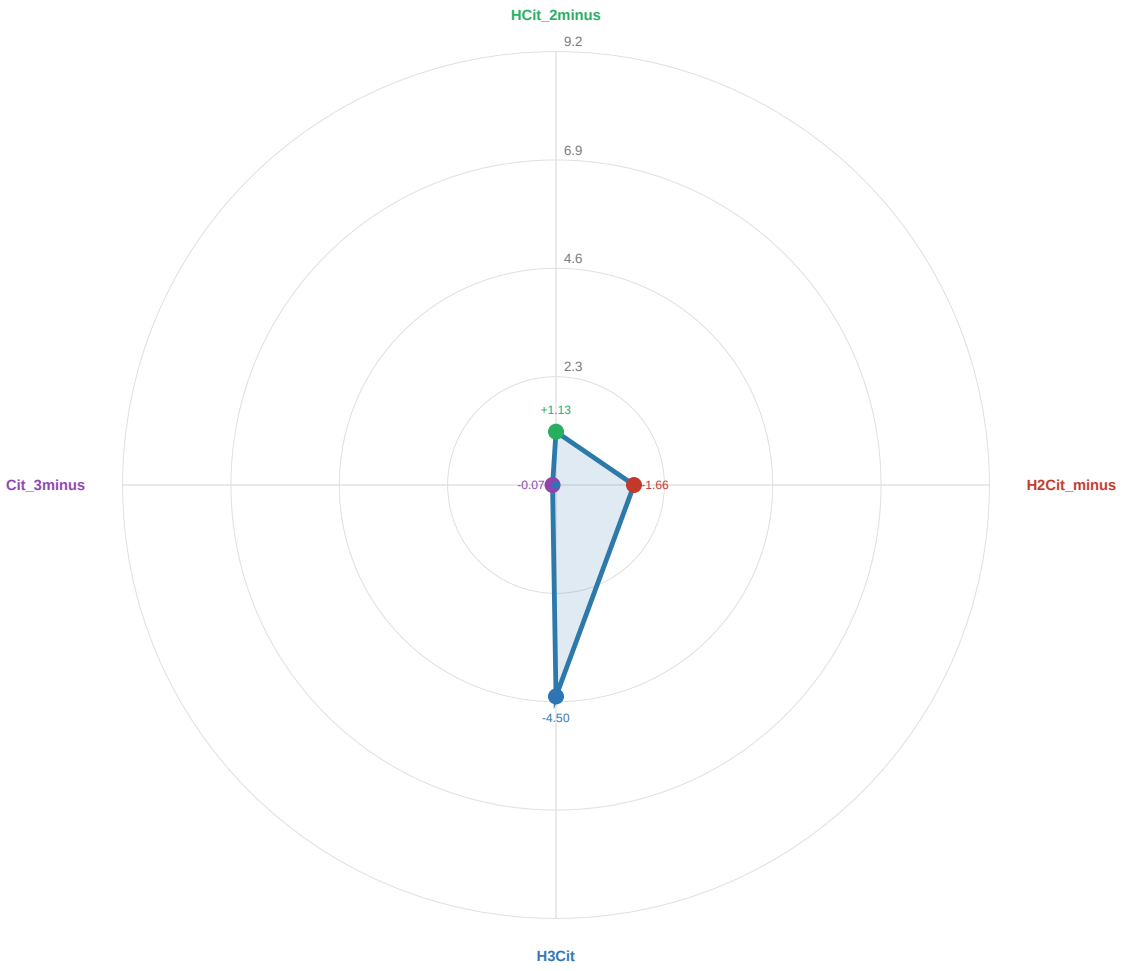
POLAR SLICE — t=87

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 88/200 | ||x||_A = 4.928



From 86: delta ||x||_A = -0.025

Phase = -159.8 deg

H3Cit=-4.50 H2Cit_=-1.66 HCit_2=+1.13 Cit_3m=-0.07

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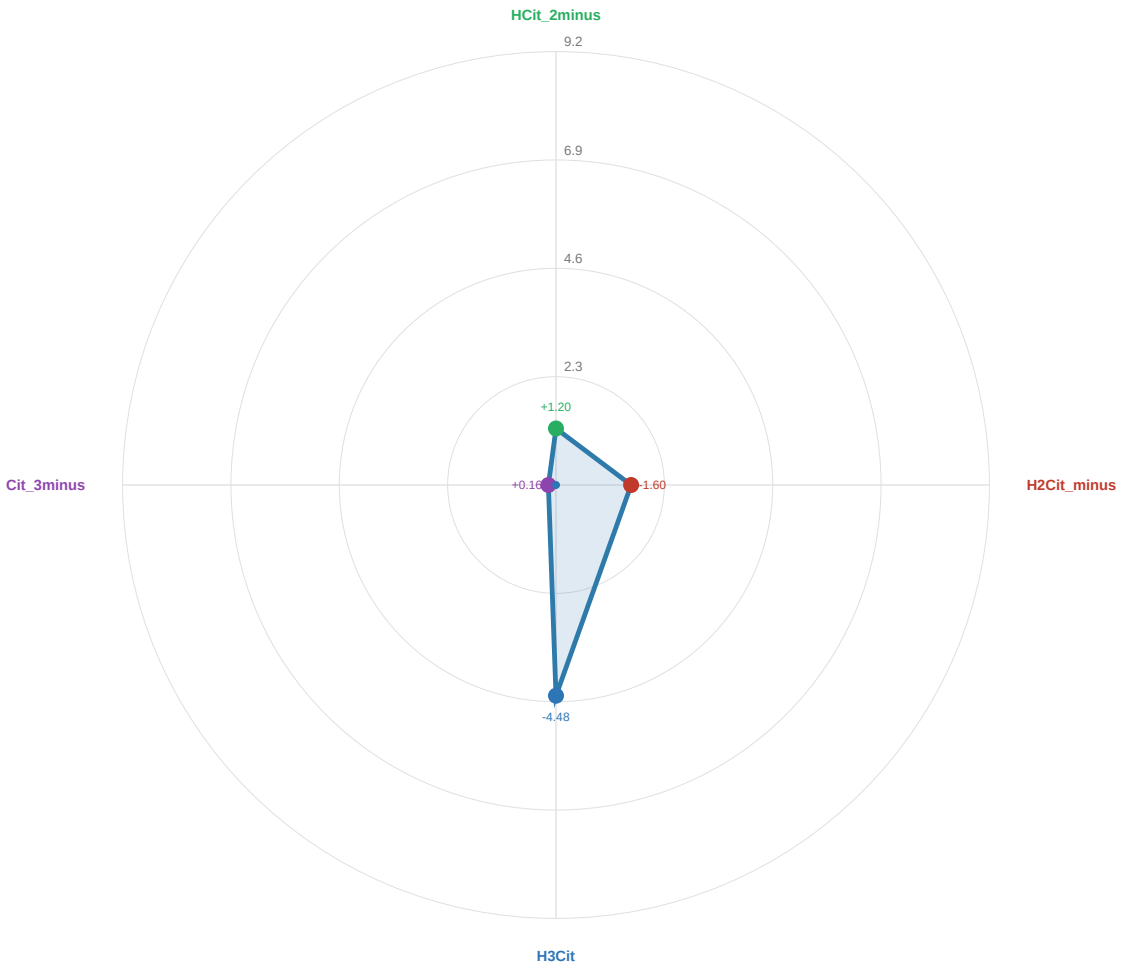
POLAR SLICE — t=88

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 89/200 | ||x||_A = 4.912



From 87: delta ||x||_A = -0.016

Phase = -160.4 deg

H3Cit=-4.48 H2Cit_=-1.60 HCit_2=+1.20 Cit_3m=+0.16

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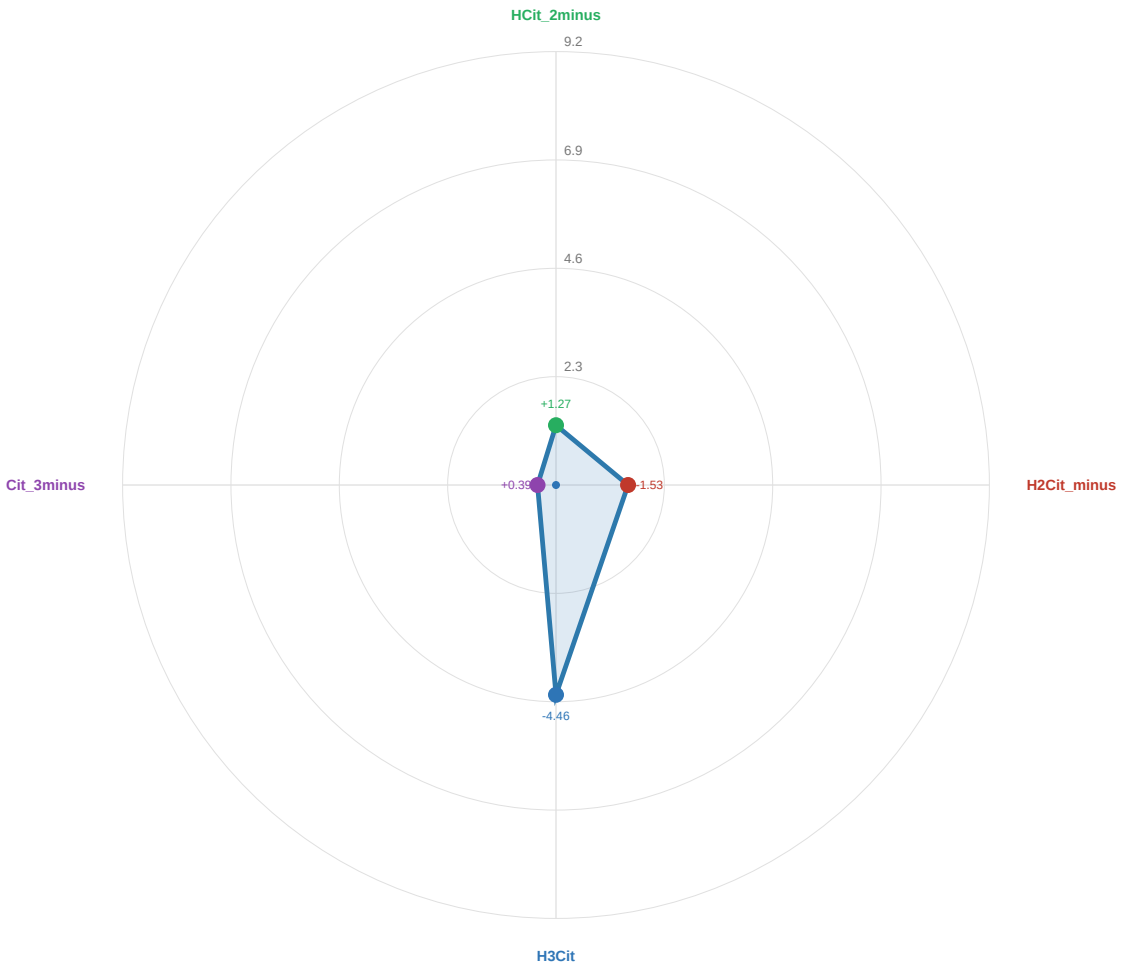
POLAR SLICE — t=89

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 90/200 | ||x||_A = 4.904



From 88: delta ||x||_A = -0.007

Phase = -161.0 deg

H3Cit=-4.46 H2Cit_=-1.53 HCit_2=+1.27 Cit_3m=+0.39

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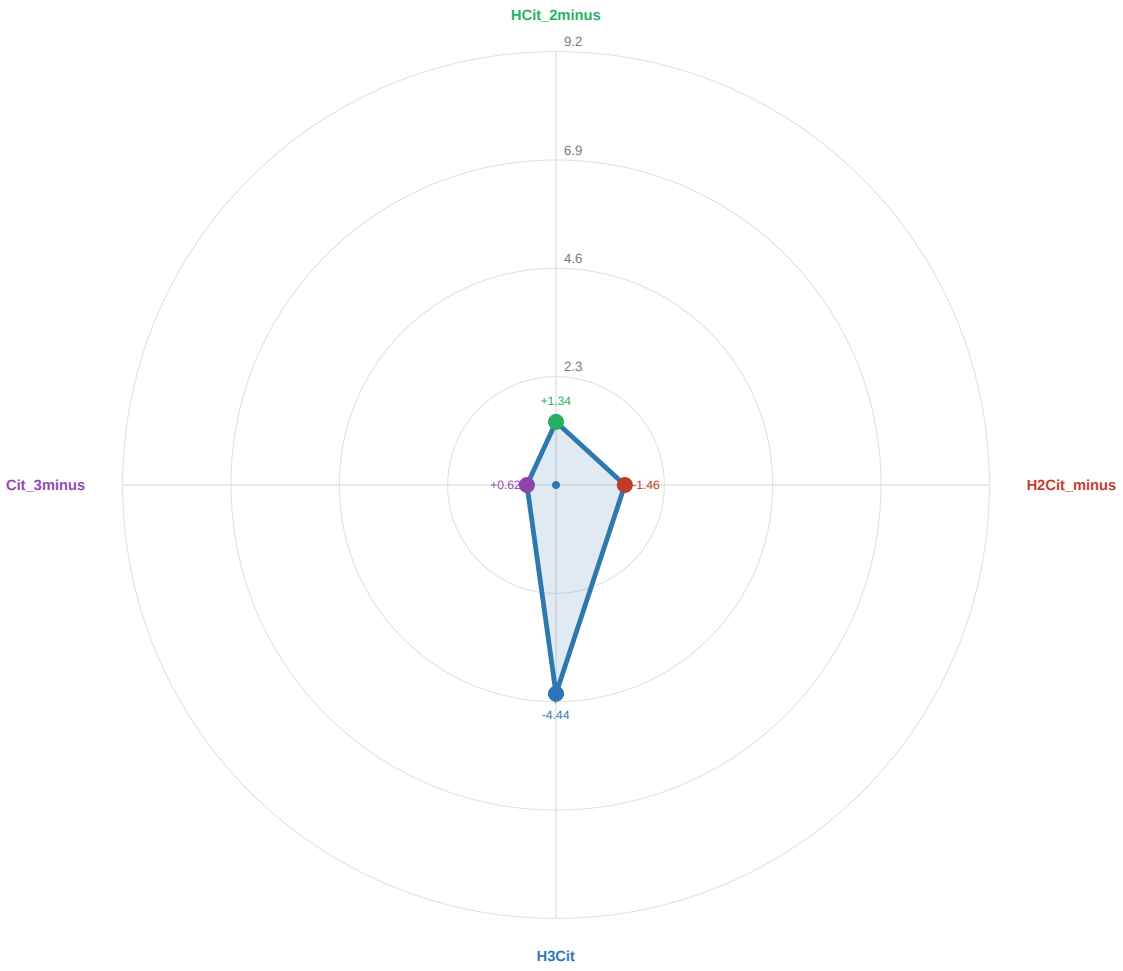
POLAR SLICE — t=90

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 91/200 | ||x||_A = 4.905



From 89: delta ||x||_A = +0.000

Phase = -161.7 deg

H3Cit=-4.44 H2Cit_=-1.46 HCit_2=+1.34 Cit_3m=+0.62

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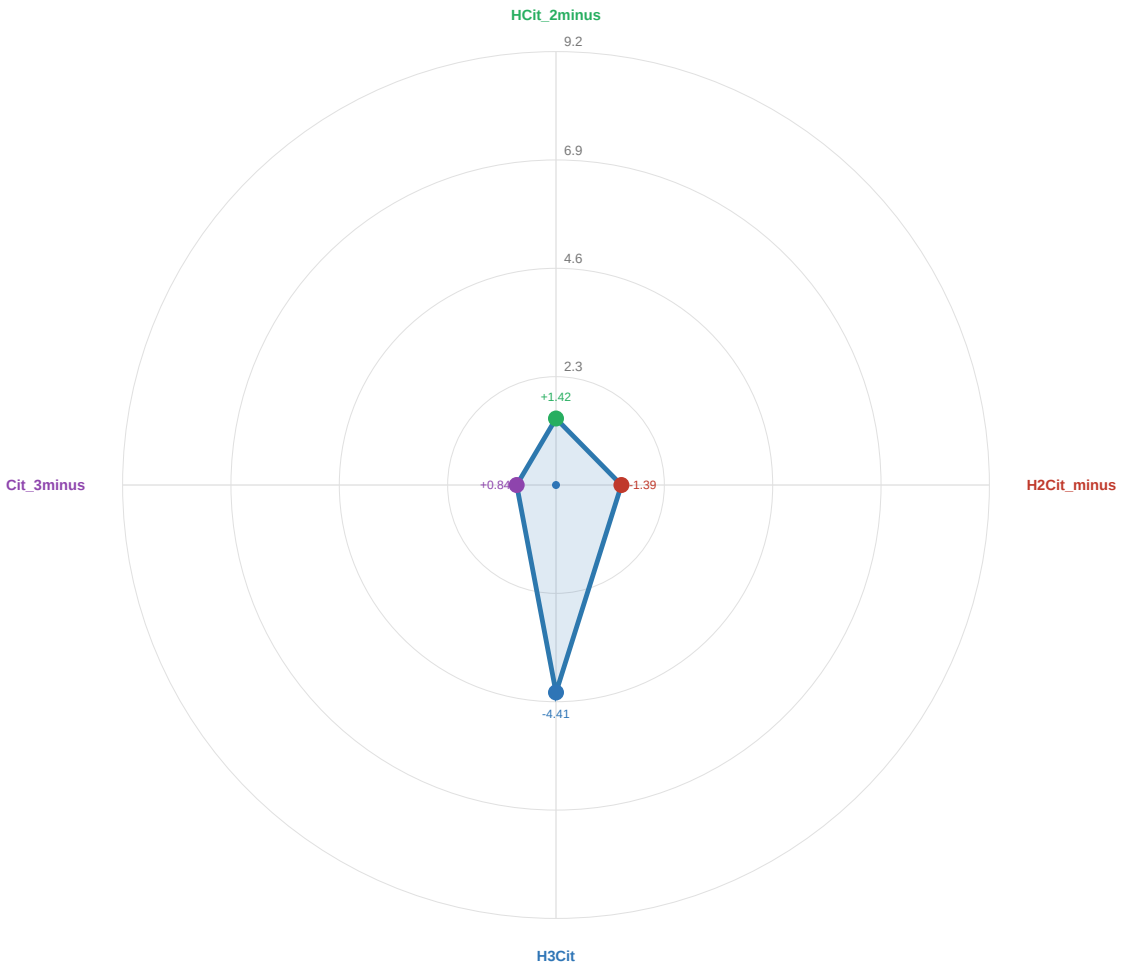
POLAR SLICE — t=91

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 92/200 | ||x||_A = 4.912



From 90: delta ||x||_A = +0.007

Phase = -162.5 deg

H3Cit=-4.41 H2Cit_=-1.39 HCit_2=+1.42 Cit_3m=+0.84

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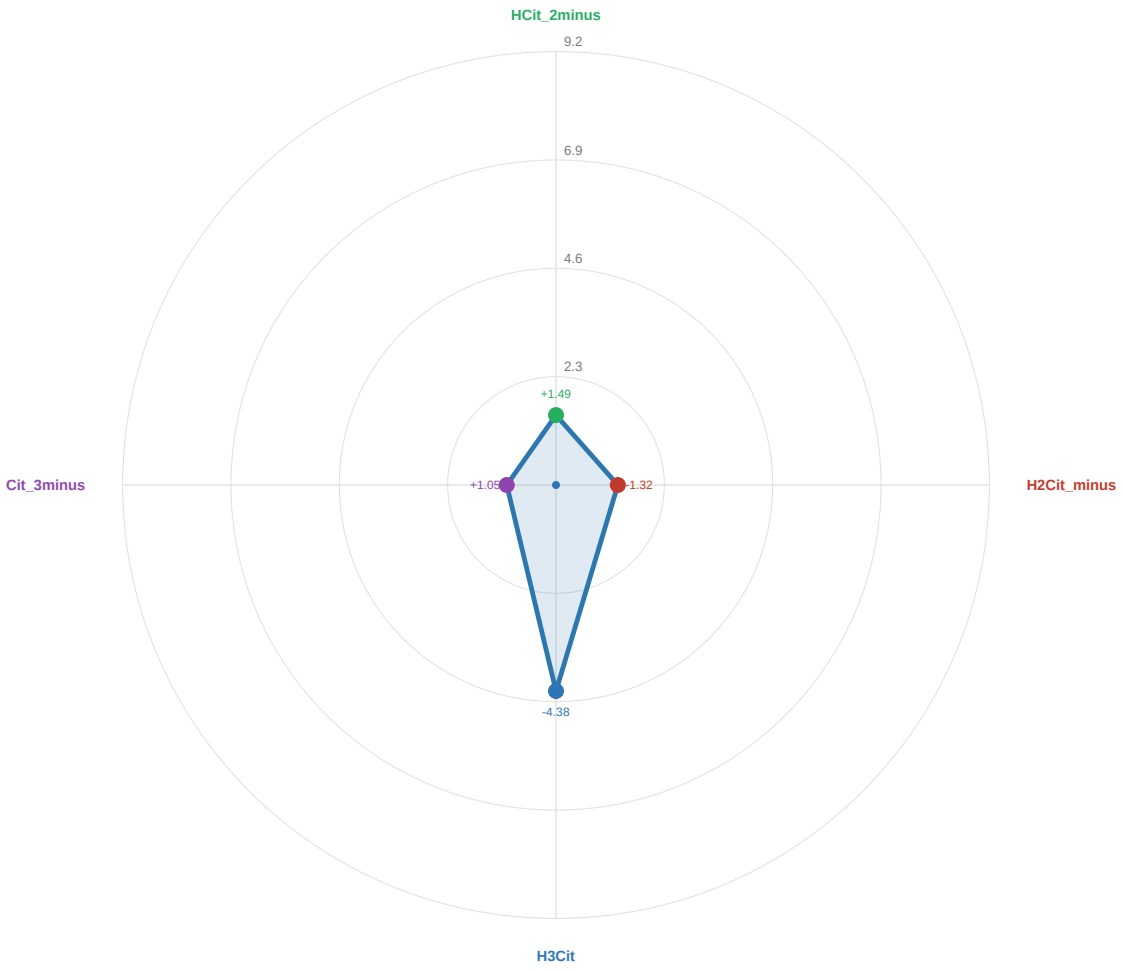
POLAR SLICE — t=92

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 93/200 | ||x||_A = 4.925



From 91: delta ||x||_A = +0.013

Phase = -163.3 deg

H3Cit=-4.38 H2Cit_=-1.32 HCit_2=+1.49 Cit_3m=+1.05

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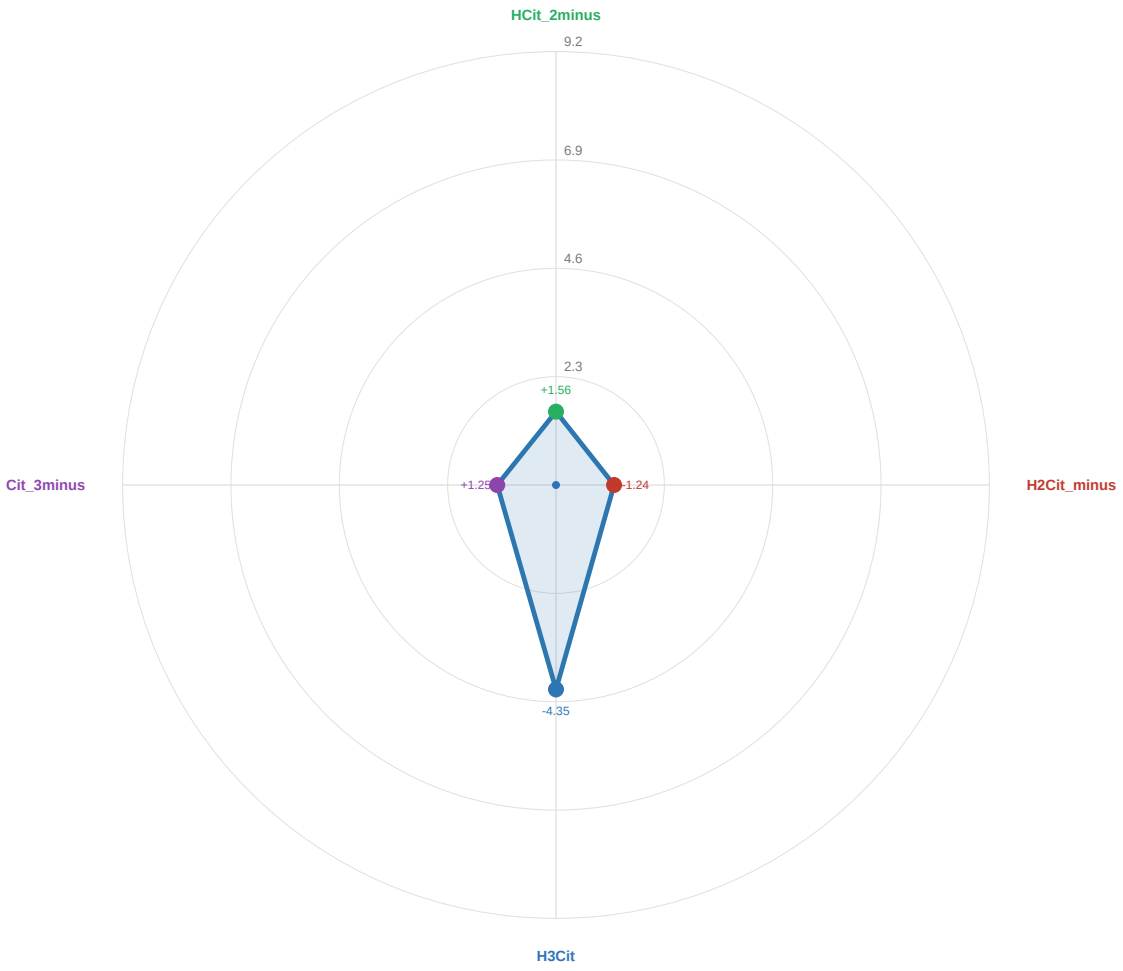
POLAR SLICE — t=93

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 94/200 | ||x||_A = 4.944



From 92: delta ||x||_A = +0.018

Phase = -164.1 deg

H3Cit=-4.35 H2Cit_=-1.24 HCit_2=+1.56 Cit_3m=+1.25

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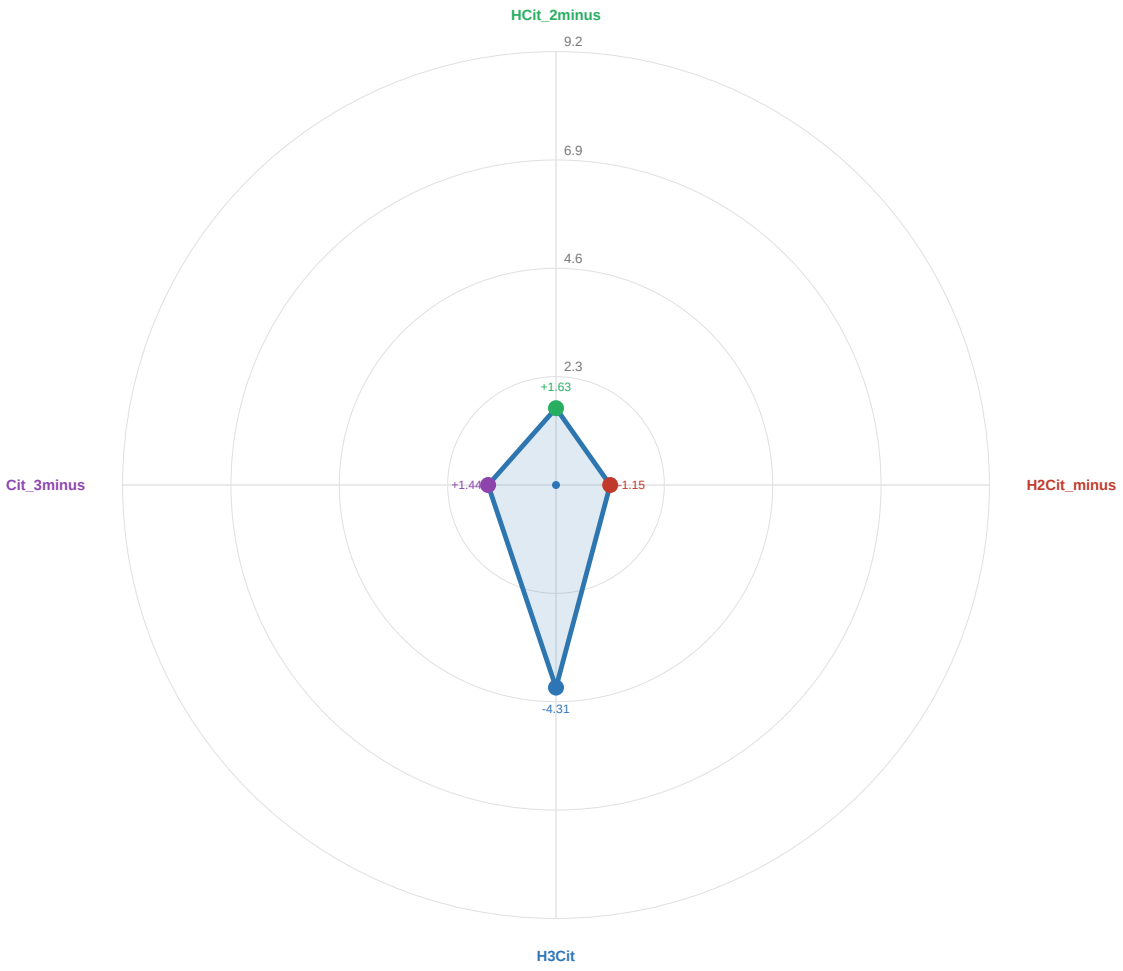
POLAR SLICE — t=94

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 95/200 | ||x||_A = 4.966



From 93: delta ||x||_A = +0.022

Phase = -165.0 deg

H3Cit=-4.31 H2Cit_=-1.15 HCit_2=+1.63 Cit_3m=+1.44

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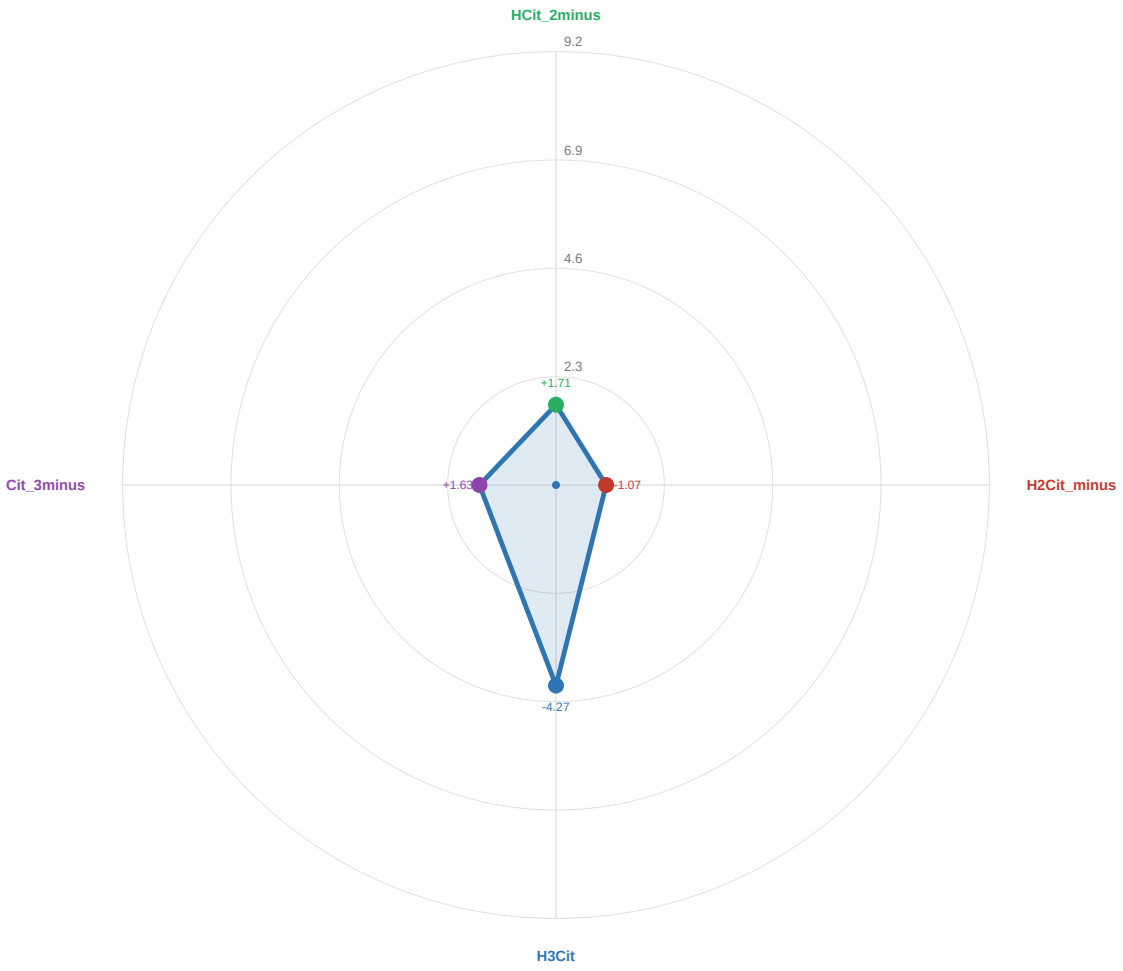
POLAR SLICE — t=95

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 96/200 | ||x||_A = 4.992



From 94: delta ||x||_A = +0.025

Phase = -166.0 deg

H3Cit=-4.27 H2Cit_=-1.07 HCit_2=+1.71 Cit_3m=+1.63

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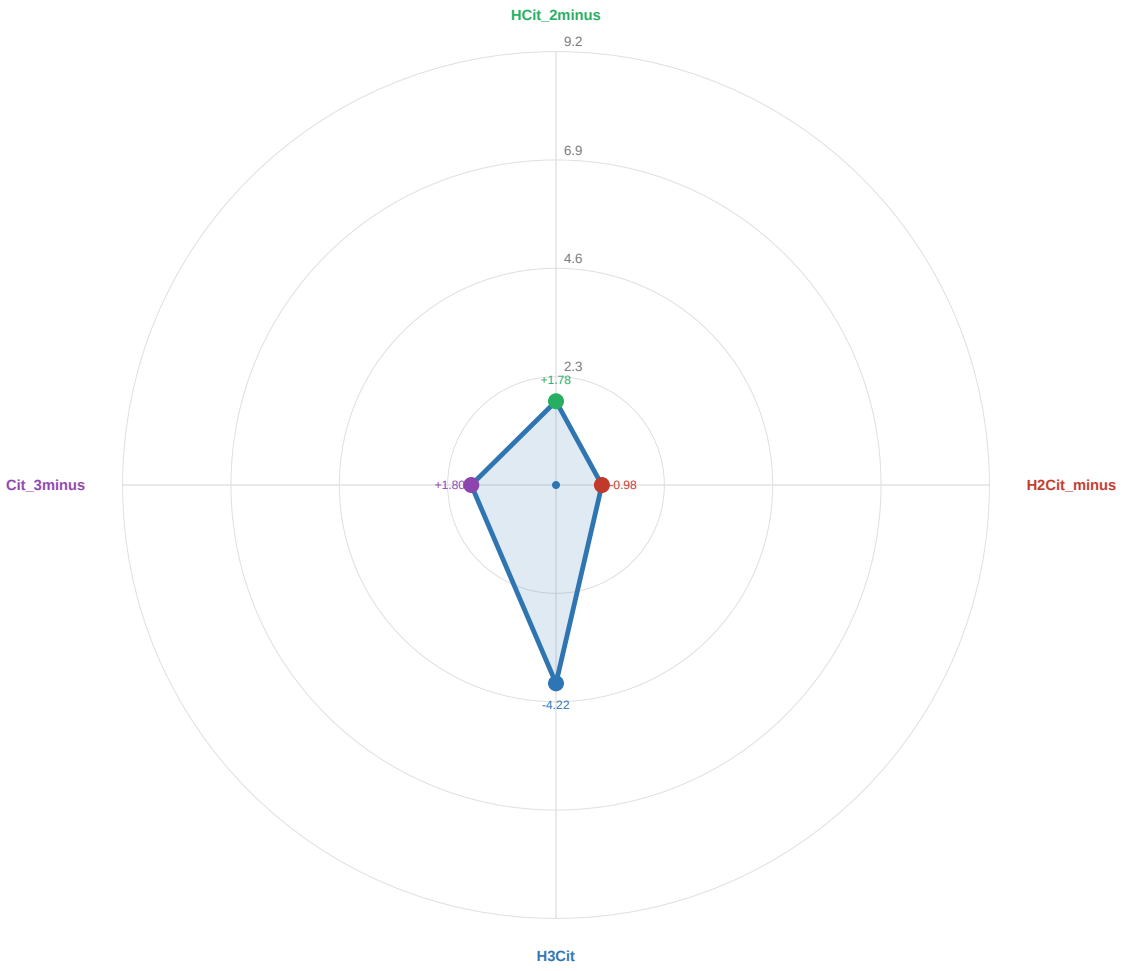
POLAR SLICE — t=96

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 97/200 | ||x||_A = 5.019



From 95: delta ||x||_A = +0.027

Phase = -167.0 deg

H3Cit=-4.22 H2Cit_=-0.98 HCit_2=+1.78 Cit_3m=+1.80

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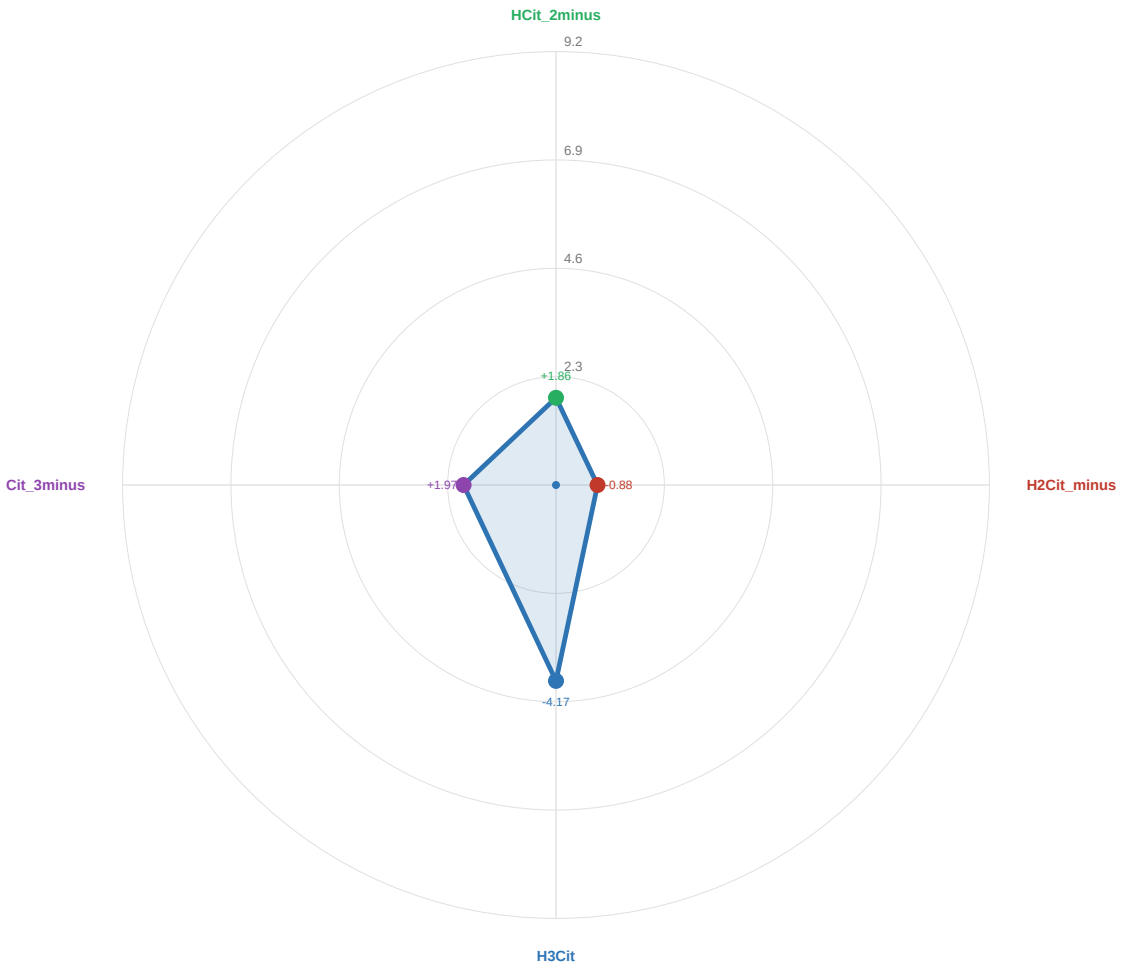
POLAR SLICE — t=97

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 98/200 | ||x||_A = 5.046



From 96: delta ||x||_A = +0.028

Phase = -168.0 deg

H3Cit=-4.17 H2Cit_=-0.88 HCit_2=+1.86 Cit_3m=+1.97

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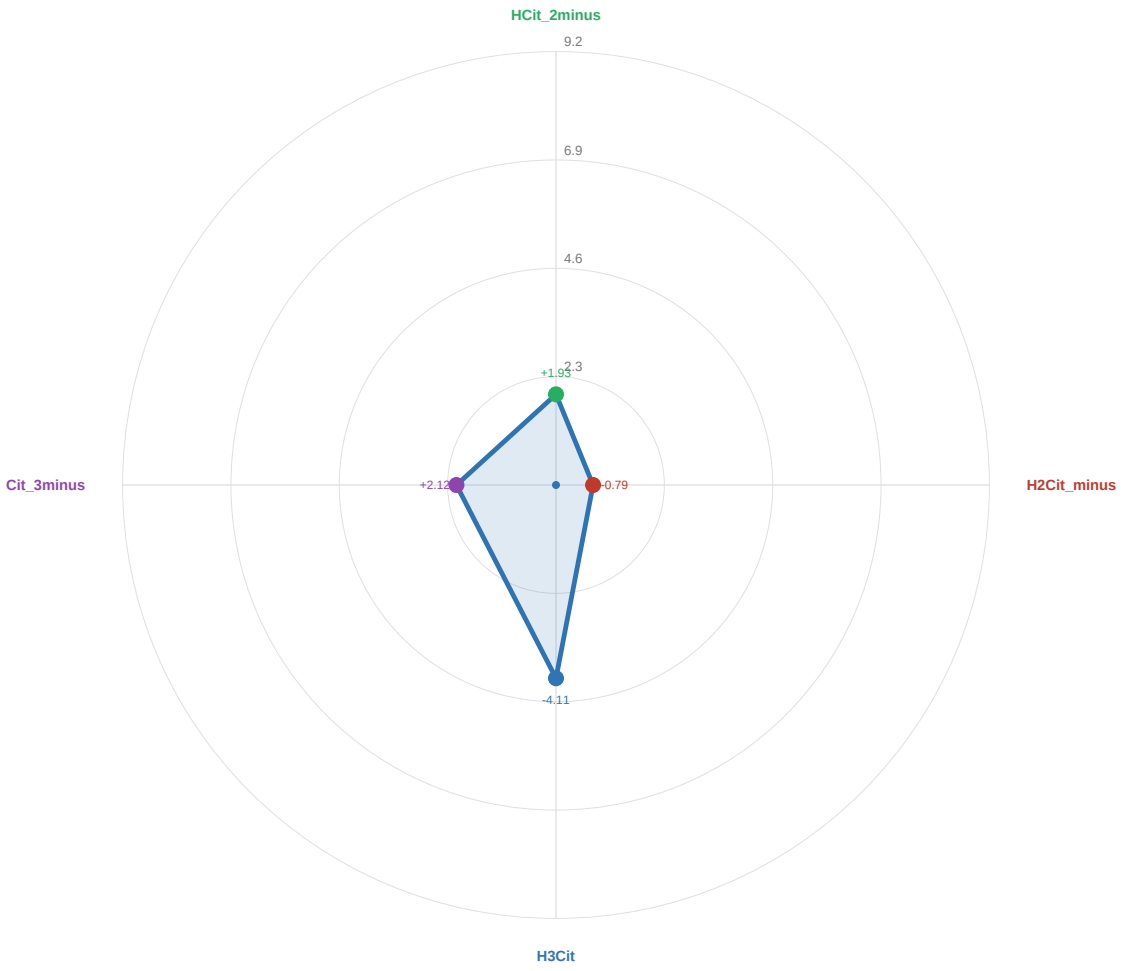
POLAR SLICE — t=98

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 99/200 | ||x||_A = 5.074



From 97: delta ||x||_A = +0.027

Phase = -169.1 deg

H3Cit=-4.11 H2Cit_=-0.79 HCit_2=+1.93 Cit_3m=+2.12

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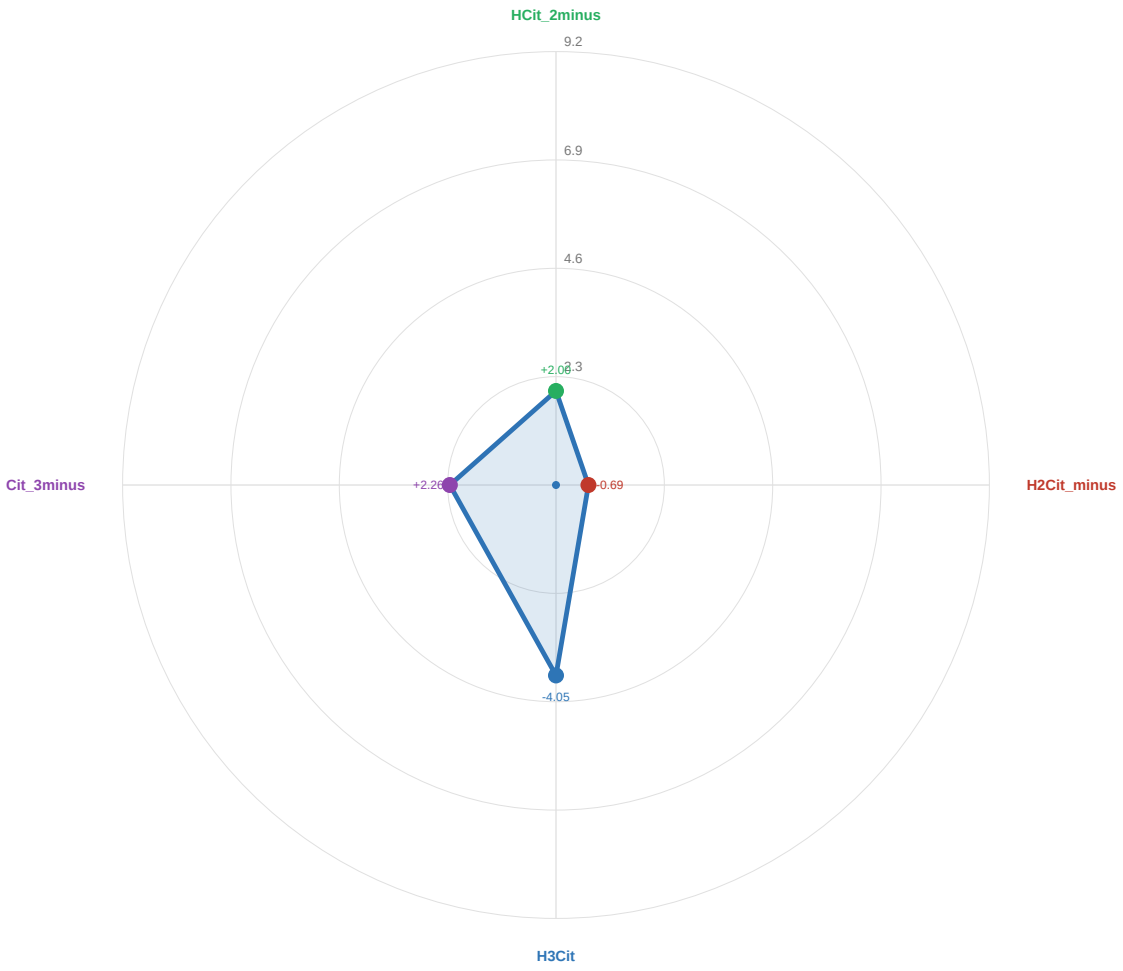
POLAR SLICE — t=99

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 100/200 | ||x||_A = 5.099



From 98: delta ||x||_A = +0.026

Phase = -170.3 deg

H3Cit=-4.05 H2Cit_=-0.69 HCit_2=+2.00 Cit_3m=+2.26

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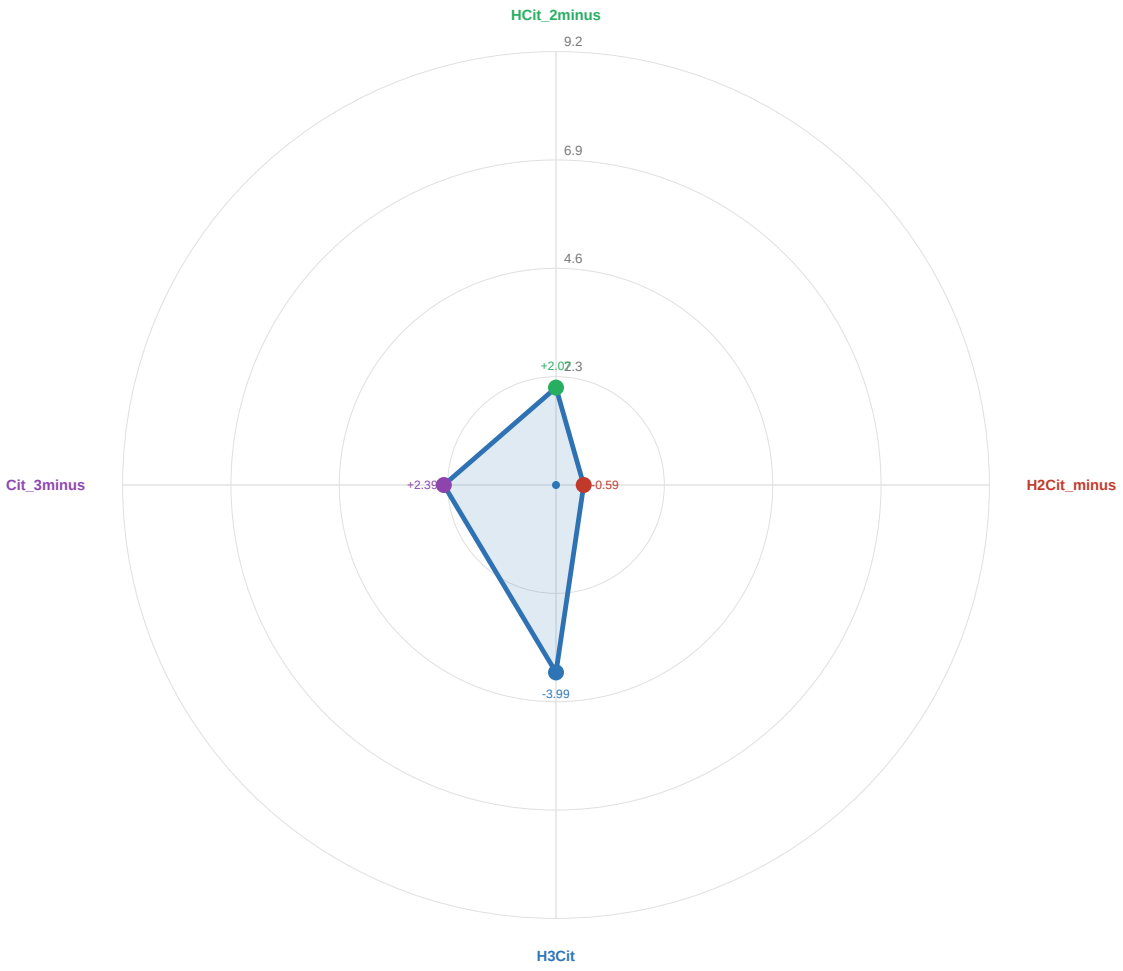
POLAR SLICE — t=100

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 101/200 | ||x||_A = 5.123



From 99: delta ||x||_A = +0.023

Phase = -171.6 deg

H3Cit=-3.99 H2Cit_=-0.59 HCit_2=+2.07 Cit_3m=+2.39

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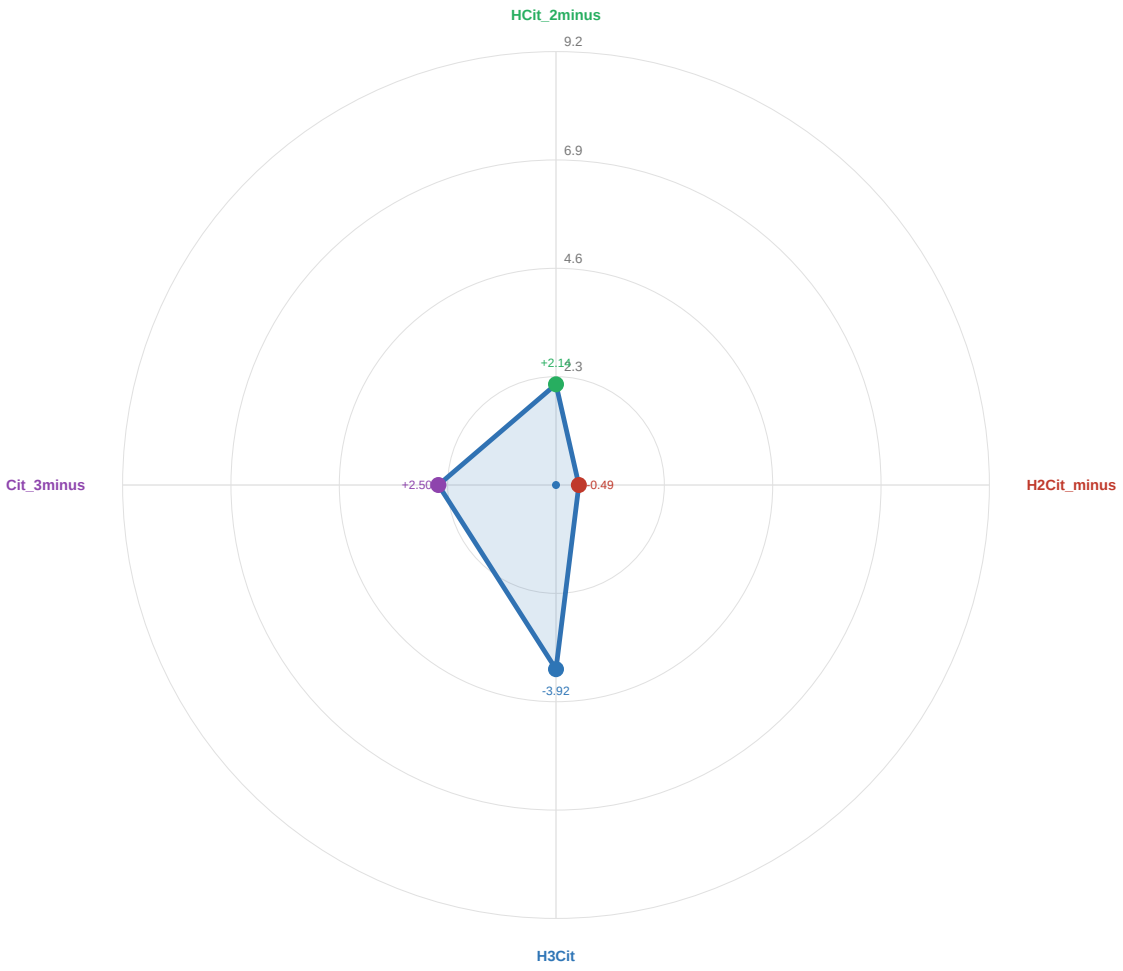
POLAR SLICE — t=101

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 102/200 | ||x||_A = 5.143



From 100: delta ||x||_A = +0.020

Phase = -172.9 deg

H3Cit=-3.92 H2Cit_=-0.49 HCit_2=+2.14 Cit_3m=+2.50

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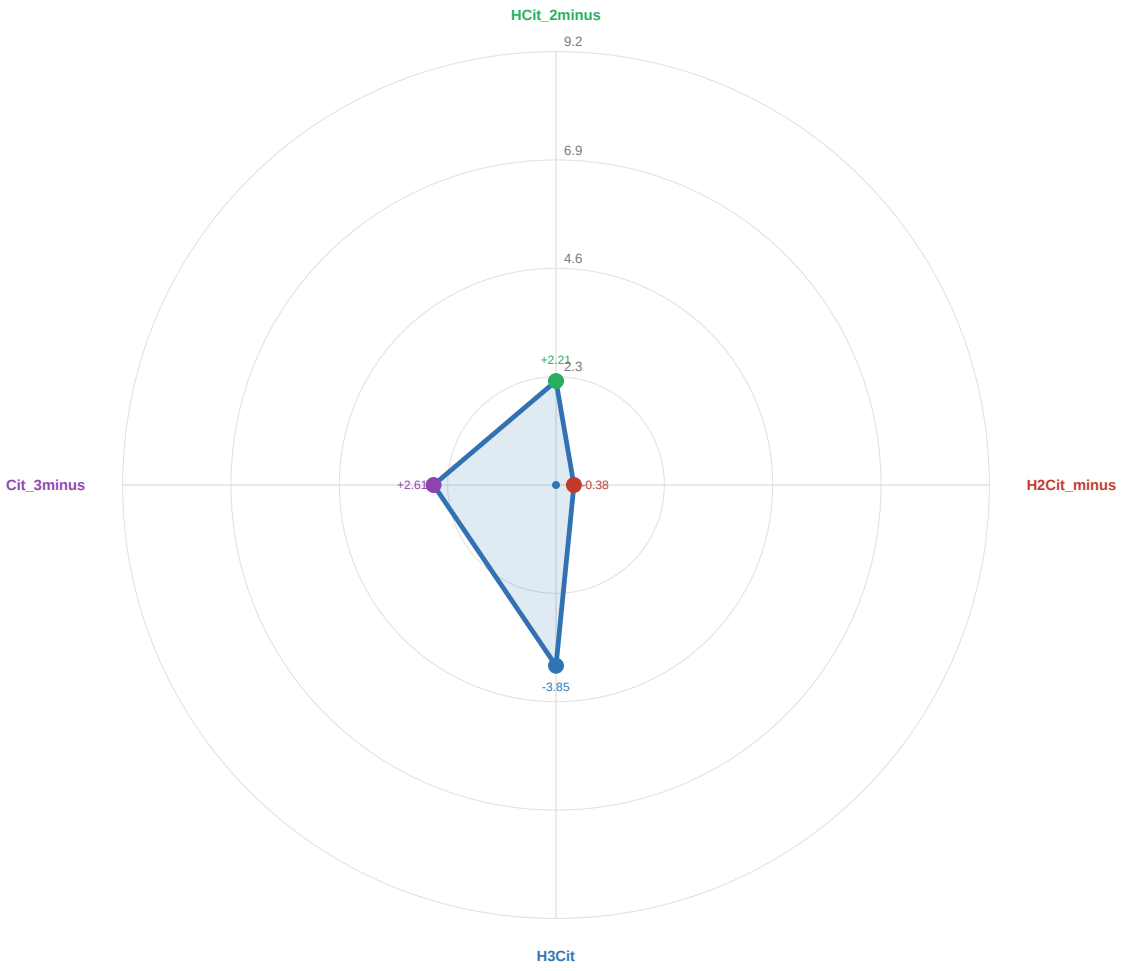
POLAR SLICE — t=102

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 103/200 | ||x||_A = 5.159



From 101: delta ||x||_A = +0.016

Phase = -174.3 deg

H3Cit=-3.85 H2Cit_=-0.38 HCit_2=+2.21 Cit_3m=+2.61

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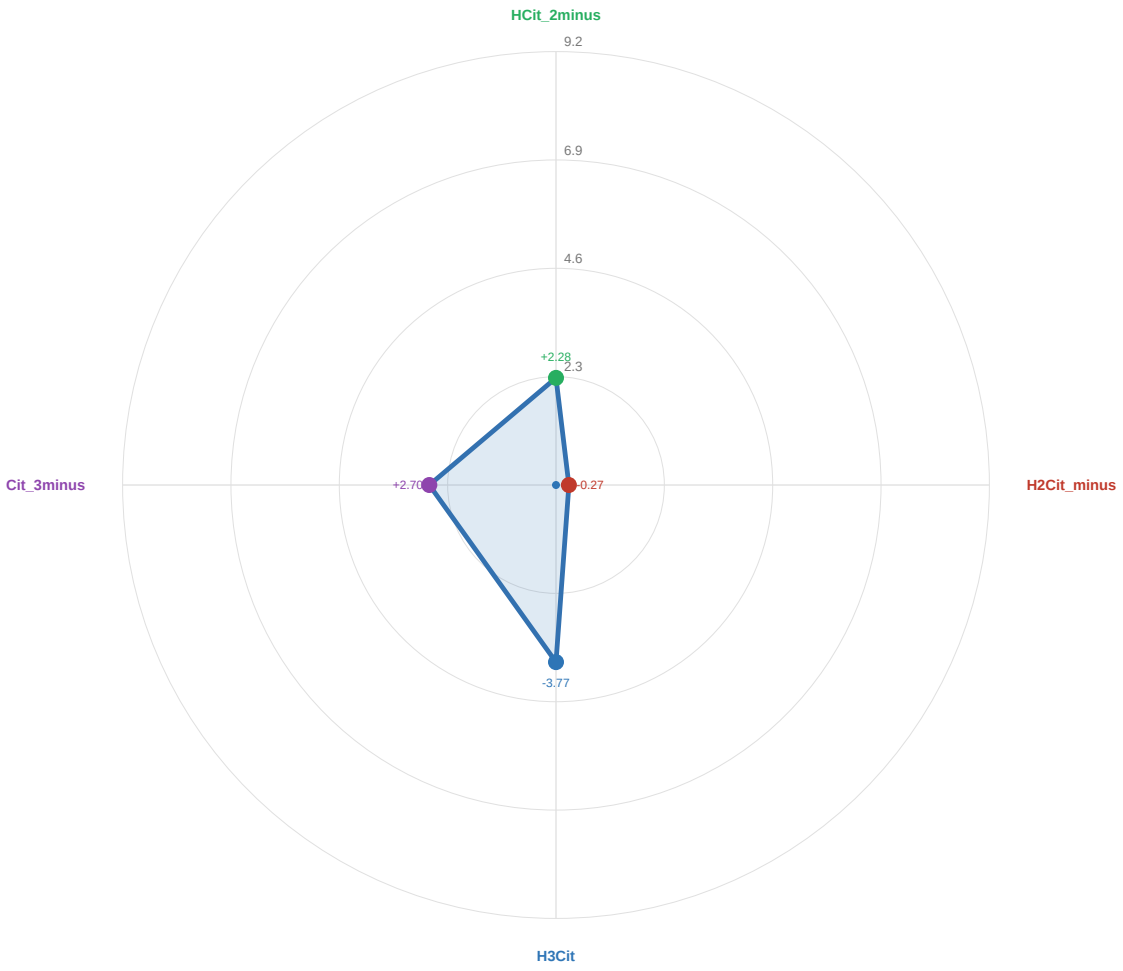
POLAR SLICE — t=103

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 104/200 | ||x||_A = 5.171



From 102: delta ||x||_A = +0.012

Phase = -175.8 deg

H3Cit=-3.77 H2Cit_=-0.27 HCit_2=+2.28 Cit_3m=+2.70

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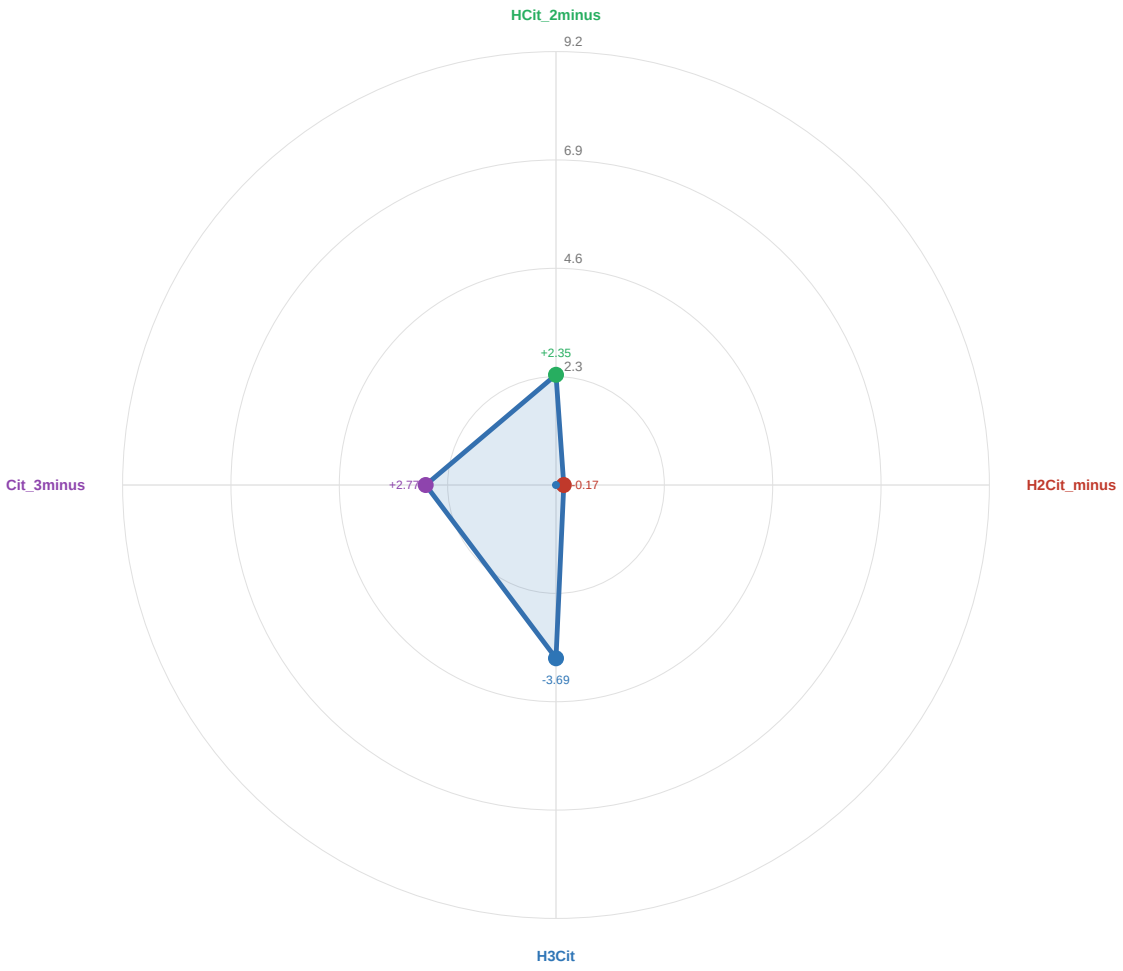
POLAR SLICE — t=104

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 105/200 | ||x||_A = 5.178



From 103: delta ||x||_A = +0.007

Phase = -177.4 deg

H3Cit=-3.69 H2Cit_=-0.17 HCit_2=+2.35 Cit_3m=+2.77

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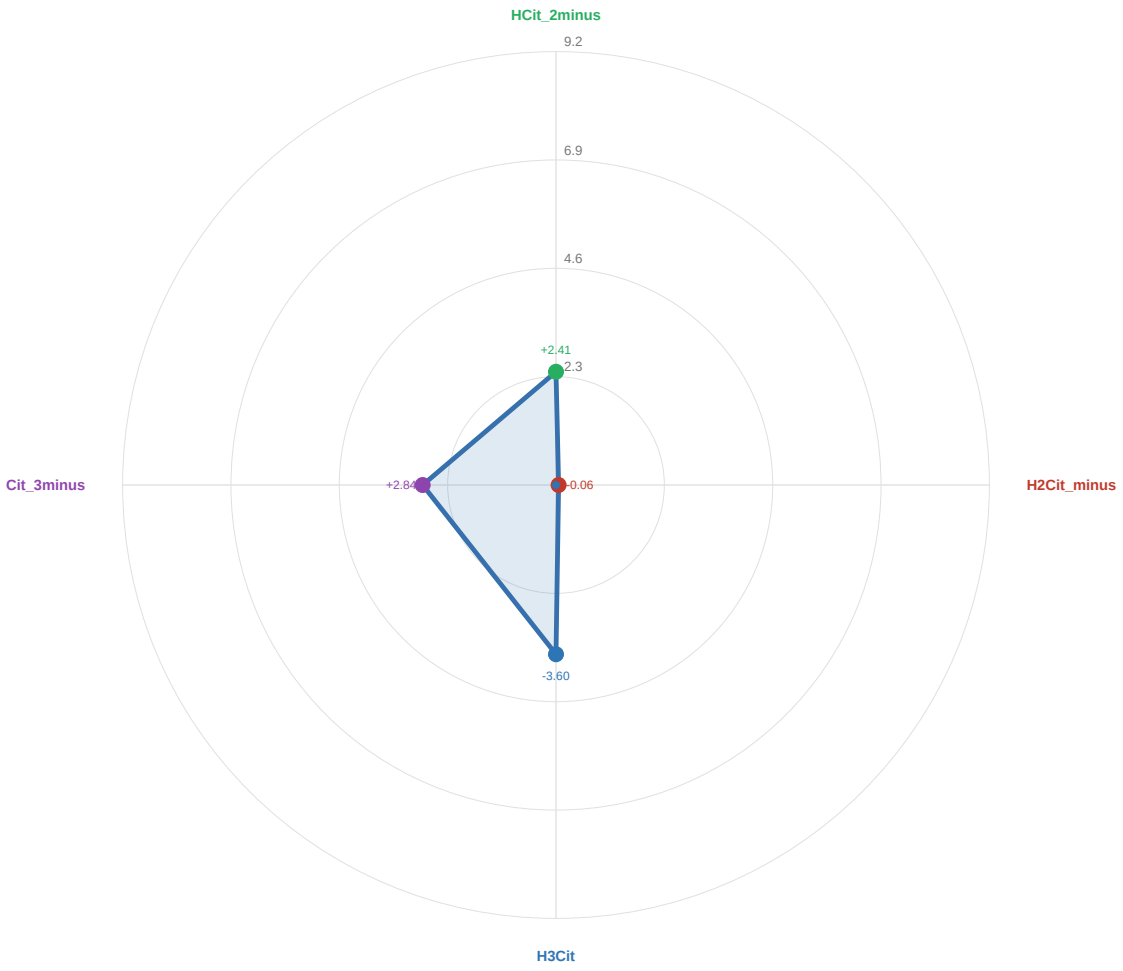
POLAR SLICE — t=105

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 106/200 | ||x||_A = 5.179



From 104: delta ||x||_A = +0.001

Phase = -179.1 deg

H3Cit=-3.60 H2Cit_=-0.06 HCit_2=+2.41 Cit_3m=+2.84

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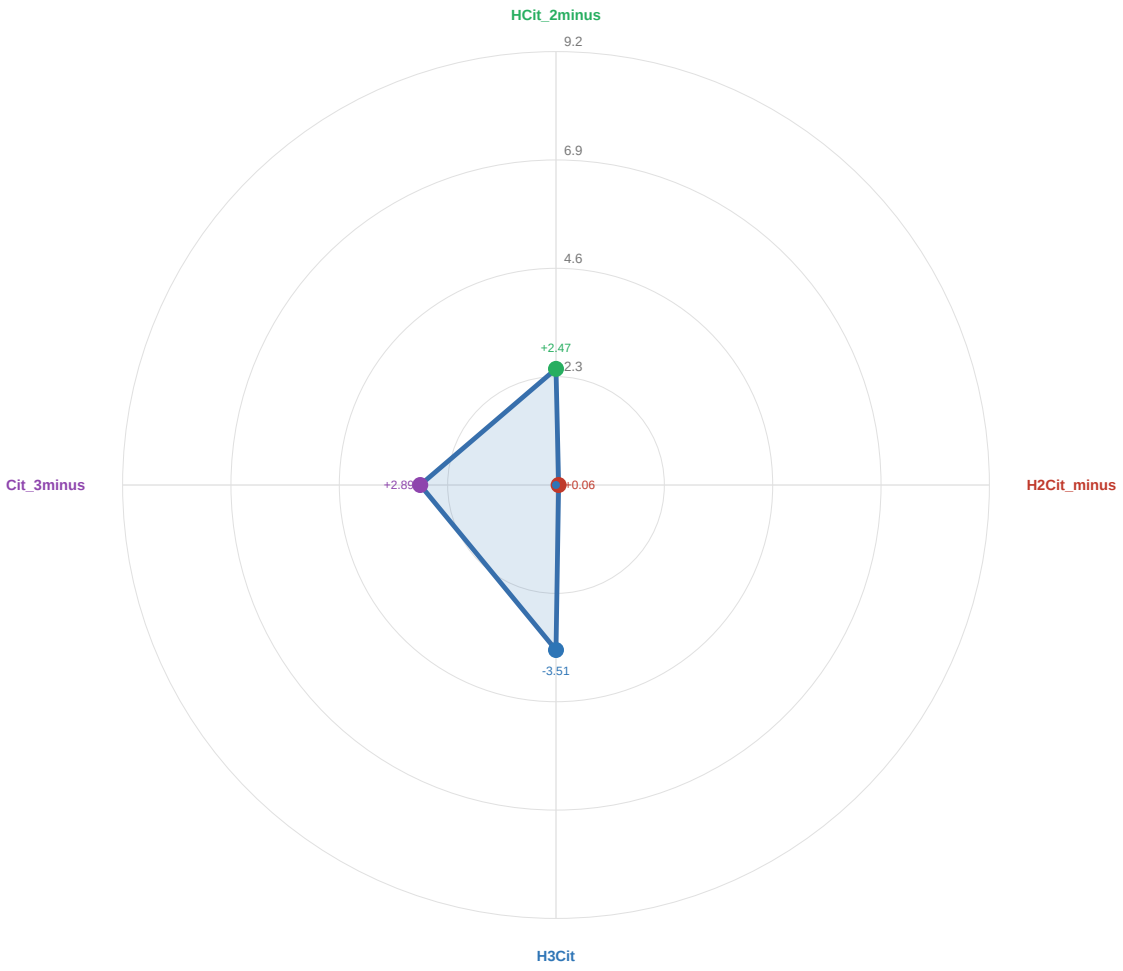
POLAR SLICE — t=106

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 107/200 | ||x||_A = 5.175



From 105: delta ||x||_A = -0.004

Phase = +179.1 deg

H3Cit=-3.51 H2Cit_=+0.06 HCit_2=+2.47 Cit_3m=+2.89

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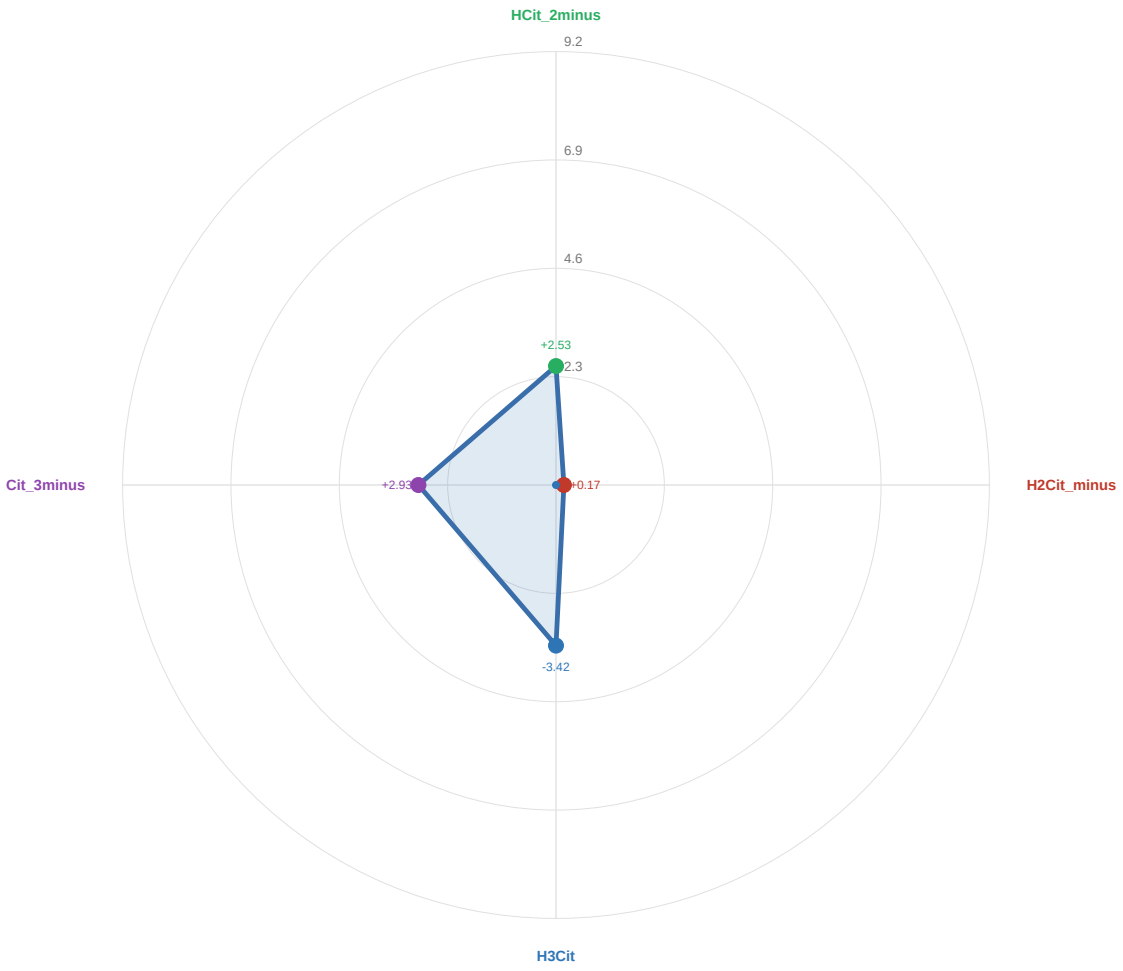
POLAR SLICE — t=107

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 108/200 | ||x||_A = 5.165



From 106: delta ||x||_A = -0.010

Phase = +177.2 deg

H3Cit=-3.42 H2Cit_=+0.17 HCit_2=+2.53 Cit_3m=+2.93

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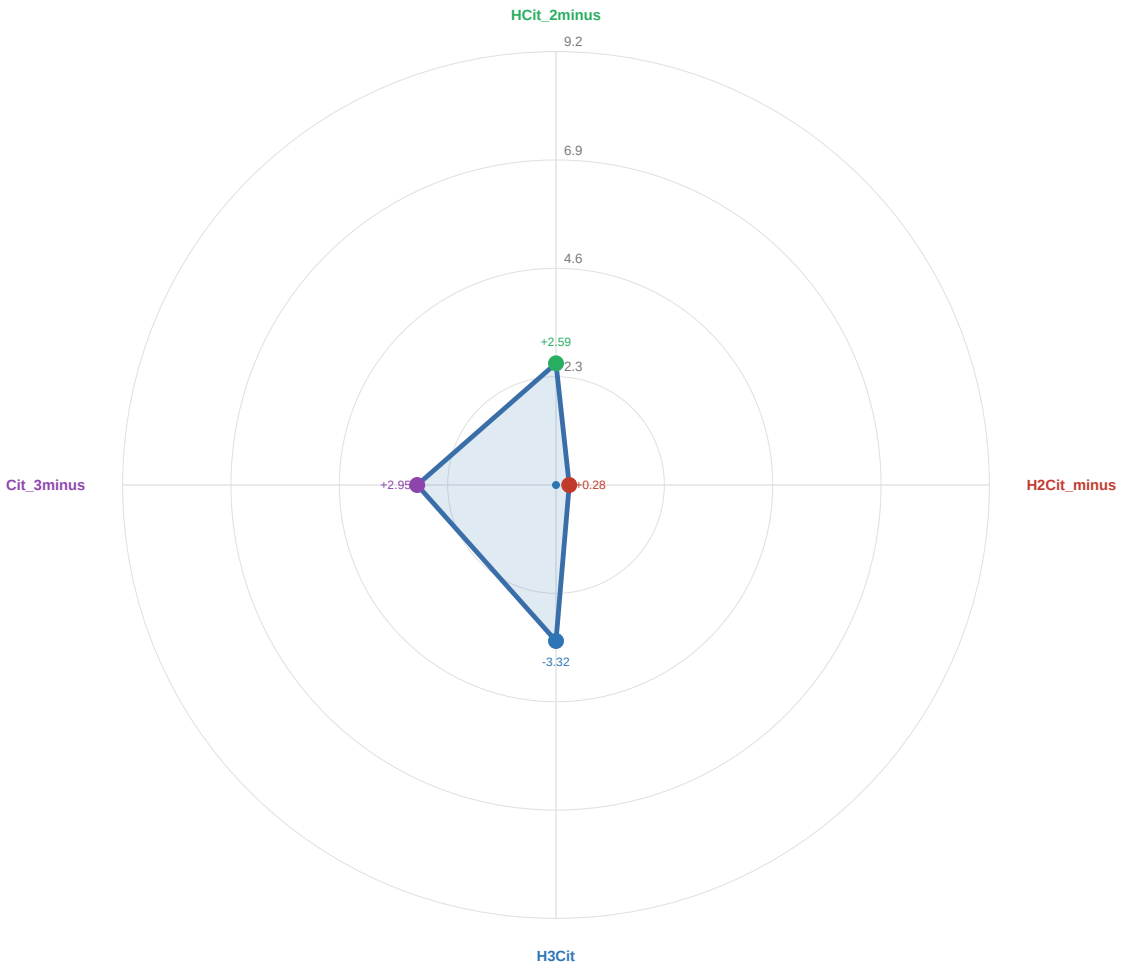
POLAR SLICE — t=108

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 109/200 | ||x||_A = 5.150



From 107: delta ||x||_A = -0.016

Phase = +175.1 deg

H3Cit=-3.32 H2Cit_=+0.28 HCit_2=+2.59 Cit_3m=+2.95

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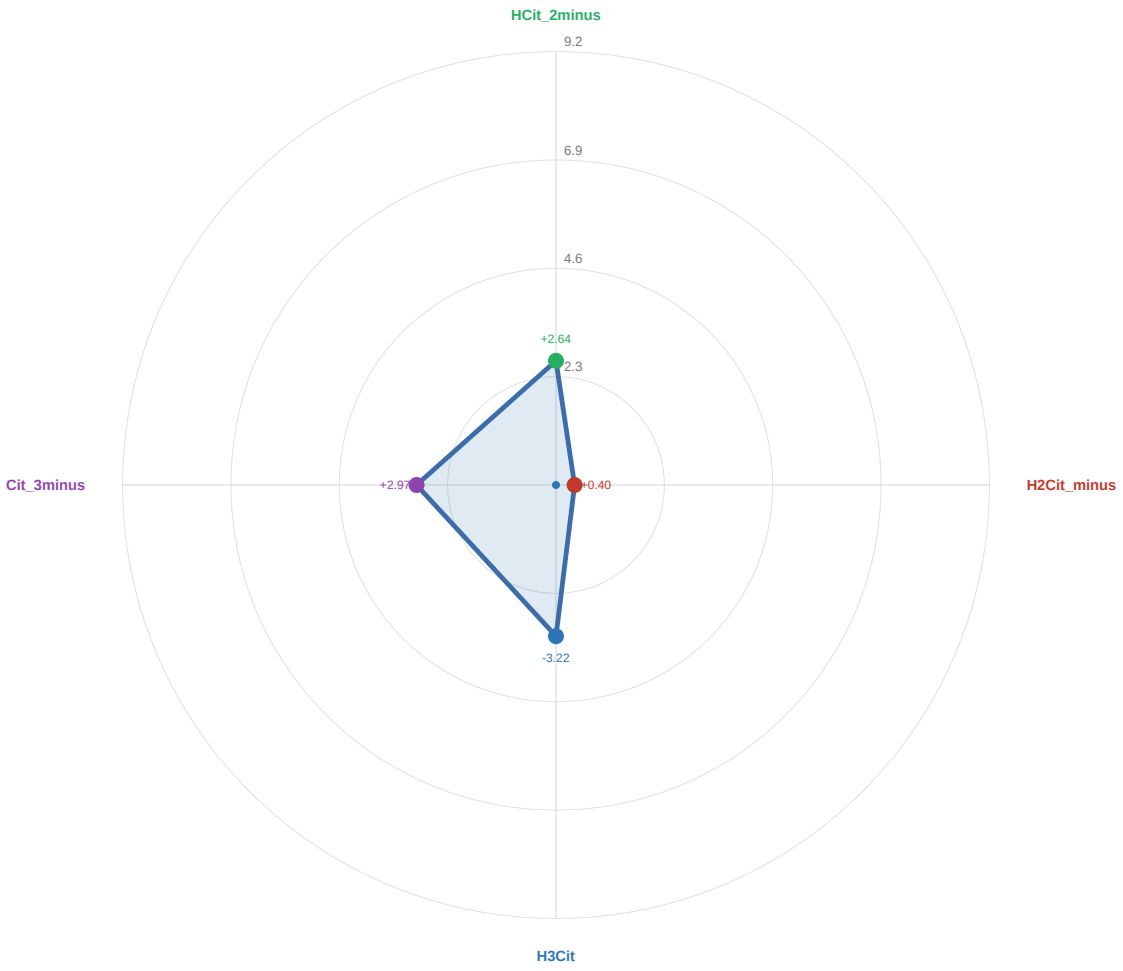
POLAR SLICE — t=109

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 110/200 | ||x||_A = 5.128



From 108: delta ||x||_A = -0.021

Phase = +173.0 deg

H3Cit=-3.22 H2Cit_=+0.40 HCit_2=+2.64 Cit_3m=+2.97

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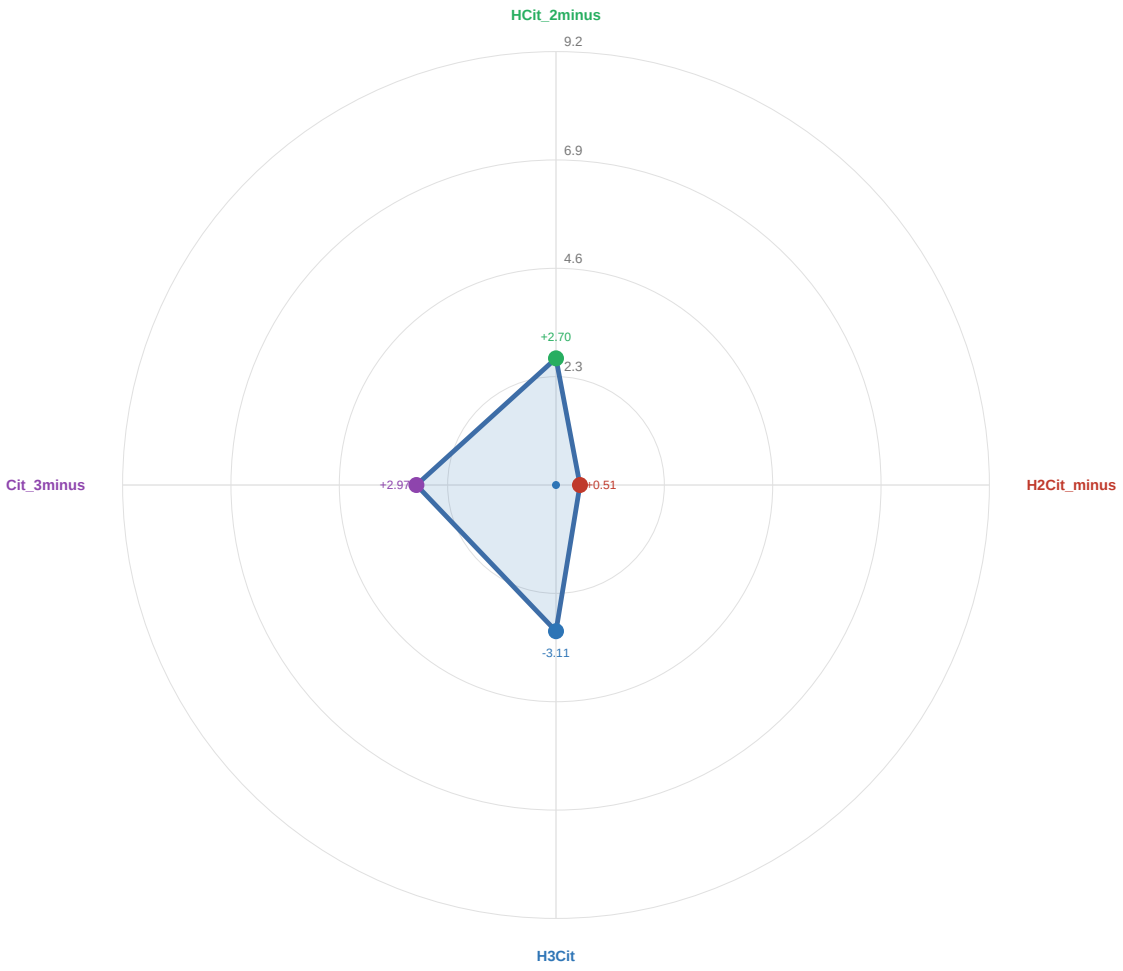
POLAR SLICE — t=110

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 111/200 | ||x||_A = 5.101



From 109: delta ||x||_A = -0.027

Phase = +170.7 deg

H3Cit=-3.11 H2Cit_=+0.51 HCit_2=+2.70 Cit_3m=+2.97

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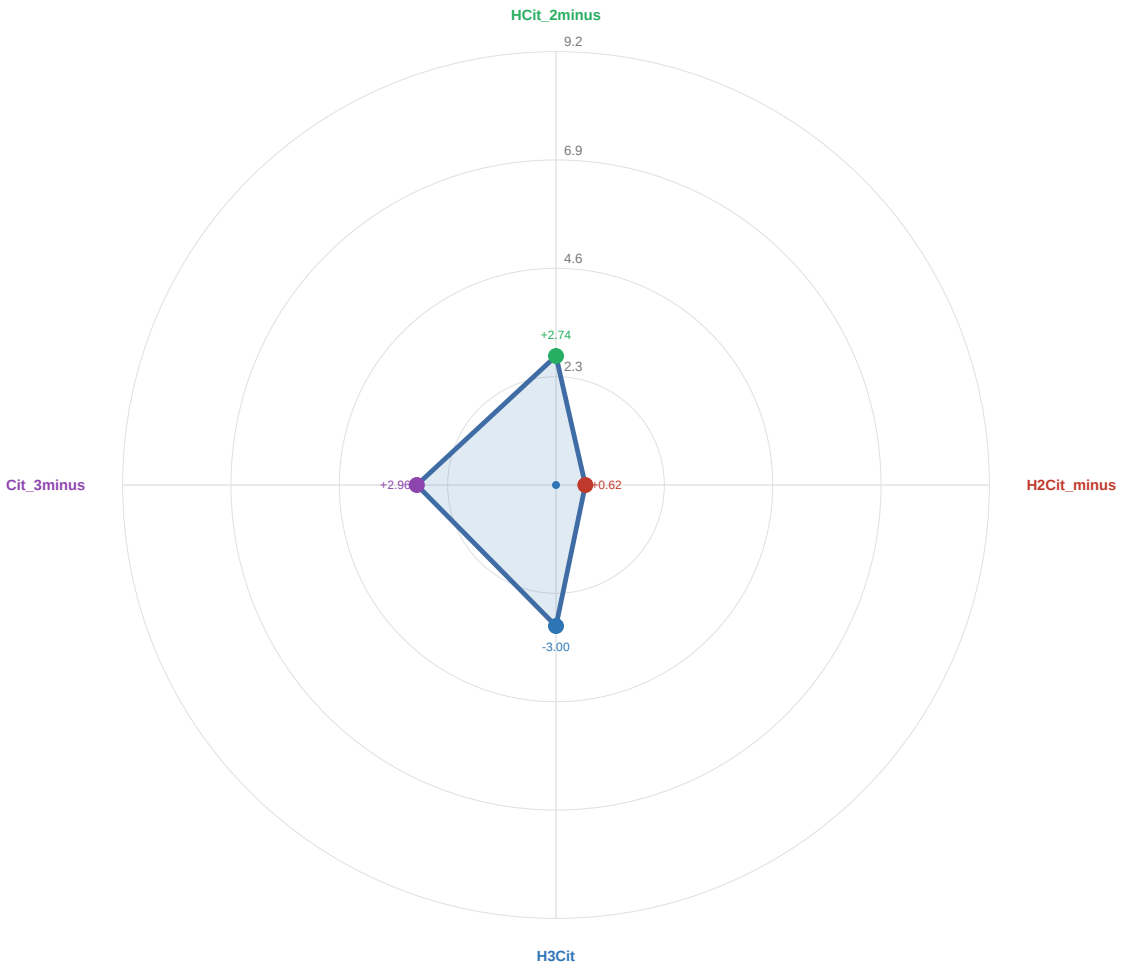
POLAR SLICE — t=111

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 112/200 | ||x||_A = 5.069



From 110: delta ||x||_A = -0.032

Phase = +168.2 deg

H3Cit=-3.00 H2Cit_=+0.62 HCit_2=+2.74 Cit_3m=+2.96

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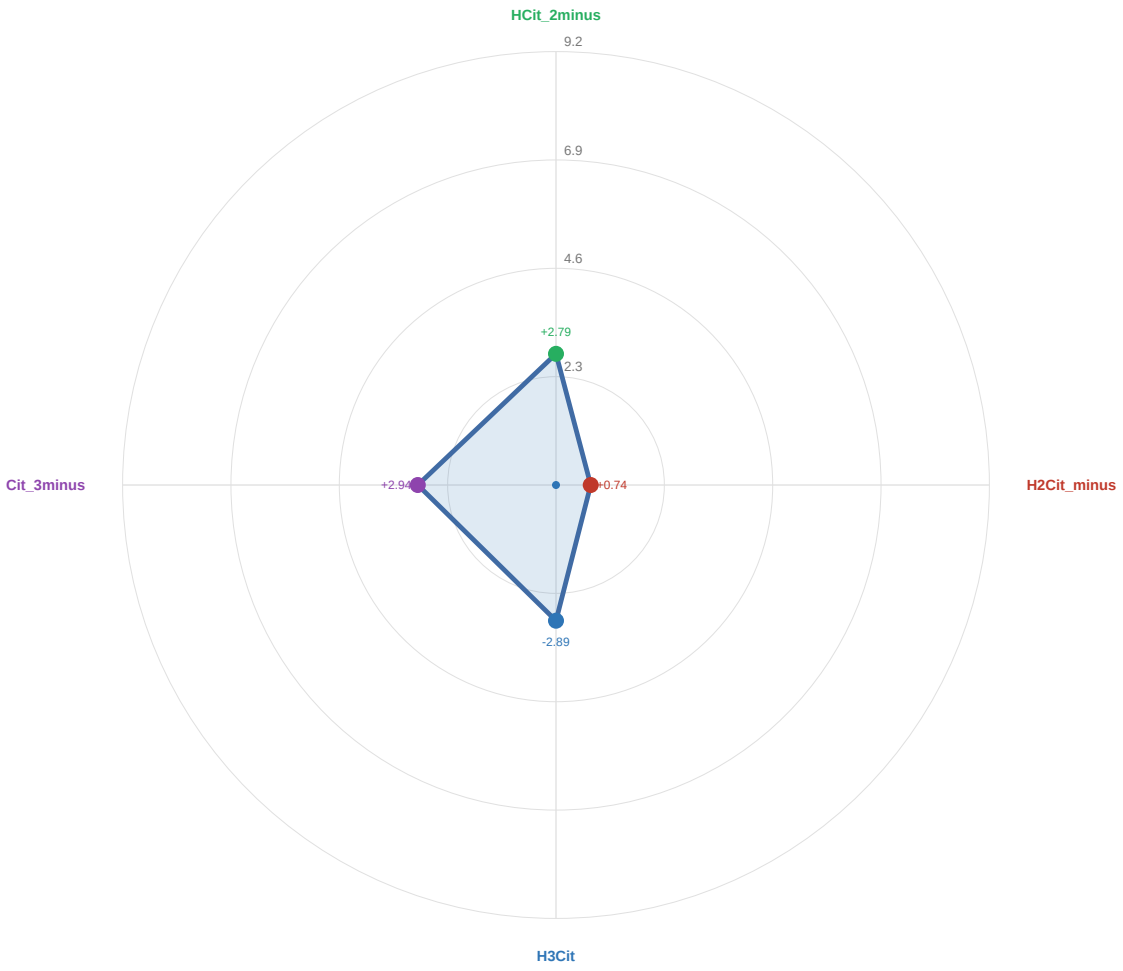
POLAR SLICE — t=112

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 113/200 | ||x||_A = 5.032



From 111: delta ||x||_A = -0.037

Phase = +165.6 deg

H3Cit=-2.89 H2Cit_=+0.74 HCit_2=+2.79 Cit_3m=+2.94

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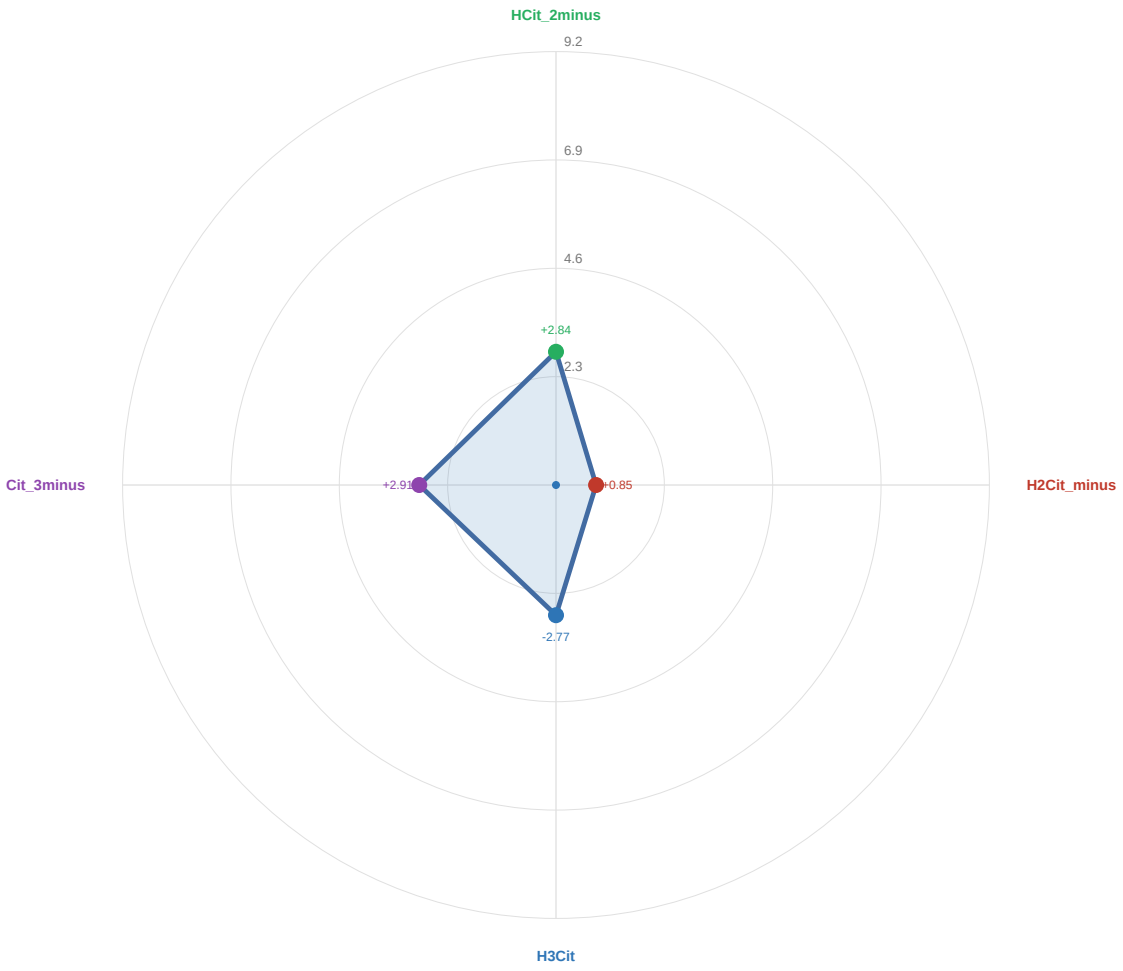
POLAR SLICE — t=113

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 114/200 | ||x||_A = 4.991



From 112: delta ||x||_A = -0.041

Phase = +162.9 deg

H3Cit=-2.77 H2Cit_=+0.85 HCit_2=+2.84 Cit_3m=+2.91

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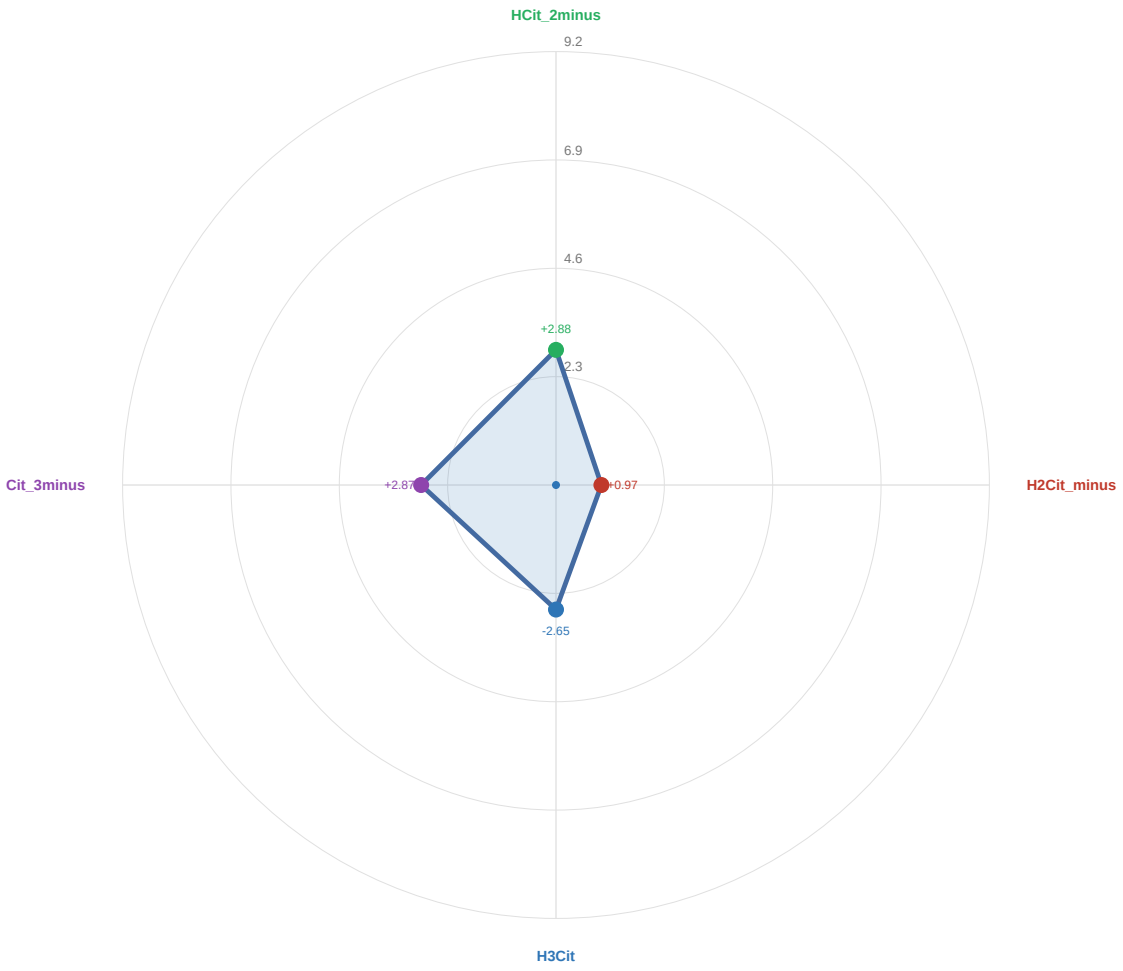
POLAR SLICE — t=114

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 115/200 | ||x||_A = 4.946



From 113: delta ||x||_A = -0.045

Phase = +160.0 deg

H3Cit=-2.65 H2Cit_=+0.97 HCit_2=+2.88 Cit_3m=+2.87

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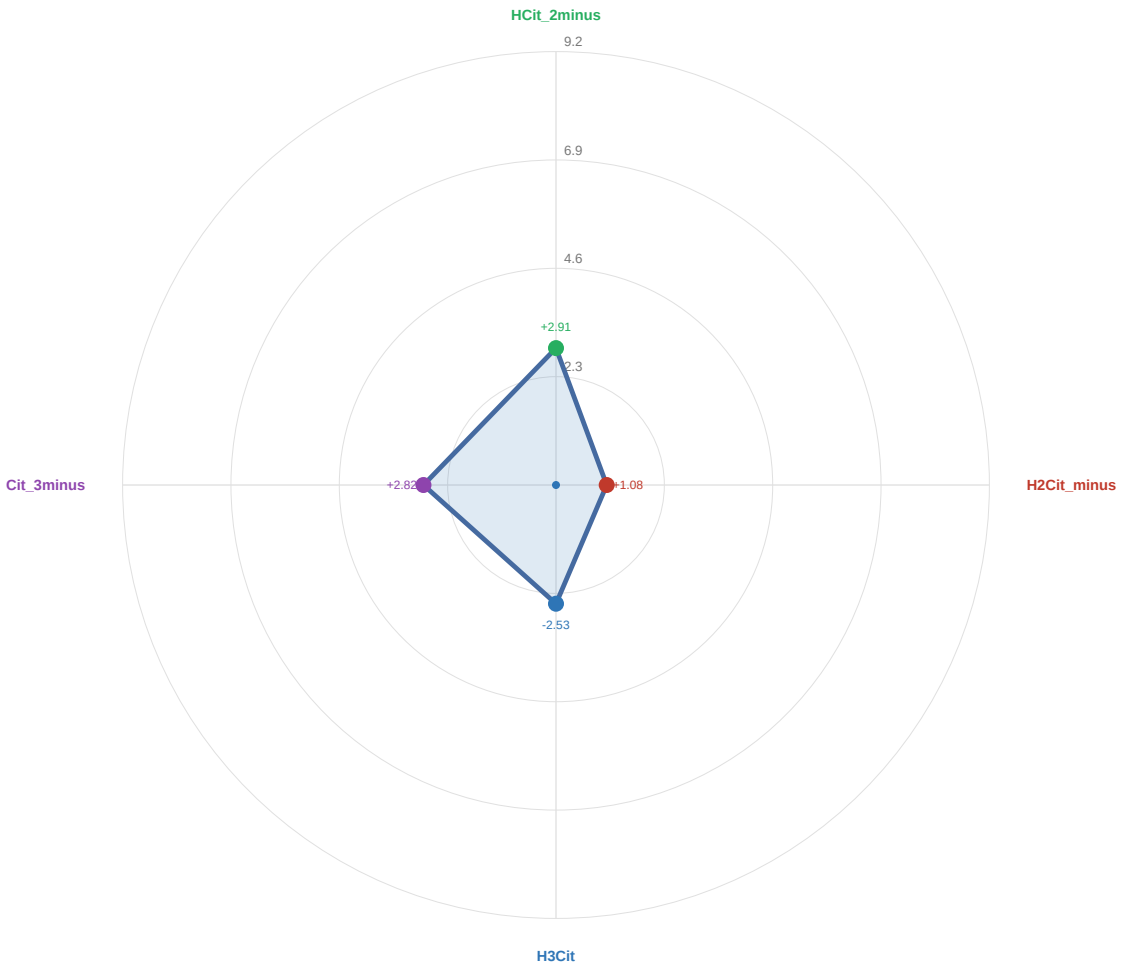
POLAR SLICE — t=115

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 116/200 | ||x||_A = 4.898



From 114: delta ||x||_A = -0.048

Phase = +156.9 deg

H3Cit=-2.53 H2Cit_=+1.08 HCit_2=+2.91 Cit_3m=+2.82

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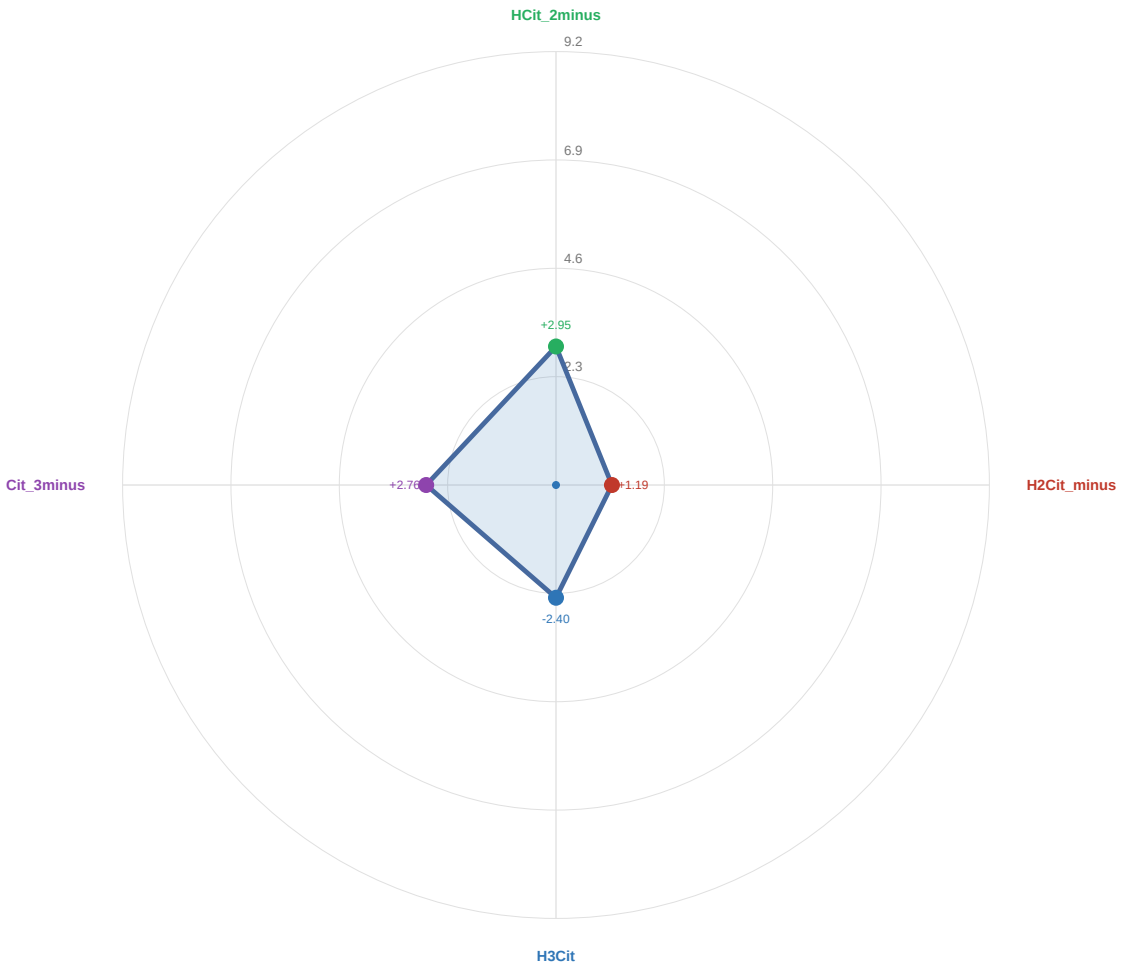
POLAR SLICE — t=116

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 117/200 | ||x||_A = 4.847



From 115: delta ||x||_A = -0.051

Phase = +153.6 deg

H3Cit=-2.40 H2Cit_=+1.19 HCit_2=+2.95 Cit_3m=+2.76

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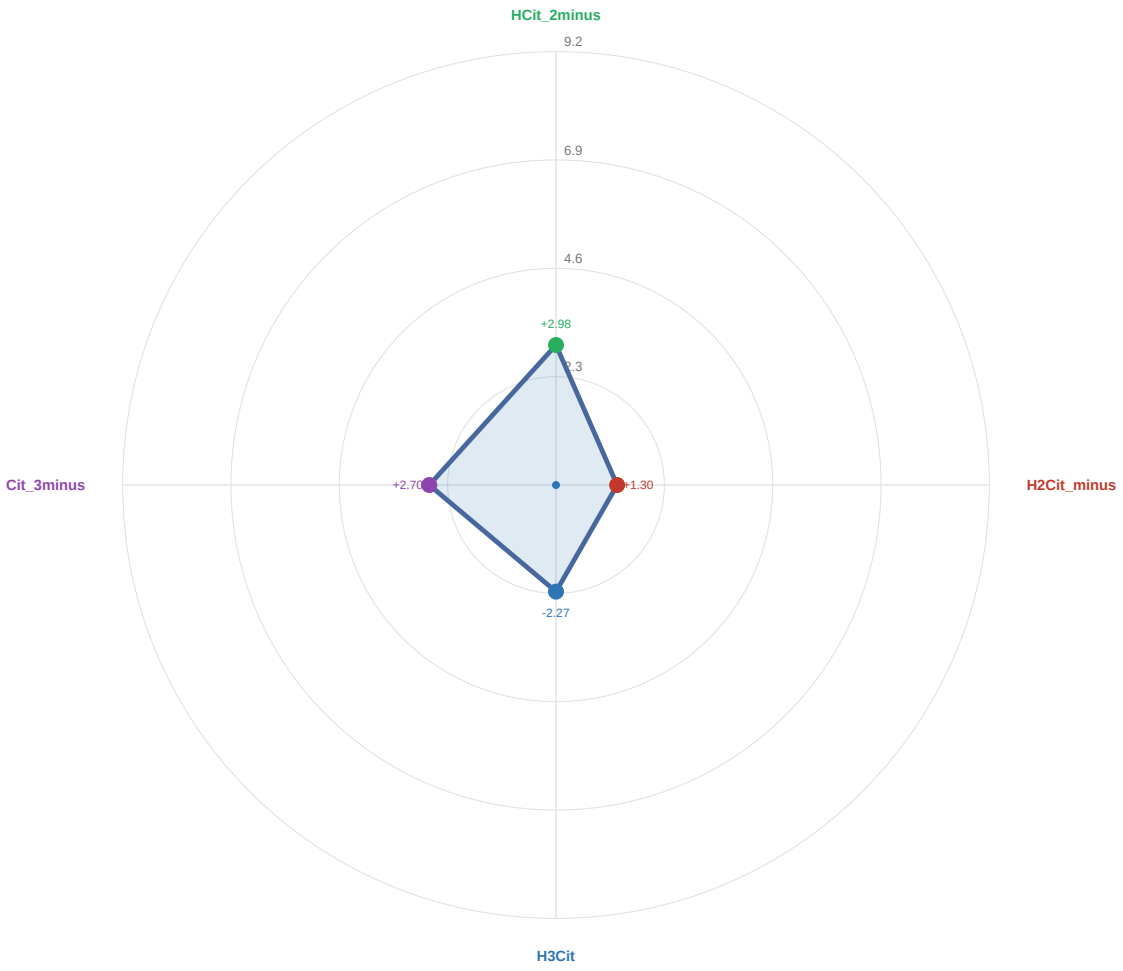
POLAR SLICE — t=117

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 118/200 | ||x||_A = 4.794



From 116: delta ||x||_A = -0.053

Phase = +150.2 deg

H3Cit=-2.27 H2Cit_=+1.30 HCit_2=+2.98 Cit_3m=+2.70

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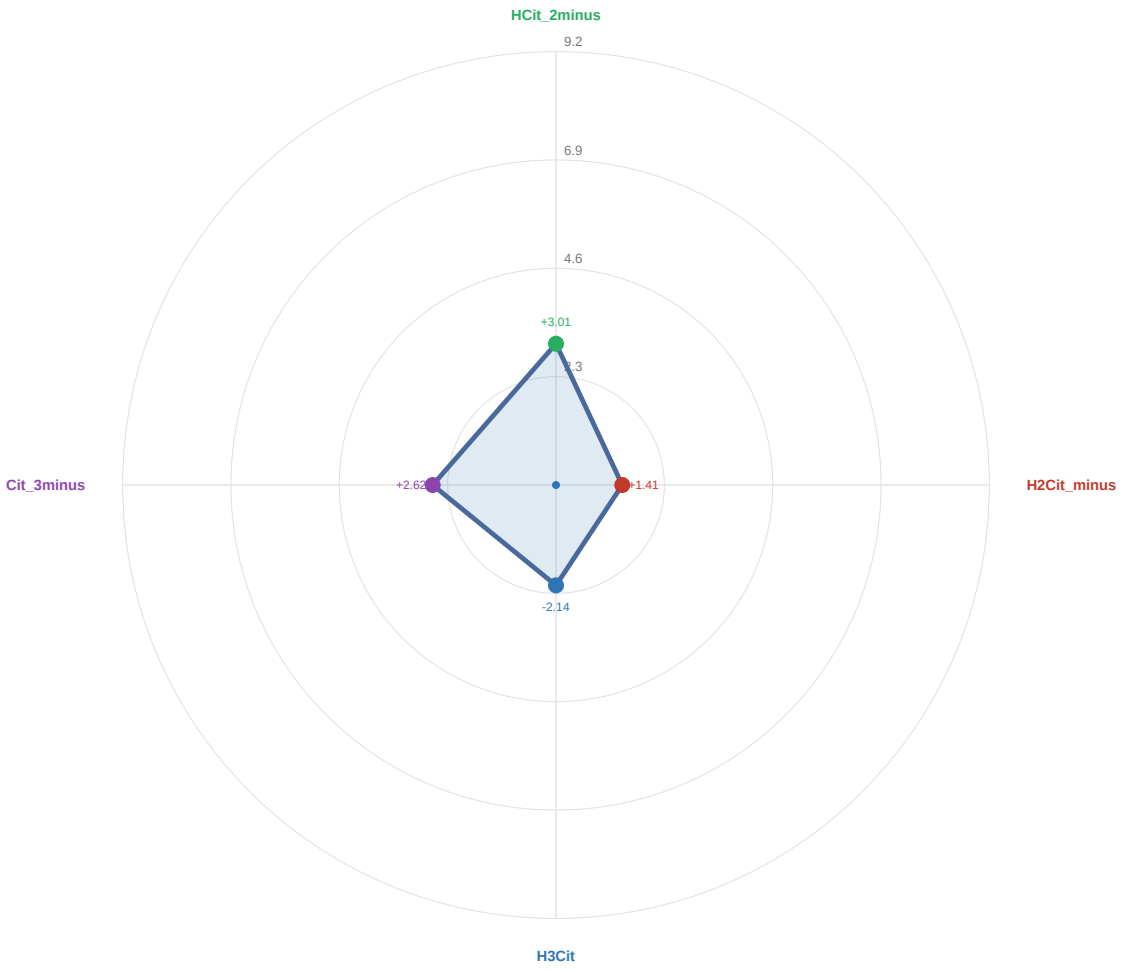
POLAR SLICE — t=118

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 119/200 | ||x||_A = 4.741



From 117: delta ||x||_A = -0.053

Phase = +146.6 deg

H3Cit=-2.14 H2Cit_=+1.41 HCit_2=+3.01 Cit_3m=+2.62

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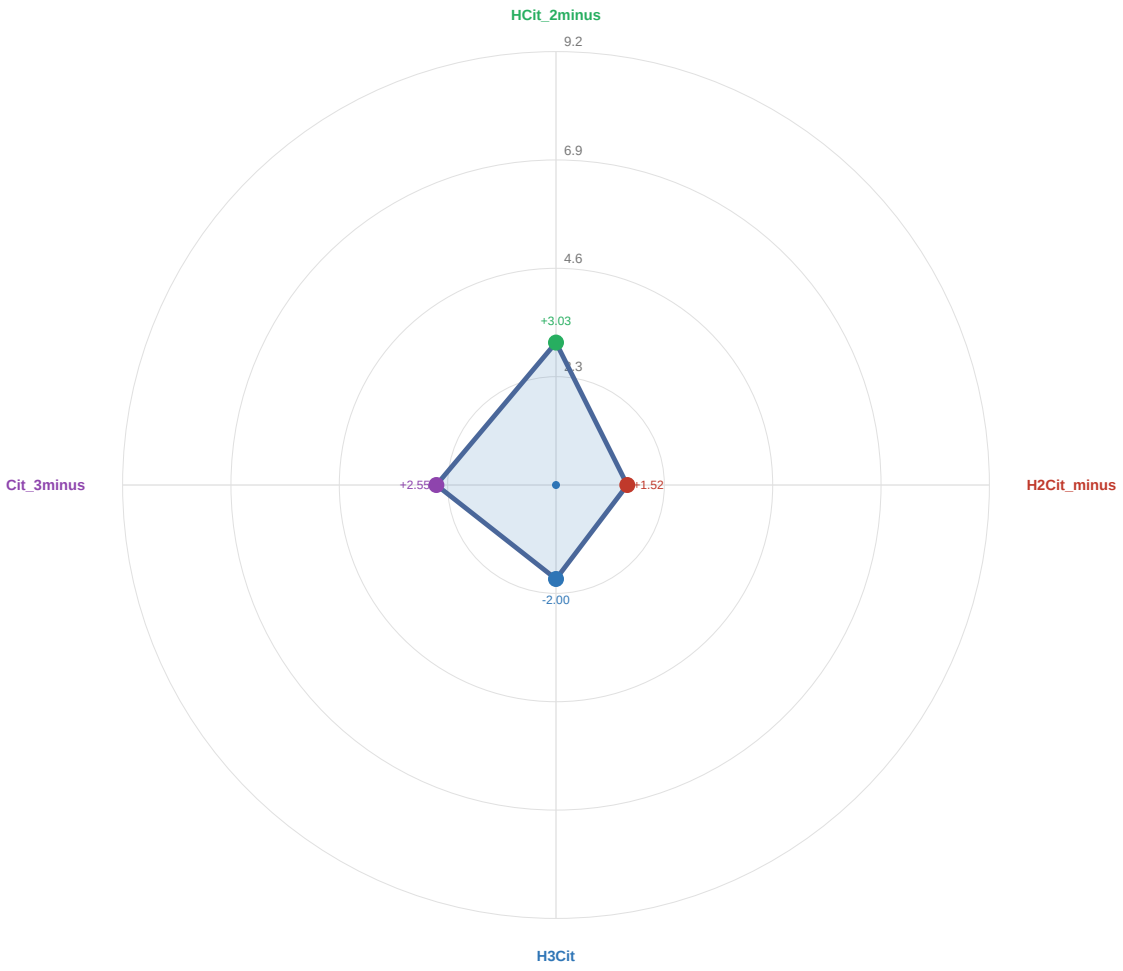
POLAR SLICE — t=119

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 120/200 | ||x||_A = 4.687



From 118: delta ||x||_A = -0.054

Phase = +142.8 deg

H3Cit=-2.00 H2Cit_=+1.52 HCit_2=+3.03 Cit_3m=+2.55

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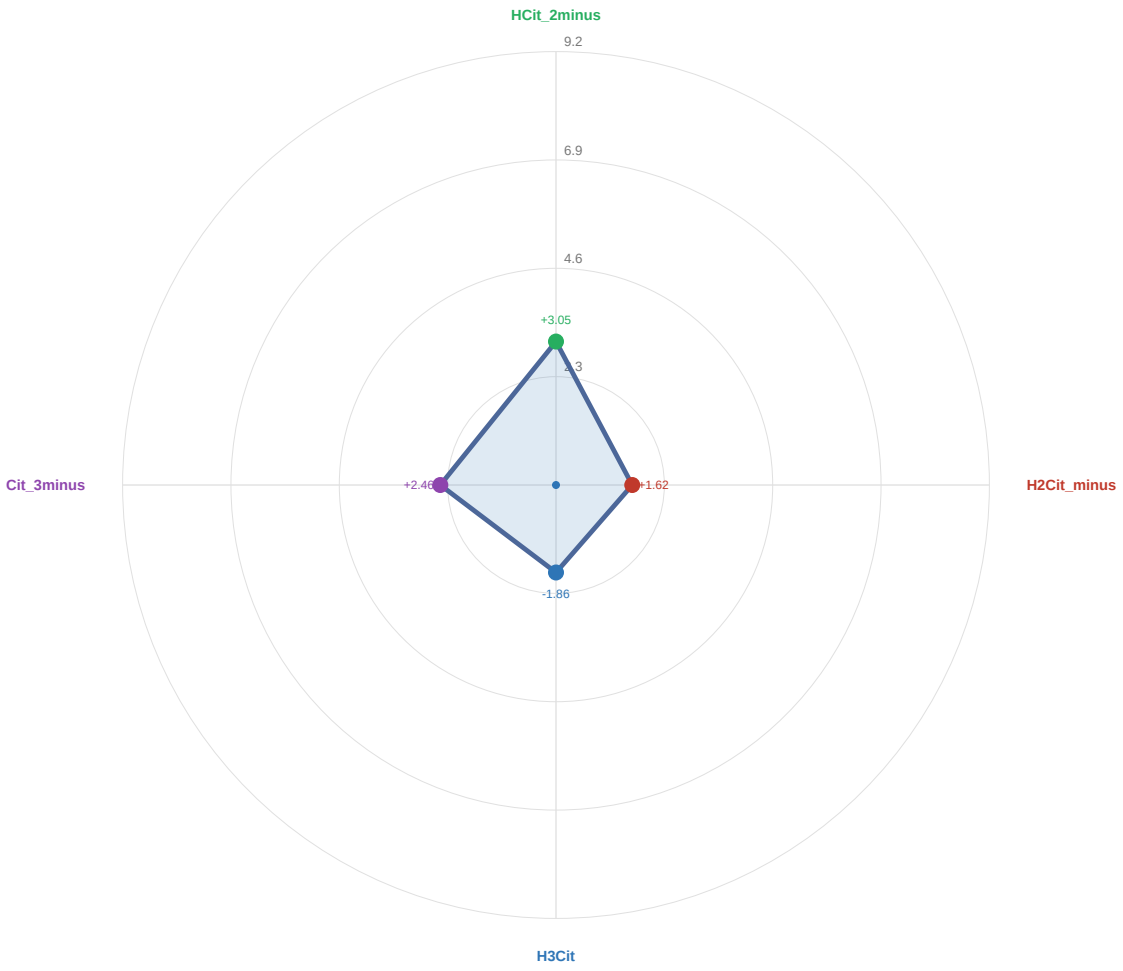
POLAR SLICE — t=120

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 121/200 | ||x||_A = 4.634



From 119: delta ||x||_A = -0.053

Phase = +138.9 deg

H3Cit=-1.86 H2Cit_=+1.62 HCit_2=+3.05 Cit_3m=+2.46

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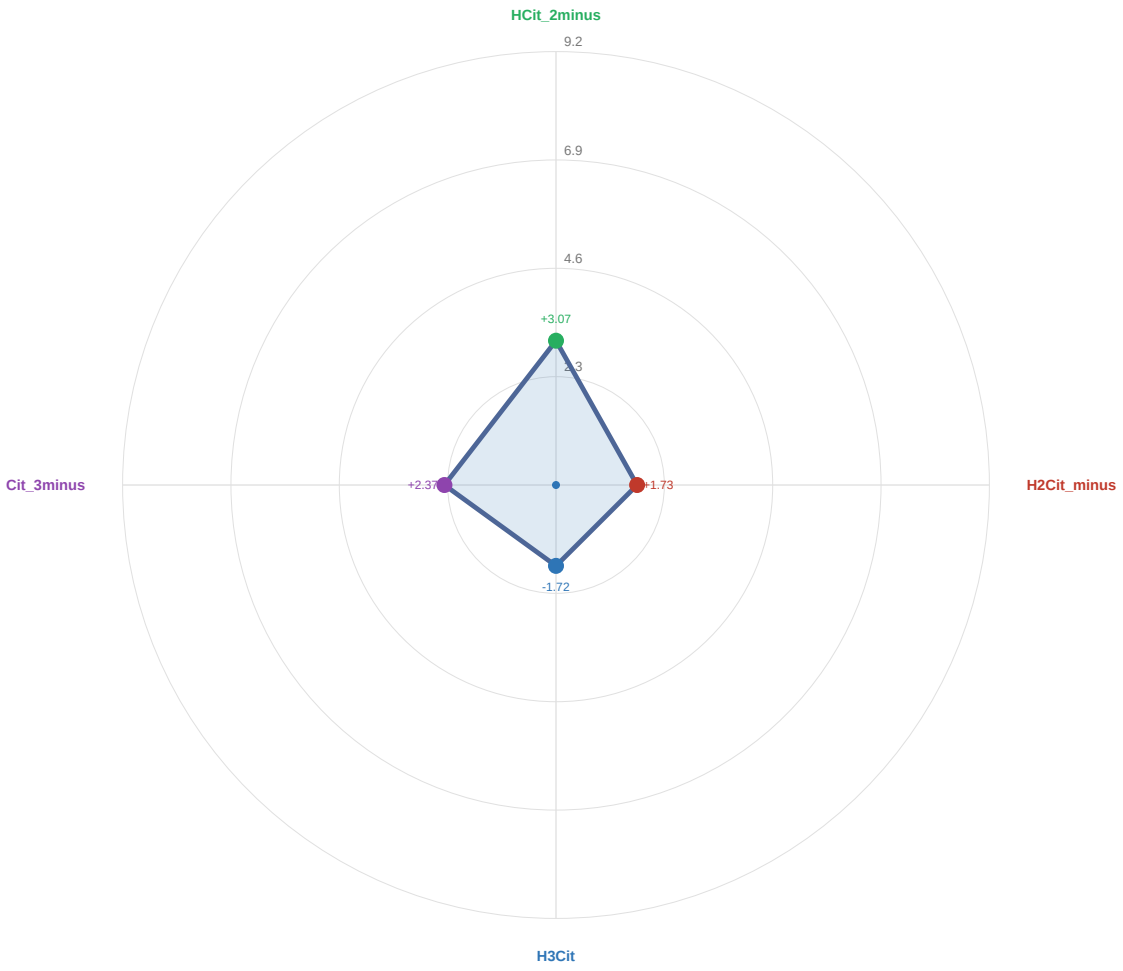
POLAR SLICE — t=121

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 122/200 | ||x||_A = 4.583



From 120: delta ||x||_A = -0.051

Phase = +134.9 deg

H3Cit=-1.72 H2Cit_=+1.73 HCit_2=+3.07 Cit_3m=+2.37

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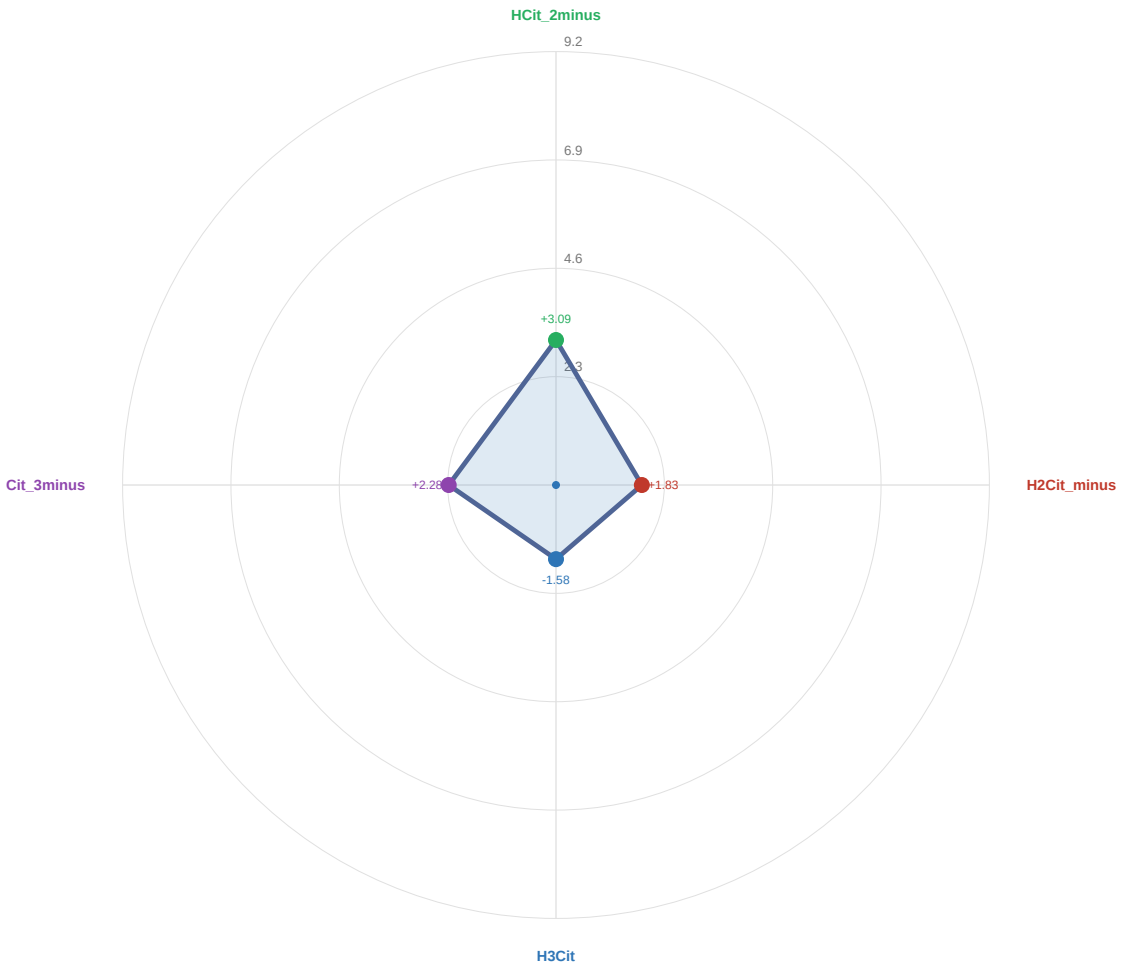
POLAR SLICE — t=122

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 123/200 | ||x||_A = 4.534



From 121: delta ||x||_A = -0.049

Phase = +130.8 deg

H3Cit=-1.58 H2Cit_=+1.83 HCit_2=+3.09 Cit_3m=+2.28

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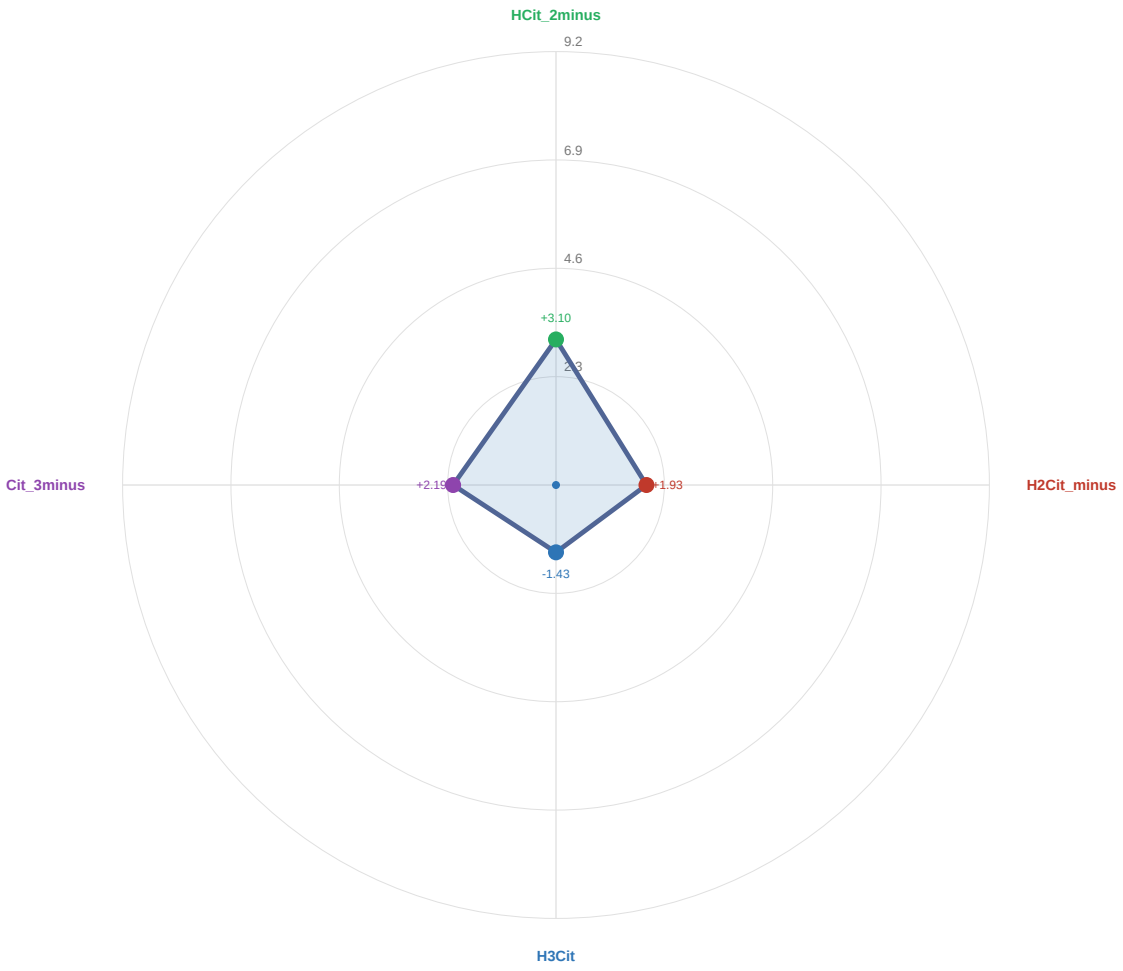
POLAR SLICE — t=123

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 124/200 | ||x||_A = 4.489



From 122: delta ||x||_A = -0.046

Phase = +126.6 deg

H3Cit=-1.43 H2Cit_=+1.93 HCit_2=+3.10 Cit_3m=+2.19

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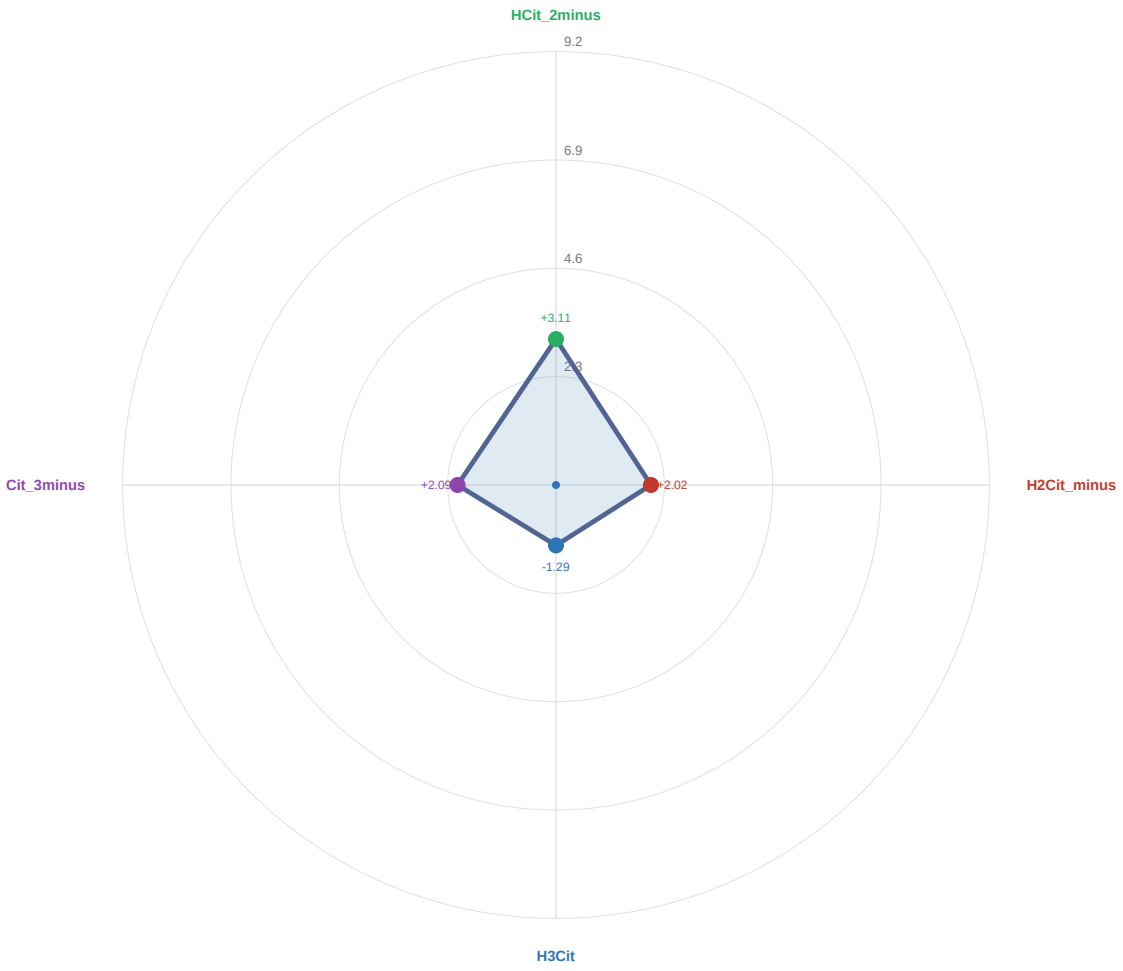
POLAR SLICE — t=124

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 125/200 | ||x||_A = 4.447



From 123: delta ||x||_A = -0.042

Phase = +122.5 deg

H3Cit=-1.29 H2Cit_=+2.02 HCit_2=+3.11 Cit_3m=+2.09

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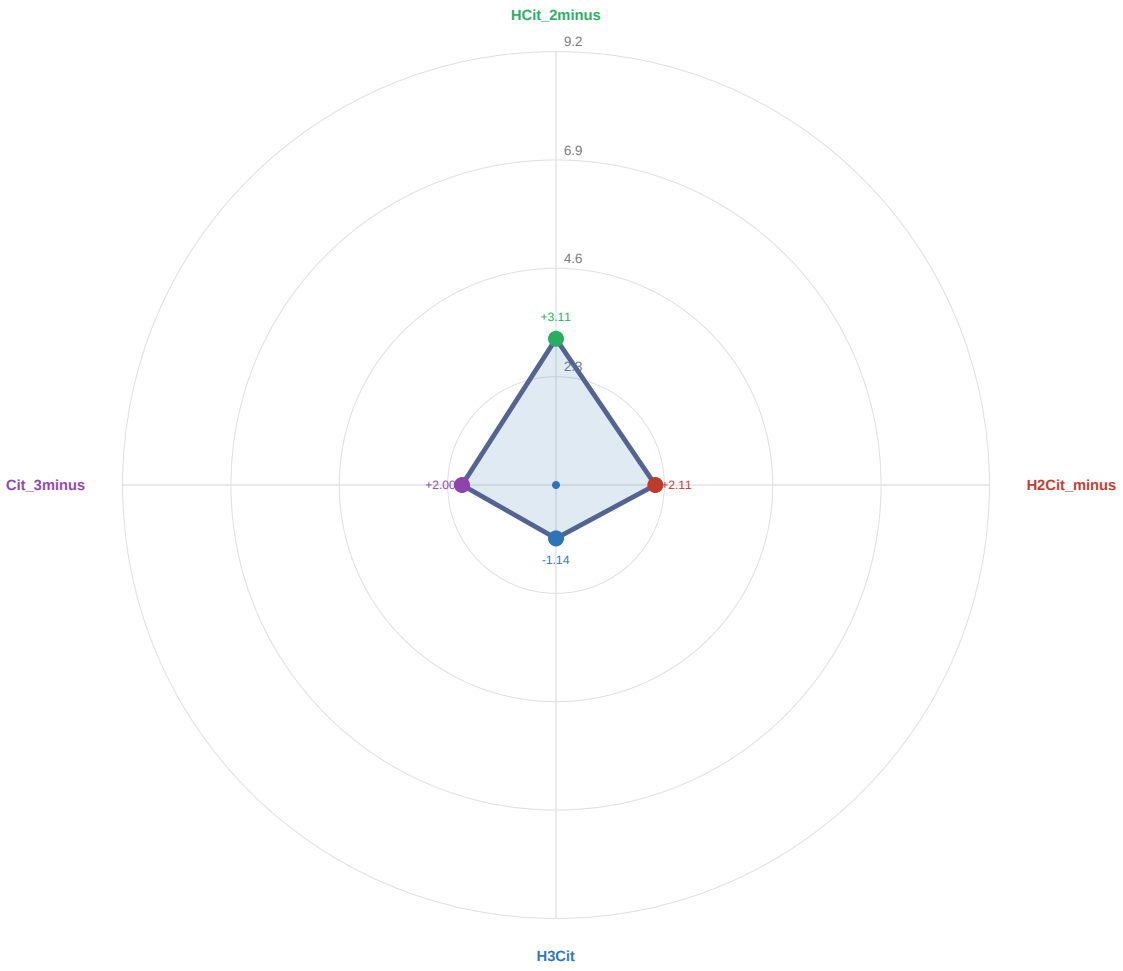
POLAR SLICE — t=125

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 126/200 | ||x||_A = 4.410



From 124: delta ||x||_A = -0.037

Phase = +118.3 deg

H3Cit=-1.14 H2Cit_=+2.11 HCit_2=+3.11 Cit_3m=+2.00

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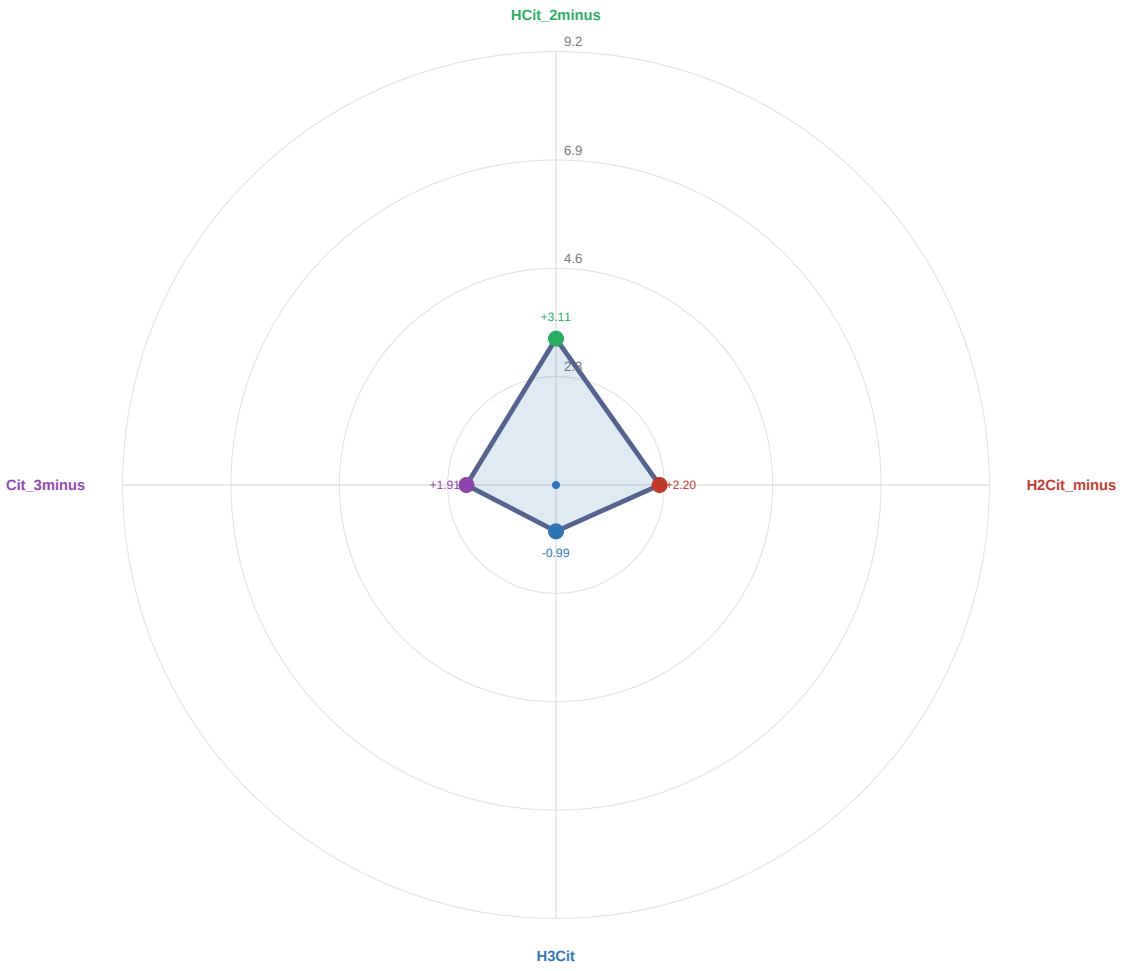
POLAR SLICE — t=126

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 127/200 | ||x||_A = 4.377



From 125: delta ||x||_A = -0.032

Phase = +114.1 deg

H3Cit=-0.99 H2Cit_=+2.20 HCit_2=+3.11 Cit_3m=+1.91

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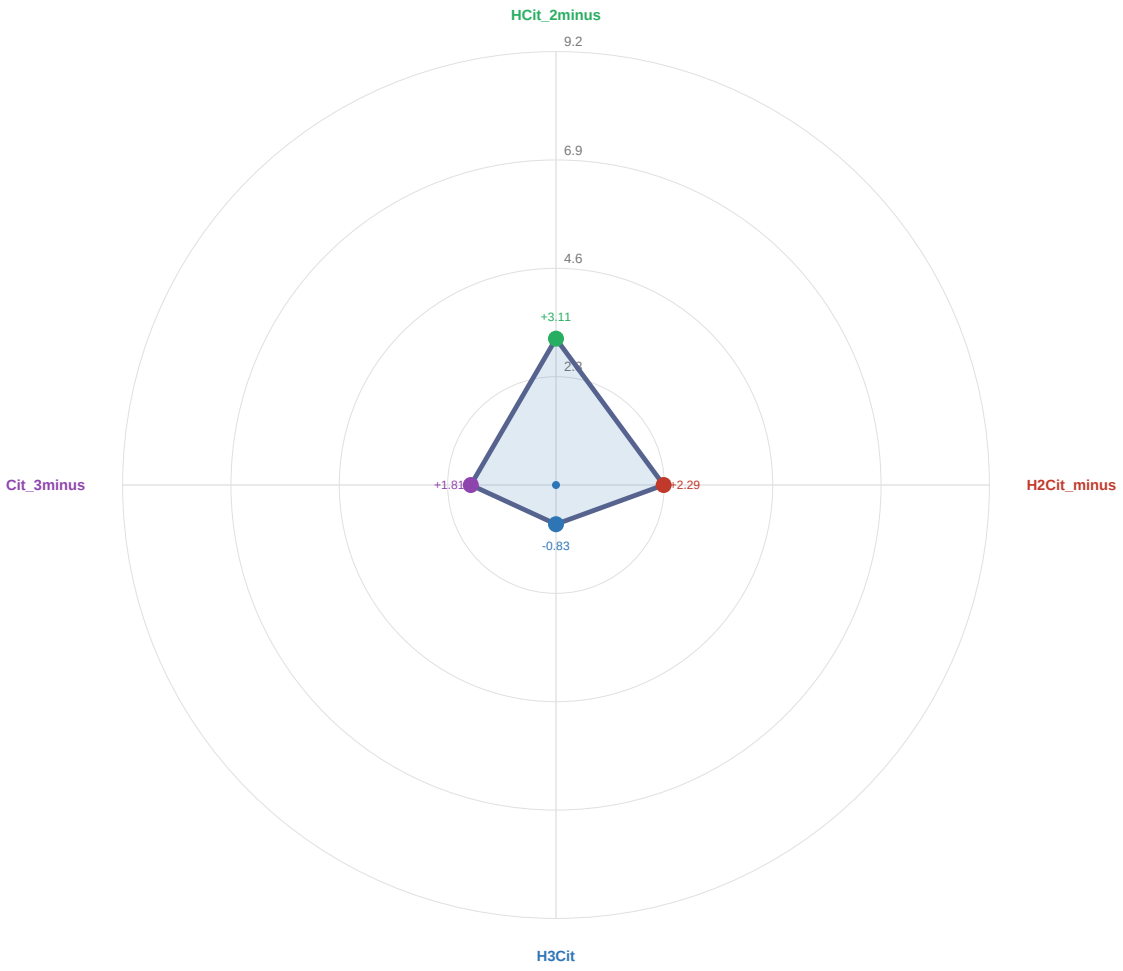
POLAR SLICE — t=127

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 128/200 | ||x||_A = 4.351



From 126: delta ||x||_A = -0.027

Phase = +110.0 deg

H3Cit=-0.83 H2Cit_=+2.29 HCit_2=+3.11 Cit_3m=+1.81

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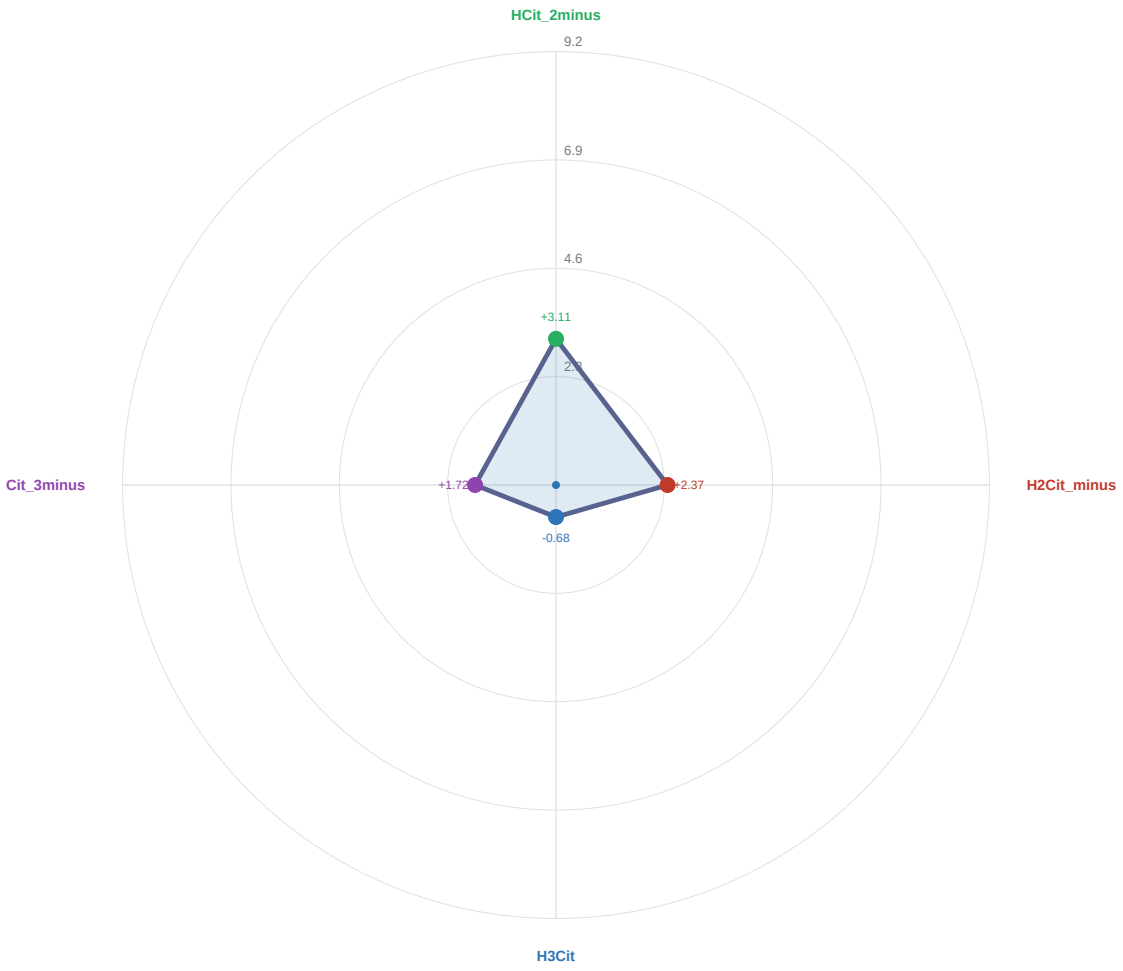
POLAR SLICE — t=128

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 129/200 | ||x||_A = 4.330



From 127: delta ||x||_A = -0.021

Phase = +106.0 deg

H3Cit=-0.68 H2Cit_=+2.37 HCit_2=+3.11 Cit_3m=+1.72

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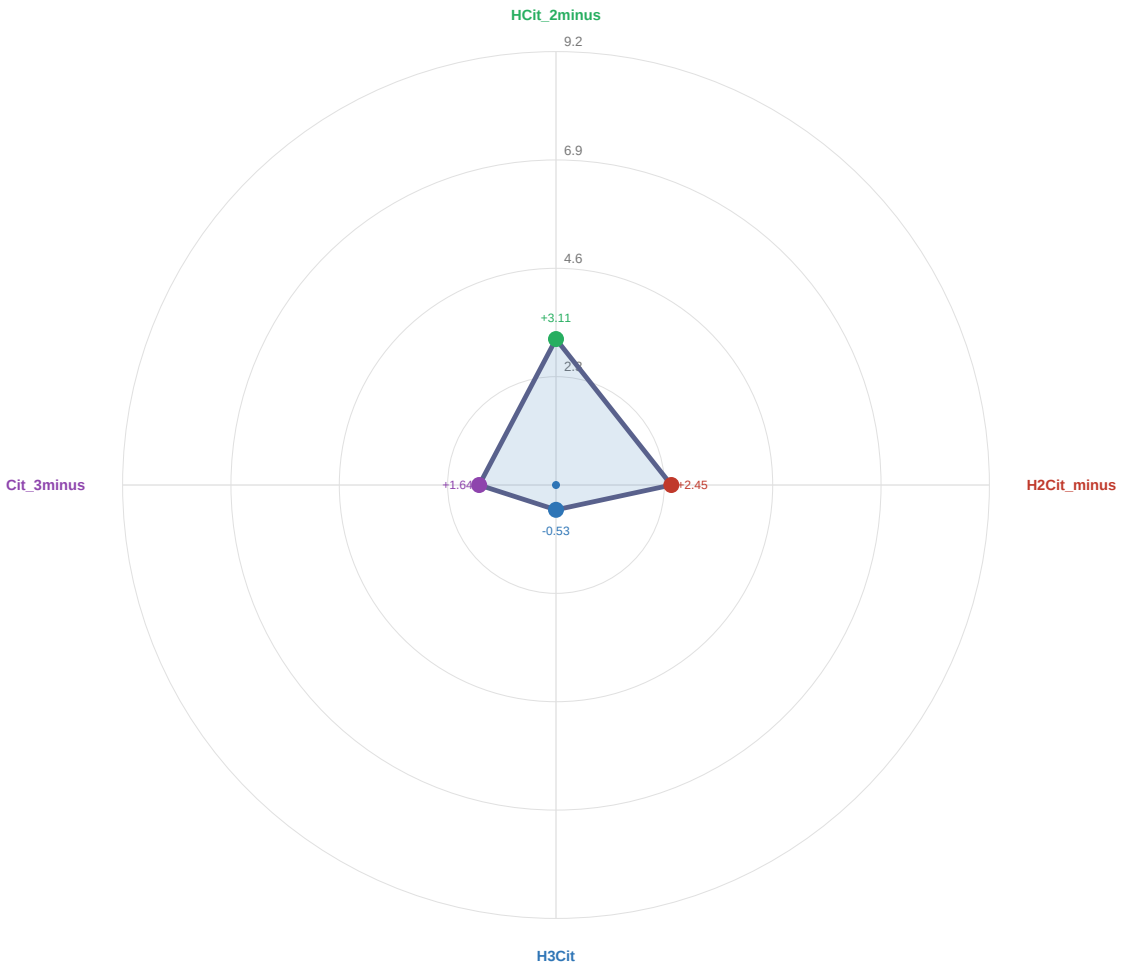
POLAR SLICE — t=129

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 130/200 | ||x||_A = 4.316



From 128: delta ||x||_A = -0.014

Phase = +102.1 deg

H3Cit=-0.53 H2Cit_=+2.45 HCit_2=+3.11 Cit_3m=+1.64

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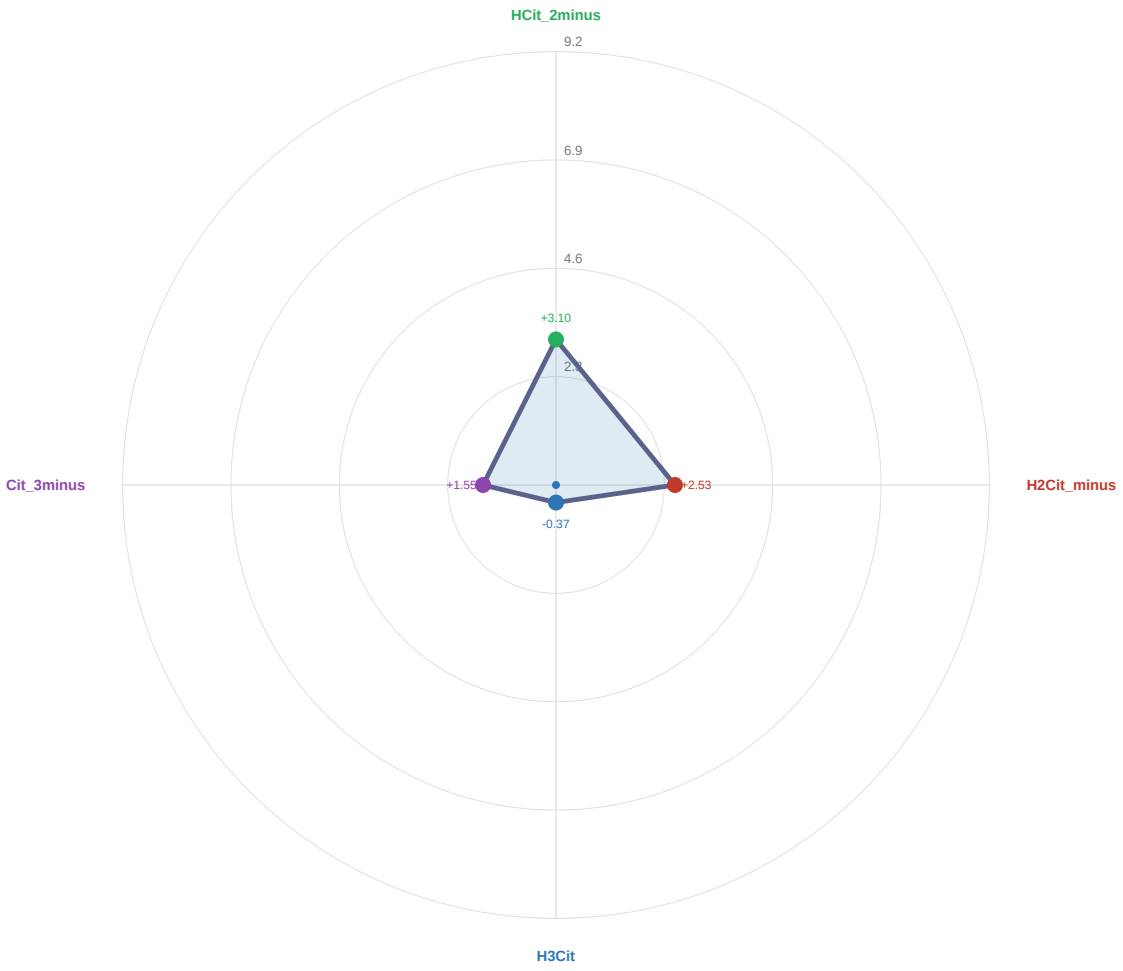
POLAR SLICE — t=130

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 131/200 | ||x||_A = 4.308



From 129: delta ||x||_A = -0.008

Phase = +98.4 deg

H3Cit=-0.37 H2Cit_=+2.53 HCit_2=+3.10 Cit_3m=+1.55

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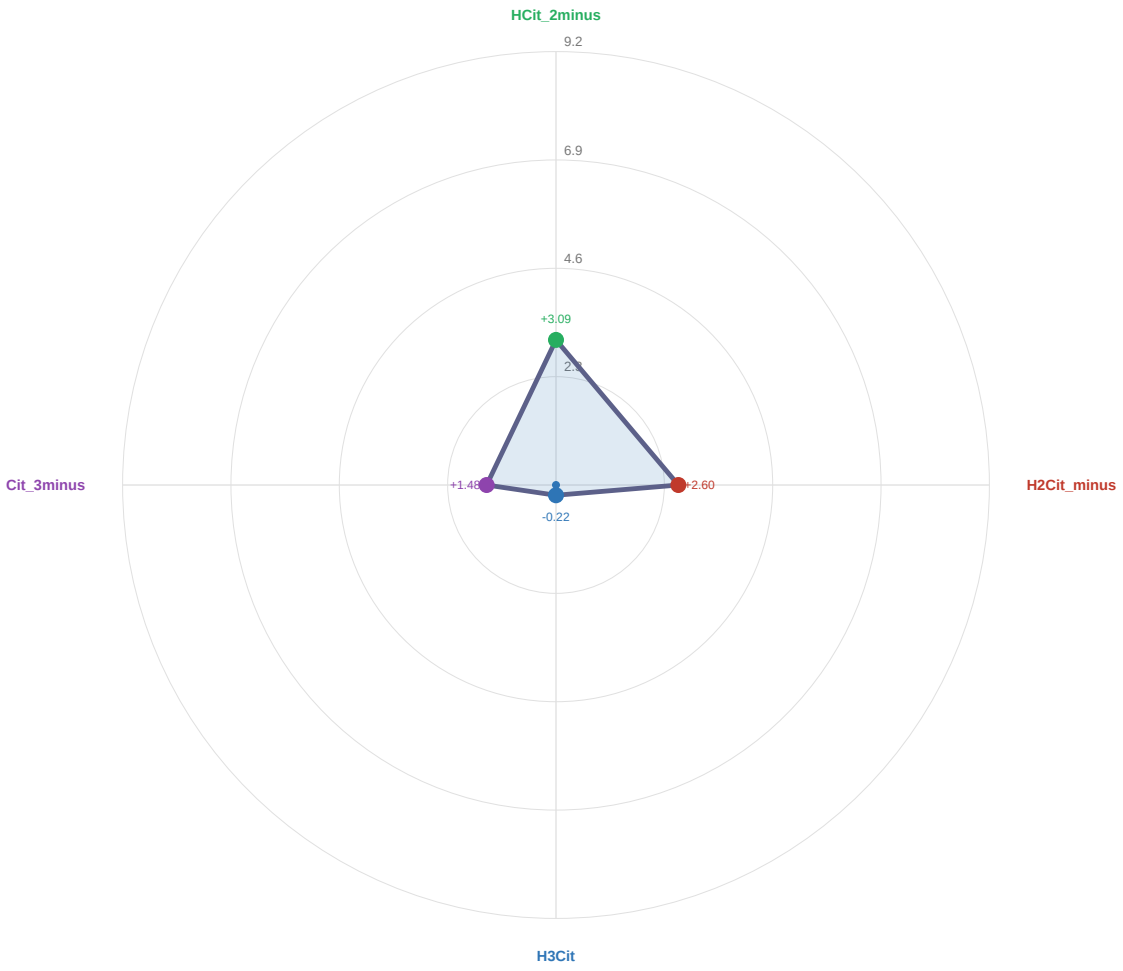
POLAR SLICE — t=131

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 132/200 | ||x||_A = 4.306



From 130: delta ||x||_A = -0.002

Phase = +94.8 deg

H3Cit=-0.22 H2Cit_=+2.60 HCit_2=+3.09 Cit_3m=+1.48

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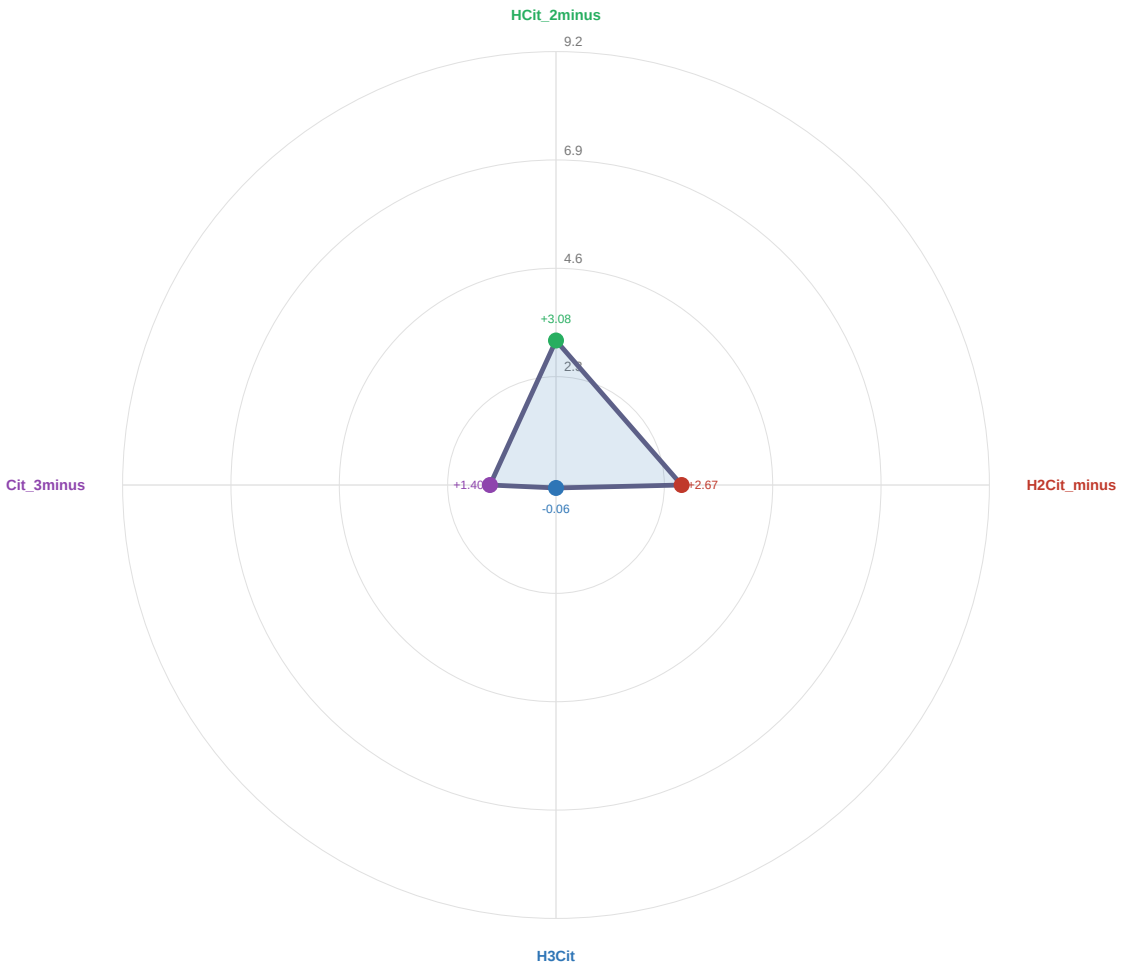
POLAR SLICE — t=132

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 133/200 | ||x||_A = 4.311



From 131: delta ||x||_A = +0.005

Phase = +91.3 deg

H3Cit=-0.06 H2Cit_=+2.67 HCit_2=+3.08 Cit_3m=+1.40

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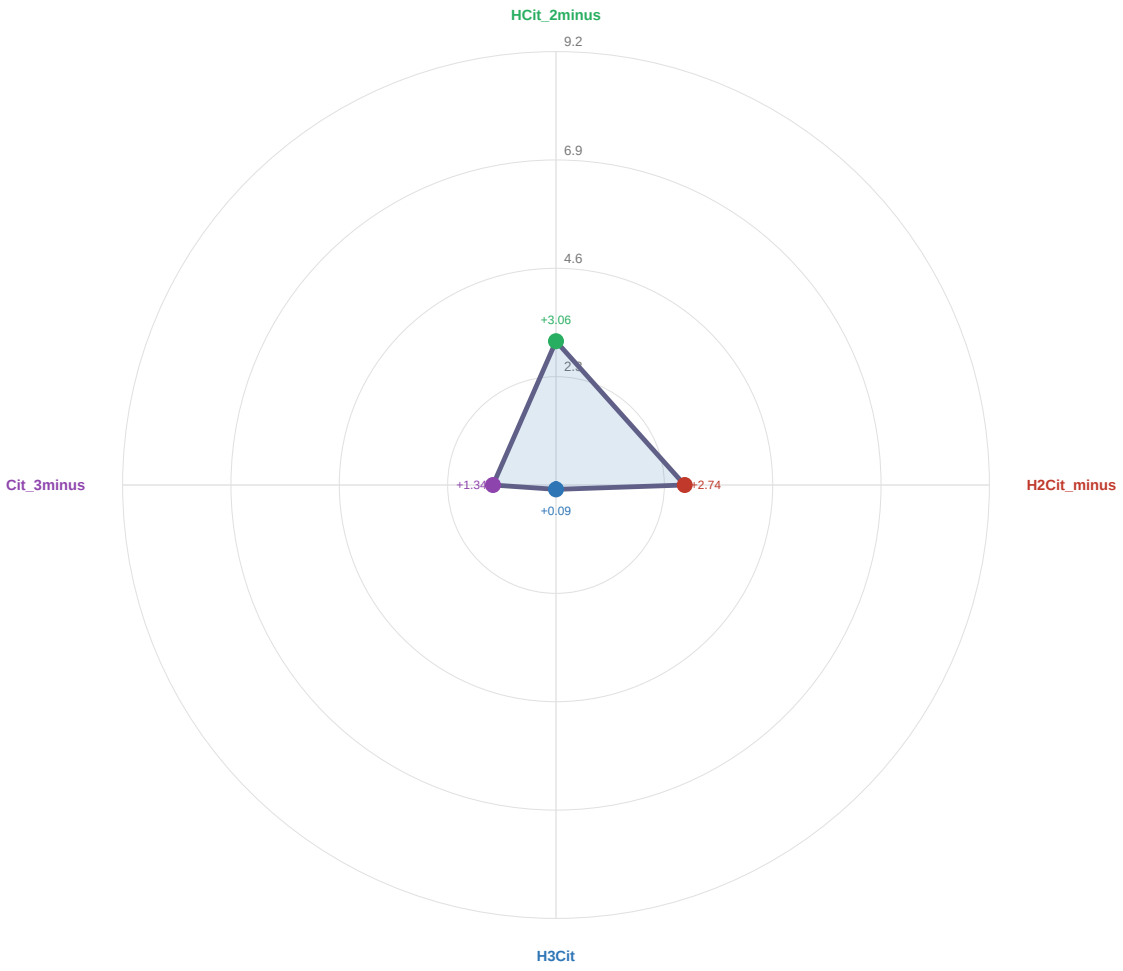
POLAR SLICE — t=133

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 134/200 | ||x||_A = 4.321



From 132: delta ||x||_A = +0.011

Phase = +88.1 deg

H3Cit=+0.09 H2Cit_=+2.74 HCit_2=+3.06 Cit_3m=+1.34

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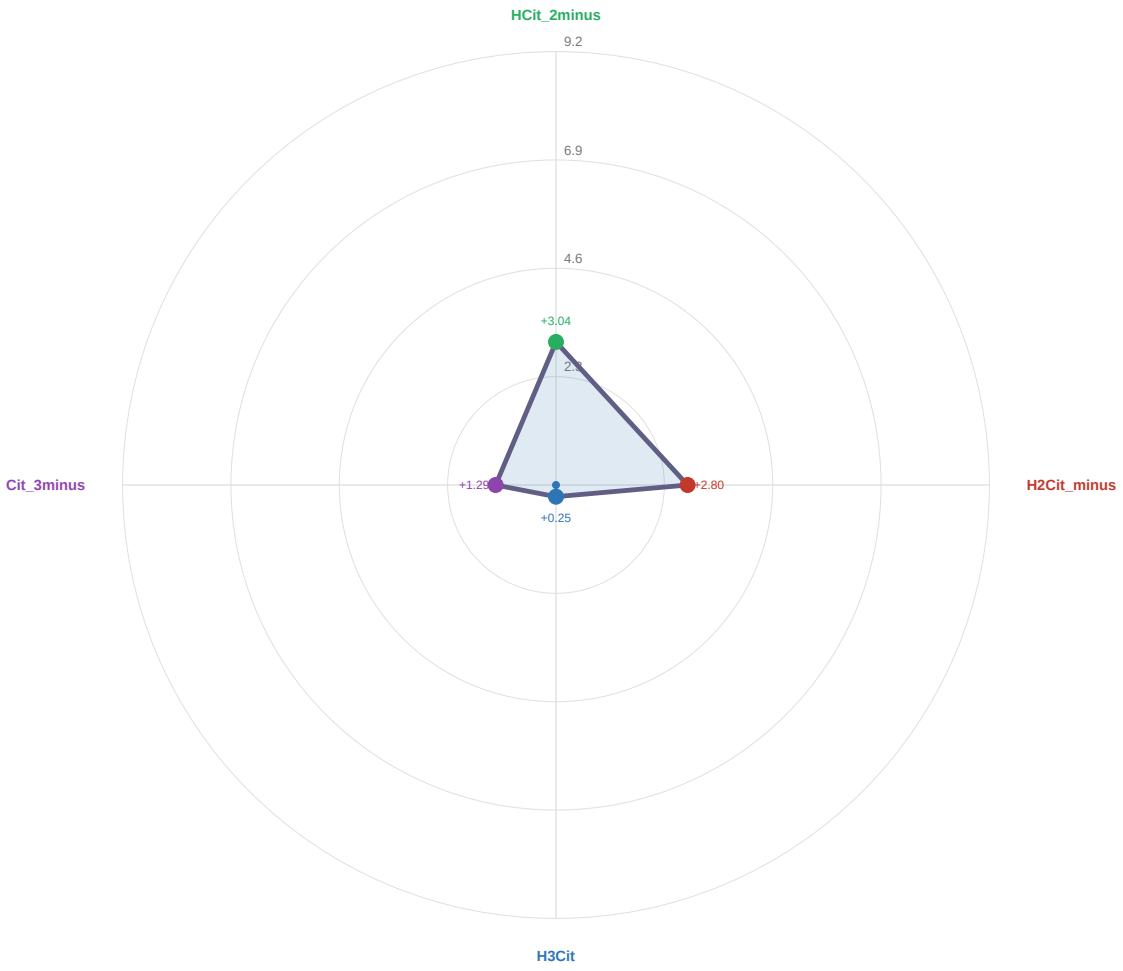
POLAR SLICE — t=134

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 135/200 | ||x||_A = 4.338



From 133: delta ||x||_A = +0.017

Phase = +84.9 deg

H3Cit=+0.25 H2Cit_=+2.80 HCit_2=+3.04 Cit_3m=+1.29

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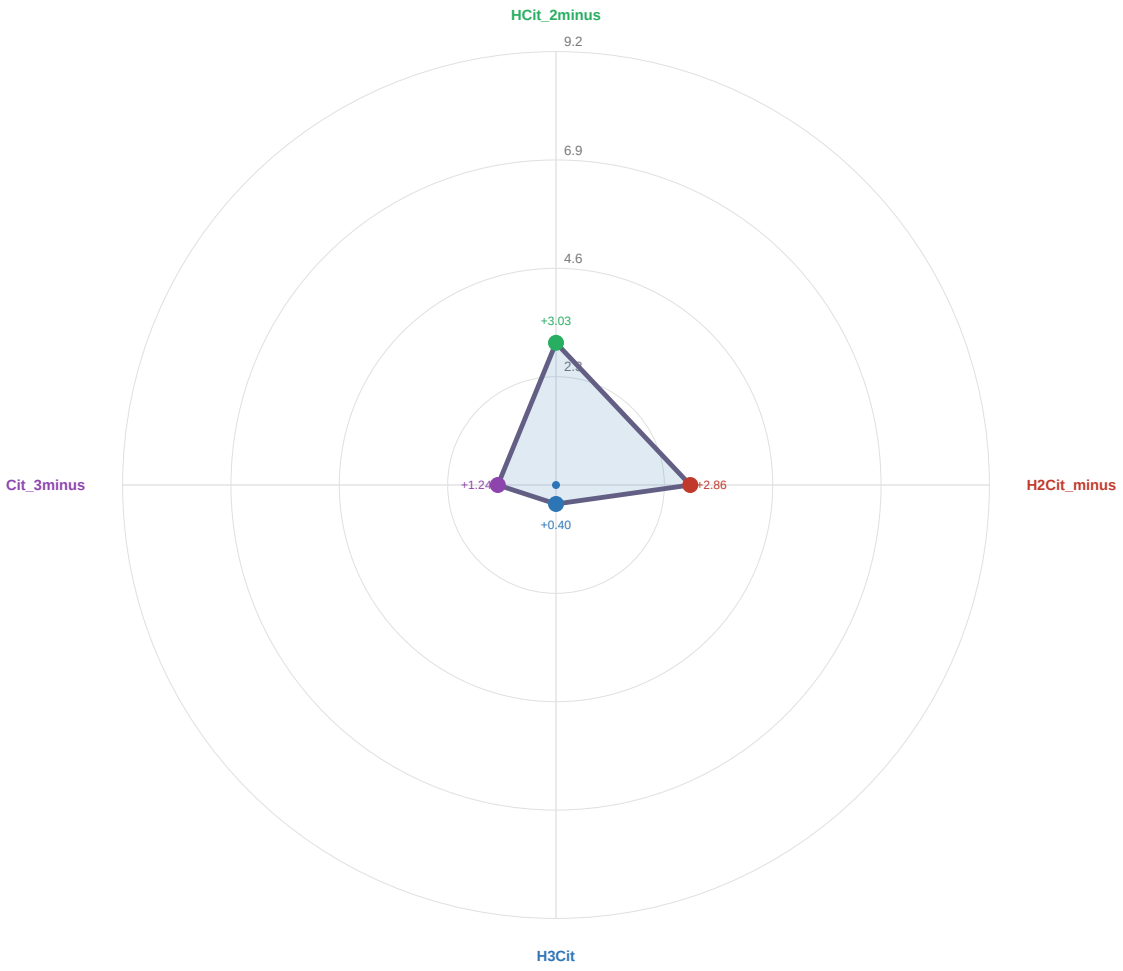
POLAR SLICE — t=135

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 136/200 | ||x||_A = 4.361



From 134: delta ||x||_A = +0.023

Phase = +82.0 deg

H3Cit=+0.40 H2Cit_=+2.86 HCit_2=+3.03 Cit_3m=+1.24

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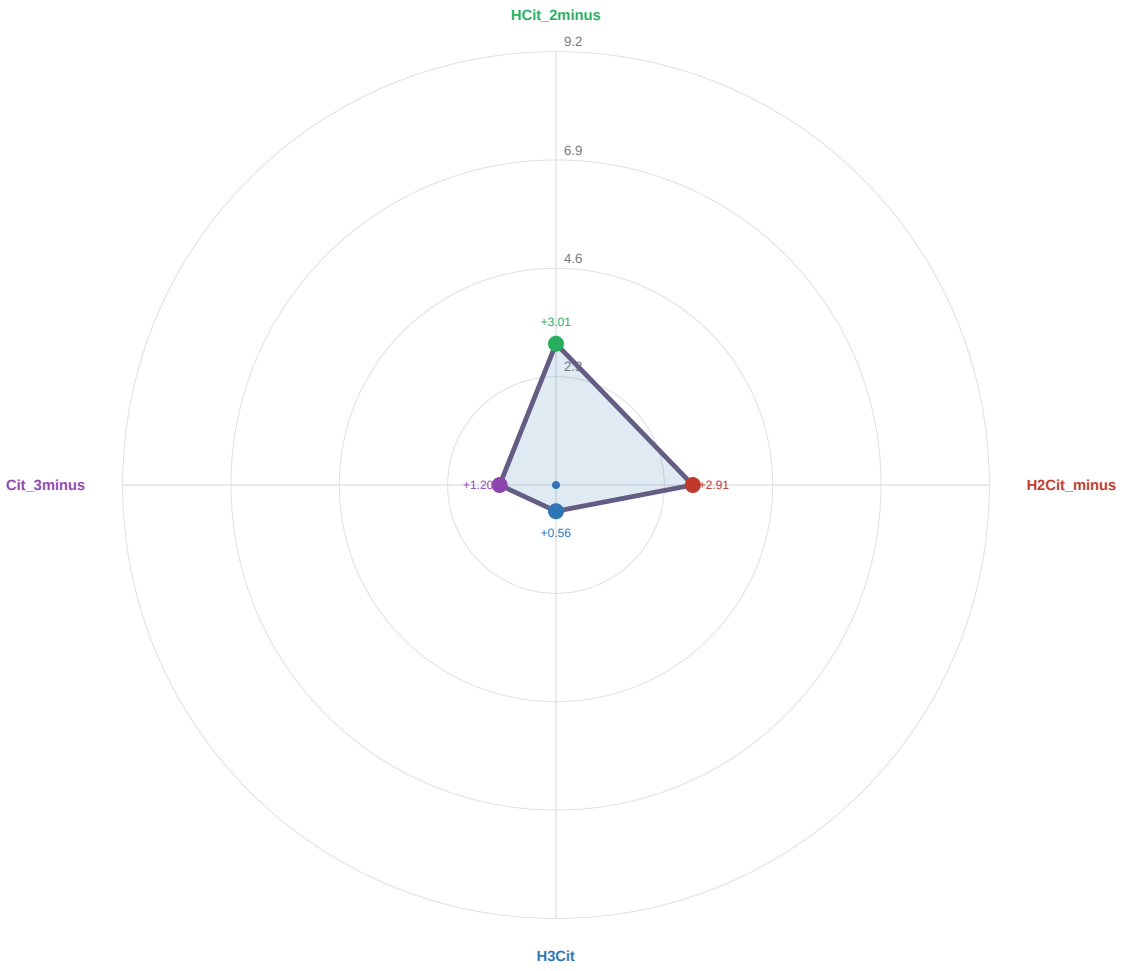
POLAR SLICE — t=136

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 137/200 | ||x||_A = 4.389



From 135: delta ||x||_A = +0.028

Phase = +79.1 deg

H3Cit=+0.56 H2Cit_=+2.91 HCit_2=+3.01 Cit_3m=+1.20

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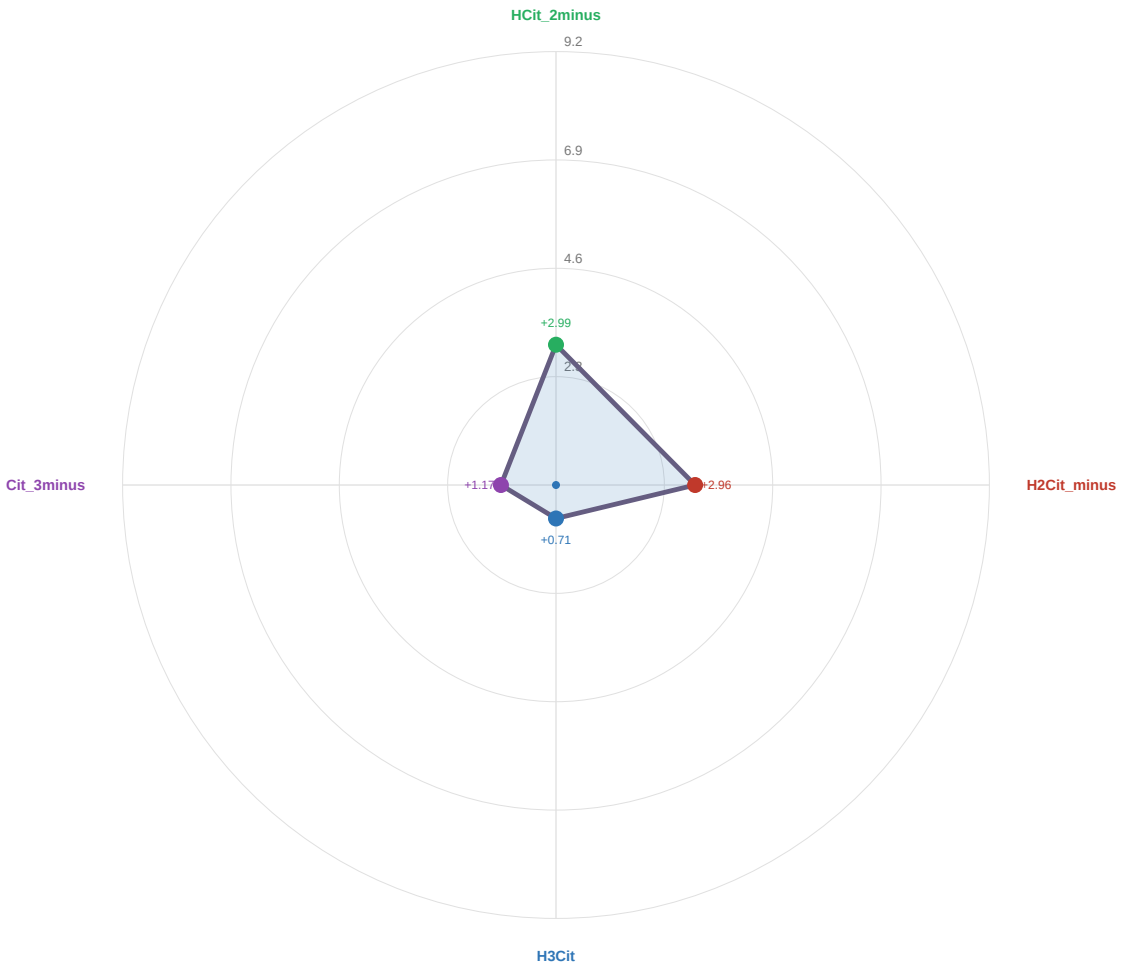
POLAR SLICE — t=137

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 138/200 | ||x||_A = 4.422



From 136: delta ||x||_A = +0.033

Phase = +76.5 deg

H3Cit=+0.71 H2Cit_=+2.96 HCit_2=+2.99 Cit_3m=+1.17

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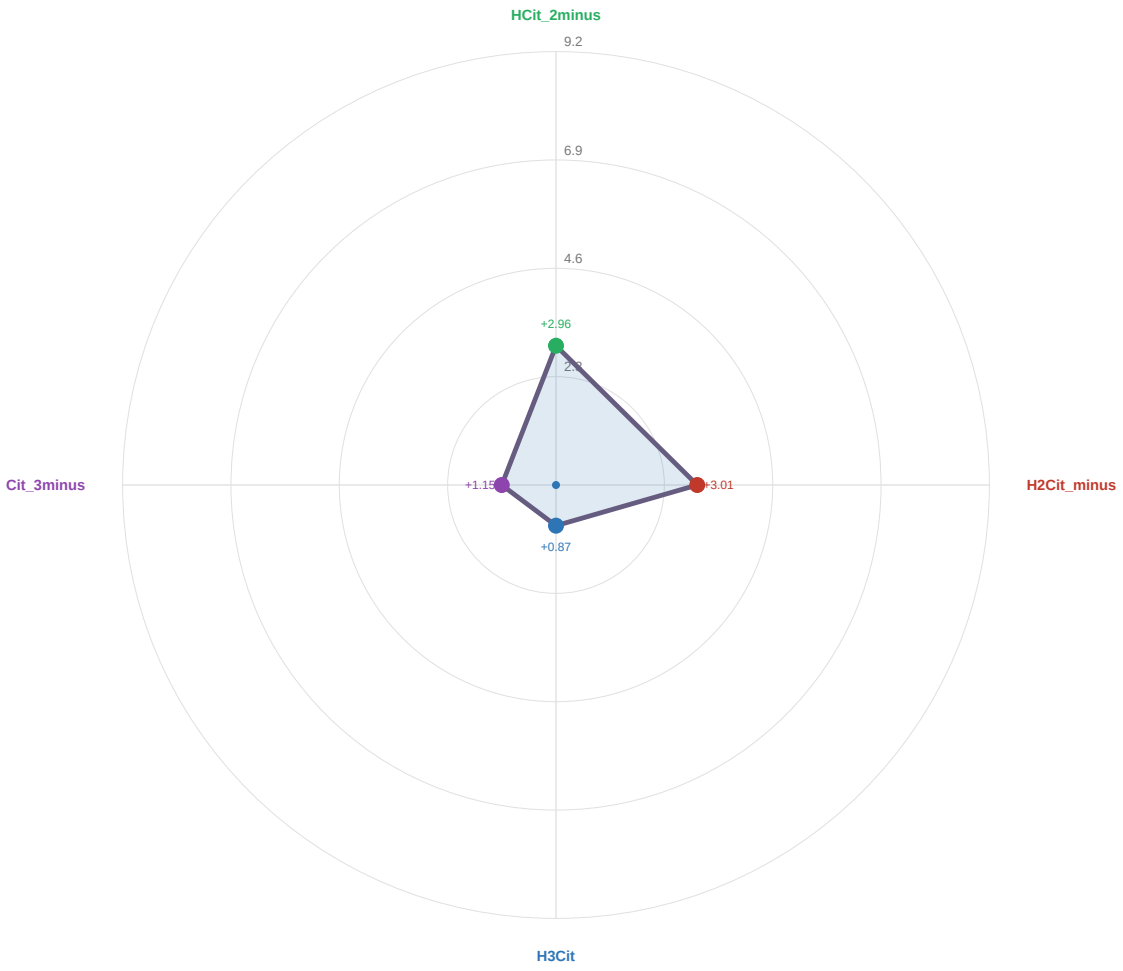
POLAR SLICE — t=138

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 139/200 | ||x||_A = 4.461



From 137: delta ||x||_A = +0.038

Phase = +73.9 deg

H3Cit=+0.87 H2Cit_=+3.01 HCit_2=+2.96 Cit_3m=+1.15

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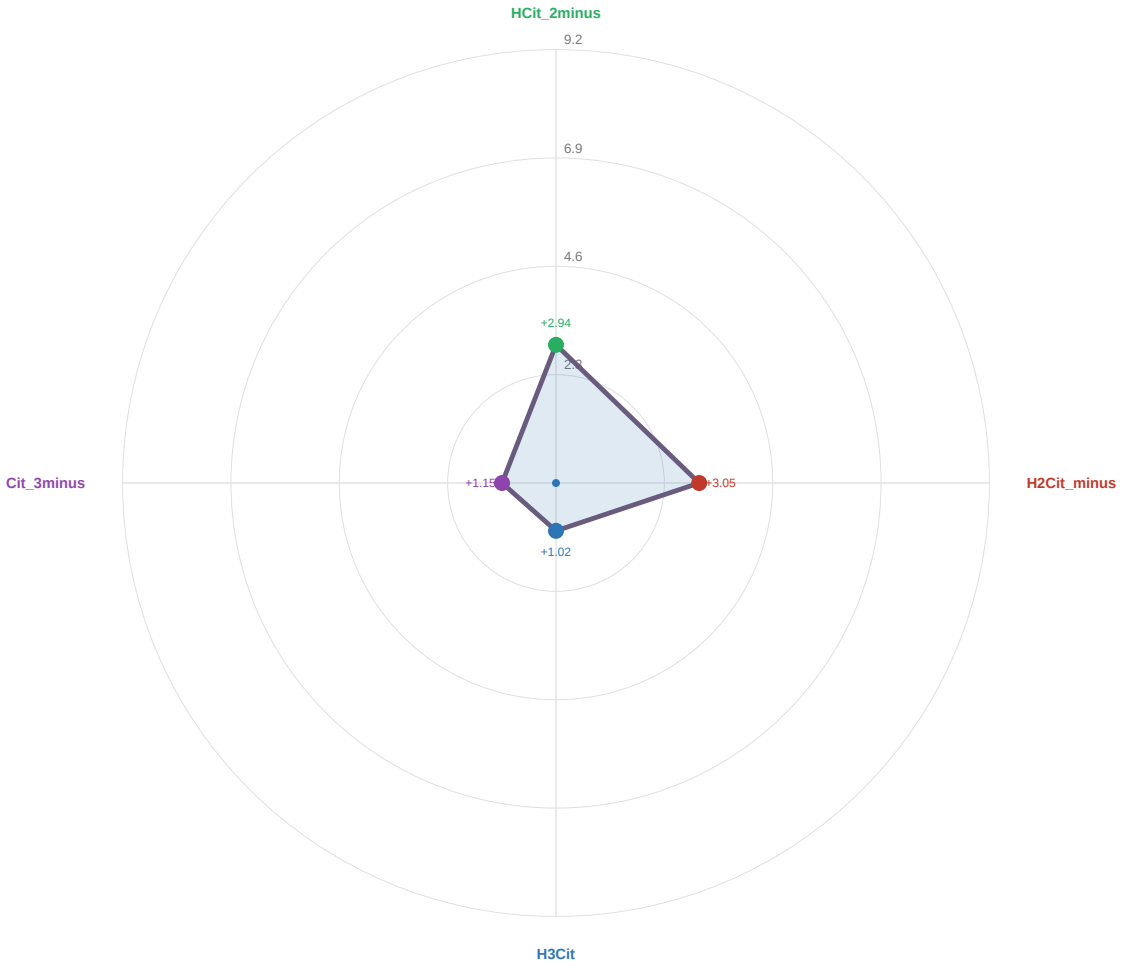
POLAR SLICE — t=139

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 140/200 | ||x||_A = 4.504



From 138: delta ||x||_A = +0.043

Phase = +71.5 deg

H3Cit=+1.02 H2Cit_=+3.05 HCit_2=+2.94 Cit_3m=+1.15

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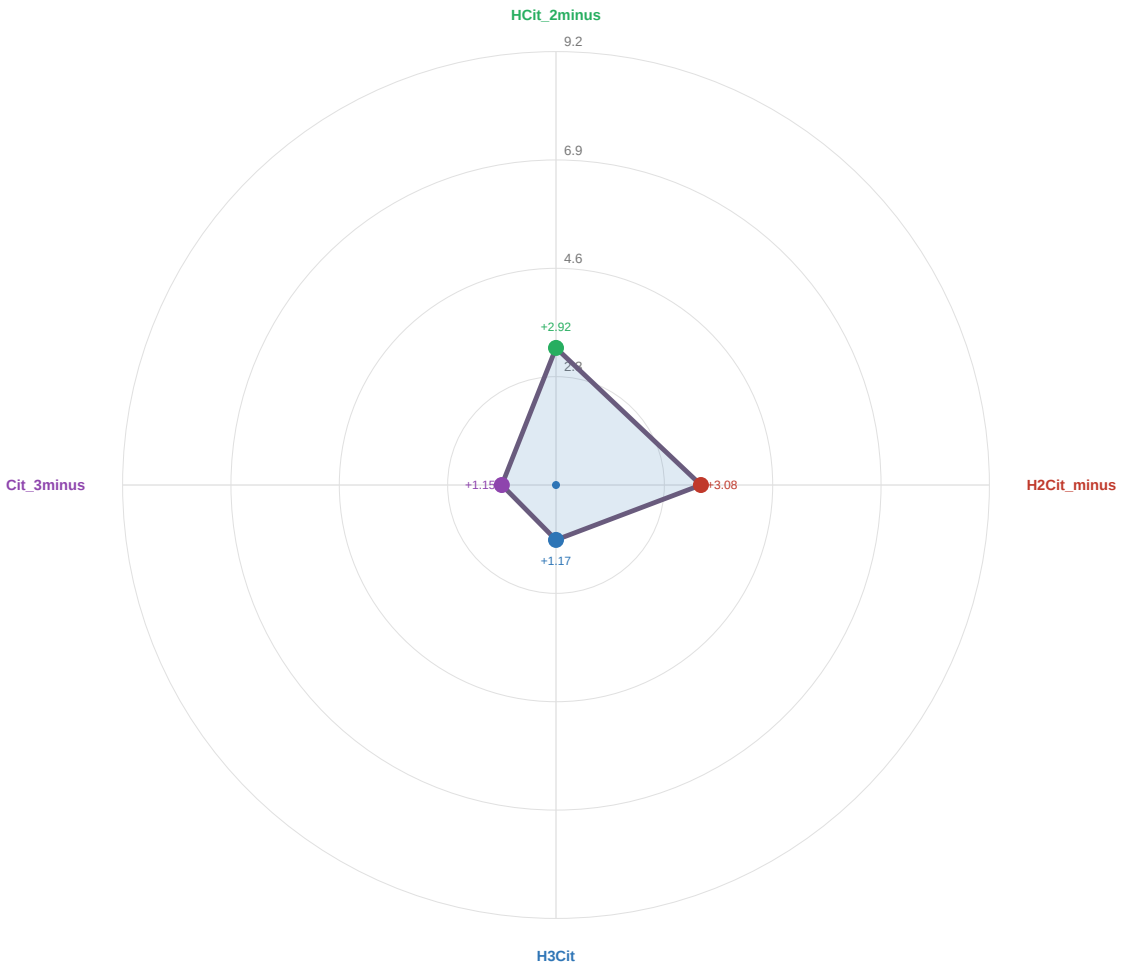
POLAR SLICE — t=140

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 141/200 | ||x||_A = 4.552



From 139: delta ||x||_A = +0.048

Phase = +69.3 deg

H3Cit=+1.17 H2Cit_=+3.08 HCit_2=+2.92 Cit_3m=+1.15

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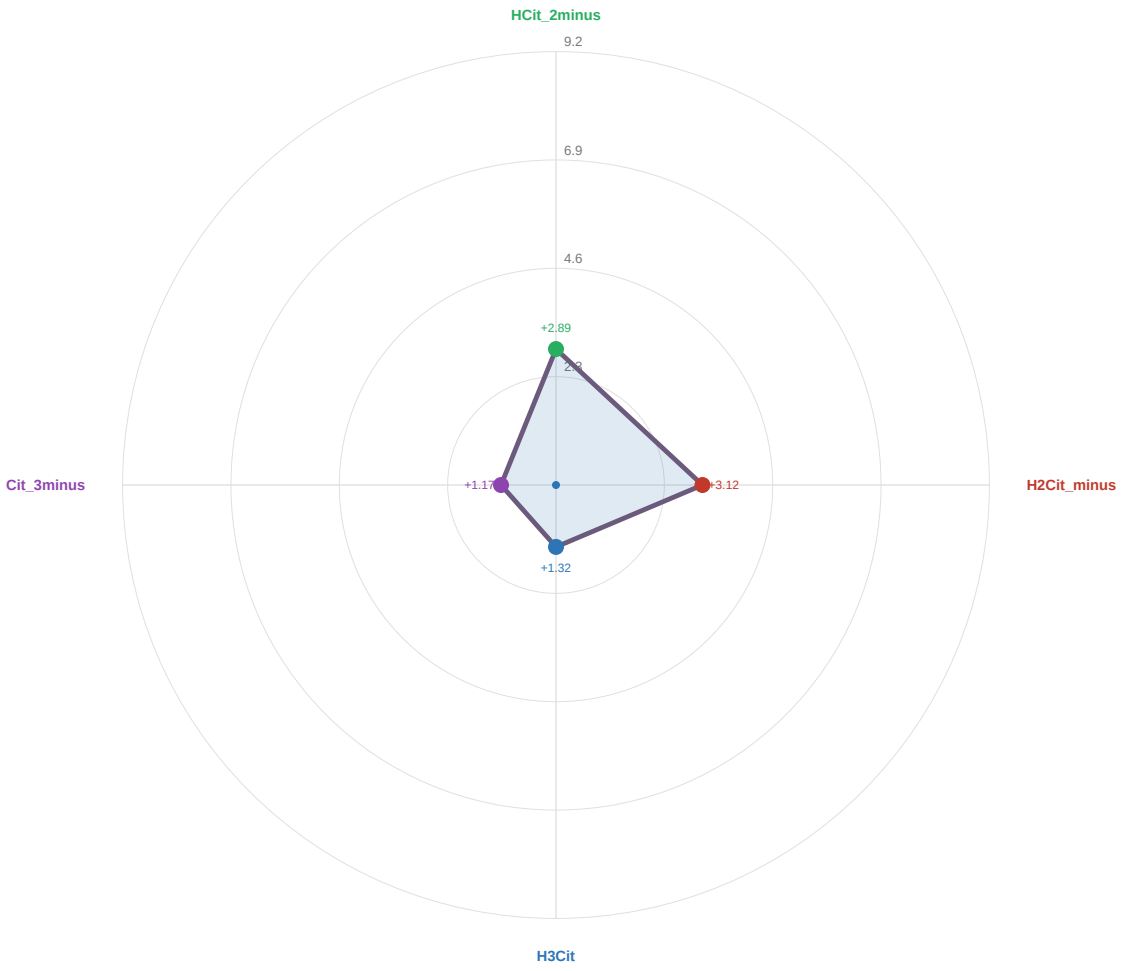
POLAR SLICE — t=141

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 142/200 | ||x||_A = 4.604



From 140: delta ||x||_A = +0.052

Phase = +67.1 deg

H3Cit=+1.32 H2Cit_=+3.12 HCit_2=+2.89 Cit_3m=+1.17

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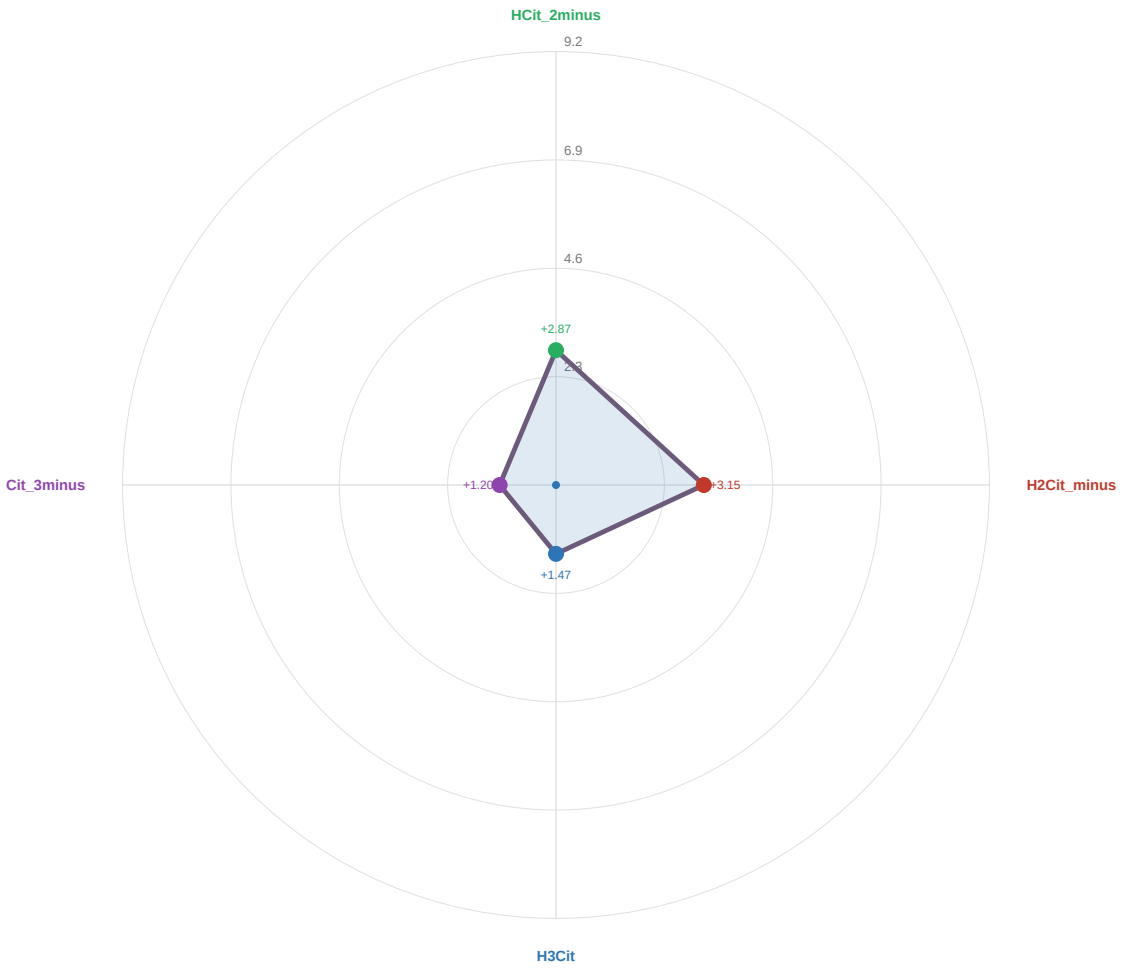
POLAR SLICE — t=142

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 143/200 | ||x||_A = 4.660



From 141: delta ||x||_A = +0.056

Phase = +65.0 deg

H3Cit=+1.47 H2Cit_=+3.15 HCit_2=+2.87 Cit_3m=+1.20

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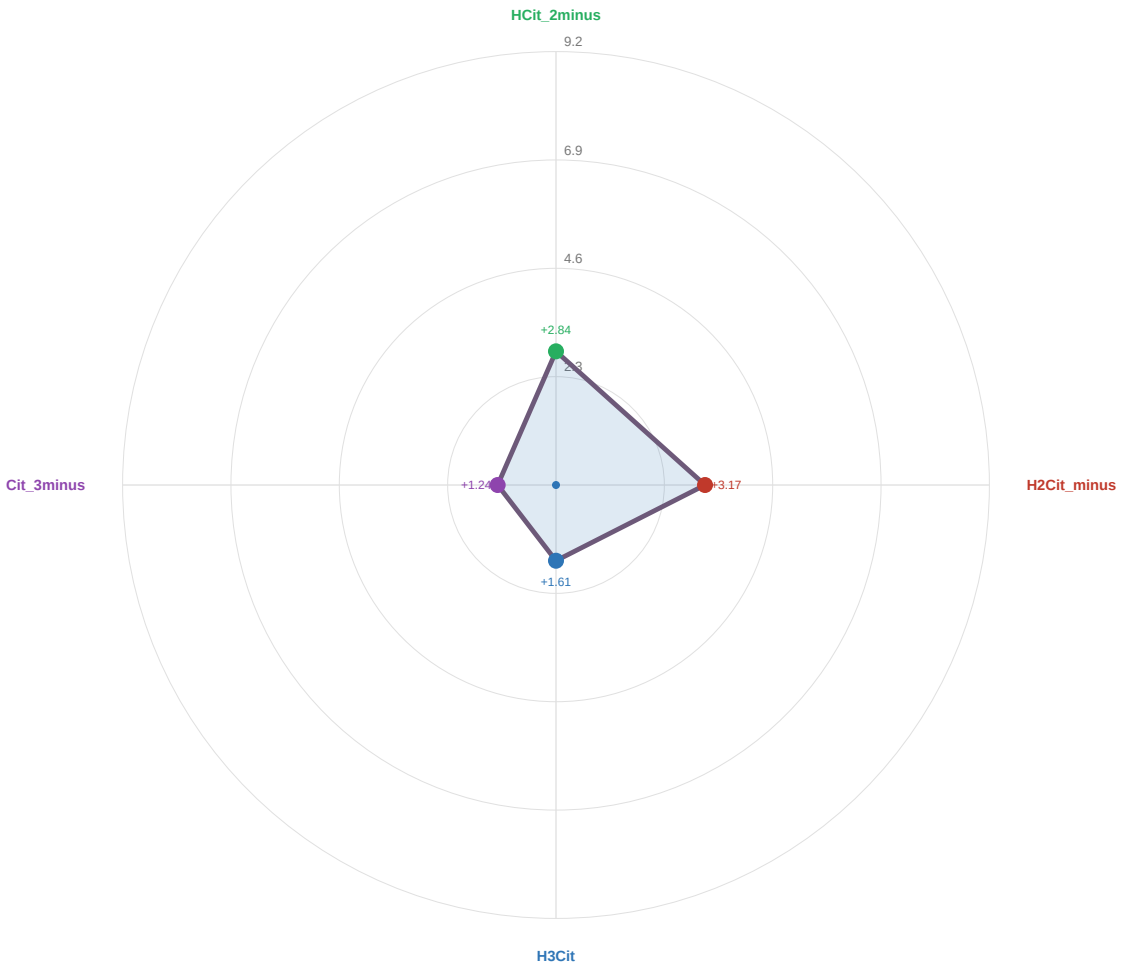
POLAR SLICE — t=143

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 144/200 | ||x||_A = 4.721



From 142: delta ||x||_A = +0.061

Phase = +63.1 deg

H3Cit=+1.61 H2Cit_=+3.17 HCit_2=+2.84 Cit_3m=+1.24

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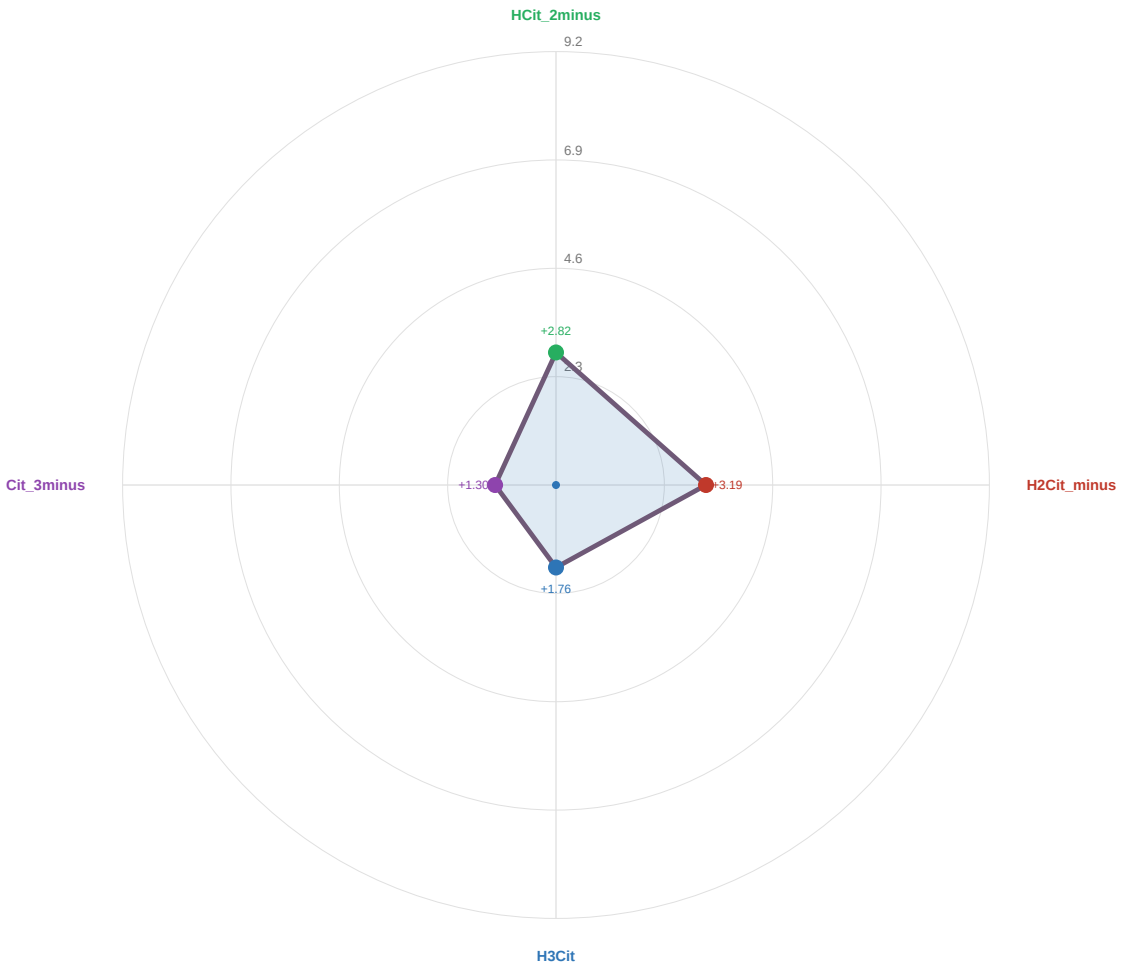
POLAR SLICE — t=144

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 145/200 | ||x||_A = 4.786



From 143: delta ||x||_A = +0.065

Phase = +61.2 deg

H3Cit=+1.76 H2Cit_=+3.19 HCit_2=+2.82 Cit_3m=+1.30

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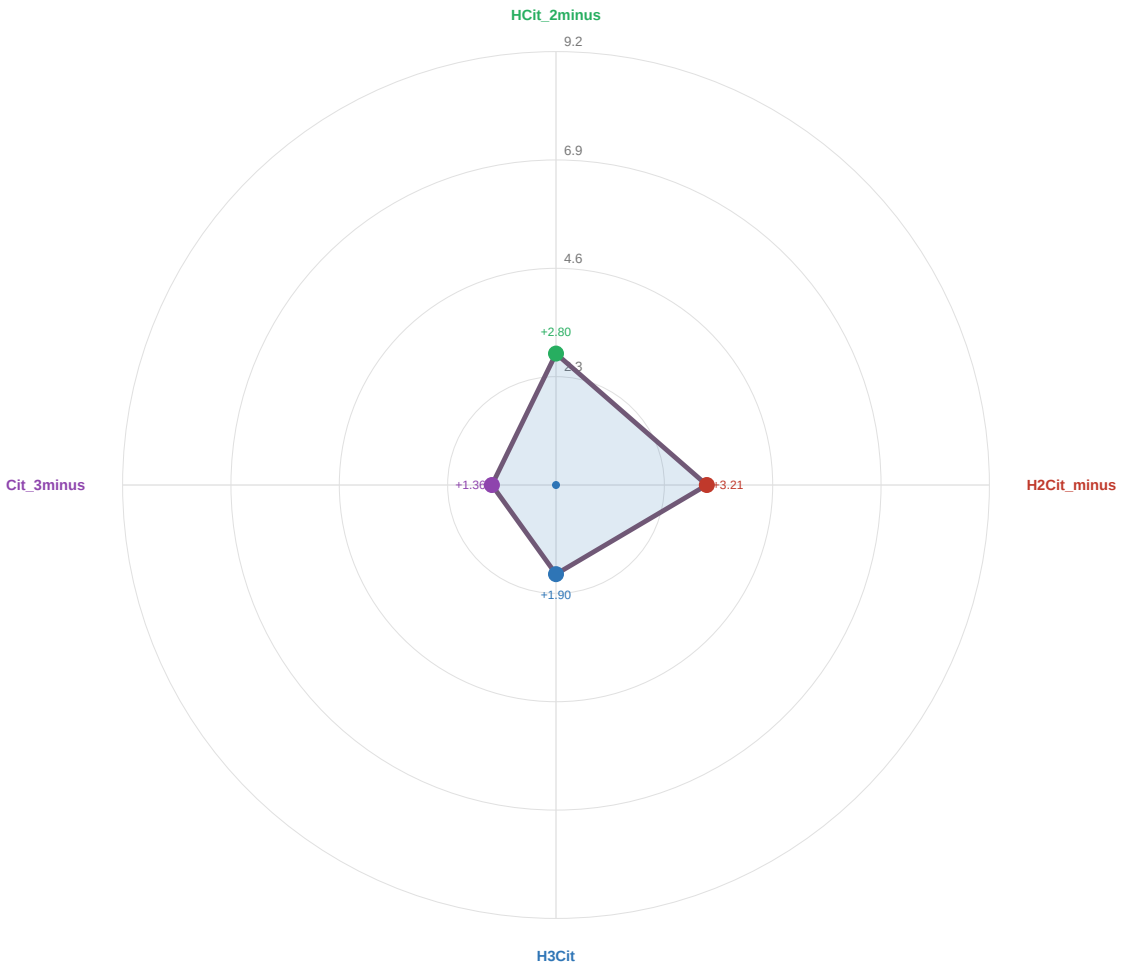
POLAR SLICE — t=145

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 146/200 | ||x||_A = 4.856



From 144: delta ||x||_A = +0.070

Phase = +59.4 deg

H3Cit=+1.90 H2Cit_=+3.21 HCit_2=+2.80 Cit_3m=+1.36

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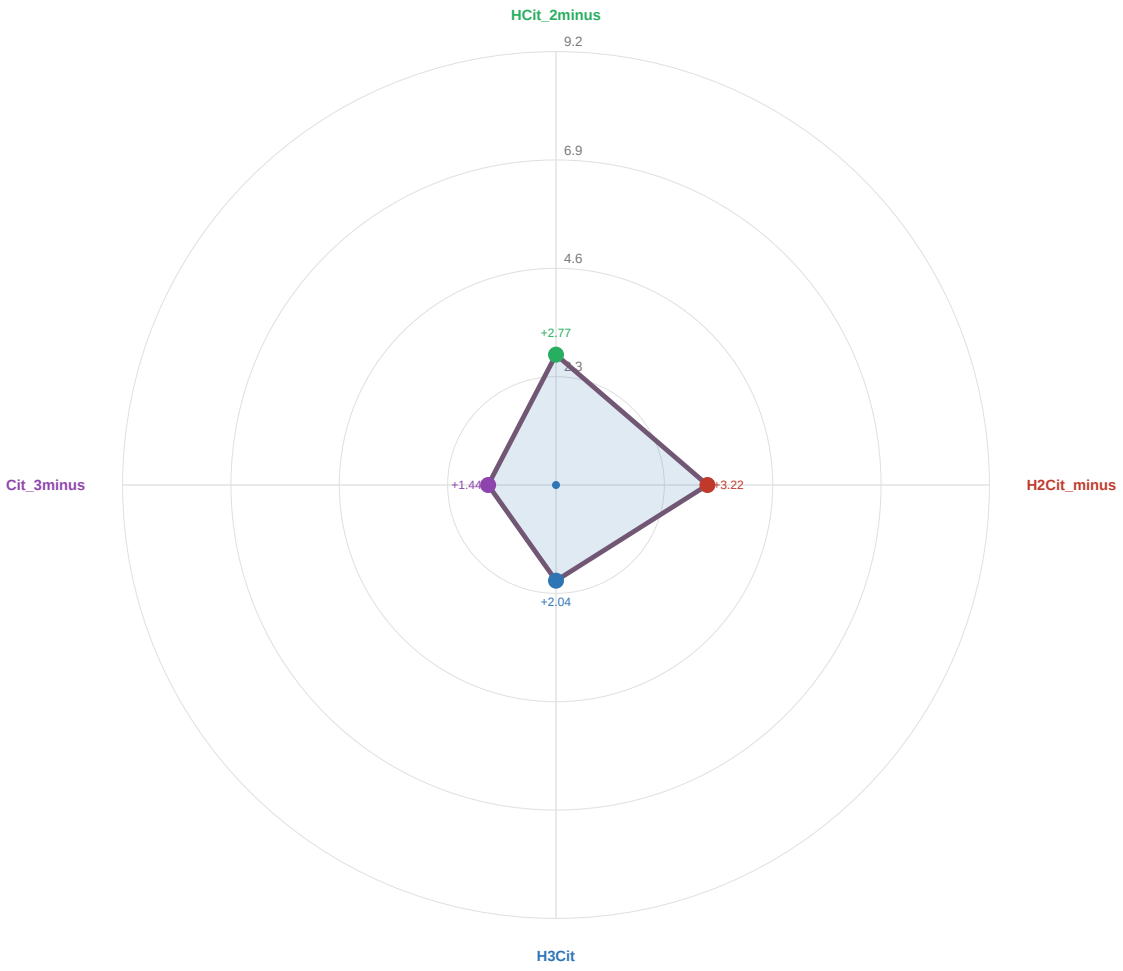
POLAR SLICE — t=146

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 147/200 | ||x||_A = 4.931



From 145: delta ||x||_A = +0.075

Phase = +57.7 deg

H3Cit=+2.04 H2Cit_=+3.22 HCit_2=+2.77 Cit_3m=+1.44

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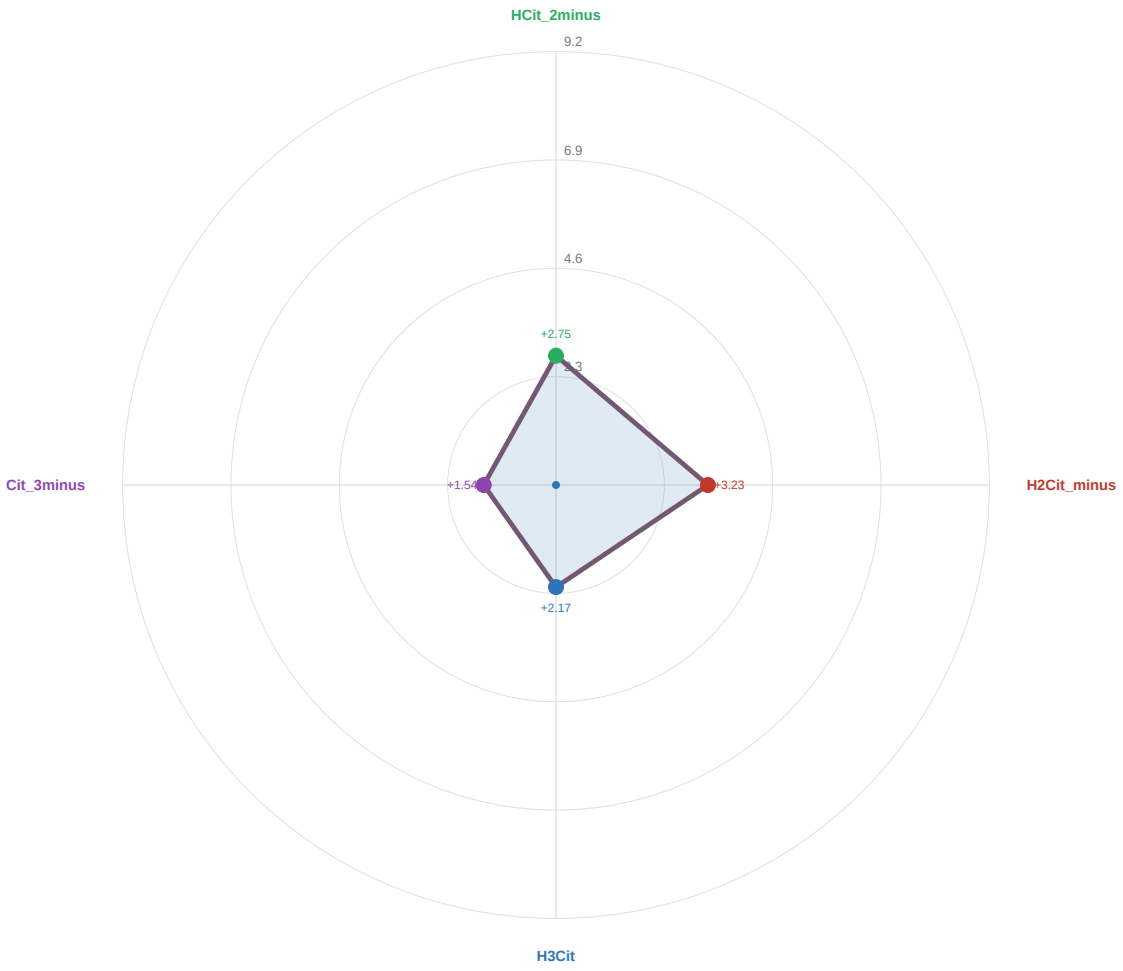
POLAR SLICE — t=147

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 148/200 | ||x||_A = 5.011



From 146: delta ||x||_A = +0.080

Phase = +56.1 deg

H3Cit=+2.17 H2Cit_=+3.23 HCit_2=+2.75 Cit_3m=+1.54

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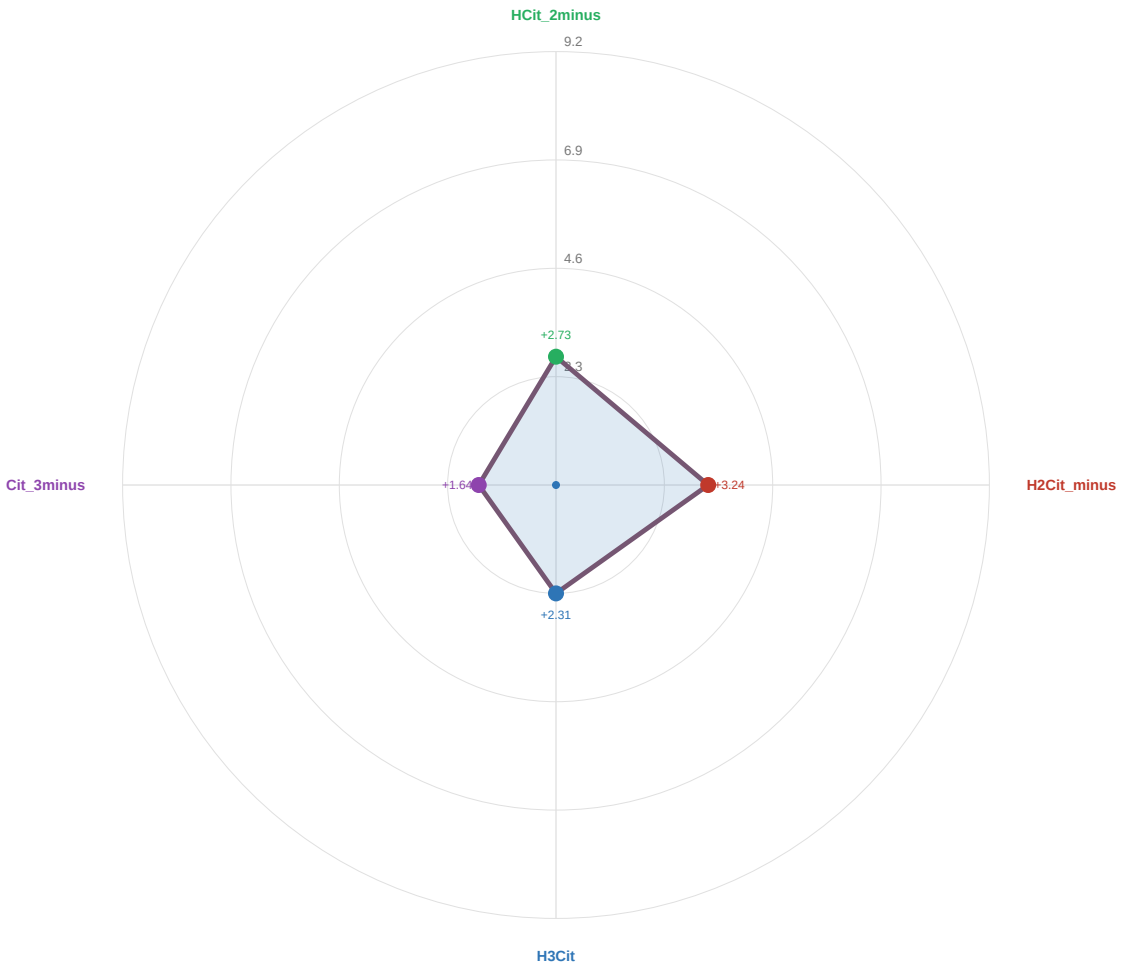
POLAR SLICE — t=148

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 149/200 | ||x||_A = 5.096



From 147: delta ||x||_A = +0.085

Phase = +54.5 deg

H3Cit=+2.31 H2Cit_=+3.24 HCit_2=+2.73 Cit_3m=+1.64

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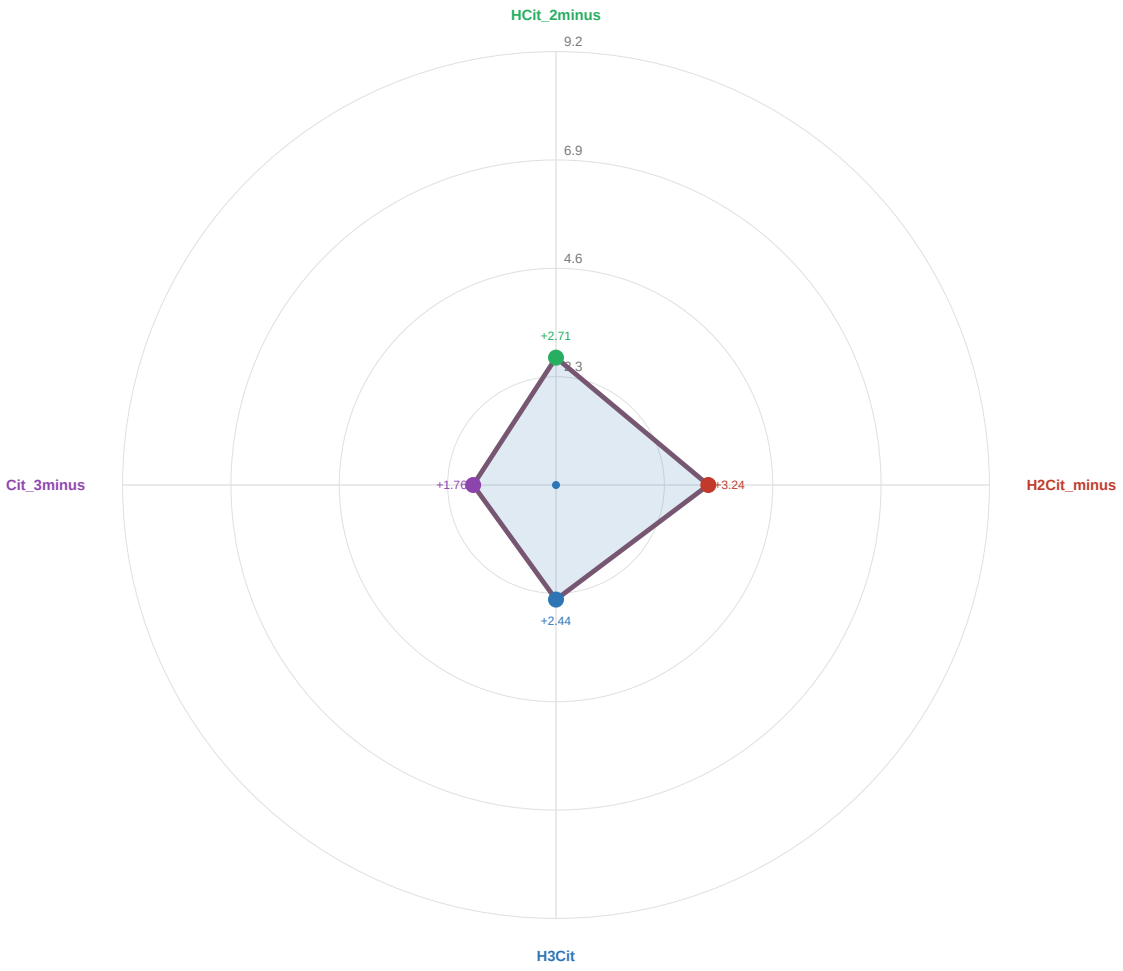
POLAR SLICE — t=149

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 150/200 | ||x||_A = 5.187



From 148: delta ||x||_A = +0.091

Phase = +53.1 deg

H3Cit=+2.44 H2Cit_=+3.24 HCit_2=+2.71 Cit_3m=+1.76

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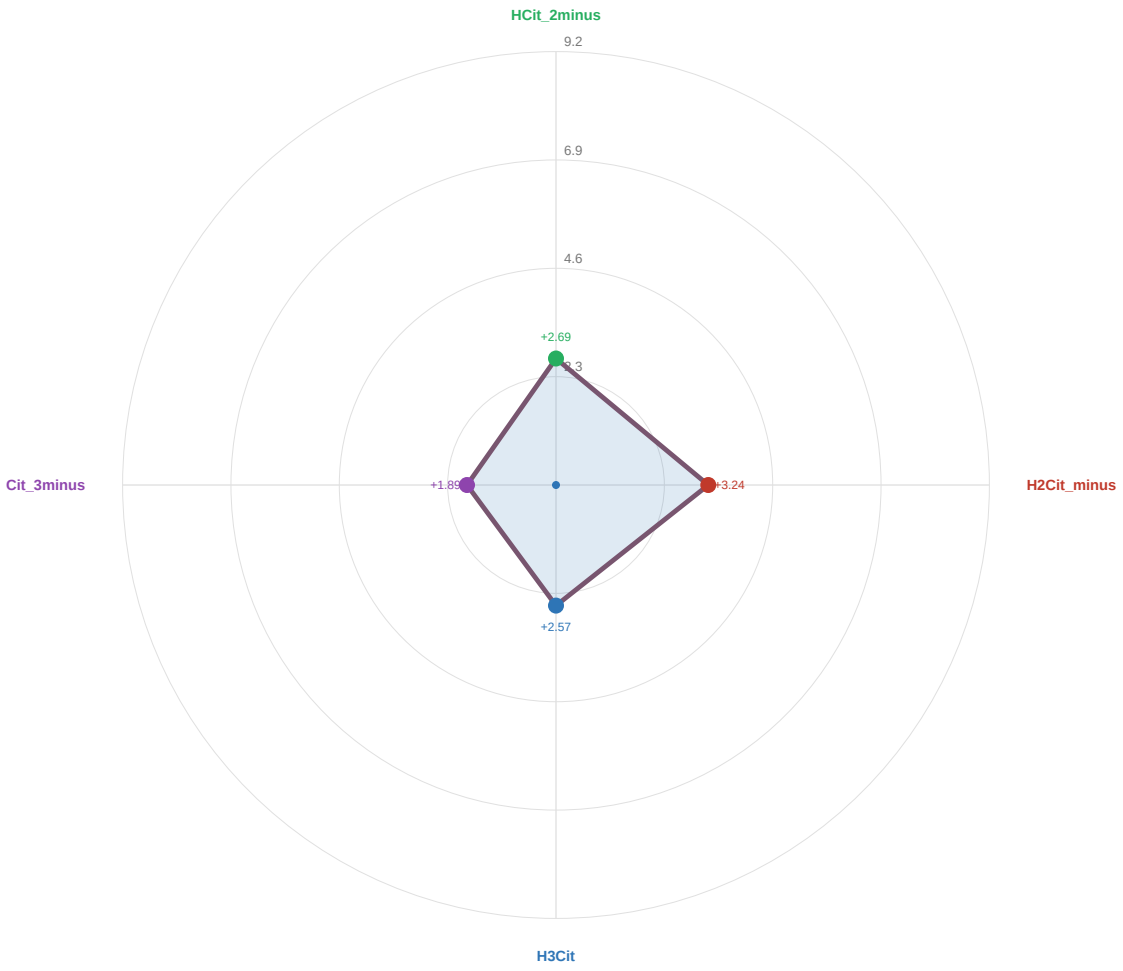
POLAR SLICE — t=150

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 151/200 | ||x||_A = 5.284



From 149: delta ||x||_A = +0.097

Phase = +51.6 deg

H3Cit=+2.57 H2Cit_=+3.24 HCit_2=+2.69 Cit_3m=+1.89

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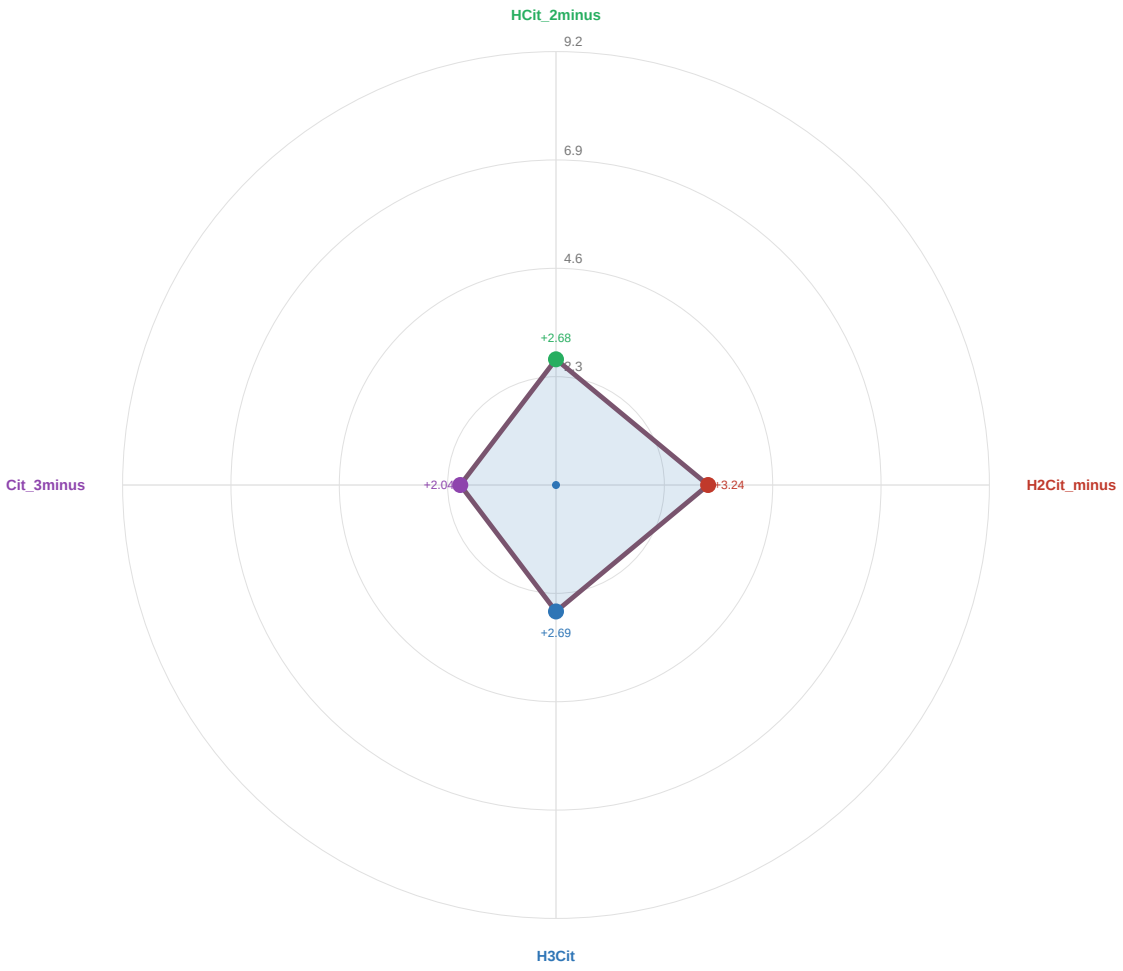
POLAR SLICE — t=151

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 152/200 | ||x||_A = 5.387



From 150: delta ||x||_A = +0.103

Phase = +50.3 deg

H3Cit=+2.69 H2Cit_=+3.24 HCit_2=+2.68 Cit_3m=+2.04

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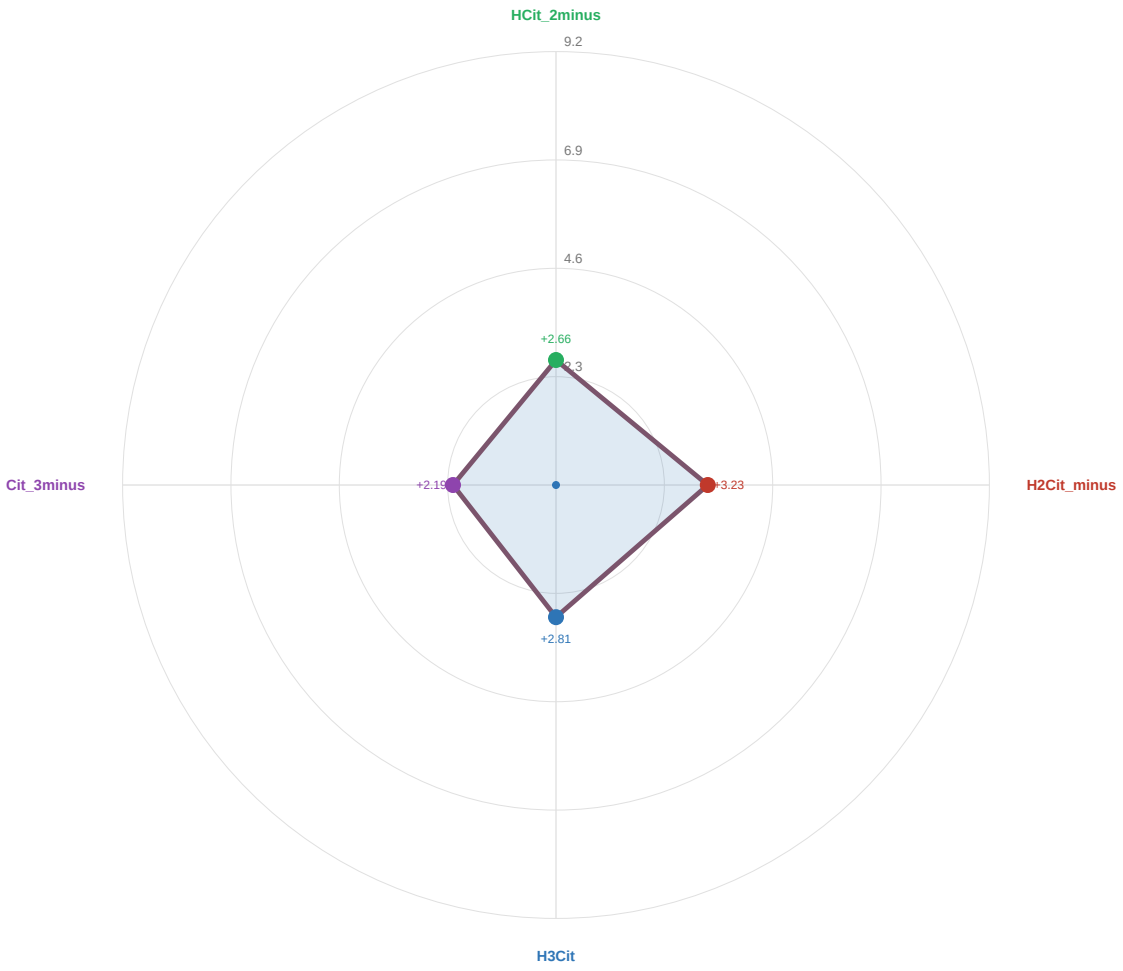
POLAR SLICE — t=152

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 153/200 | ||x||_A = 5.497



From 151: delta ||x||_A = +0.110

Phase = +49.0 deg

H3Cit=+2.81 H2Cit_=+3.23 HCit_2=+2.66 Cit_3m=+2.19

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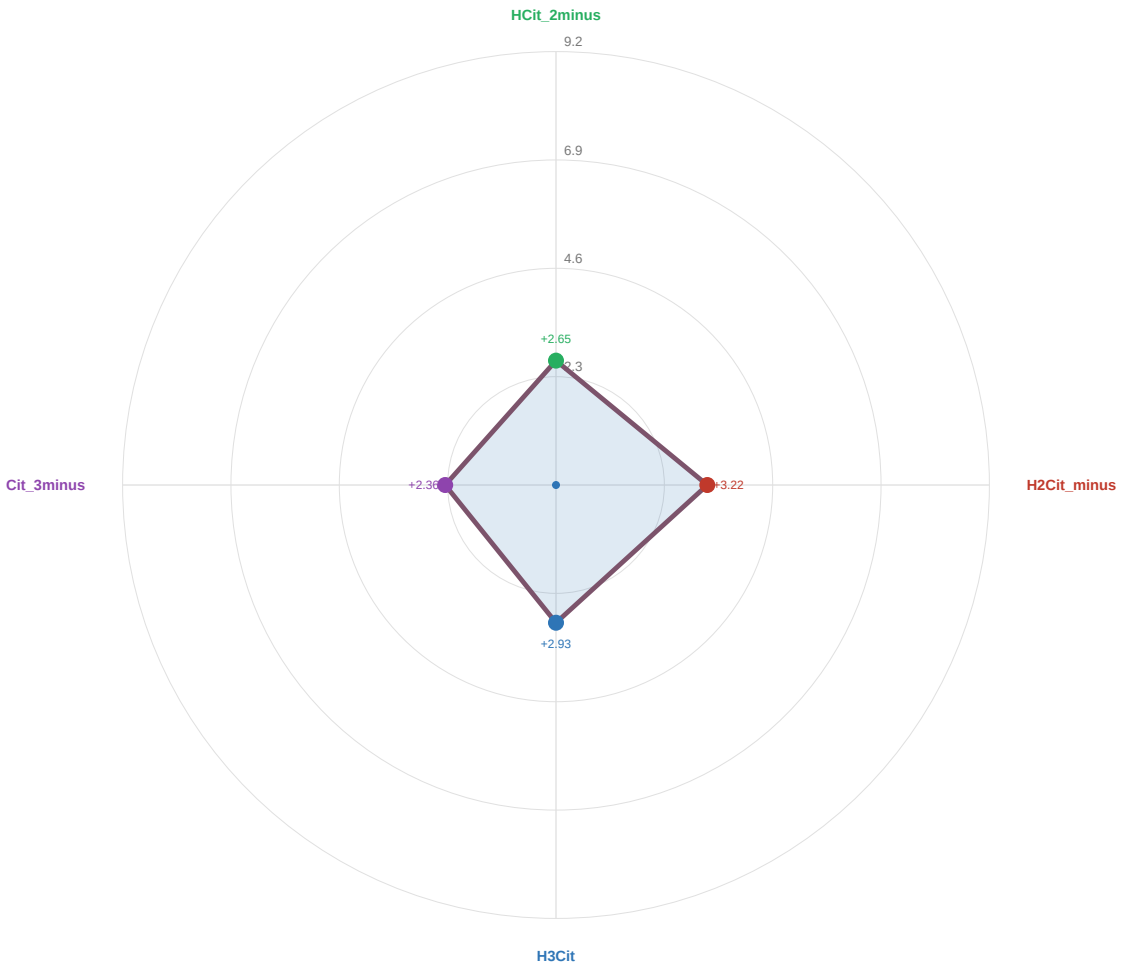
POLAR SLICE — t=153

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 154/200 | ||x||_A = 5.614



From 152: delta ||x||_A = +0.117

Phase = +47.7 deg

H3Cit=+2.93 H2Cit_=+3.22 HCit_2=+2.65 Cit_3m=+2.36

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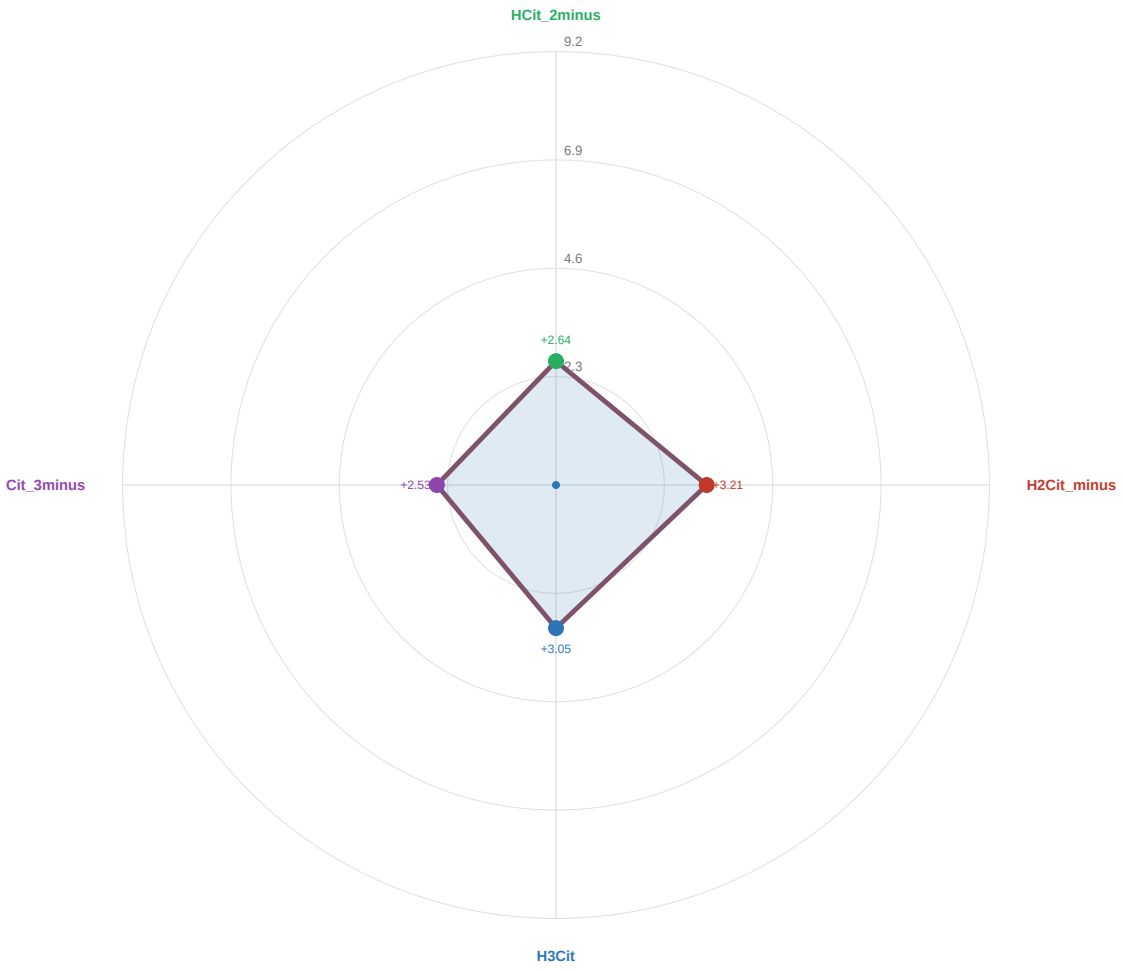
POLAR SLICE — t=154

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 155/200 | ||x||_A = 5.739



From 153: delta ||x||_A = +0.124

Phase = +46.5 deg

H3Cit=+3.05 H2Cit_=+3.21 HCit_2=+2.64 Cit_3m=+2.53

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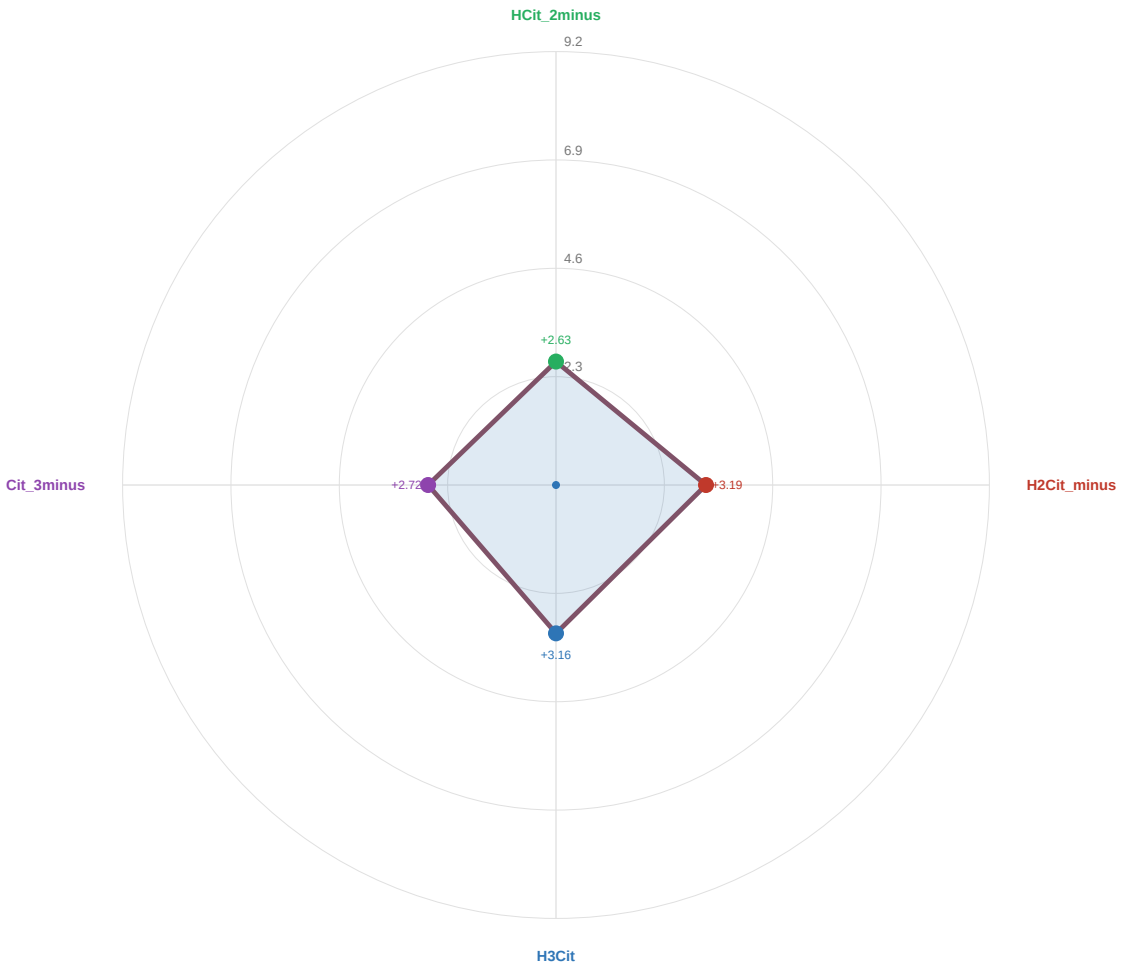
POLAR SLICE — t=155

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 156/200 | ||x||_A = 5.870



From 154: delta ||x||_A = +0.132

Phase = +45.3 deg

H3Cit=+3.16 H2Cit_=+3.19 HCit_2=+2.63 Cit_3m=+2.72

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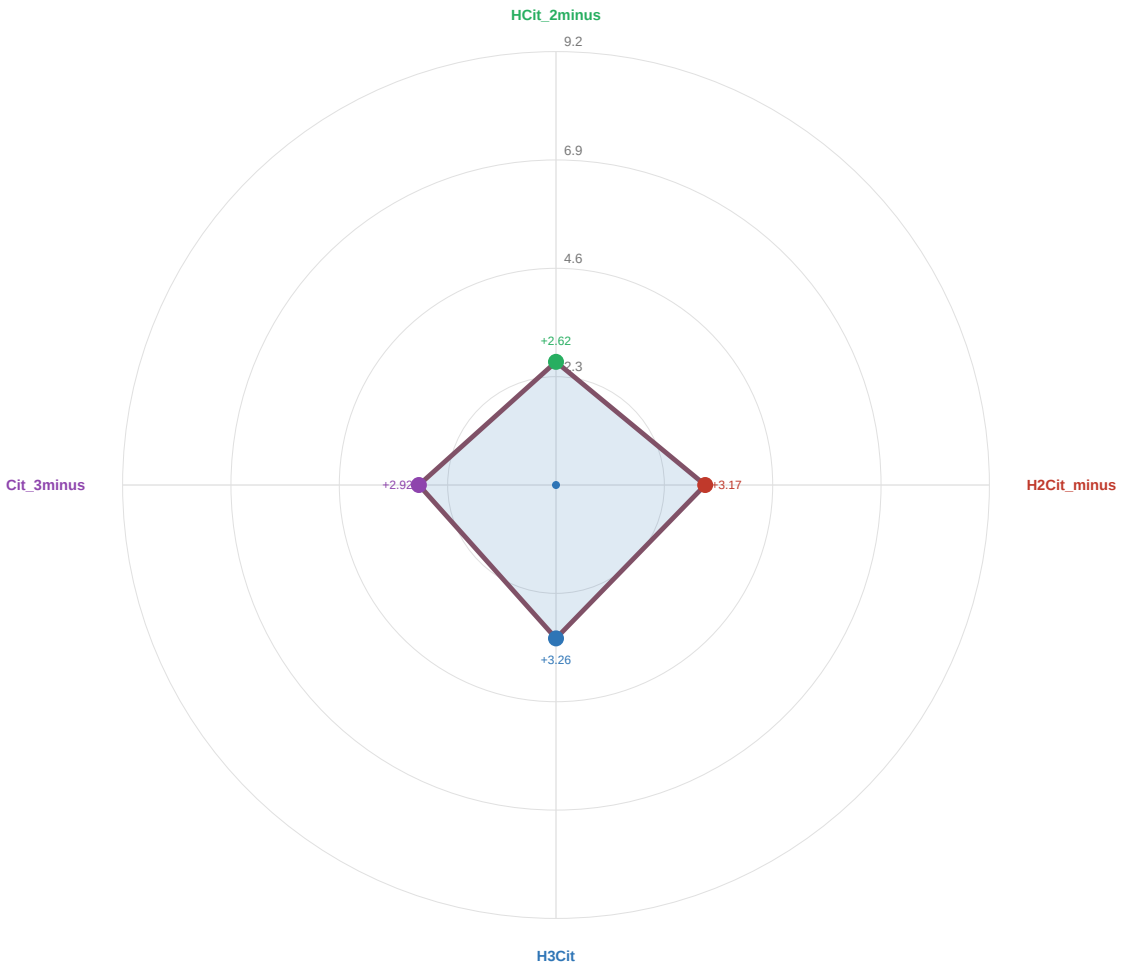
POLAR SLICE — t=156

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 157/200 | ||x||_A = 6.009



From 155: delta ||x||_A = +0.139

Phase = +44.2 deg

H3Cit=+3.26 H2Cit_=+3.17 HCit_2=+2.62 Cit_3m=+2.92

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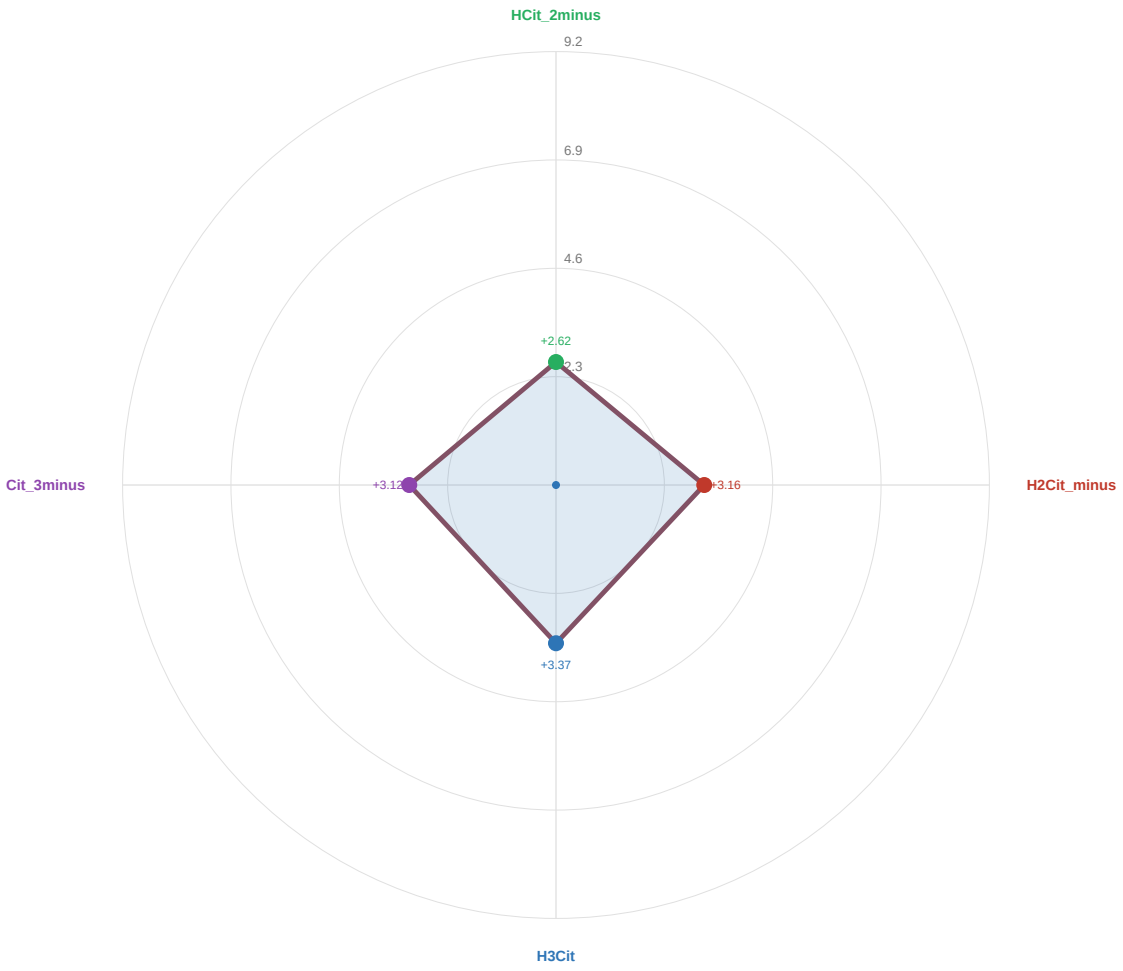
POLAR SLICE — t=157

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 158/200 | ||x||_A = 6.156



From 156: delta ||x||_A = +0.146

Phase = +43.1 deg

H3Cit=+3.37 H2Cit_=+3.16 HCit_2=+2.62 Cit_3m=+3.12

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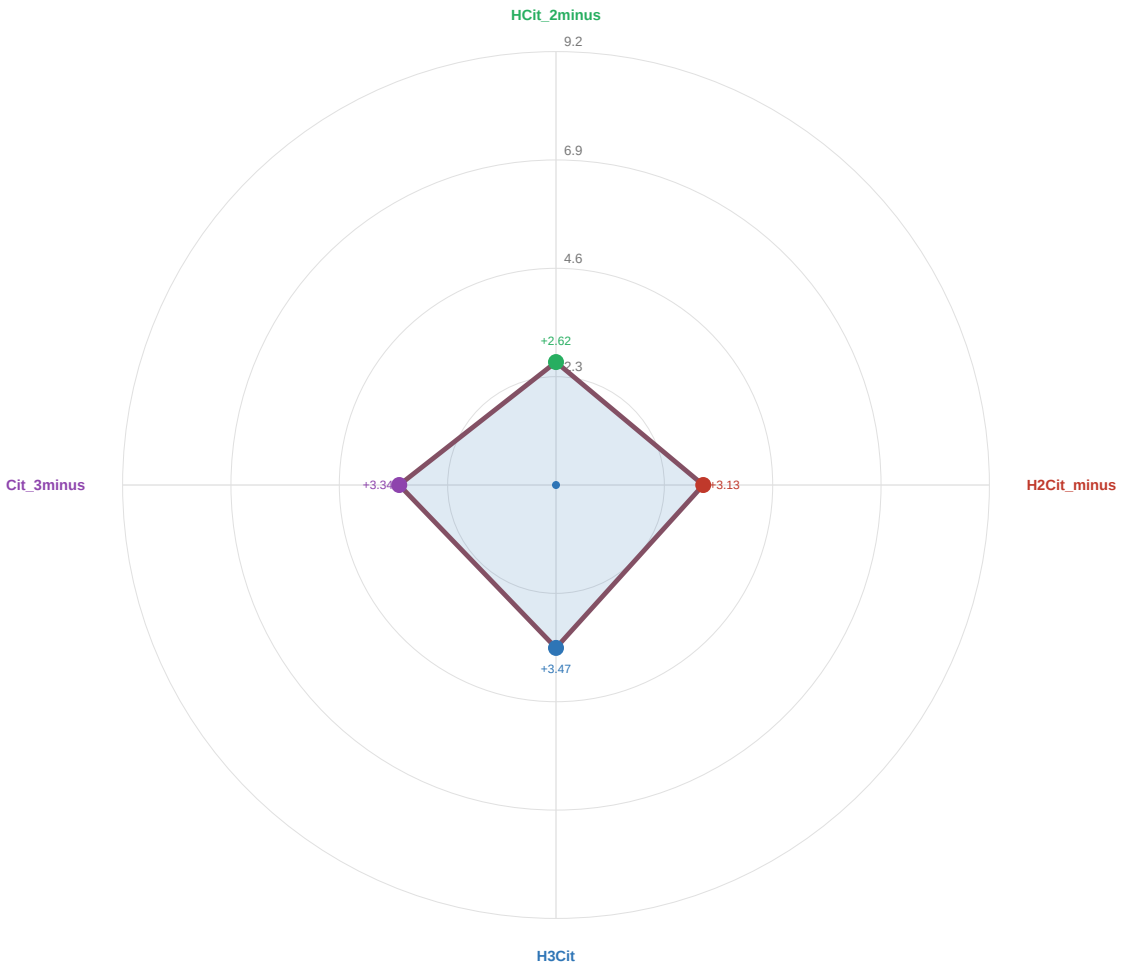
POLAR SLICE — t=158

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 159/200 | ||x||_A = 6.309



From 157: delta ||x||_A = +0.154

Phase = +42.1 deg

H3Cit=+3.47 H2Cit_=+3.13 HCit_2=+2.62 Cit_3m=+3.34

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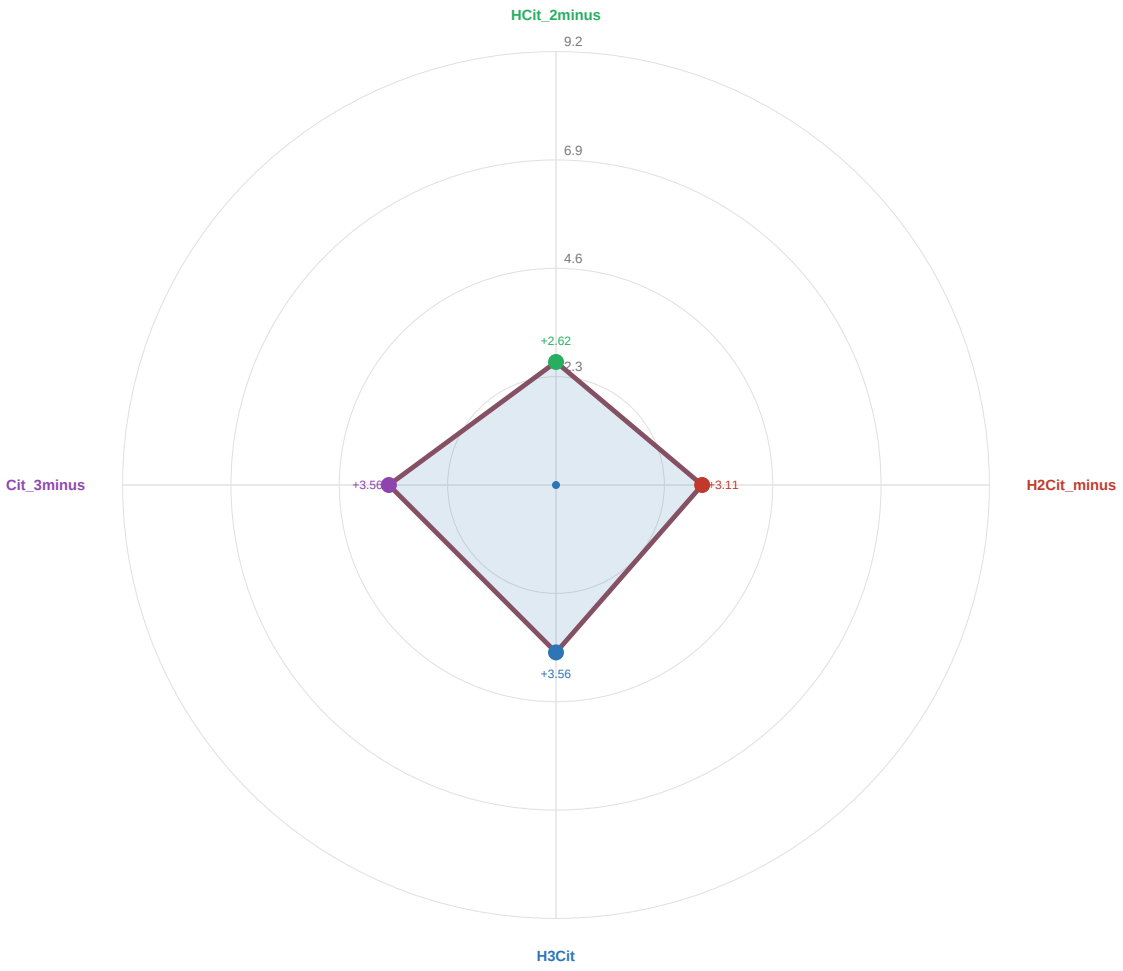
POLAR SLICE — t=159

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 160/200 | ||x||_A = 6.470



From 158: delta ||x||_A = +0.161

Phase = +41.1 deg

H3Cit=+3.56 H2Cit_=+3.11 HCit_2=+2.62 Cit_3m=+3.56

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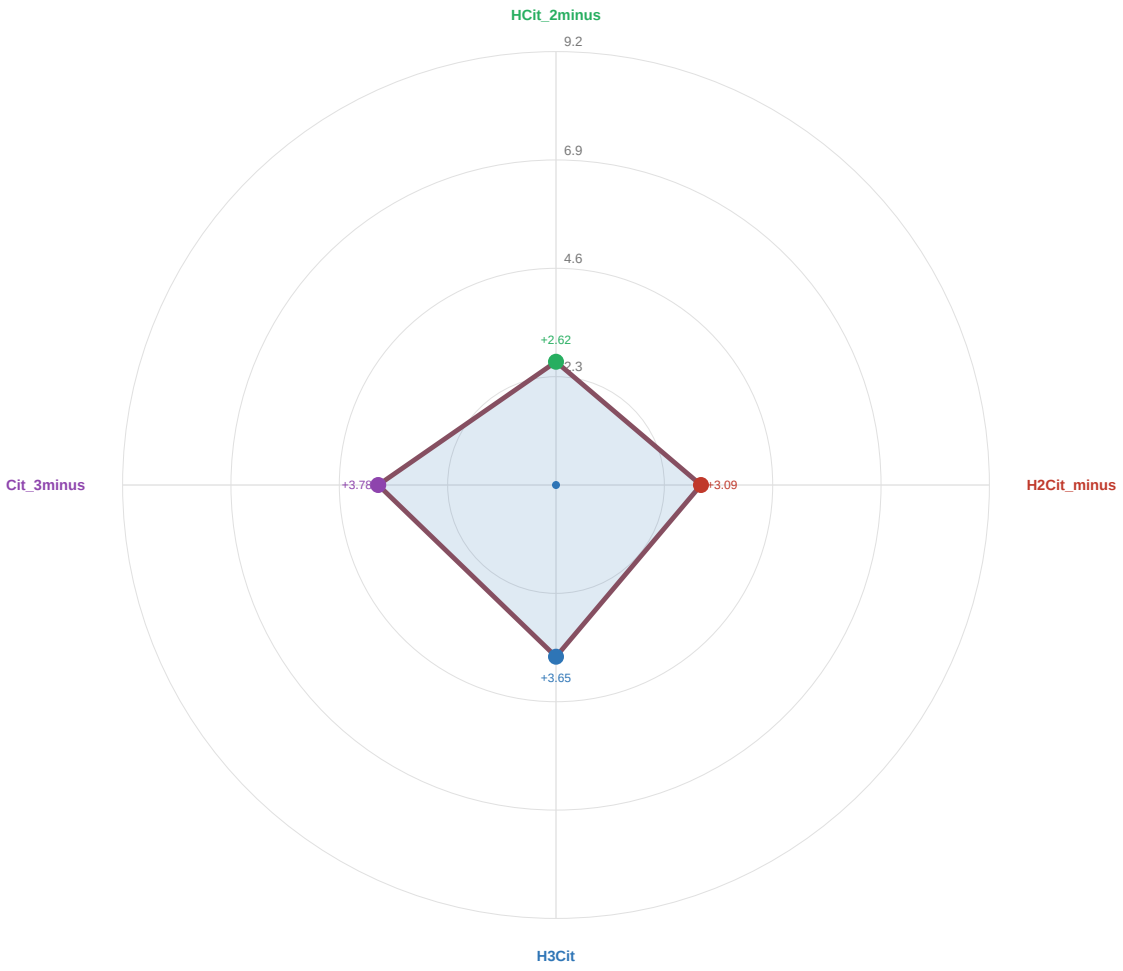
POLAR SLICE — t=160

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 161/200 | ||x||_A = 6.637



From 159: delta ||x||_A = +0.167

Phase = +40.2 deg

H3Cit=+3.65 H2Cit_=+3.09 HCit_2=+2.62 Cit_3m=+3.78

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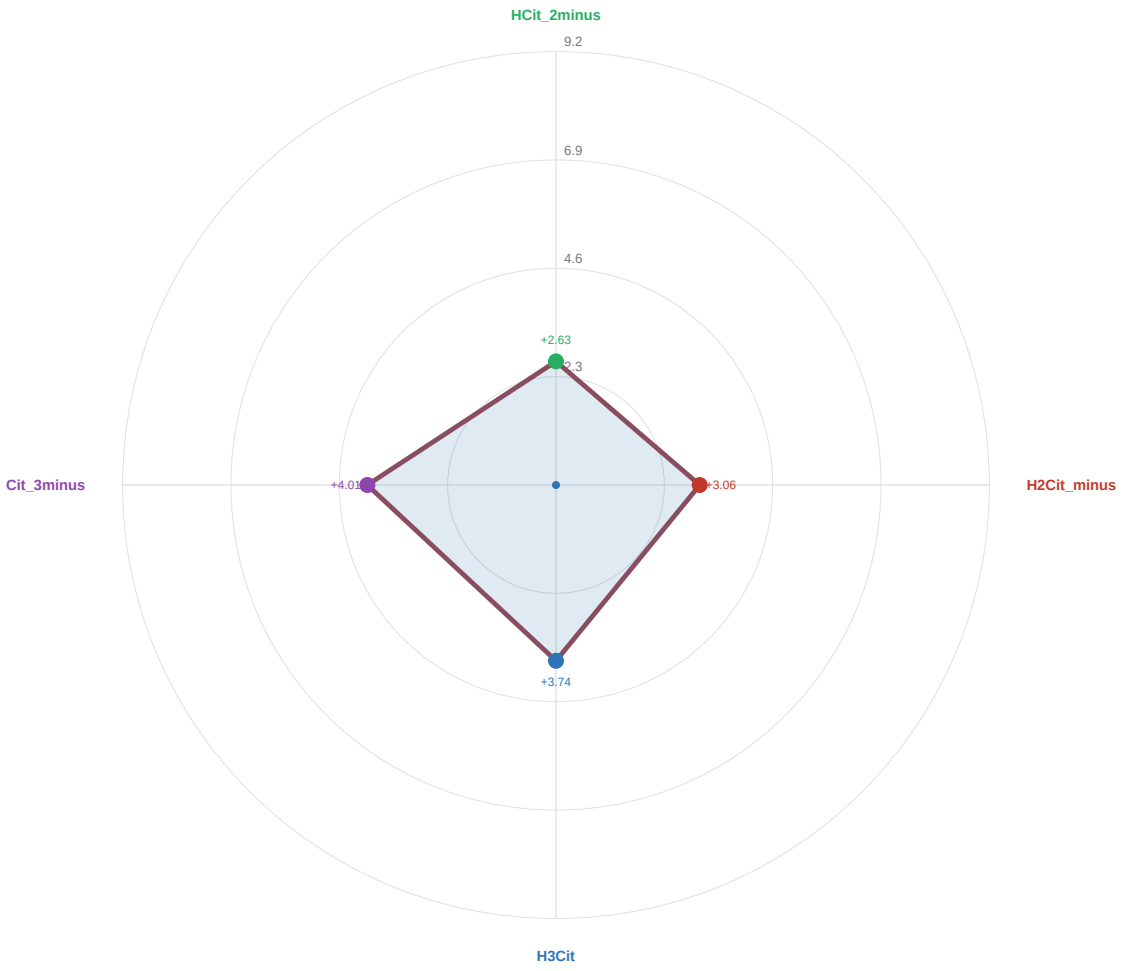
POLAR SLICE — t=161

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 162/200 | ||x||_A = 6.811



From 160: delta ||x||_A = +0.174

Phase = +39.3 deg

H3Cit=+3.74 H2Cit_=+3.06 HCit_2=+2.63 Cit_3m=+4.01

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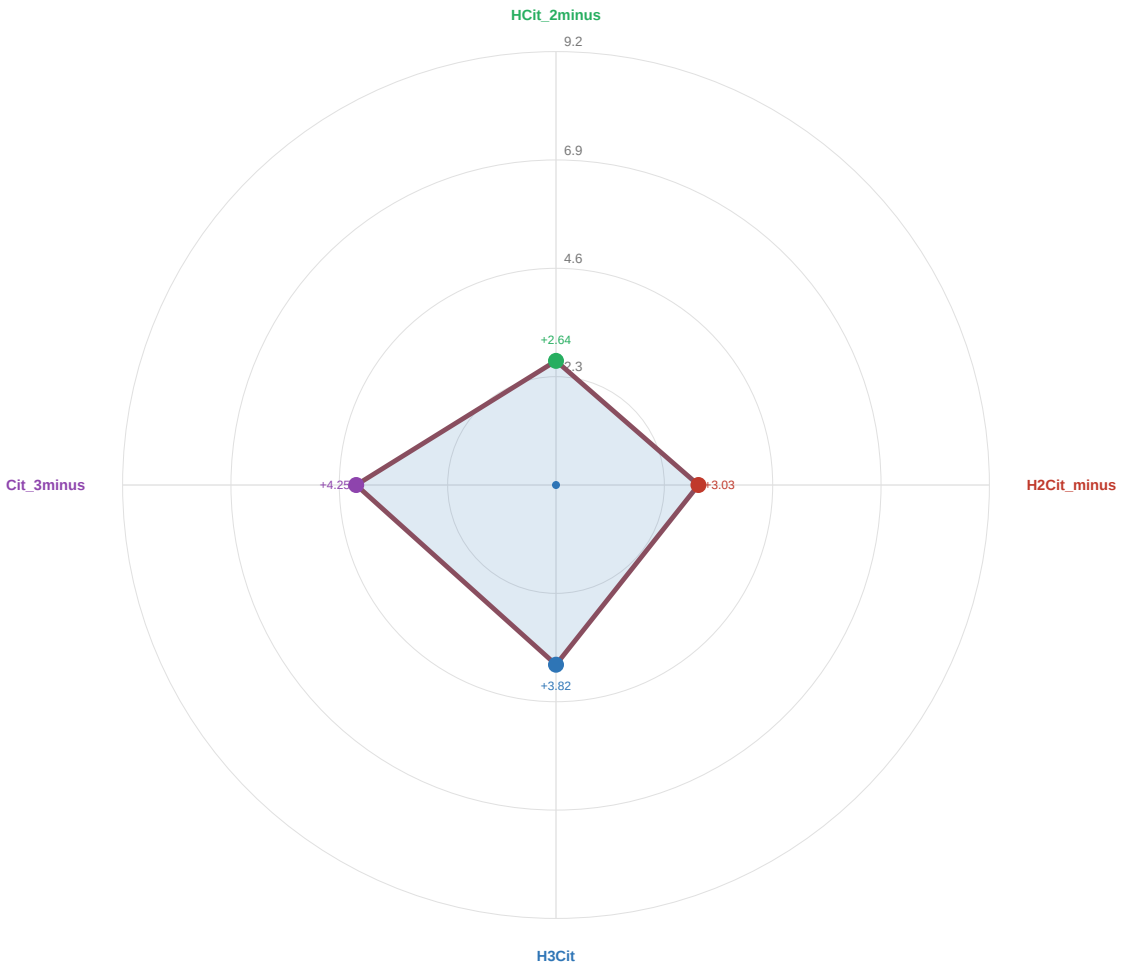
POLAR SLICE — t=162

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 163/200 | ||x||_A = 6.990



From 161: delta ||x||_A = +0.179

Phase = +38.4 deg

H3Cit=+3.82 H2Cit_=+3.03 HCit_2=+2.64 Cit_3m=+4.25

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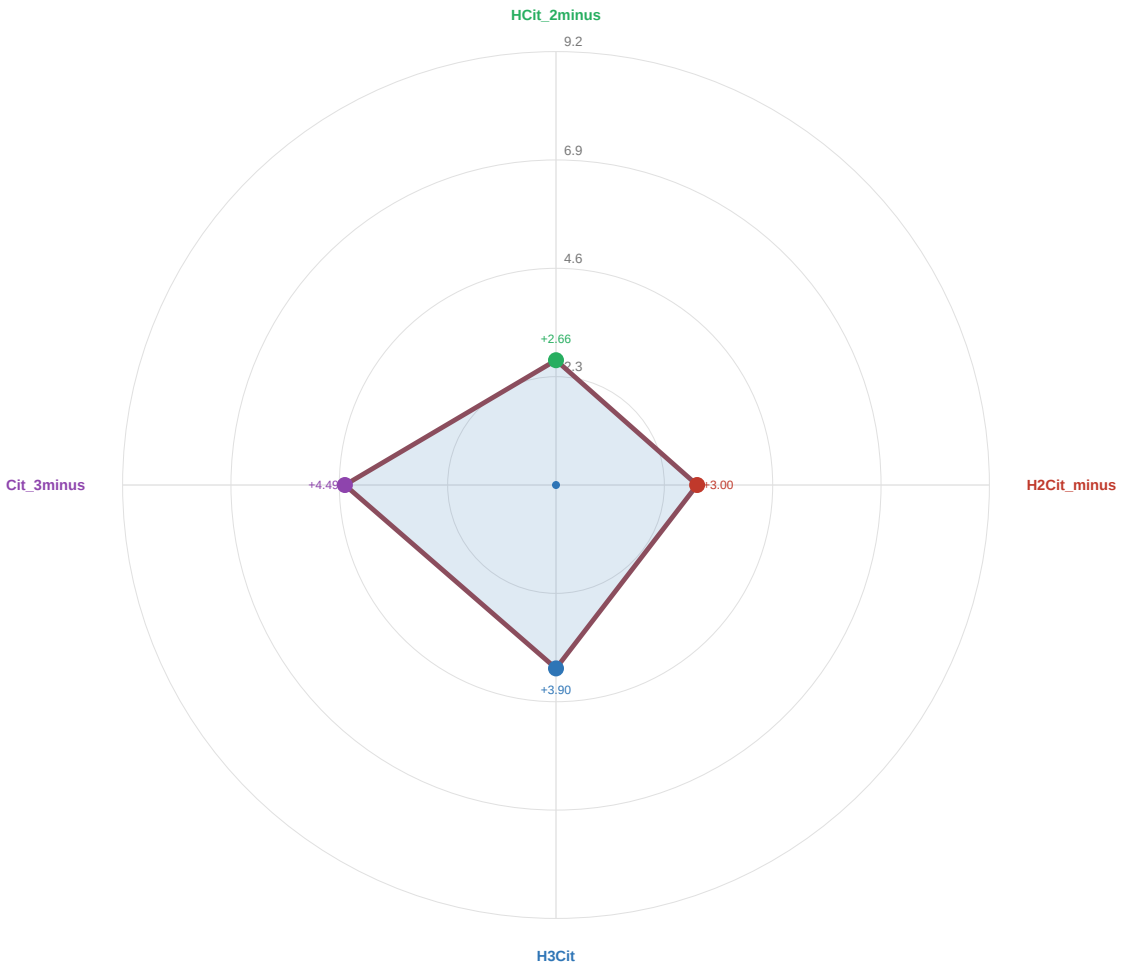
POLAR SLICE — t=163

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 164/200 | ||x||_A = 7.175



From 162: delta ||x||_A = +0.185

Phase = +37.6 deg

H3Cit=+3.90 H2Cit_=+3.00 HCit_2=+2.66 Cit_3m=+4.49

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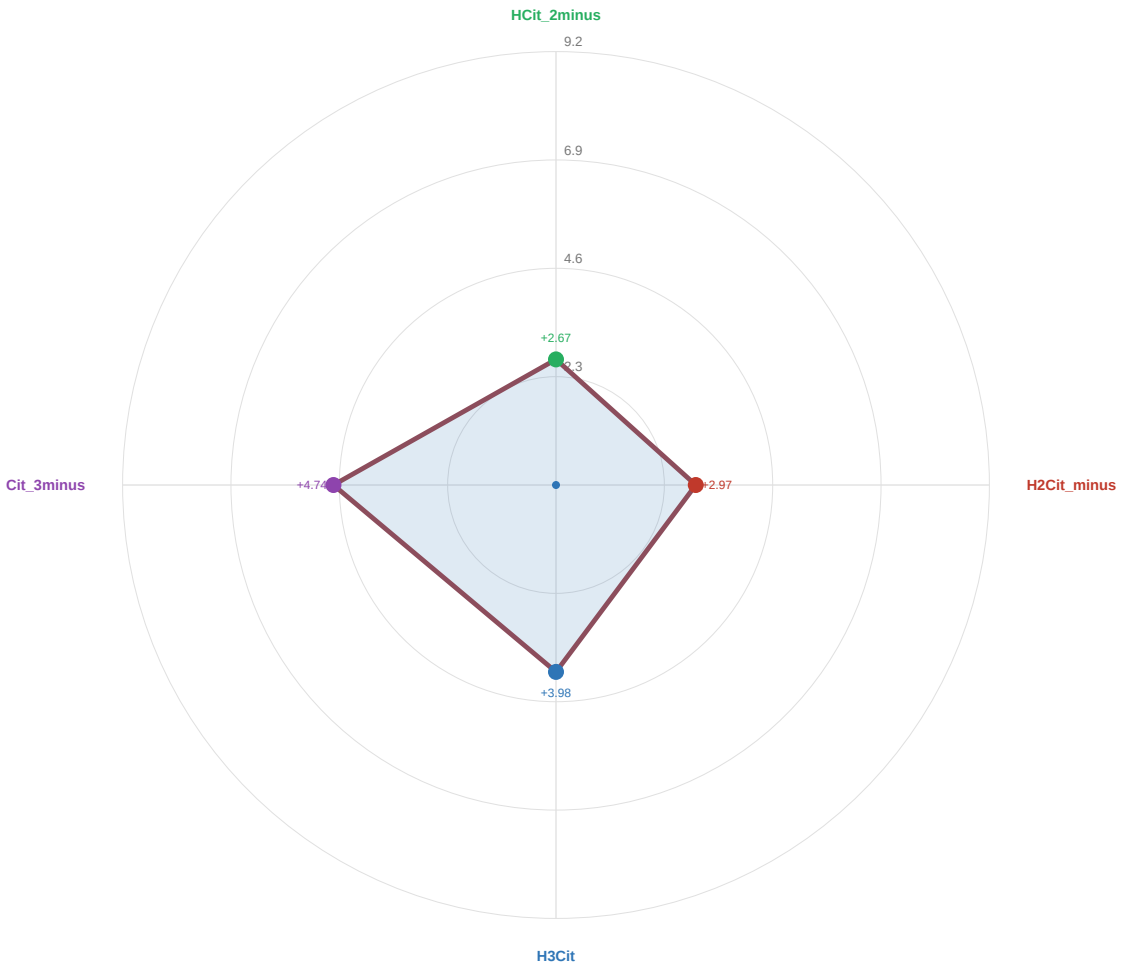
POLAR SLICE — t=164

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 165/200 | ||x||_A = 7.364



From 163: delta ||x||_A = +0.189

Phase = +36.8 deg

H3Cit=+3.98 H2Cit_=+2.97 HCit_2=+2.67 Cit_3m=+4.74

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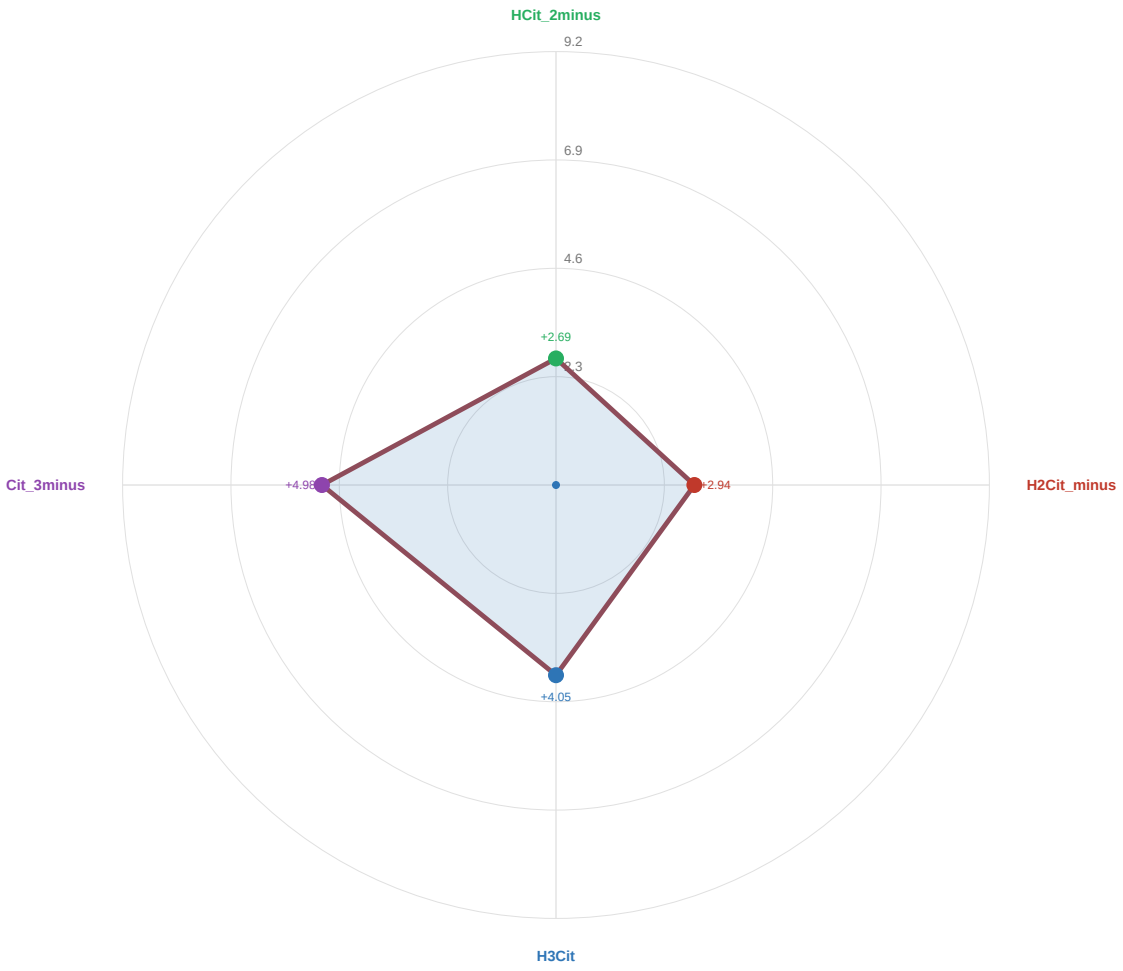
POLAR SLICE — t=165

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 166/200 | ||x||_A = 7.557



From 164: delta ||x||_A = +0.193

Phase = +36.0 deg

H3Cit=+4.05 H2Cit_=+2.94 HCit_2=+2.69 Cit_3m=+4.98

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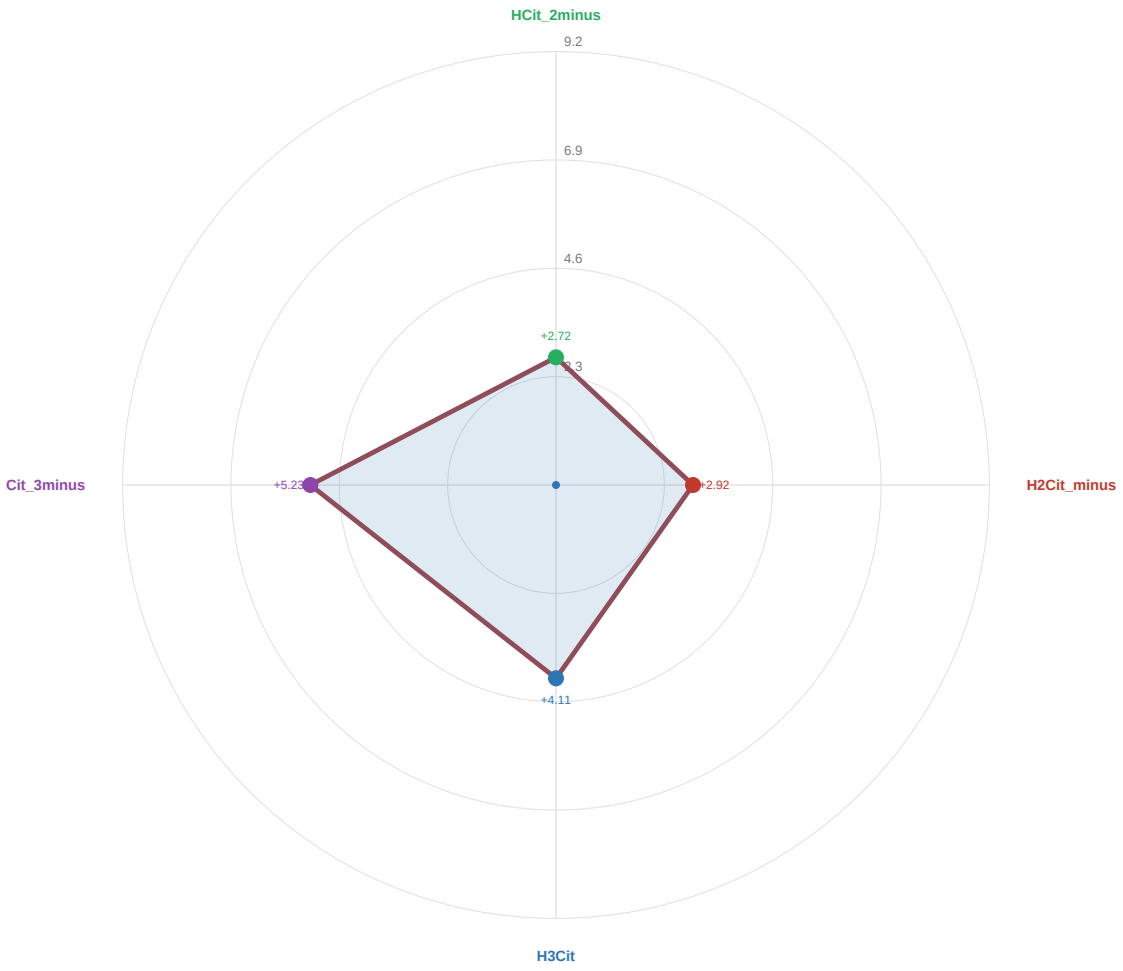
POLAR SLICE — t=166

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 167/200 | ||x||_A = 7.753



From 165: delta ||x||_A = +0.196

Phase = +35.3 deg

H3Cit=+4.11 H2Cit_=+2.92 HCit_2=+2.72 Cit_3m=+5.23

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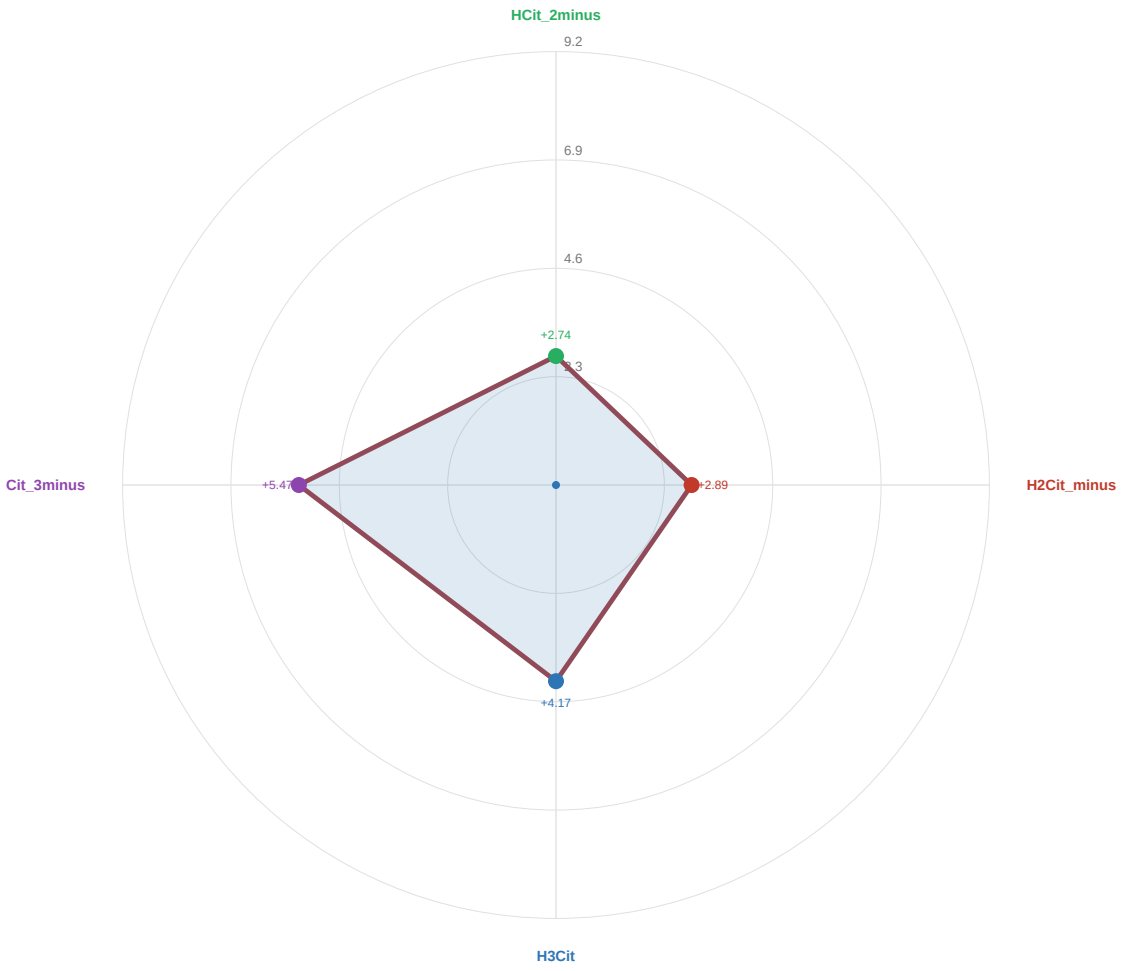
POLAR SLICE — t=167

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 168/200 | ||x||_A = 7.952



From 166: delta ||x||_A = +0.198

Phase = +34.7 deg

H3Cit=+4.17 H2Cit_=+2.89 HCit_2=+2.74 Cit_3m=+5.47

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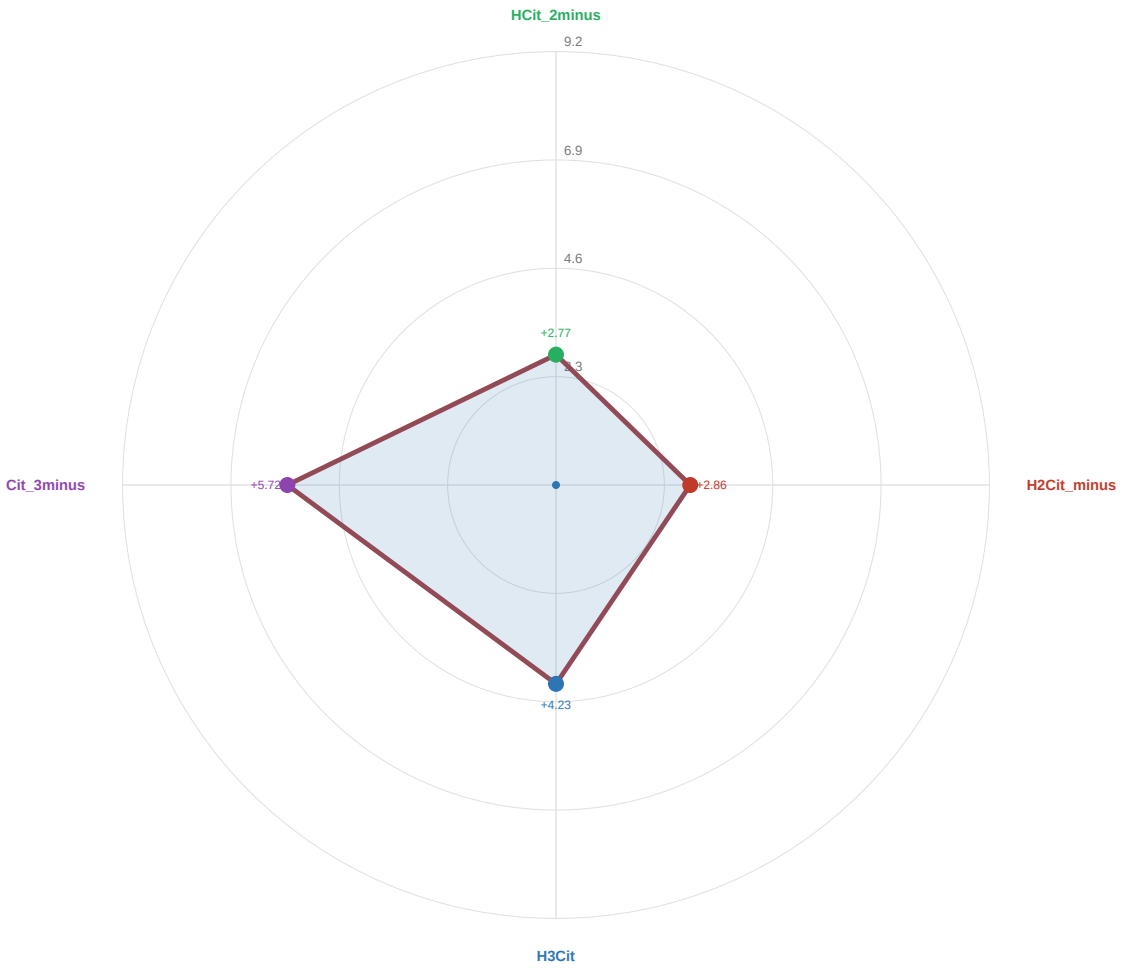
POLAR SLICE — t=168

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 169/200 | ||x||_A = 8.151



From 167: delta ||x||_A = +0.200

Phase = +34.0 deg

H3Cit=+4.23 H2Cit_=+2.86 HCit_2=+2.77 Cit_3m=+5.72

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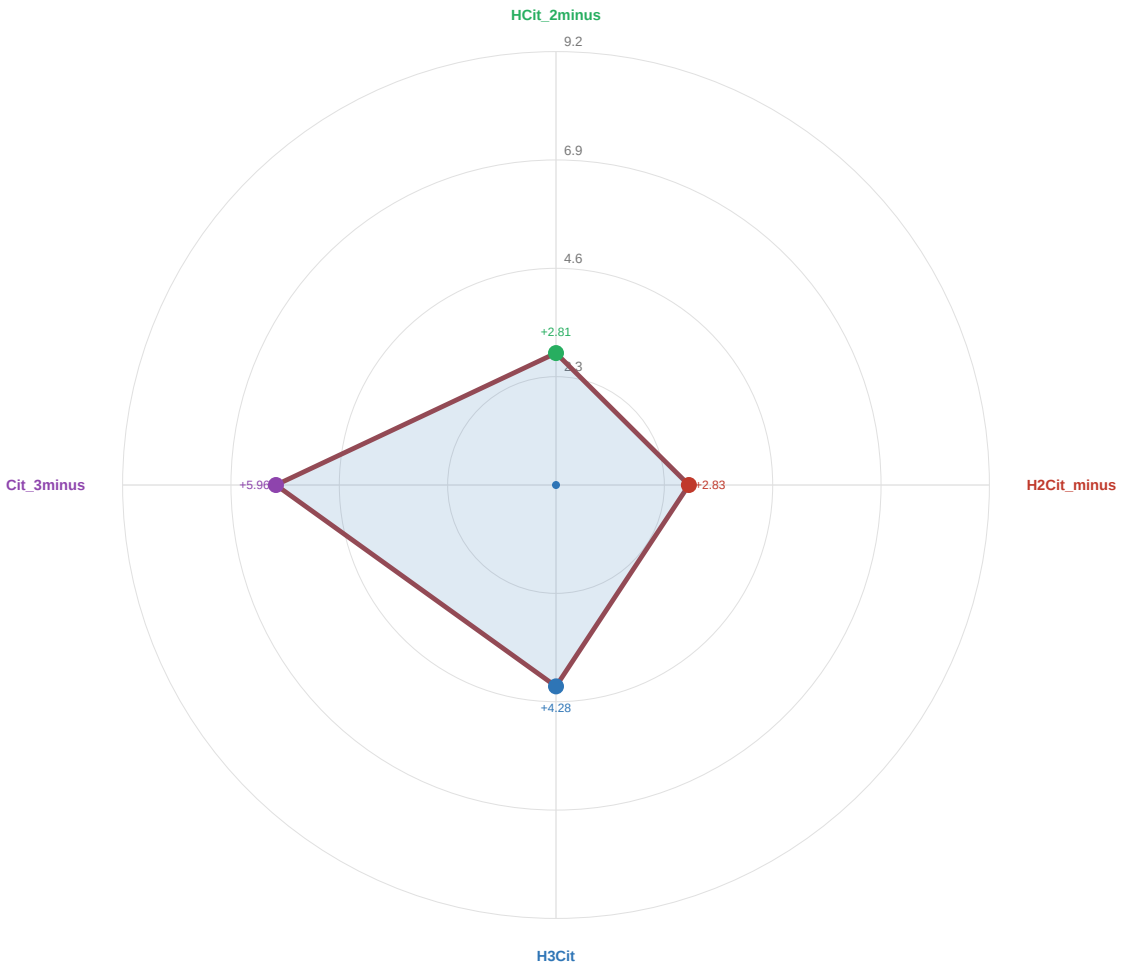
POLAR SLICE — t=169

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 170/200 | ||x||_A = 8.352



From 168: delta ||x||_A = +0.200

Phase = +33.4 deg

H3Cit=+4.28 H2Cit_=+2.83 HCit_2=+2.81 Cit_3m=+5.96

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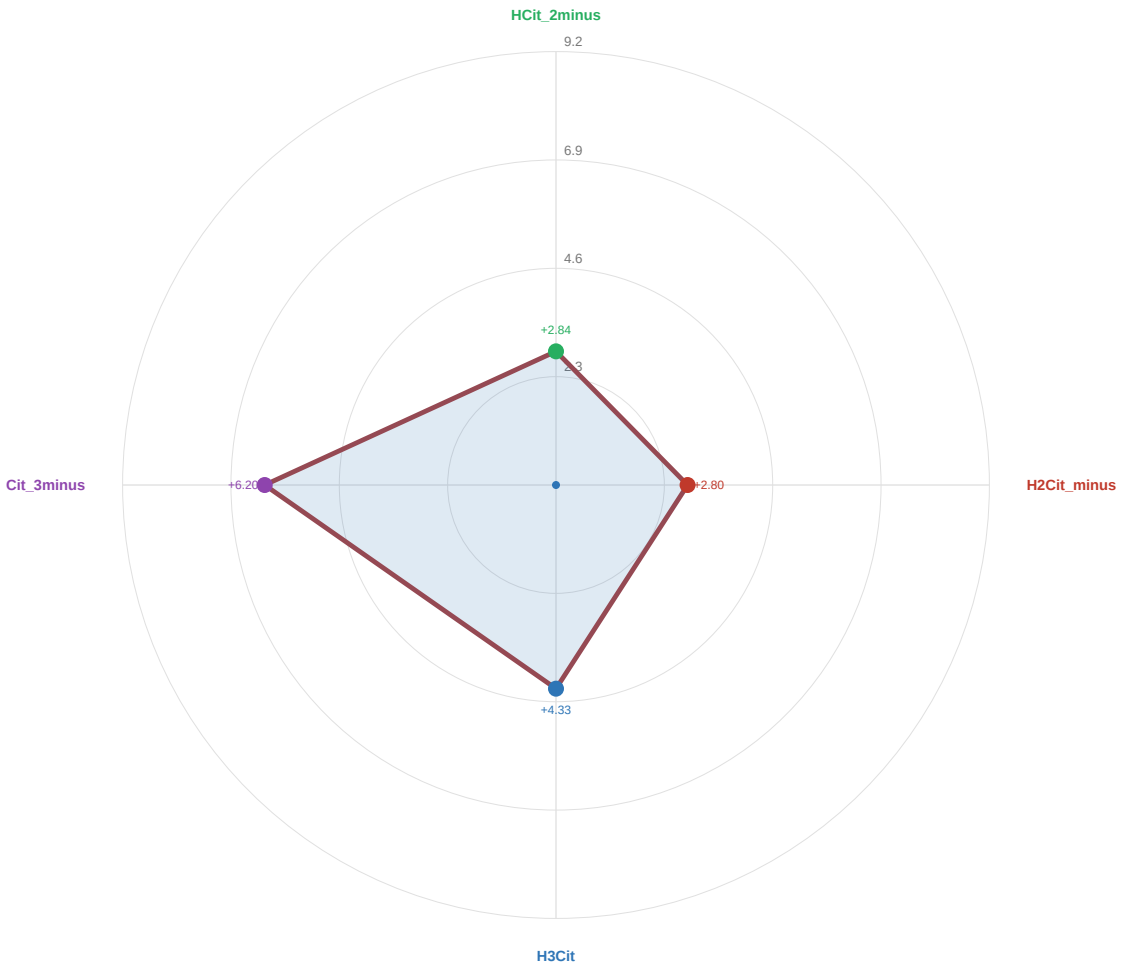
POLAR SLICE — t=170

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 171/200 | ||x||_A = 8.552



From 169: delta ||x||_A = +0.200

Phase = +32.9 deg

H3Cit=+4.33 H2Cit_=+2.80 HCit_2=+2.84 Cit_3m=+6.20

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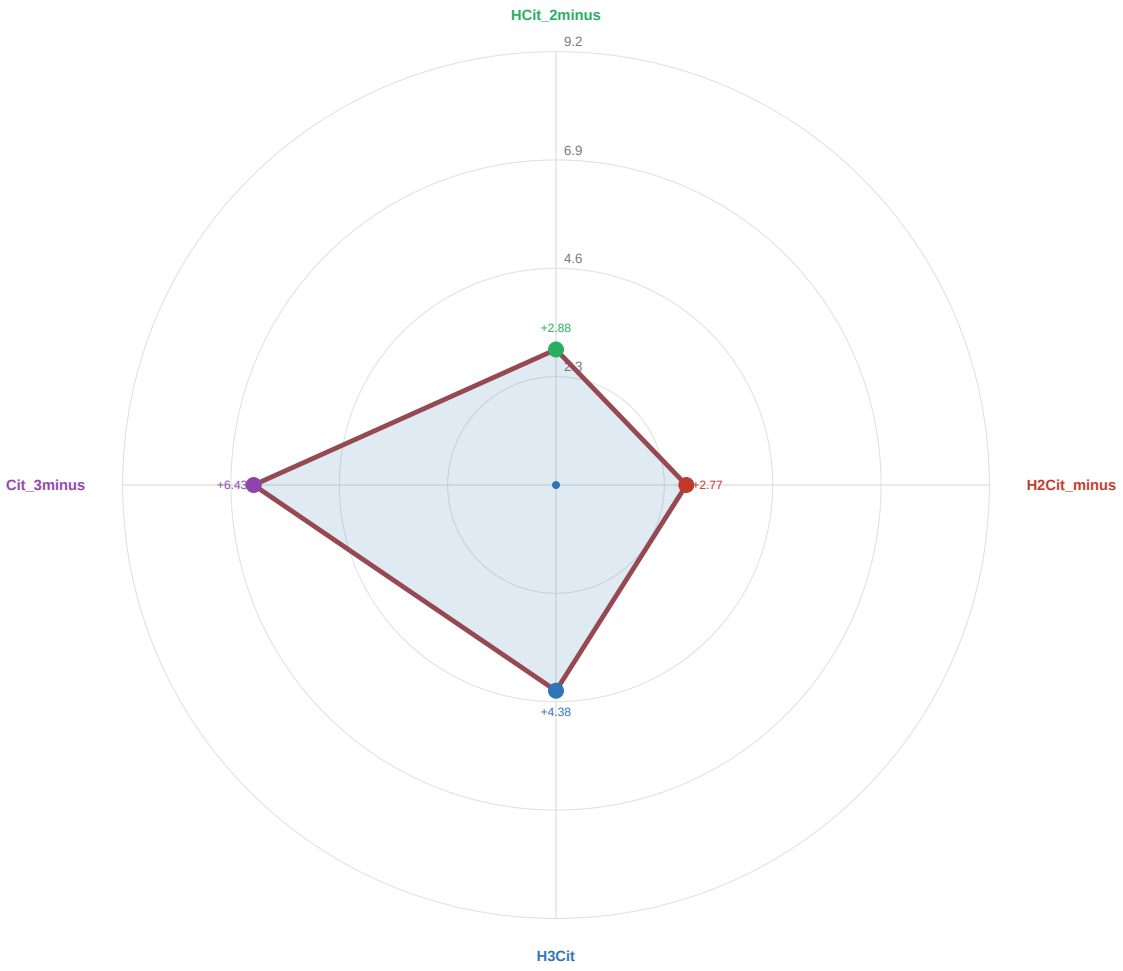
POLAR SLICE — t=171

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 172/200 | ||x||_A = 8.750



From 170: delta ||x||_A = +0.199

Phase = +32.3 deg

H3Cit=+4.38 H2Cit_=+2.77 HCit_2=+2.88 Cit_3m=+6.43

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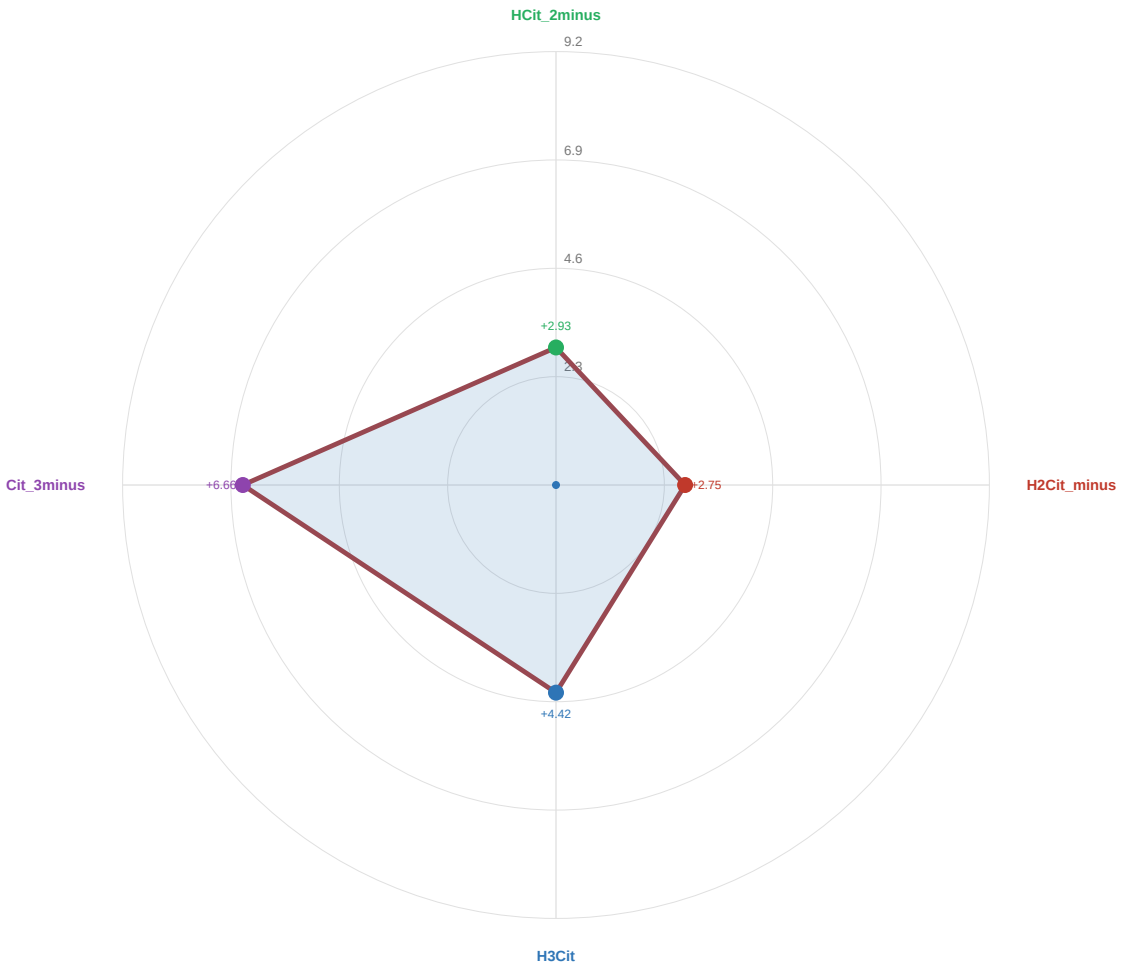
POLAR SLICE — t=172

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 173/200 | ||x||_A = 8.947



From 171: delta ||x||_A = +0.197

Phase = +31.9 deg

H3Cit=+4.42 H2Cit_=+2.75 HCit_2=+2.93 Cit_3m=+6.66

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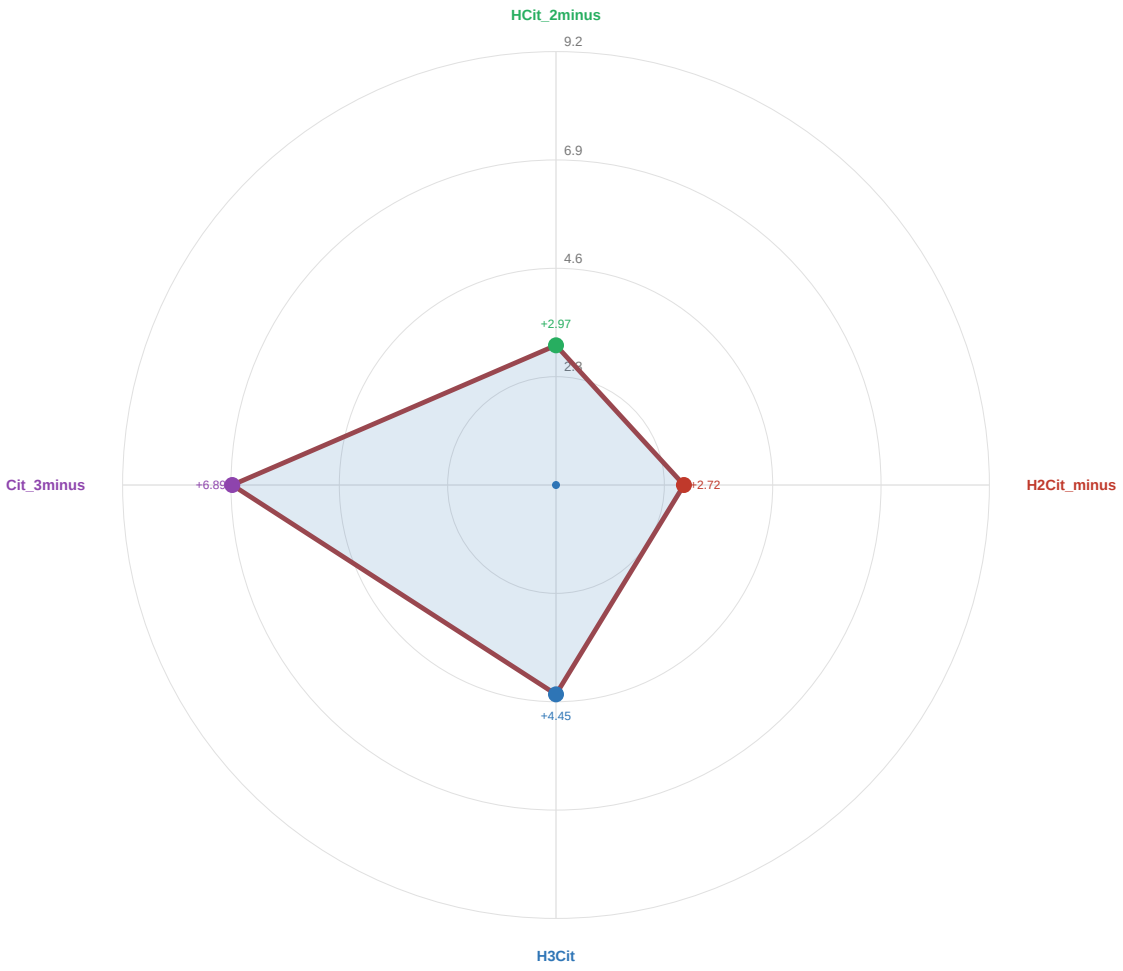
POLAR SLICE — t=173

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 174/200 | ||x||_A = 9.141



From 172: delta ||x||_A = +0.194

Phase = +31.4 deg

H3Cit=+4.45 H2Cit_=+2.72 HCit_2=+2.97 Cit_3m=+6.89

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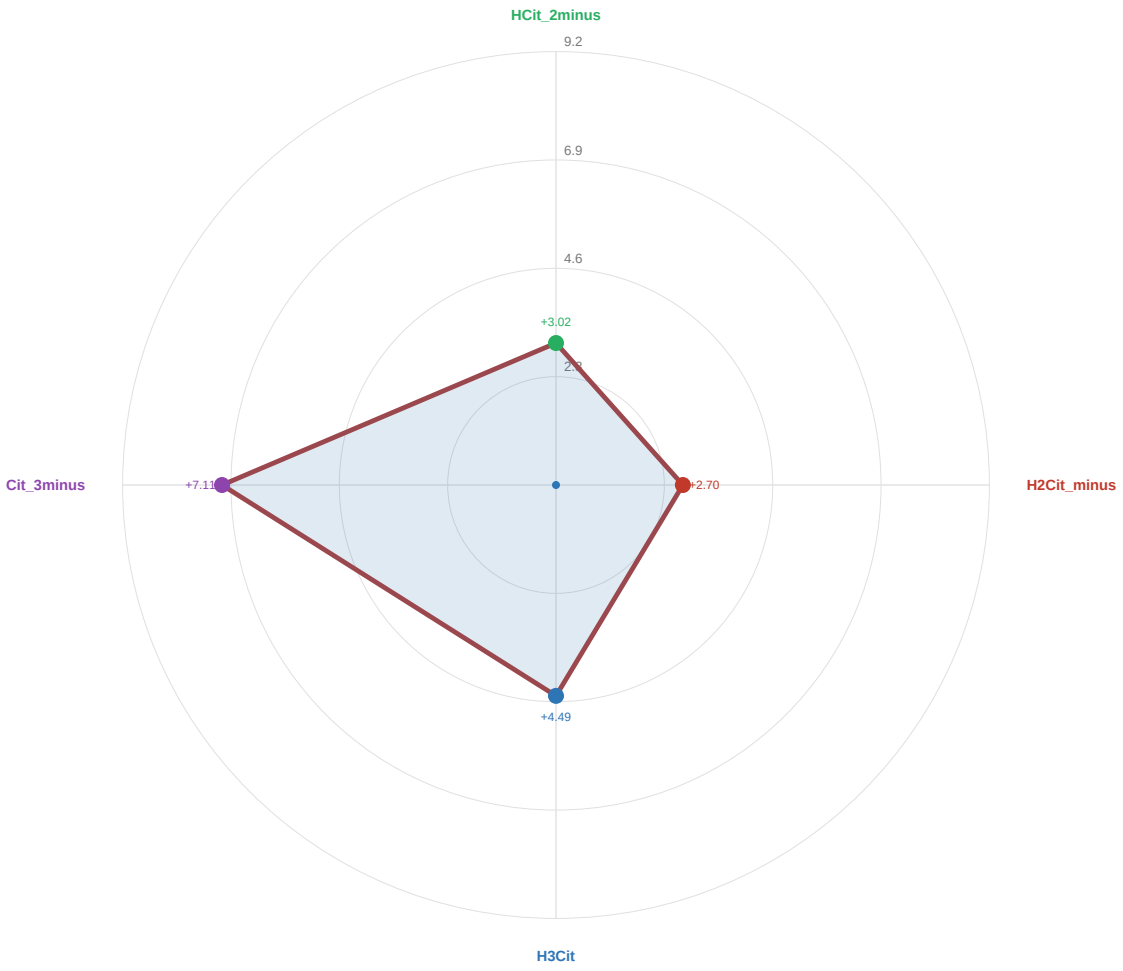
POLAR SLICE — t=174

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 175/200 | ||x||_A = 9.330



From 173: delta ||x||_A = +0.190

Phase = +31.0 deg

H3Cit=+4.49 H2Cit_=+2.70 HCit_2=+3.02 Cit_3m=+7.11

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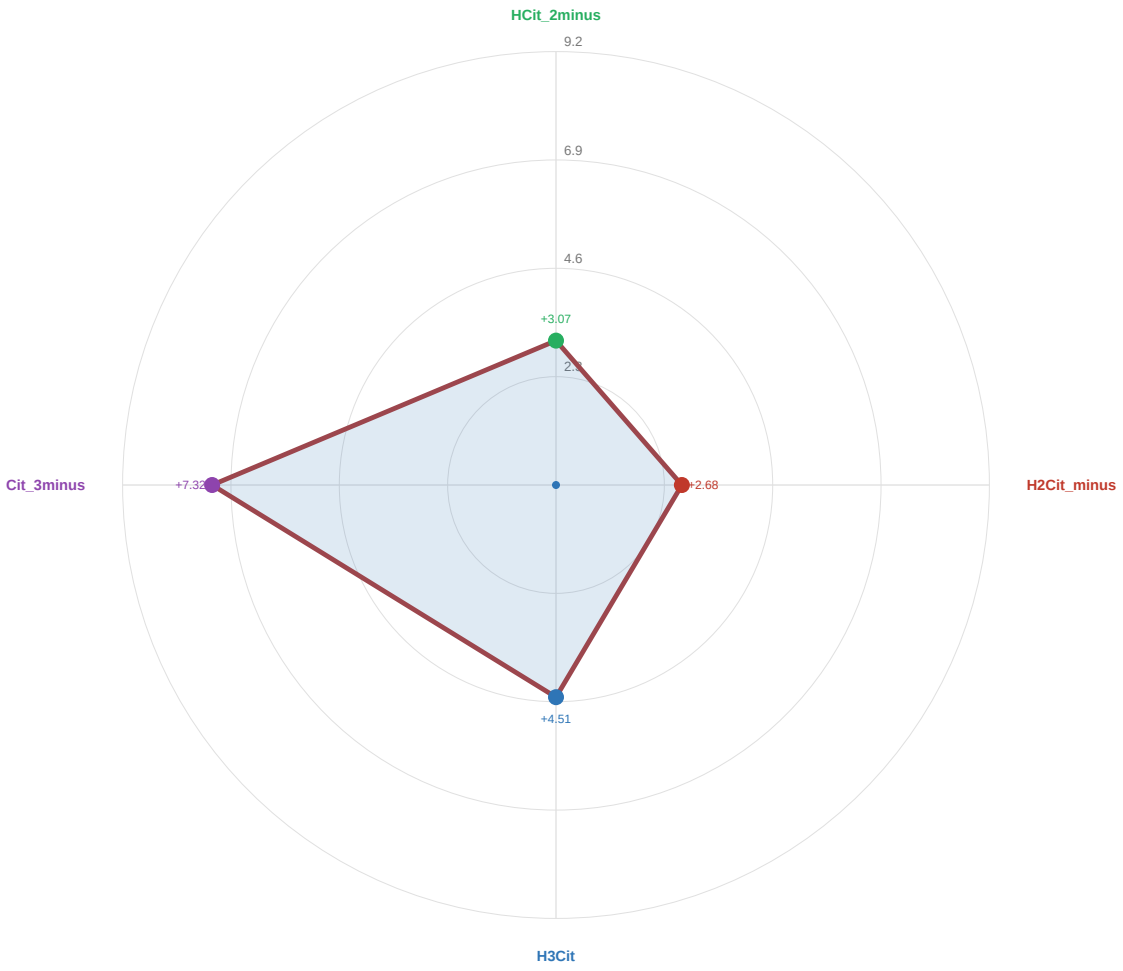
POLAR SLICE — t=175

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 176/200 | ||x||_A = 9.515



From 174: delta ||x||_A = +0.185

Phase = +30.7 deg

H3Cit=+4.51 H2Cit_=+2.68 HCit_2=+3.07 Cit_3m=+7.32

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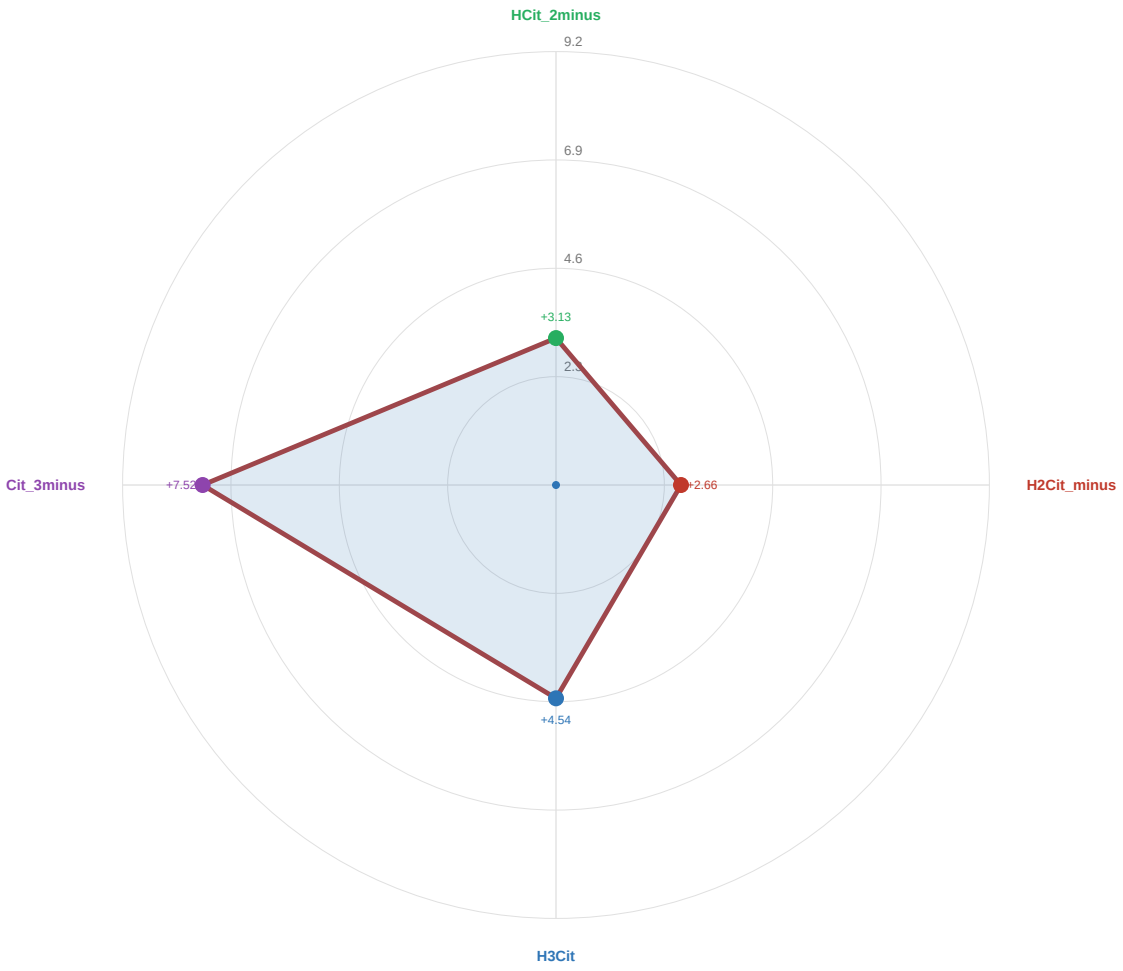
POLAR SLICE — t=176

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 177/200 | ||x||_A = 9.695



From 175: delta ||x||_A = +0.180

Phase = +30.4 deg

H3Cit=+4.54 H2Cit_=+2.66 HCit_2=+3.13 Cit_3m=+7.52

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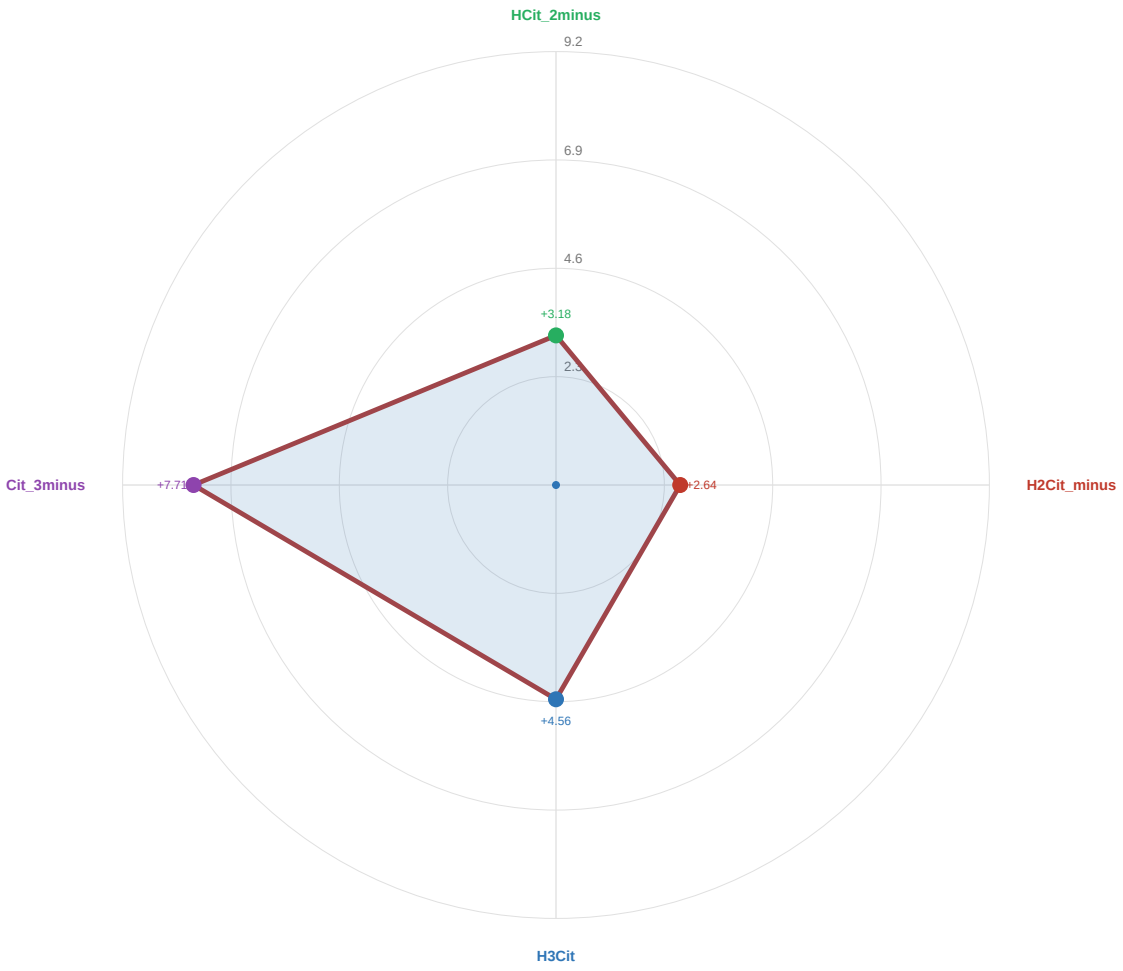
POLAR SLICE — t=177

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 178/200 | ||x||_A = 9.869



From 176: delta ||x||_A = +0.174

Phase = +30.1 deg

H3Cit=+4.56 H2Cit_=+2.64 HCit_2=+3.18 Cit_3m=+7.71

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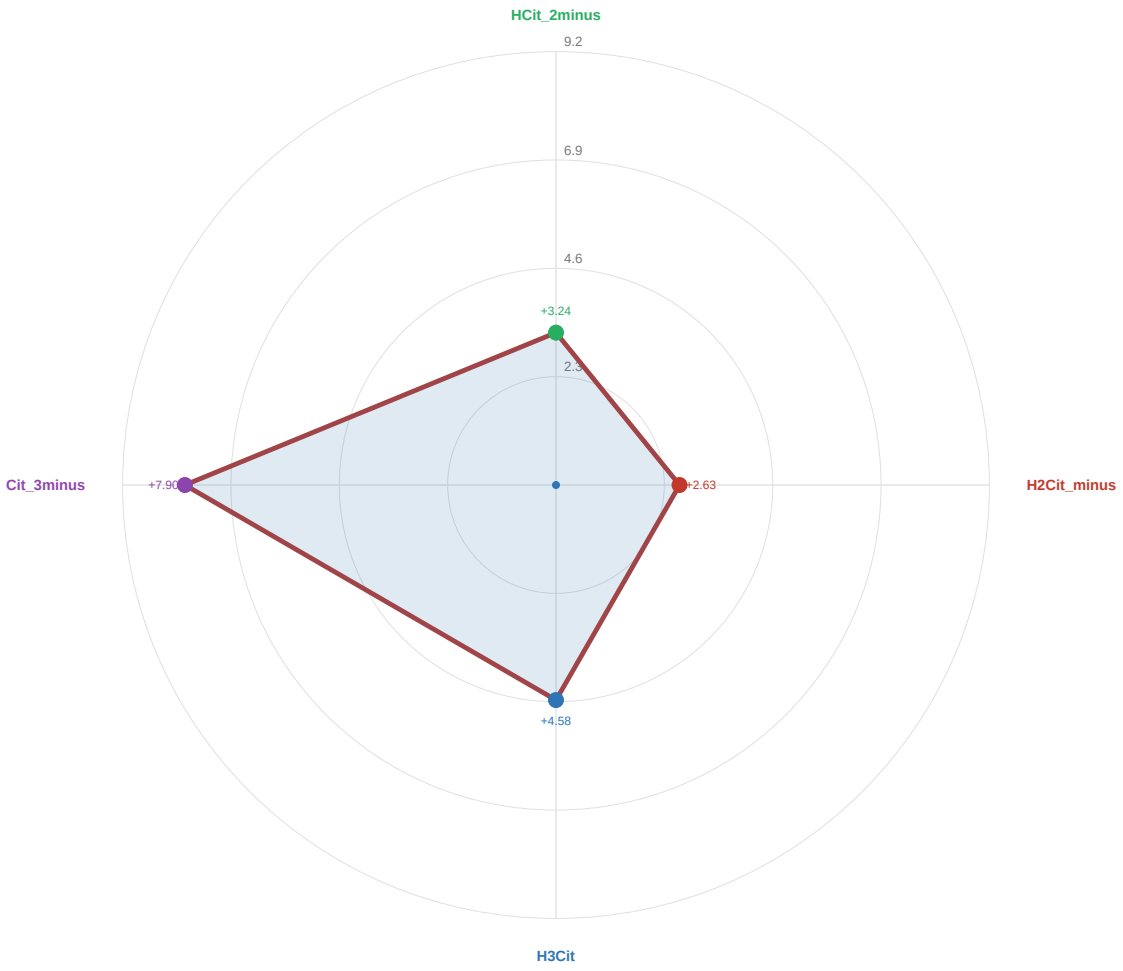
POLAR SLICE — t=178

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 179/200 | ||x||_A = 10.036



From 177: delta ||x||_A = +0.167

Phase = +29.9 deg

H3Cit=+4.58 H2Cit_=+2.63 HCit_2=+3.24 Cit_3m=+7.90

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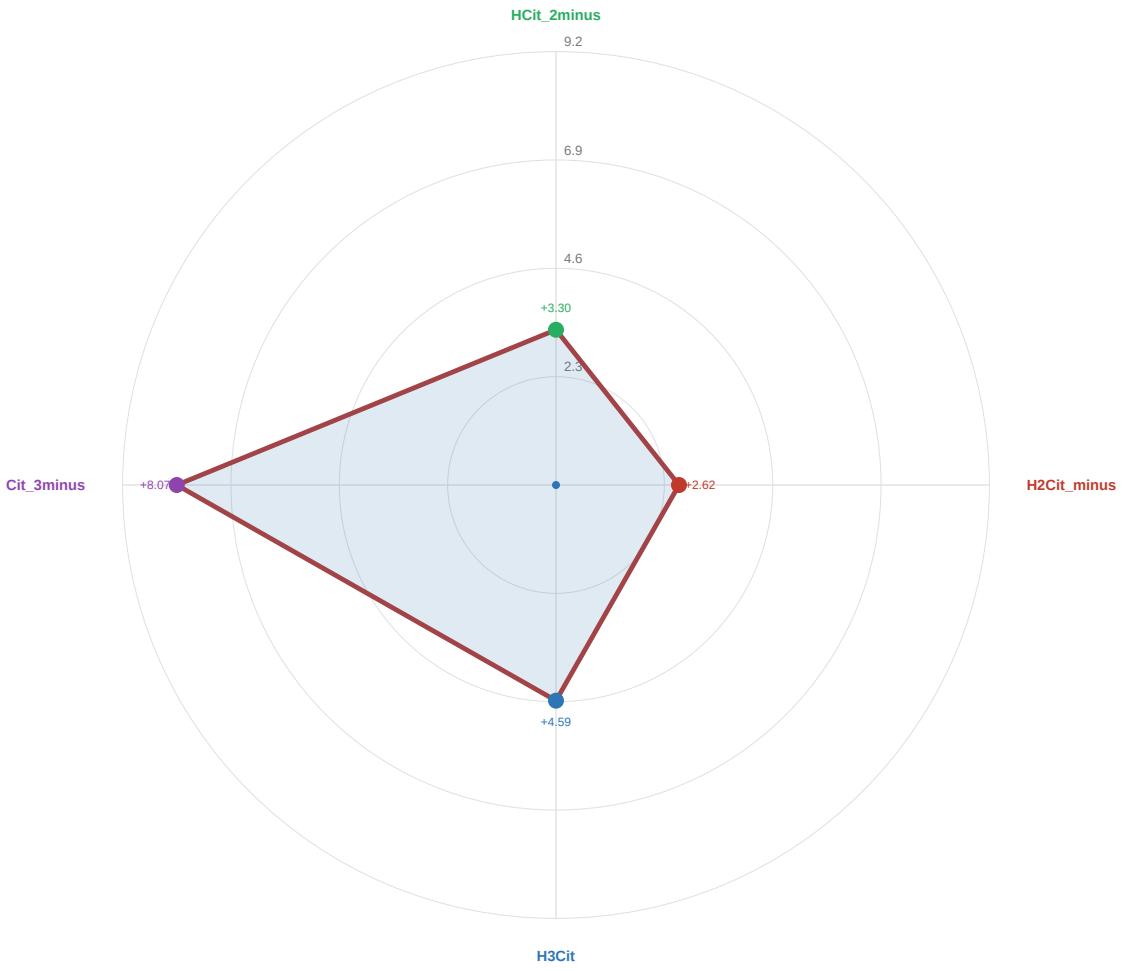
POLAR SLICE — t=179

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 180/200 | ||x||_A = 10.195



From 178: delta ||x||_A = +0.159

Phase = +29.7 deg

H3Cit=+4.59 H2Cit_=+2.62 HCit_2=+3.30 Cit_3m=+8.07

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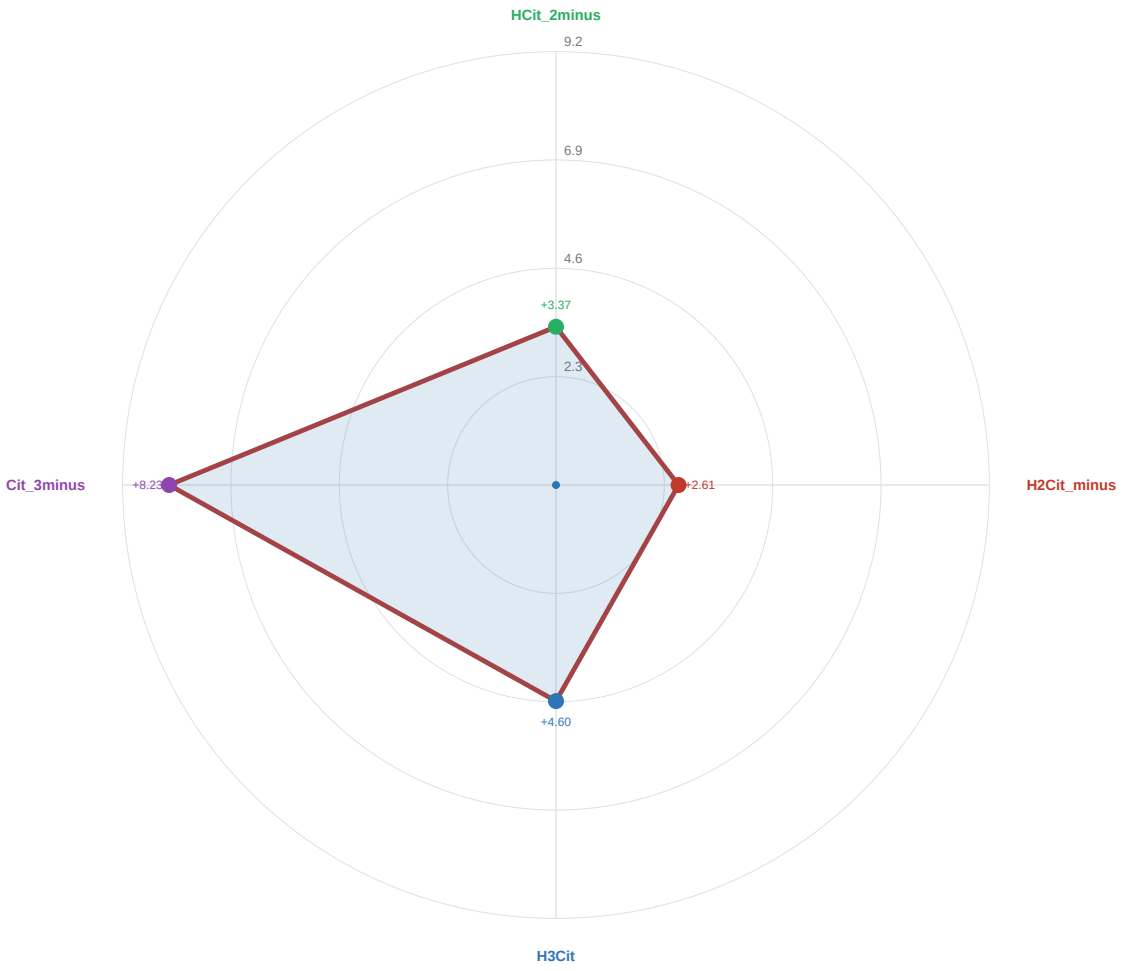
POLAR SLICE — t=180

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 181/200 | ||x||_A = 10.346



From 179: delta ||x||_A = +0.151

Phase = +29.6 deg

H3Cit=+4.60 H2Cit_=+2.61 HCit_2=+3.37 Cit_3m=+8.23

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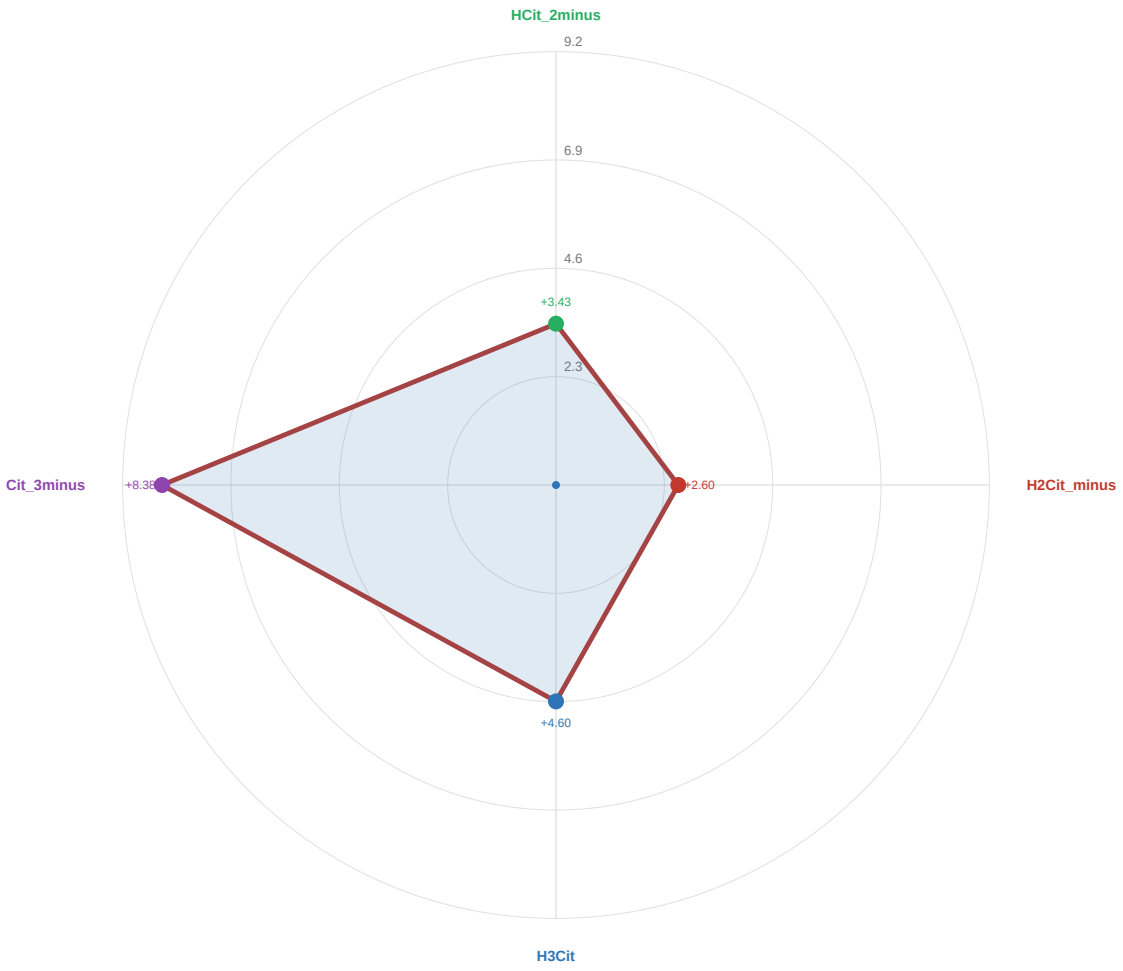
POLAR SLICE — t=181

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 182/200 | ||x||_A = 10.489



From 180: delta ||x||_A = +0.143

Phase = +29.5 deg

H3Cit=+4.60 H2Cit_=+2.60 HCit_2=+3.43 Cit_3m=+8.38

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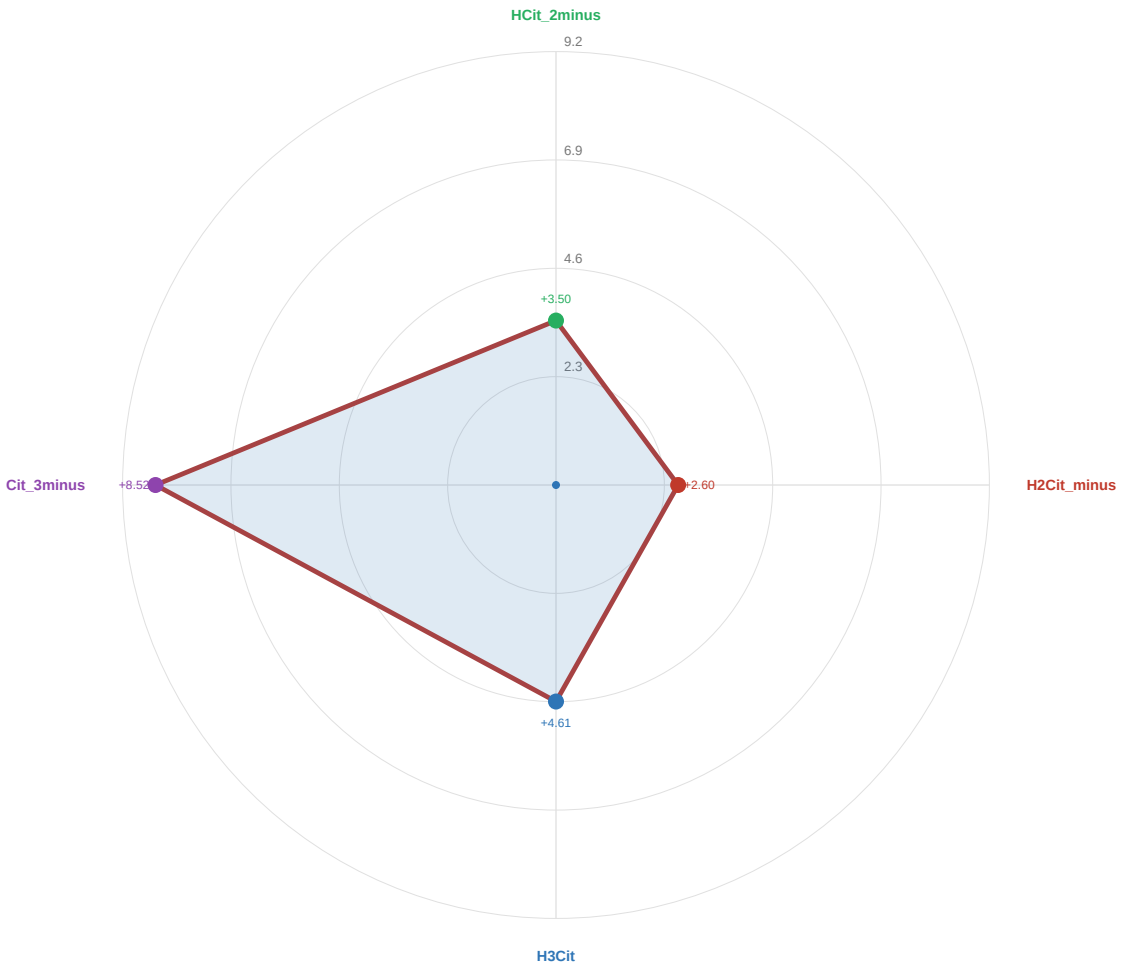
POLAR SLICE — t=182

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 183/200 | ||x||_A = 10.624



From 181: delta ||x||_A = +0.134

Phase = +29.4 deg

H3Cit=+4.61 H2Cit_=+2.60 HCit_2=+3.50 Cit_3m=+8.52

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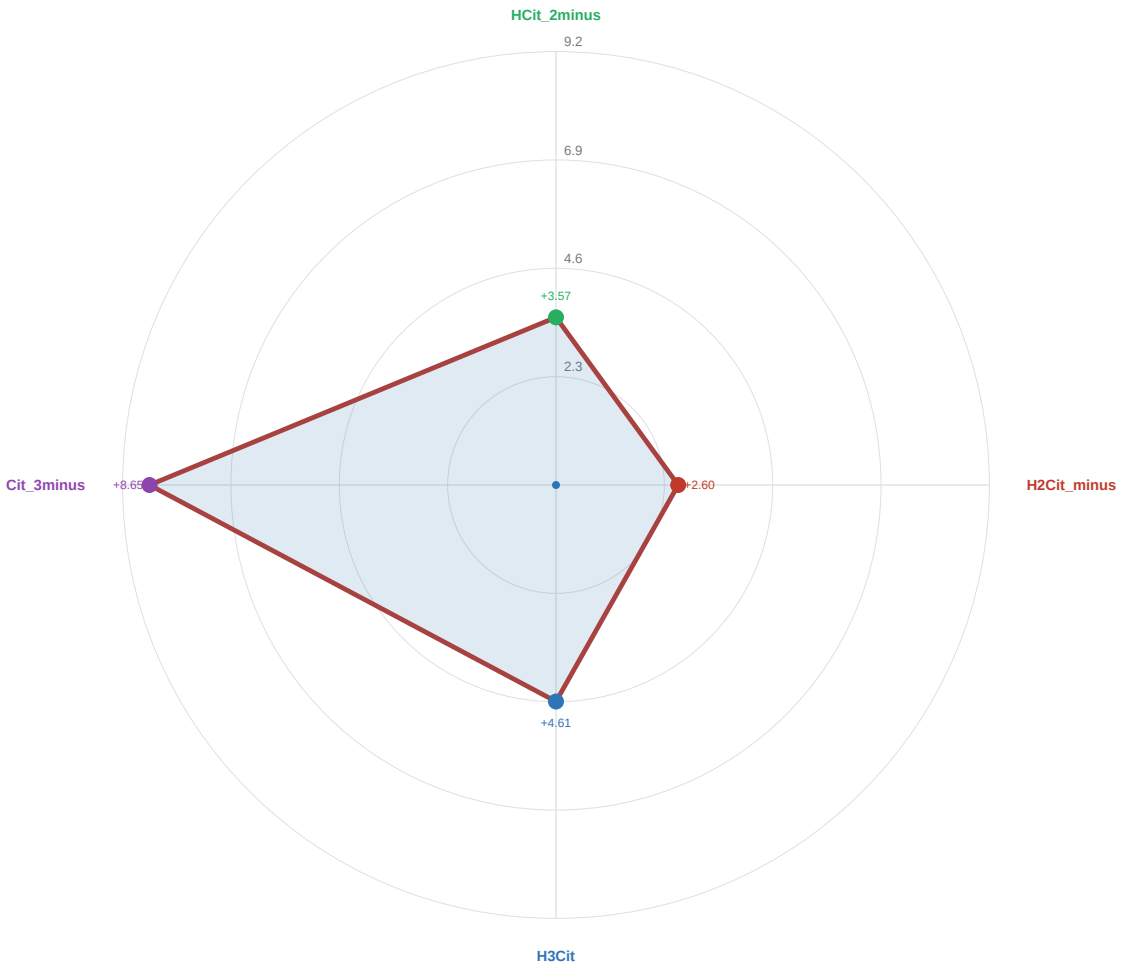
POLAR SLICE — t=183

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 184/200 | ||x||_A = 10.748



From 182: delta ||x||_A = +0.125

Phase = +29.4 deg

H3Cit=+4.61 H2Cit_=+2.60 HCit_2=+3.57 Cit_3m=+8.65

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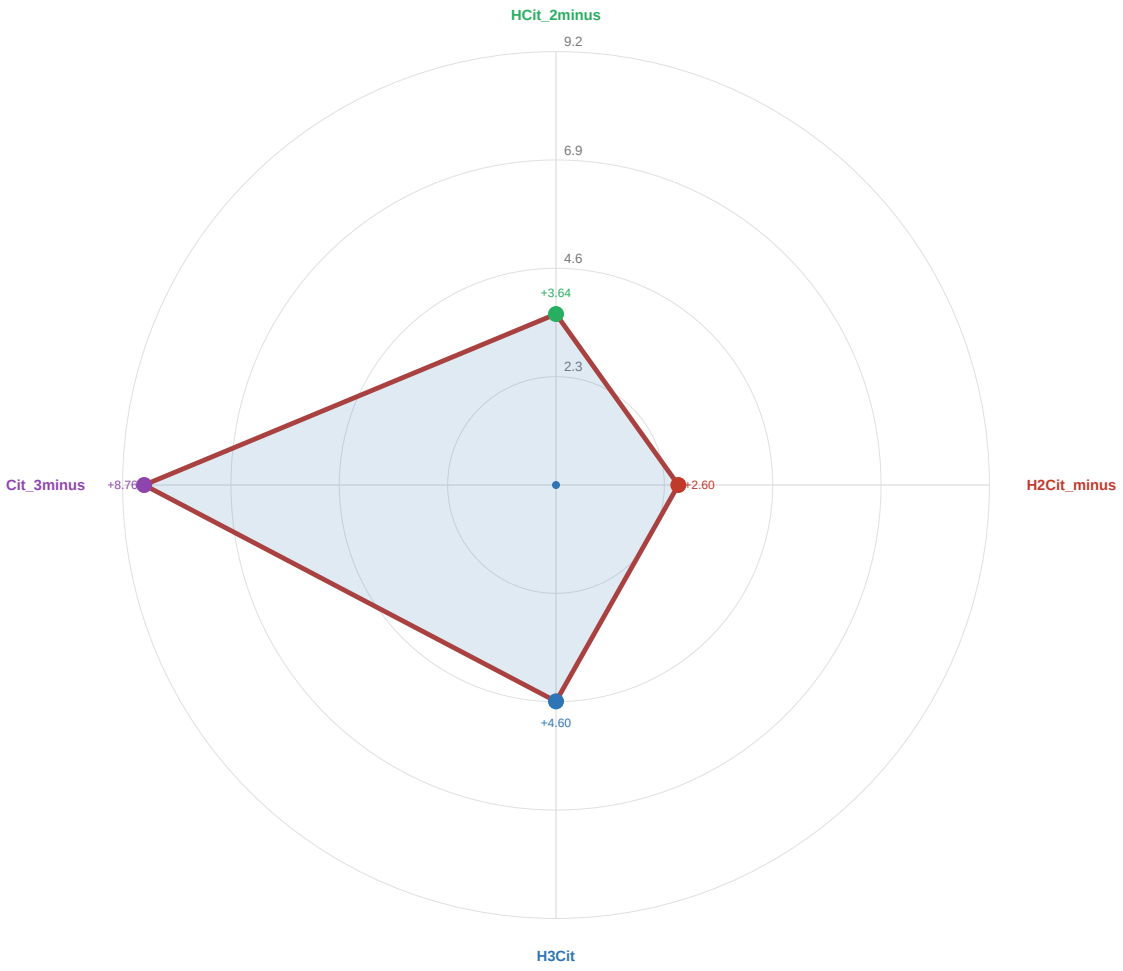
POLAR SLICE — t=184

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 185/200 | ||x||_A = 10.864



From 183: delta ||x||_A = +0.115

Phase = +29.5 deg

H3Cit=+4.60 H2Cit_=+2.60 HCit_2=+3.64 Cit_3m=+8.76

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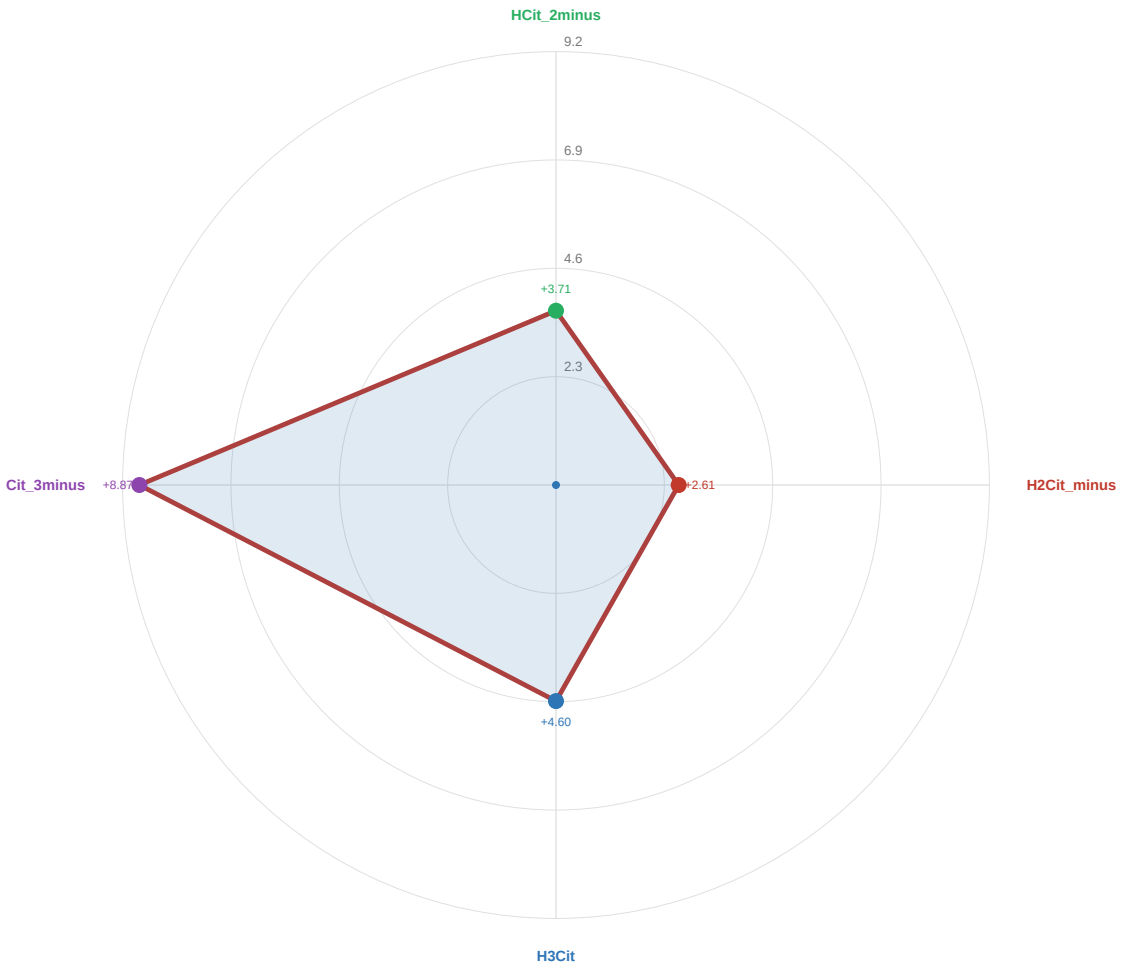
POLAR SLICE — t=185

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 186/200 | ||x||_A = 10.969



From 184: delta ||x||_A = +0.105

Phase = +29.6 deg

H3Cit=+4.60 H2Cit_=+2.61 HCit_2=+3.71 Cit_3m=+8.87

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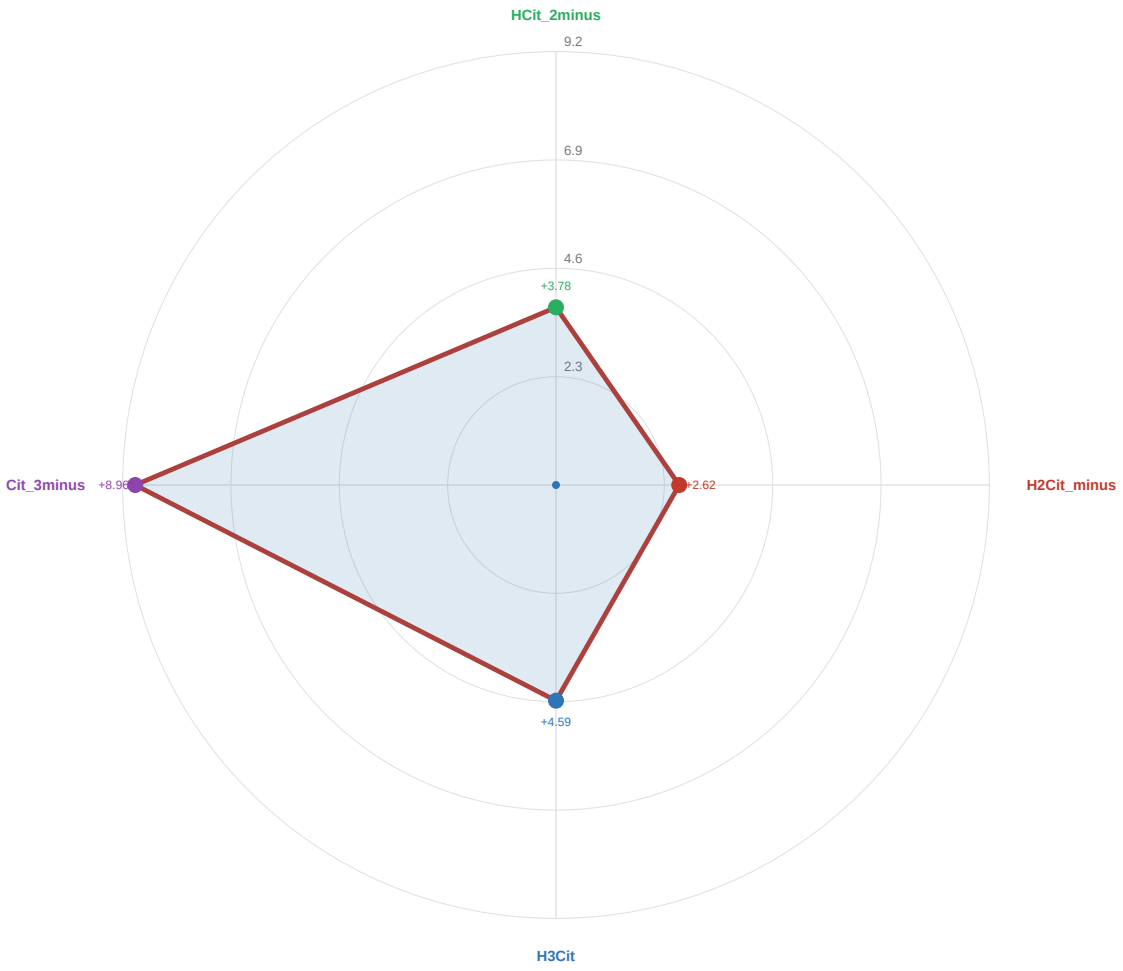
POLAR SLICE — t=186

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 187/200 | ||x||_A = 11.065



From 185: delta ||x||_A = +0.096

Phase = +29.7 deg

H3Cit=+4.59 H2Cit_=+2.62 HCit_2=+3.78 Cit_3m=+8.96

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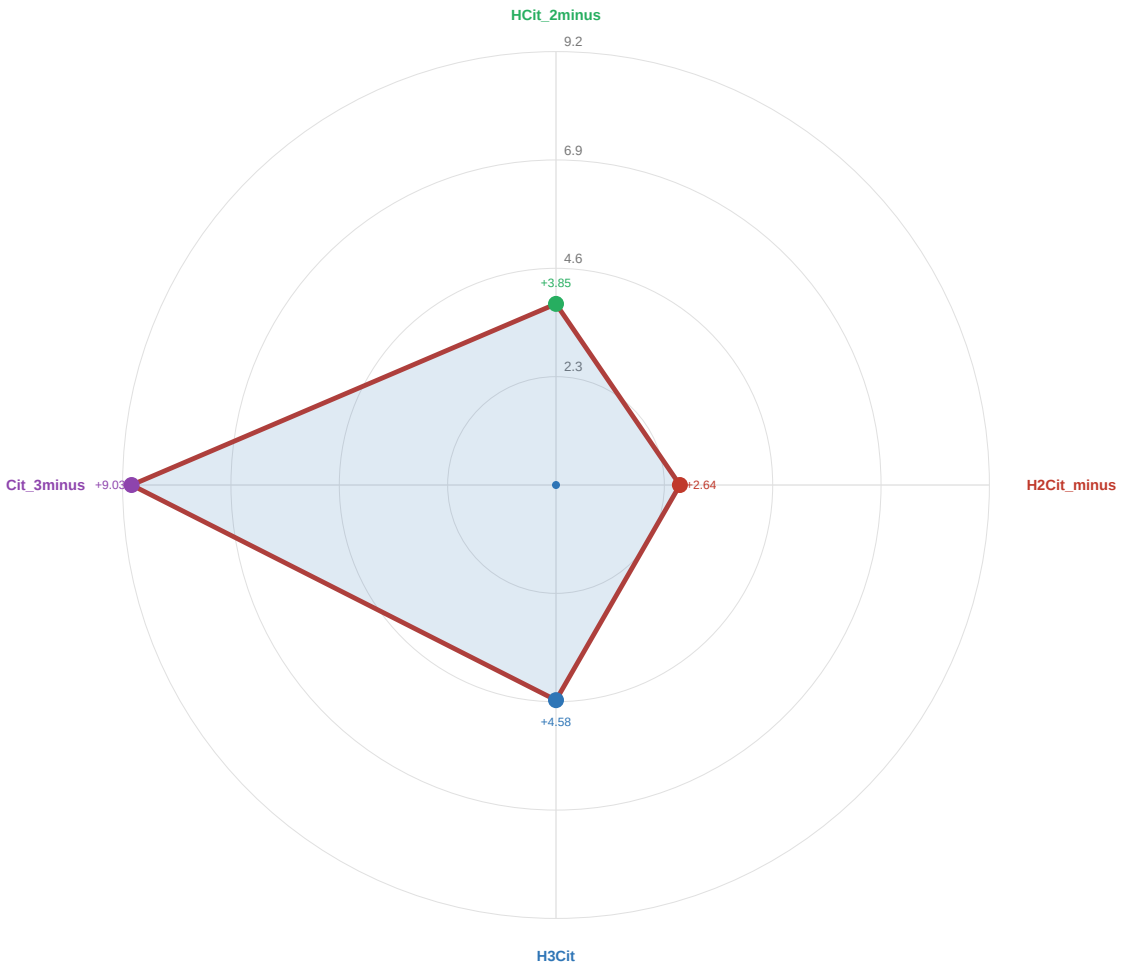
POLAR SLICE — t=187

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 188/200 | ||x||_A = 11.150



From 186: delta ||x||_A = +0.086

Phase = +29.9 deg

H3Cit=+4.58 H2Cit_=+2.64 HCit_2=+3.85 Cit_3m=+9.03

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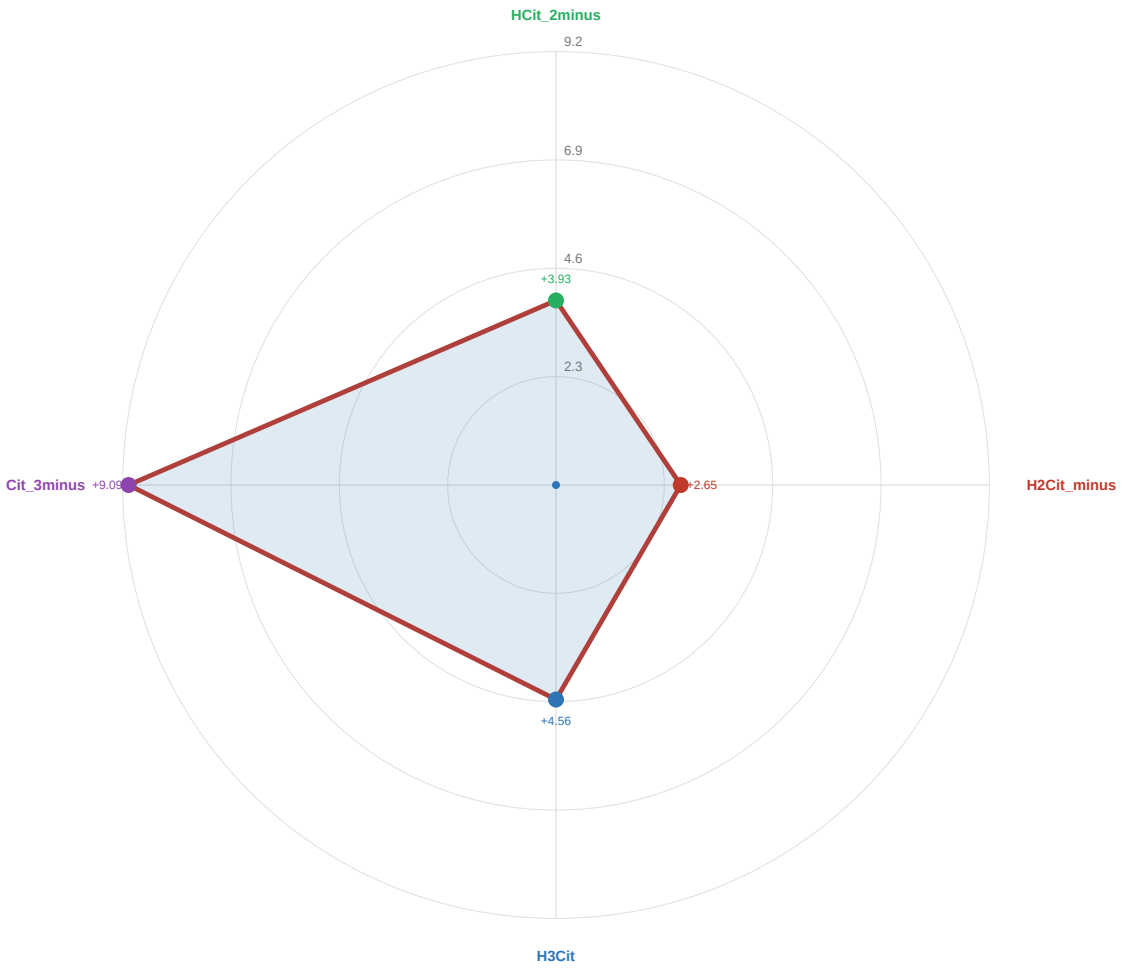
POLAR SLICE — t=188

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 189/200 | ||x||_A = 11.226



From 187: delta ||x||_A = +0.076

Phase = +30.2 deg

H3Cit=+4.56 H2Cit_=+2.65 HCit_2=+3.93 Cit_3m=+9.09

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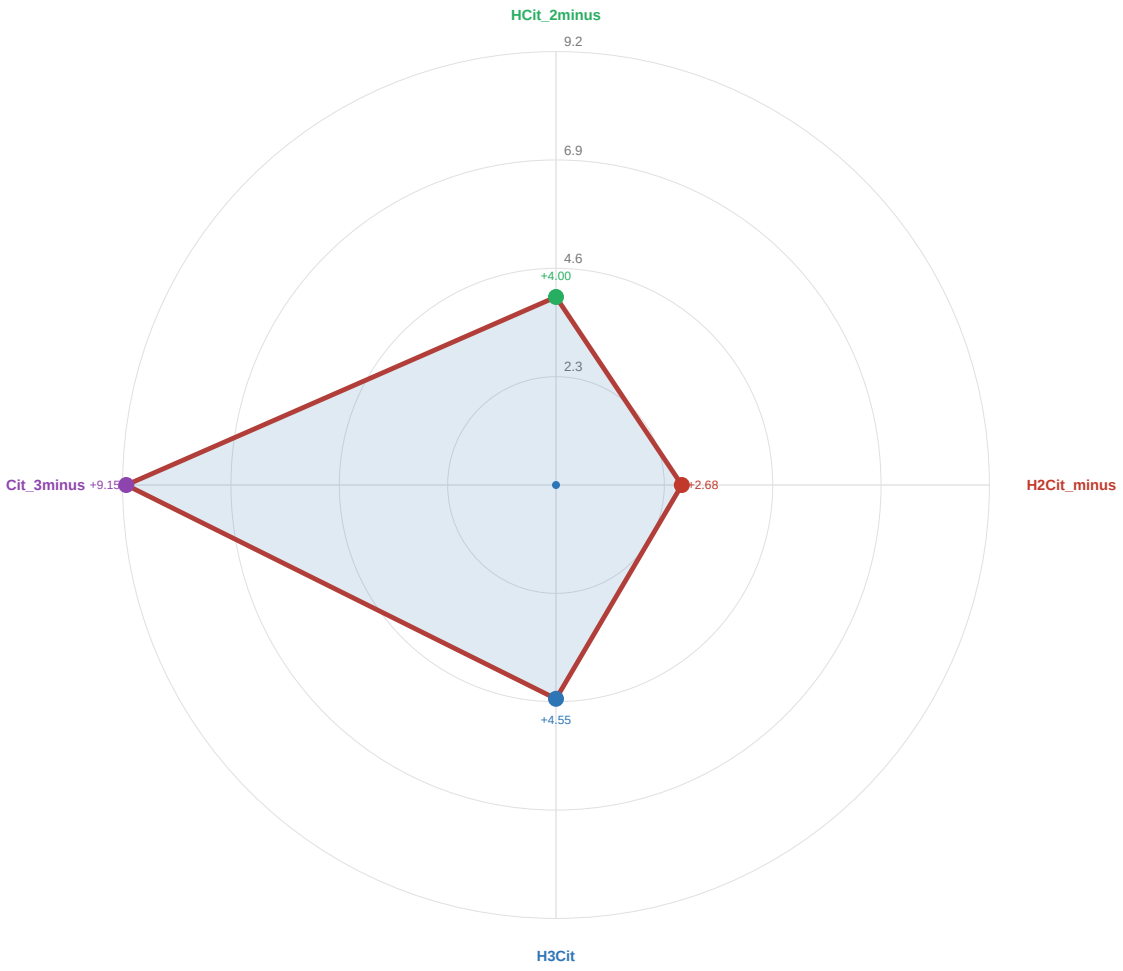
POLAR SLICE — t=189

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 190/200 | ||x||_A = 11.292



From 188: delta ||x||_A = +0.066

Phase = +30.5 deg

H3Cit=+4.55 H2Cit_=+2.68 HCit_2=+4.00 Cit_3m=+9.15

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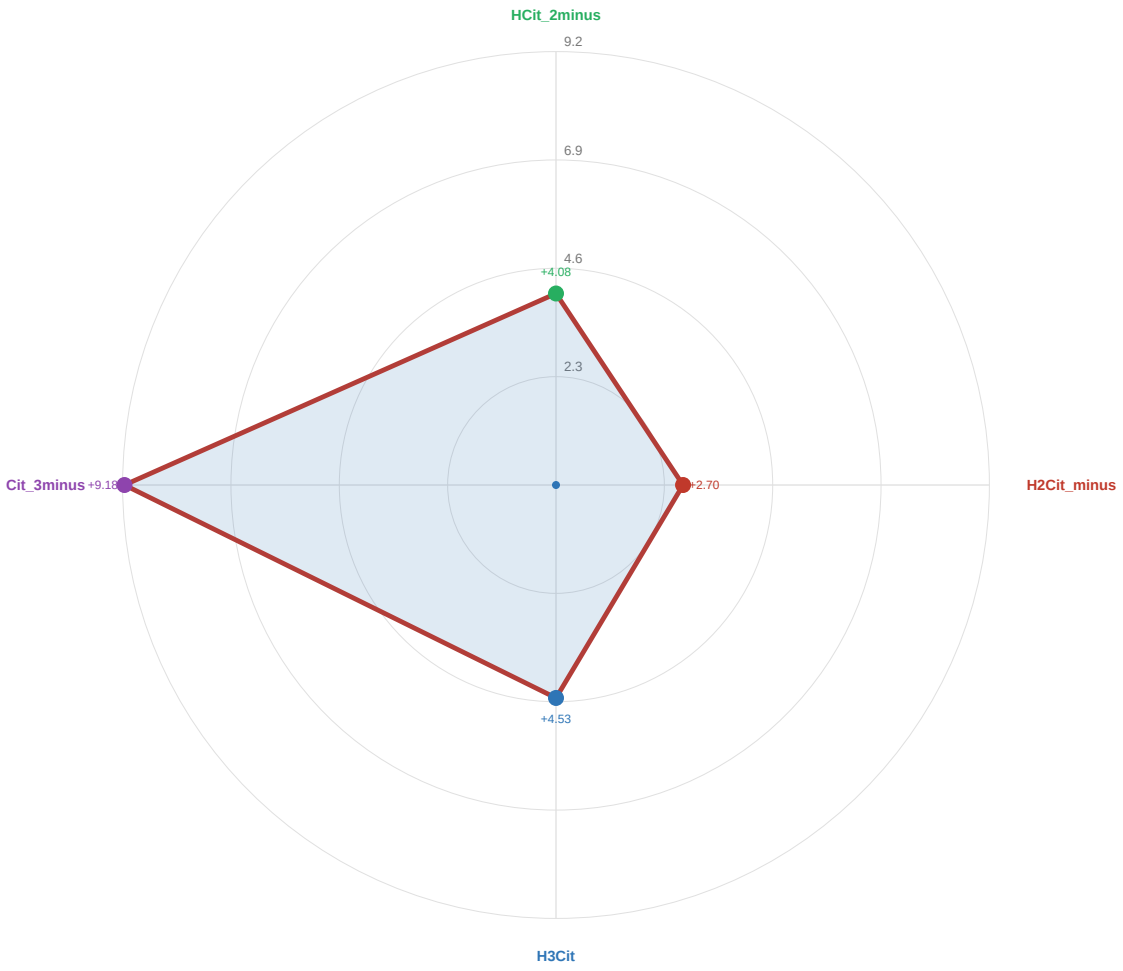
POLAR SLICE — t=190

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 191/200 | ||x||_A = 11.348



From 189: delta ||x||_A = +0.056

Phase = +30.8 deg

H3Cit=+4.53 H2Cit_=+2.70 HCit_2=+4.08 Cit_3m=+9.18

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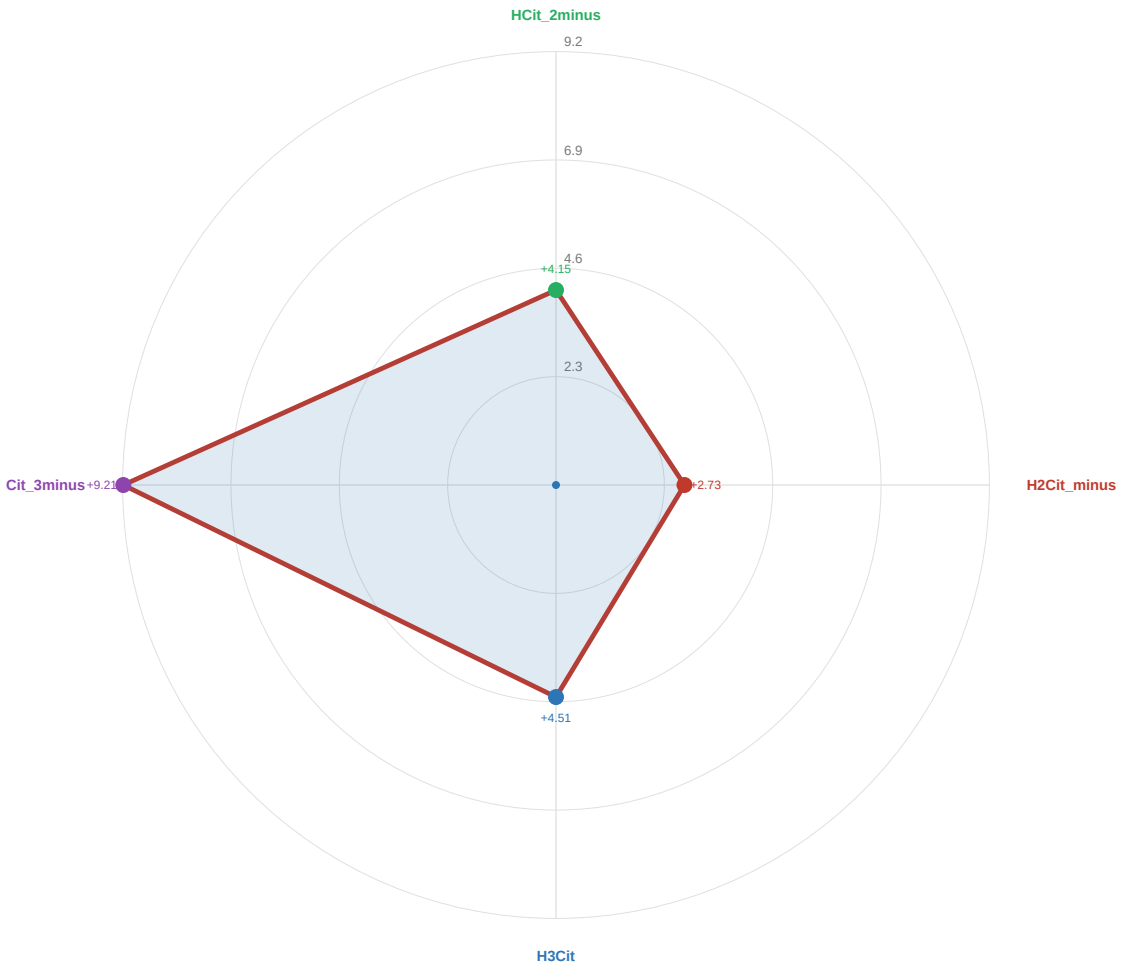
POLAR SLICE — t=191

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 192/200 | ||x||_A = 11.394



From 190: delta ||x||_A = +0.047

Phase = +31.2 deg

H3Cit=+4.51 H2Cit_=+2.73 HCit_2=+4.15 Cit_3m=+9.21

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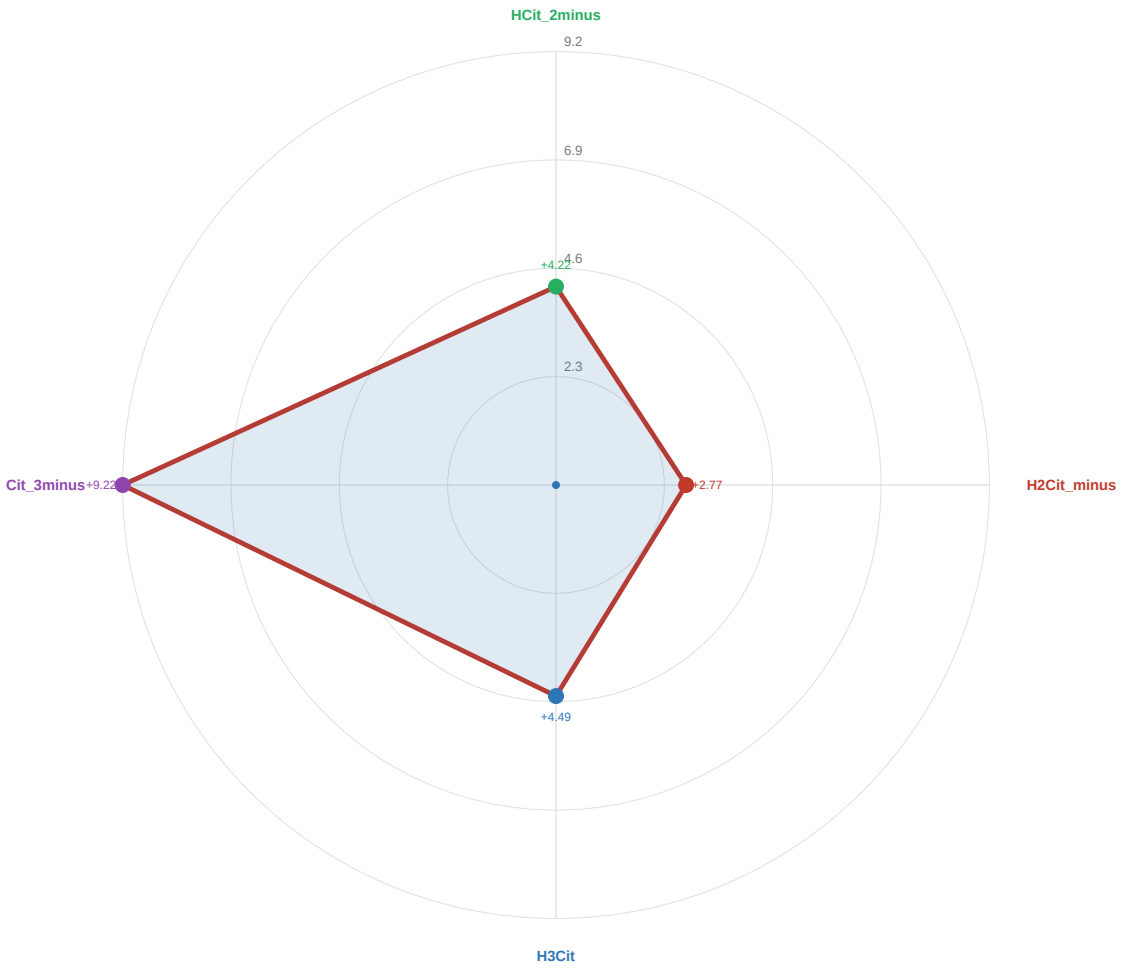
POLAR SLICE — t=192

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 193/200 | ||x||_A = 11.432



From 191: delta ||x||_A = +0.038

Phase = +31.7 deg

H3Cit=+4.49 H2Cit_=+2.77 HCit_2=+4.22 Cit_3m=+9.22

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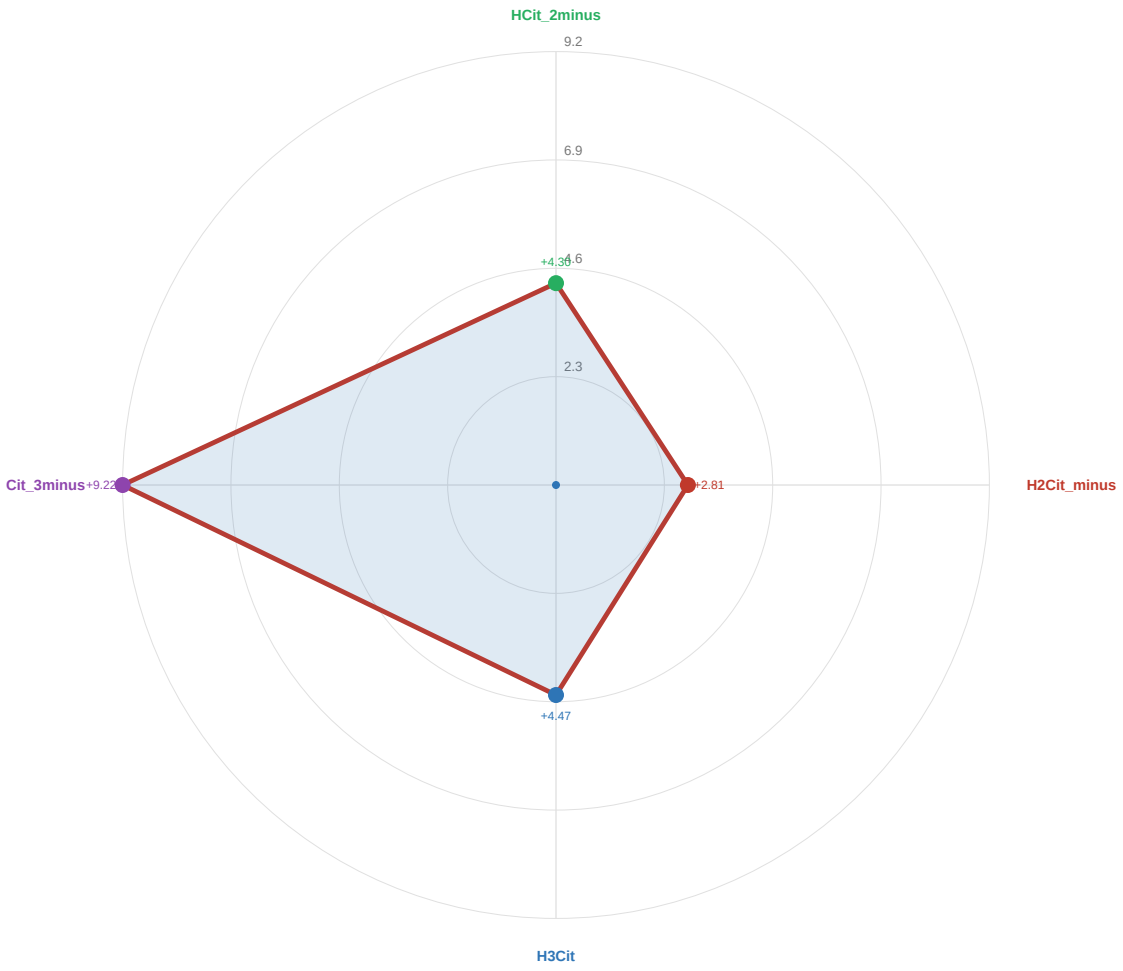
POLAR SLICE — t=193

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 194/200 | ||x||_A = 11.461



From 192: delta ||x||_A = +0.029

Phase = +32.1 deg

H3Cit=+4.47 H2Cit_=+2.81 HCit_2=+4.30 Cit_3m=+9.22

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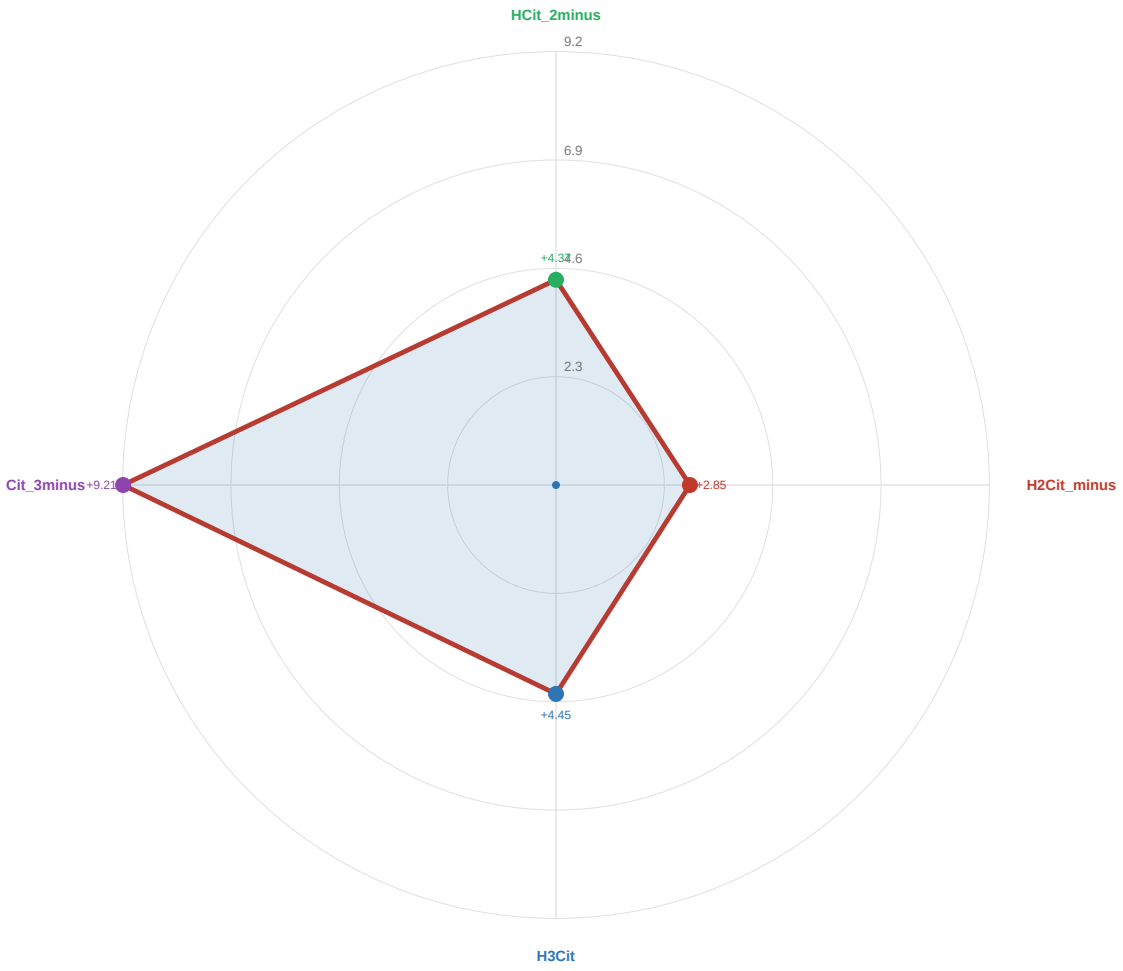
POLAR SLICE — t=194

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 195/200 | ||x||_A = 11.481



From 193: delta ||x||_A = +0.020

Phase = +32.7 deg

H3Cit=+4.45 H2Cit_=+2.85 HCit_2=+4.37 Cit_3m=+9.21

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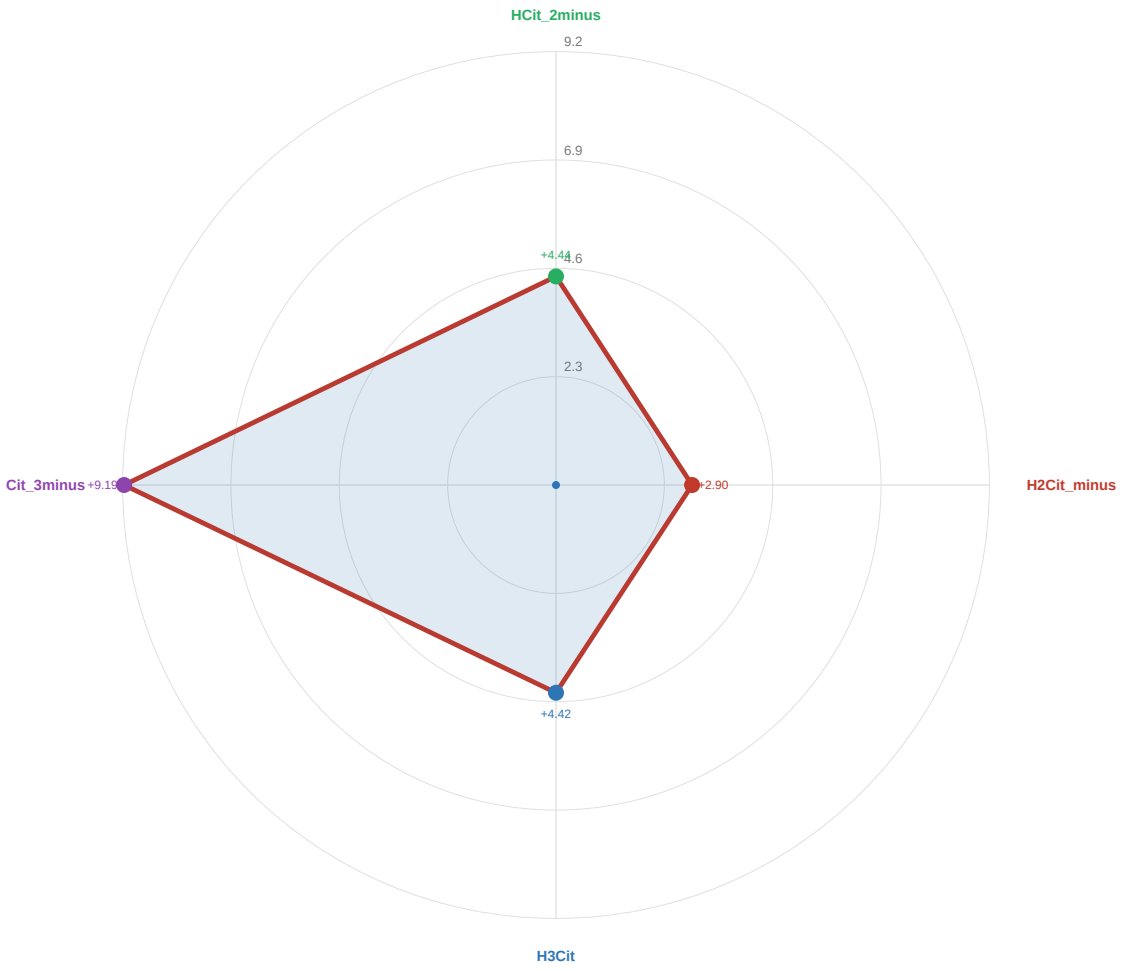
POLAR SLICE — t=195

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 196/200 | ||x||_A = 11.494



From 194: delta ||x||_A = +0.013

Phase = +33.2 deg

H3Cit=+4.42 H2Cit_=+2.90 HCit_2=+4.44 Cit_3m=+9.19

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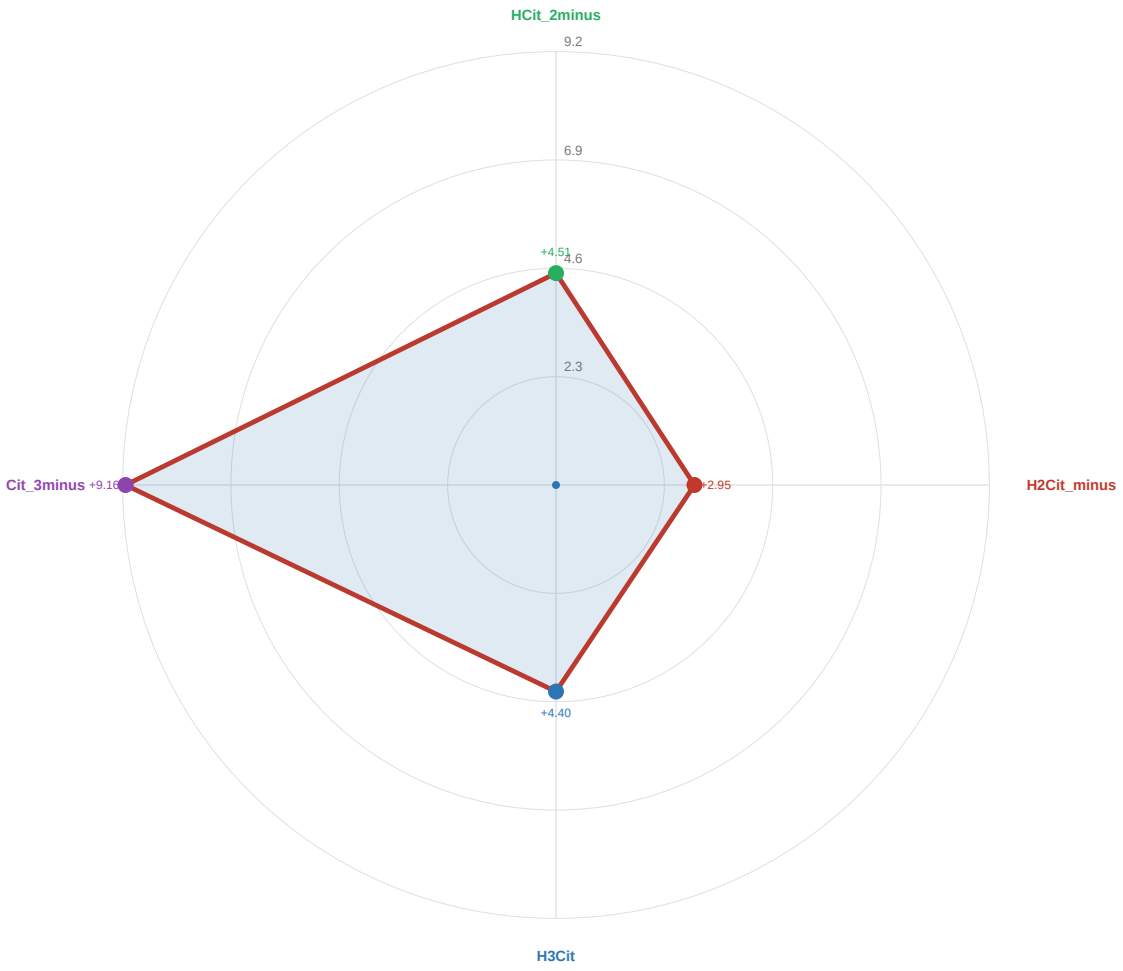
POLAR SLICE — t=196

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 197/200 | ||x||_A = 11.499



From 195: delta ||x||_A = +0.005

Phase = +33.8 deg

H3Cit=+4.40 H2Cit_=+2.95 HCit_2=+4.51 Cit_3m=+9.16

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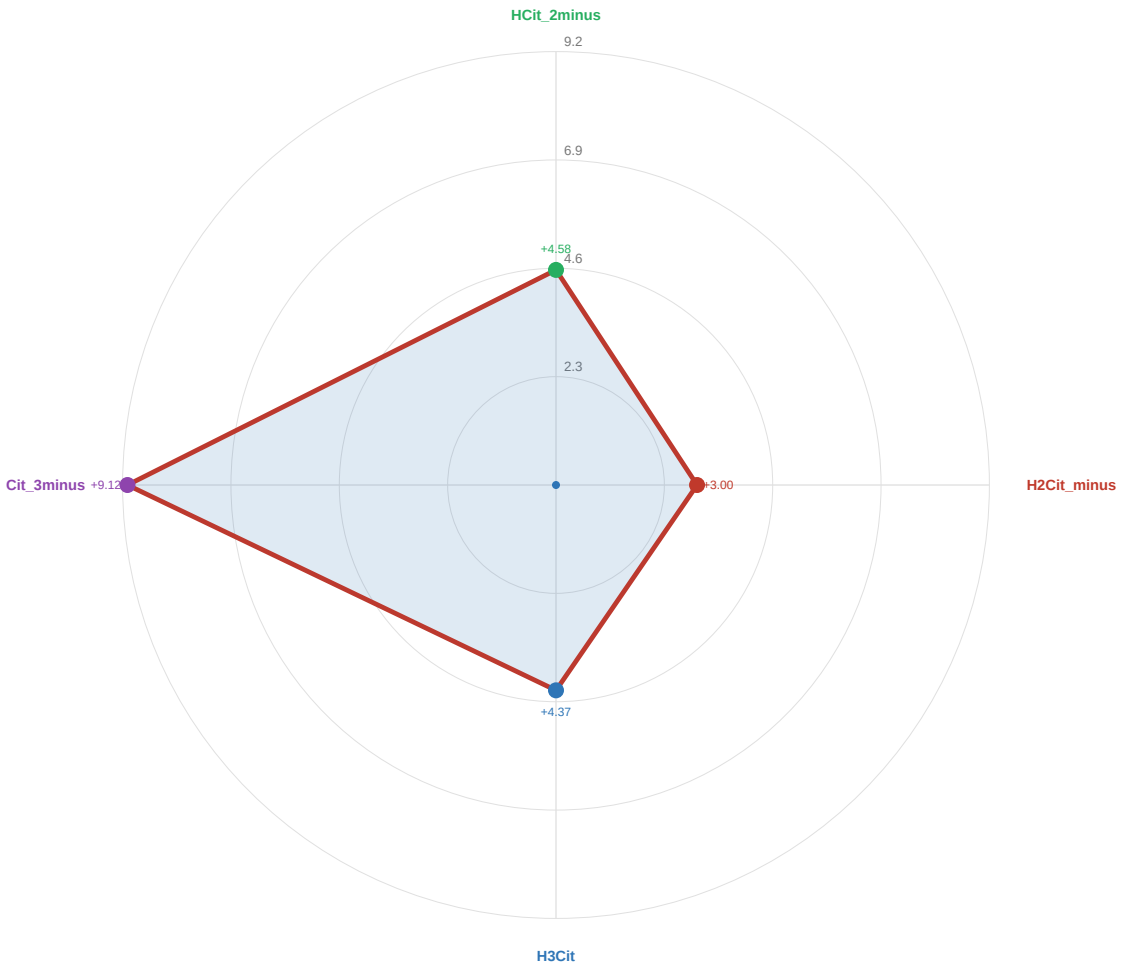
POLAR SLICE — t=197

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 198/200 | ||x||_A = 11.498



From 196: delta ||x||_A = -0.001

Phase = +34.5 deg

H3Cit=+4.37 H2Cit_=+3.00 HCit_2=+4.58 Cit_3m=+9.12

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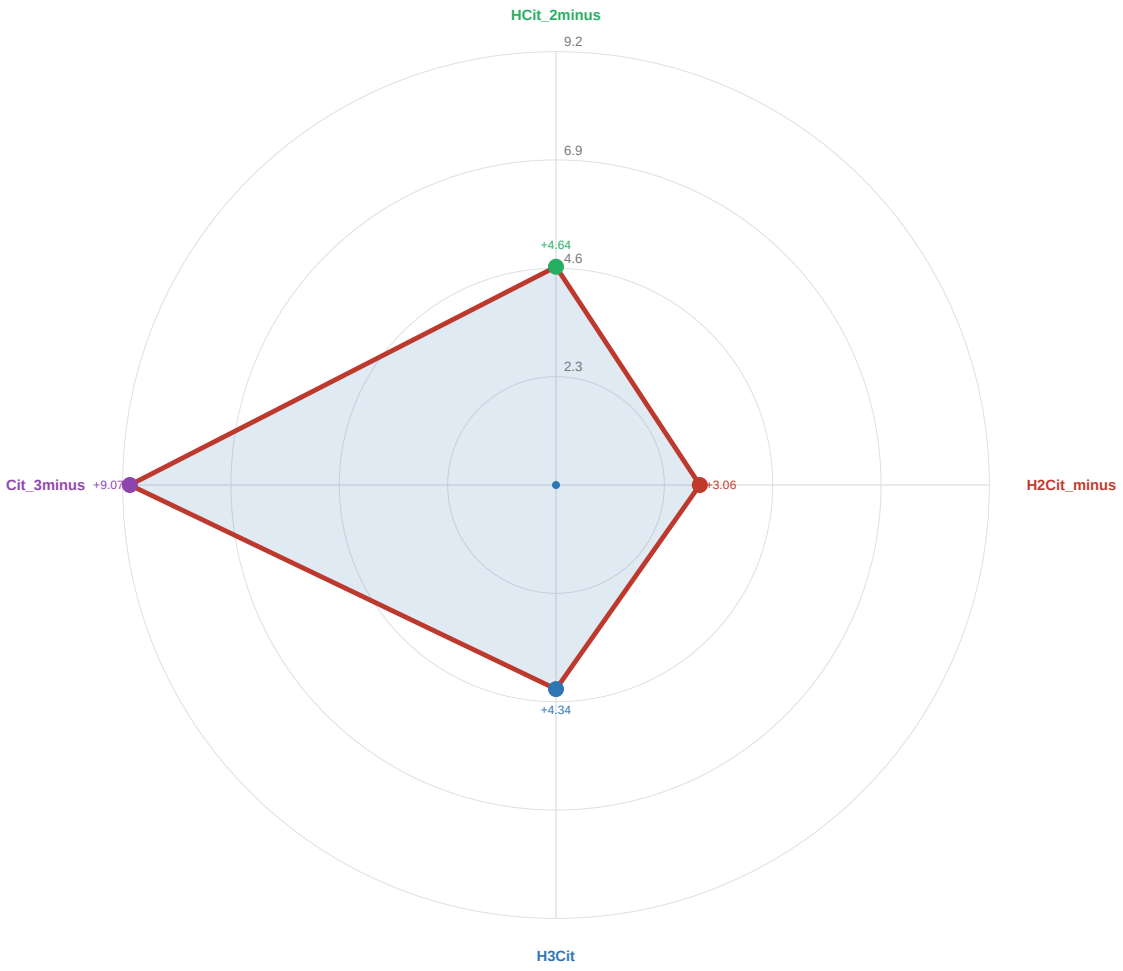
POLAR SLICE — t=198

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 199/200 | ||x||_A = 11.490



From 197: delta ||x||_A = -0.007

Phase = +35.2 deg

H3Cit=+4.34 H2Cit_=+3.06 HCit_2=+4.64 Cit_3m=+9.07

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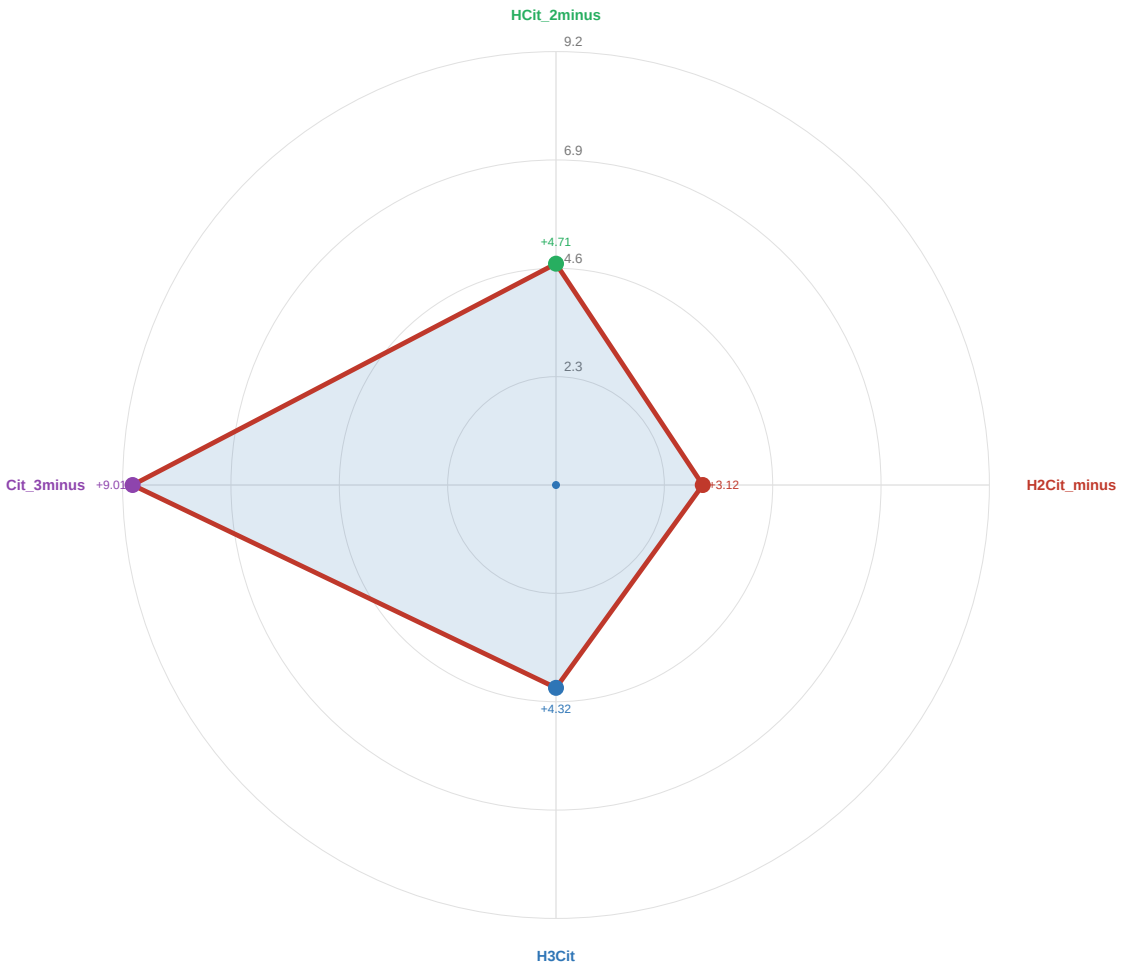
POLAR SLICE — t=199

CITRIC_ACID_TITRATIO | Citric Acid Titration | Interval 200/200 | ||x||_A = 11.478



From 198: delta ||x||_A = -0.013

Phase = +35.9 deg

H3Cit=+4.32 H2Cit_=+3.12 HCit_2=+4.71 Cit_3m=+9.01

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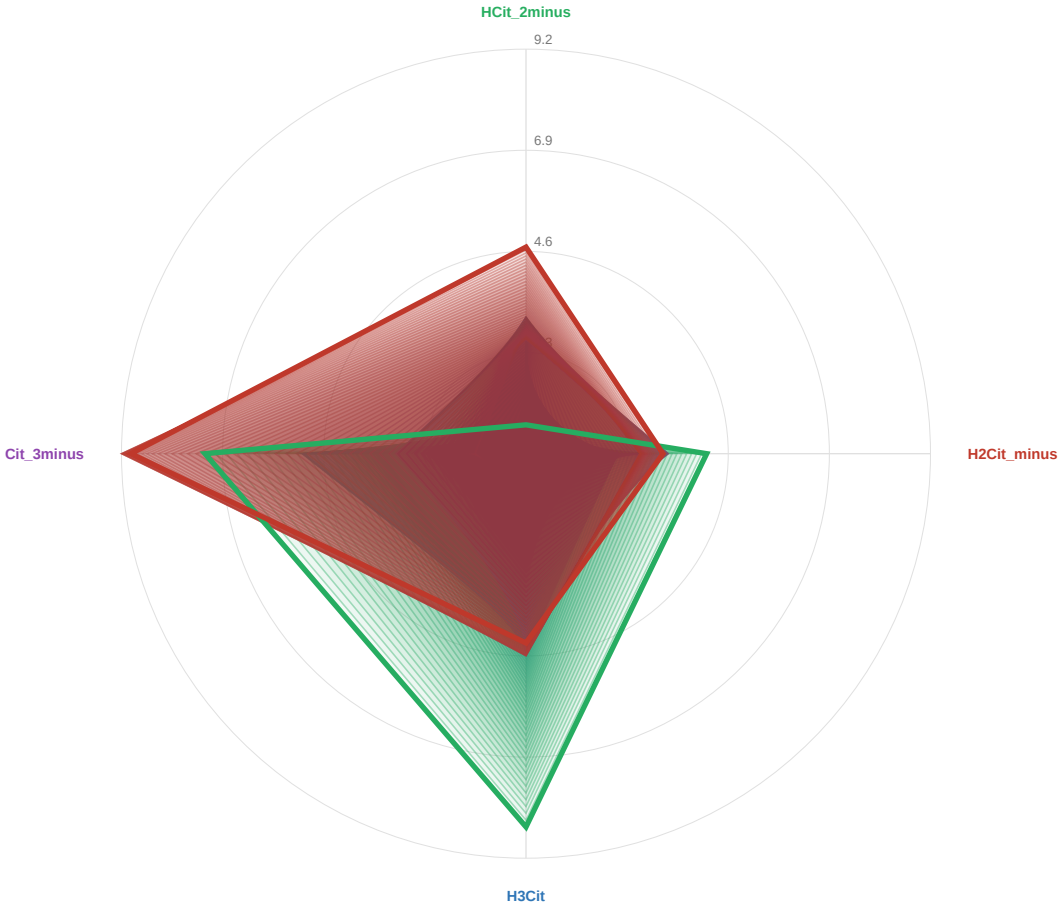
GHOST STACK — ALL INTERVALS OVERLAID

CITRIC_ACID_TITRATIO | Citric Acid Titration | 200 slices composited



All 200 polar slices overlaid. Bold outlines: first (0, green) and last (199, red).

Density reveals where the manifold concentrates. Shape migration shows structural change over time.

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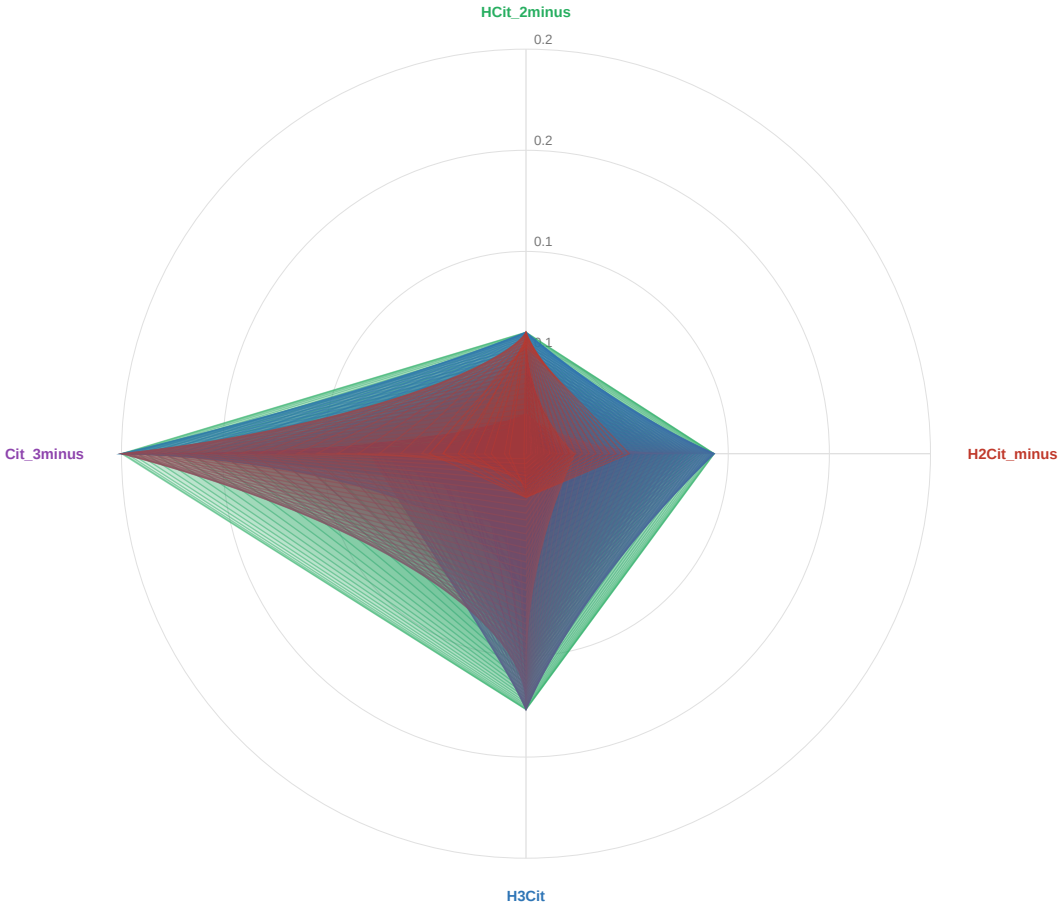
DIFFERENCE POLAR — YEAR-OVER-YEAR CLR CHANGE

CITRIC_ACID_TITRATIO | Citric Acid Titration | 199 delta intervals



Year-over-year CLR differences (delta_CLR_j = CLR_j(t) - CLR_j(t-1)). Large polygons = rapid change. Small = stability.

Max delta: 0.25 | Mean delta: 0.15

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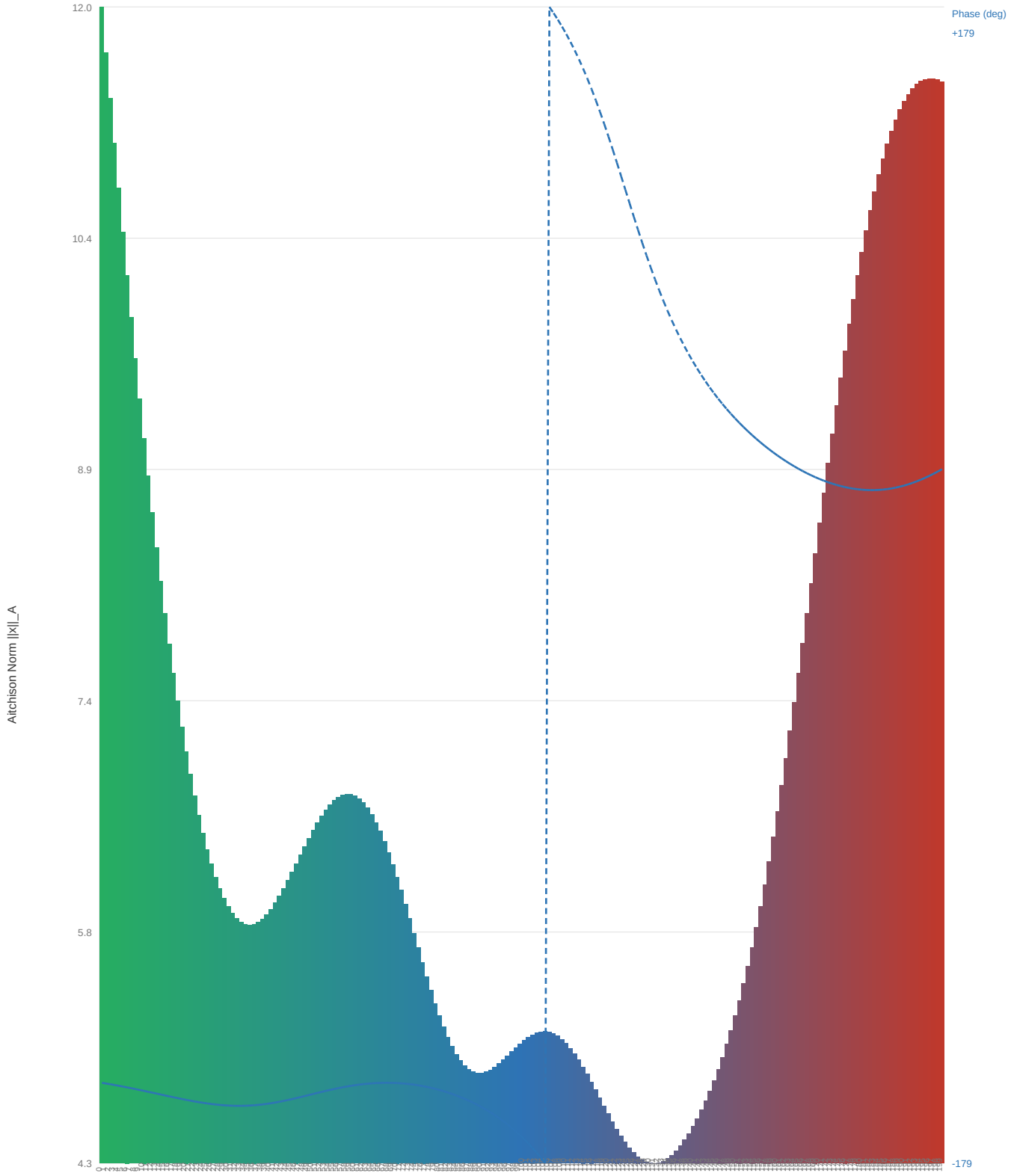
AITCHISON NORM PROFILE

CITRIC_ACID_TITRATIO | Citric Acid Titration | 200 intervals



Min ||x|| = 4.31 | Max ||x|| = 11.97 | Mean = 6.76 | Range = 7.67

Bars = Aitchison norm per interval. Dashed line = CLR phase angle (secondary axis).

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VALUE TABLE — CLR COORDINATES PER INTERVAL

CITRIC_ACID_TITRATIO | Citric Acid Titration | Rows 1-32 of 200

Year	‖x‖_A	Phase	H3Cit	H2Cit_m	HCit_2m	Cit_3mi
0	11.97	-154.2	-8.51	-4.12	-0.66	-7.32
1	11.67	-154.4	-8.35	-4.00	-0.58	-7.07
2	11.37	-154.6	-8.20	-3.89	-0.51	-6.83
3	11.07	-154.8	-8.04	-3.78	-0.44	-6.59
4	10.77	-155.1	-7.89	-3.66	-0.36	-6.35
5	10.48	-155.3	-7.74	-3.55	-0.29	-6.11
6	10.20	-155.6	-7.58	-3.44	-0.22	-5.88
7	9.91	-155.9	-7.43	-3.33	-0.15	-5.65
8	9.64	-156.1	-7.28	-3.22	-0.09	-5.43
9	9.37	-156.4	-7.13	-3.12	-0.02	-5.22
10	9.12	-156.7	-6.99	-3.01	+0.05	-5.02
11	8.86	-157.0	-6.84	-2.91	+0.11	-4.82
12	8.62	-157.2	-6.70	-2.81	+0.17	-4.64
13	8.39	-157.5	-6.56	-2.71	+0.23	-4.46
14	8.17	-157.8	-6.42	-2.62	+0.28	-4.30
15	7.95	-158.1	-6.29	-2.53	+0.33	-4.15
16	7.75	-158.4	-6.16	-2.44	+0.38	-4.00
17	7.55	-158.7	-6.03	-2.36	+0.43	-3.88
18	7.37	-158.9	-5.90	-2.27	+0.48	-3.76
19	7.20	-159.2	-5.78	-2.20	+0.52	-3.66
20	7.04	-159.5	-5.66	-2.12	+0.56	-3.56
21	6.89	-159.7	-5.54	-2.05	+0.59	-3.49
22	6.75	-159.9	-5.43	-1.98	+0.62	-3.42
23	6.62	-160.2	-5.32	-1.92	+0.65	-3.37
24	6.50	-160.4	-5.22	-1.86	+0.68	-3.33
25	6.39	-160.6	-5.12	-1.81	+0.70	-3.30
26	6.29	-160.7	-5.02	-1.75	+0.72	-3.29
27	6.20	-160.9	-4.93	-1.71	+0.73	-3.28
28	6.13	-161.0	-4.84	-1.66	+0.75	-3.29
29	6.06	-161.1	-4.75	-1.62	+0.76	-3.31
30	6.01	-161.2	-4.67	-1.59	+0.76	-3.34
31	5.96	-161.3	-4.60	-1.56	+0.77	-3.38

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VALUE TABLE — CLR COORDINATES PER INTERVAL

CITRIC_ACID_TITRATIO | Citric Acid Titration | Rows 33-64 of 200

Year	‖x‖_A	Phase	H3Cit	H2Cit_m	HCit_2m	Cit_3mi
32	5.93	-161.3	-4.53	-1.53	+0.77	-3.43
33	5.91	-161.3	-4.46	-1.51	+0.76	-3.49
34	5.89	-161.3	-4.40	-1.49	+0.76	-3.55
35	5.89	-161.2	-4.34	-1.48	+0.75	-3.62
36	5.89	-161.1	-4.28	-1.46	+0.74	-3.70
37	5.91	-161.0	-4.23	-1.46	+0.73	-3.78
38	5.93	-160.9	-4.19	-1.45	+0.72	-3.87
39	5.96	-160.7	-4.14	-1.45	+0.70	-3.96
40	5.99	-160.5	-4.11	-1.45	+0.68	-4.05
41	6.03	-160.3	-4.07	-1.46	+0.66	-4.15
42	6.08	-160.0	-4.04	-1.47	+0.64	-4.24
43	6.13	-159.8	-4.02	-1.48	+0.62	-4.34
44	6.18	-159.5	-4.00	-1.49	+0.60	-4.43
45	6.24	-159.2	-3.98	-1.51	+0.57	-4.52
46	6.29	-158.9	-3.97	-1.53	+0.55	-4.61
47	6.35	-158.6	-3.95	-1.55	+0.53	-4.69
48	6.41	-158.3	-3.95	-1.57	+0.50	-4.77
49	6.46	-158.0	-3.94	-1.60	+0.48	-4.84
50	6.52	-157.6	-3.94	-1.62	+0.45	-4.91
51	6.56	-157.3	-3.94	-1.65	+0.43	-4.96
52	6.61	-157.0	-3.95	-1.68	+0.41	-5.01
53	6.65	-156.7	-3.96	-1.71	+0.39	-5.05
54	6.68	-156.4	-3.97	-1.74	+0.37	-5.08
55	6.71	-156.1	-3.98	-1.76	+0.35	-5.10
56	6.73	-155.8	-4.00	-1.79	+0.33	-5.10
57	6.75	-155.6	-4.01	-1.82	+0.31	-5.10
58	6.75	-155.3	-4.03	-1.85	+0.30	-5.08
59	6.75	-155.1	-4.05	-1.88	+0.29	-5.06
60	6.74	-154.9	-4.07	-1.91	+0.28	-5.02
61	6.72	-154.7	-4.10	-1.93	+0.27	-4.96
62	6.70	-154.6	-4.12	-1.96	+0.27	-4.89
63	6.66	-154.4	-4.15	-1.98	+0.27	-4.82

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VALUE TABLE — CLR COORDINATES PER INTERVAL

CITRIC_ACID_TITRATIO | Citric Acid Titration | Rows 65-96 of 200

Year	‖x‖_A	Phase	H3Cit	H2Cit_m	HCit_2m	Cit_3mi
64	6.62	-154.3	-4.17	-2.01	+0.27	-4.72
65	6.57	-154.3	-4.20	-2.03	+0.27	-4.62
66	6.51	-154.2	-4.23	-2.04	+0.28	-4.50
67	6.44	-154.2	-4.25	-2.06	+0.29	-4.37
68	6.37	-154.2	-4.28	-2.07	+0.30	-4.23
69	6.29	-154.2	-4.30	-2.08	+0.32	-4.07
70	6.21	-154.3	-4.33	-2.09	+0.34	-3.91
71	6.12	-154.3	-4.36	-2.09	+0.36	-3.73
72	6.02	-154.4	-4.38	-2.09	+0.39	-3.55
73	5.93	-154.6	-4.40	-2.09	+0.42	-3.35
74	5.83	-154.8	-4.42	-2.09	+0.45	-3.15
75	5.74	-154.9	-4.44	-2.08	+0.49	-2.93
76	5.64	-155.2	-4.46	-2.06	+0.53	-2.71
77	5.55	-155.4	-4.48	-2.05	+0.57	-2.49
78	5.45	-155.7	-4.49	-2.03	+0.61	-2.26
79	5.37	-156.0	-4.50	-2.00	+0.66	-2.02
80	5.29	-156.4	-4.51	-1.97	+0.71	-1.78
81	5.21	-156.8	-4.52	-1.94	+0.76	-1.54
82	5.14	-157.2	-4.52	-1.90	+0.82	-1.29
83	5.08	-157.6	-4.53	-1.86	+0.88	-1.05
84	5.03	-158.1	-4.52	-1.82	+0.94	-0.80
85	4.99	-158.6	-4.52	-1.77	+1.00	-0.56
86	4.95	-159.2	-4.51	-1.72	+1.07	-0.31
87	4.93	-159.8	-4.50	-1.66	+1.13	-0.07
88	4.91	-160.4	-4.48	-1.60	+1.20	+0.16
89	4.90	-161.0	-4.46	-1.53	+1.27	+0.39
90	4.90	-161.7	-4.44	-1.46	+1.34	+0.62
91	4.91	-162.5	-4.41	-1.39	+1.42	+0.84
92	4.93	-163.3	-4.38	-1.32	+1.49	+1.05
93	4.94	-164.1	-4.35	-1.24	+1.56	+1.25
94	4.97	-165.0	-4.31	-1.15	+1.63	+1.44
95	4.99	-166.0	-4.27	-1.07	+1.71	+1.63

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VALUE TABLE — CLR COORDINATES PER INTERVAL

CITRIC_ACID_TITRATIO | Citric Acid Titration | Rows 97-128 of 200

Year	‖x‖_A	Phase	H3Cit	H2Cit_m	HCit_2m	Cit_3mi
96	5.02	-167.0	-4.22	-0.98	+1.78	+1.80
97	5.05	-168.0	-4.17	-0.88	+1.86	+1.97
98	5.07	-169.1	-4.11	-0.79	+1.93	+2.12
99	5.10	-170.3	-4.05	-0.69	+2.00	+2.26
100	5.12	-171.6	-3.99	-0.59	+2.07	+2.39
101	5.14	-172.9	-3.92	-0.49	+2.14	+2.50
102	5.16	-174.3	-3.85	-0.38	+2.21	+2.61
103	5.17	-175.8	-3.77	-0.27	+2.28	+2.70
104	5.18	-177.4	-3.69	-0.17	+2.35	+2.77
105	5.18	-179.1	-3.60	-0.06	+2.41	+2.84
106	5.18	+179.1	-3.51	+0.06	+2.47	+2.89
107	5.17	+177.2	-3.42	+0.17	+2.53	+2.93
108	5.15	+175.1	-3.32	+0.28	+2.59	+2.95
109	5.13	+173.0	-3.22	+0.40	+2.64	+2.97
110	5.10	+170.7	-3.11	+0.51	+2.70	+2.97
111	5.07	+168.2	-3.00	+0.62	+2.74	+2.96
112	5.03	+165.6	-2.89	+0.74	+2.79	+2.94
113	4.99	+162.9	-2.77	+0.85	+2.84	+2.91
114	4.95	+160.0	-2.65	+0.97	+2.88	+2.87
115	4.90	+156.9	-2.53	+1.08	+2.91	+2.82
116	4.85	+153.6	-2.40	+1.19	+2.95	+2.76
117	4.79	+150.2	-2.27	+1.30	+2.98	+2.70
118	4.74	+146.6	-2.14	+1.41	+3.01	+2.62
119	4.69	+142.8	-2.00	+1.52	+3.03	+2.55
120	4.63	+138.9	-1.86	+1.62	+3.05	+2.46
121	4.58	+134.9	-1.72	+1.73	+3.07	+2.37
122	4.53	+130.8	-1.58	+1.83	+3.09	+2.28
123	4.49	+126.6	-1.43	+1.93	+3.10	+2.19
124	4.45	+122.5	-1.29	+2.02	+3.11	+2.09
125	4.41	+118.3	-1.14	+2.11	+3.11	+2.00
126	4.38	+114.1	-0.99	+2.20	+3.11	+1.91
127	4.35	+110.0	-0.83	+2.29	+3.11	+1.81

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VALUE TABLE — CLR COORDINATES PER INTERVAL

CITRIC_ACID_TITRATIO | Citric Acid Titration | Rows 129-160 of 200

Year	‖x‖_A	Phase	H3Cit	H2Cit_m	HCit_2m	Cit_3mi
128	4.33	+106.0	-0.68	+2.37	+3.11	+1.72
129	4.32	+102.1	-0.53	+2.45	+3.11	+1.64
130	4.31	+98.4	-0.37	+2.53	+3.10	+1.55
131	4.31	+94.8	-0.22	+2.60	+3.09	+1.48
132	4.31	+91.3	-0.06	+2.67	+3.08	+1.40
133	4.32	+88.1	+0.09	+2.74	+3.06	+1.34
134	4.34	+84.9	+0.25	+2.80	+3.04	+1.29
135	4.36	+82.0	+0.40	+2.86	+3.03	+1.24
136	4.39	+79.1	+0.56	+2.91	+3.01	+1.20
137	4.42	+76.5	+0.71	+2.96	+2.99	+1.17
138	4.46	+73.9	+0.87	+3.01	+2.96	+1.15
139	4.50	+71.5	+1.02	+3.05	+2.94	+1.15
140	4.55	+69.3	+1.17	+3.08	+2.92	+1.15
141	4.60	+67.1	+1.32	+3.12	+2.89	+1.17
142	4.66	+65.0	+1.47	+3.15	+2.87	+1.20
143	4.72	+63.1	+1.61	+3.17	+2.84	+1.24
144	4.79	+61.2	+1.76	+3.19	+2.82	+1.30
145	4.86	+59.4	+1.90	+3.21	+2.80	+1.36
146	4.93	+57.7	+2.04	+3.22	+2.77	+1.44
147	5.01	+56.1	+2.17	+3.23	+2.75	+1.54
148	5.10	+54.5	+2.31	+3.24	+2.73	+1.64
149	5.19	+53.1	+2.44	+3.24	+2.71	+1.76
150	5.28	+51.6	+2.57	+3.24	+2.69	+1.89
151	5.39	+50.3	+2.69	+3.24	+2.68	+2.04
152	5.50	+49.0	+2.81	+3.23	+2.66	+2.19
153	5.61	+47.7	+2.93	+3.22	+2.65	+2.36
154	5.74	+46.5	+3.05	+3.21	+2.64	+2.53
155	5.87	+45.3	+3.16	+3.19	+2.63	+2.72
156	6.01	+44.2	+3.26	+3.17	+2.62	+2.92
157	6.16	+43.1	+3.37	+3.16	+2.62	+3.12
158	6.31	+42.1	+3.47	+3.13	+2.62	+3.34
159	6.47	+41.1	+3.56	+3.11	+2.62	+3.56

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VALUE TABLE — CLR COORDINATES PER INTERVAL

CITRIC_ACID_TITRATIO | Citric Acid Titration | Rows 161-192 of 200

Year	‖x‖_A	Phase	H3Cit	H2Cit_m	HCit_2m	Cit_3mi
160	6.64	+40.2	+3.65	+3.09	+2.62	+3.78
161	6.81	+39.3	+3.74	+3.06	+2.63	+4.01
162	6.99	+38.4	+3.82	+3.03	+2.64	+4.25
163	7.18	+37.6	+3.90	+3.00	+2.66	+4.49
164	7.36	+36.8	+3.98	+2.97	+2.67	+4.74
165	7.56	+36.0	+4.05	+2.94	+2.69	+4.98
166	7.75	+35.3	+4.11	+2.92	+2.72	+5.23
167	7.95	+34.7	+4.17	+2.89	+2.74	+5.47
168	8.15	+34.0	+4.23	+2.86	+2.77	+5.72
169	8.35	+33.4	+4.28	+2.83	+2.81	+5.96
170	8.55	+32.9	+4.33	+2.80	+2.84	+6.20
171	8.75	+32.3	+4.38	+2.77	+2.88	+6.43
172	8.95	+31.9	+4.42	+2.75	+2.93	+6.66
173	9.14	+31.4	+4.45	+2.72	+2.97	+6.89
174	9.33	+31.0	+4.49	+2.70	+3.02	+7.11
175	9.52	+30.7	+4.51	+2.68	+3.07	+7.32
176	9.70	+30.4	+4.54	+2.66	+3.13	+7.52
177	9.87	+30.1	+4.56	+2.64	+3.18	+7.71
178	10.04	+29.9	+4.58	+2.63	+3.24	+7.90
179	10.20	+29.7	+4.59	+2.62	+3.30	+8.07
180	10.35	+29.6	+4.60	+2.61	+3.37	+8.23
181	10.49	+29.5	+4.60	+2.60	+3.43	+8.38
182	10.62	+29.4	+4.61	+2.60	+3.50	+8.52
183	10.75	+29.4	+4.61	+2.60	+3.57	+8.65
184	10.86	+29.5	+4.60	+2.60	+3.64	+8.76
185	10.97	+29.6	+4.60	+2.61	+3.71	+8.87
186	11.06	+29.7	+4.59	+2.62	+3.78	+8.96
187	11.15	+29.9	+4.58	+2.64	+3.85	+9.03
188	11.23	+30.2	+4.56	+2.65	+3.93	+9.09
189	11.29	+30.5	+4.55	+2.68	+4.00	+9.15
190	11.35	+30.8	+4.53	+2.70	+4.08	+9.18
191	11.39	+31.2	+4.51	+2.73	+4.15	+9.21

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VALUE TABLE — CLR COORDINATES PER INTERVAL

CITRIC_ACID_TITRATIO | Citric Acid Titration | Rows 193-200 of 200

Year	‖x‖_A	Phase	H3Cit	H2Cit_m	HCit_2m	Cit_3mi
192	11.43	+31.7	+4.49	+2.77	+4.22	+9.22
193	11.46	+32.1	+4.47	+2.81	+4.30	+9.22
194	11.48	+32.7	+4.45	+2.85	+4.37	+9.21
195	11.49	+33.2	+4.42	+2.90	+4.44	+9.19
196	11.50	+33.8	+4.40	+2.95	+4.51	+9.16
197	11.50	+34.5	+4.37	+3.00	+4.58	+9.12
198	11.49	+35.2	+4.34	+3.06	+4.64	+9.07
199	11.48	+35.9	+4.32	+3.12	+4.71	+9.01

V(t) Shape: — | Total Variance: —

Path Length: — | Net Distance: — | Path Efficiency: —

Amalgamation Pass: — | Carriers: 4 | Observations: 200

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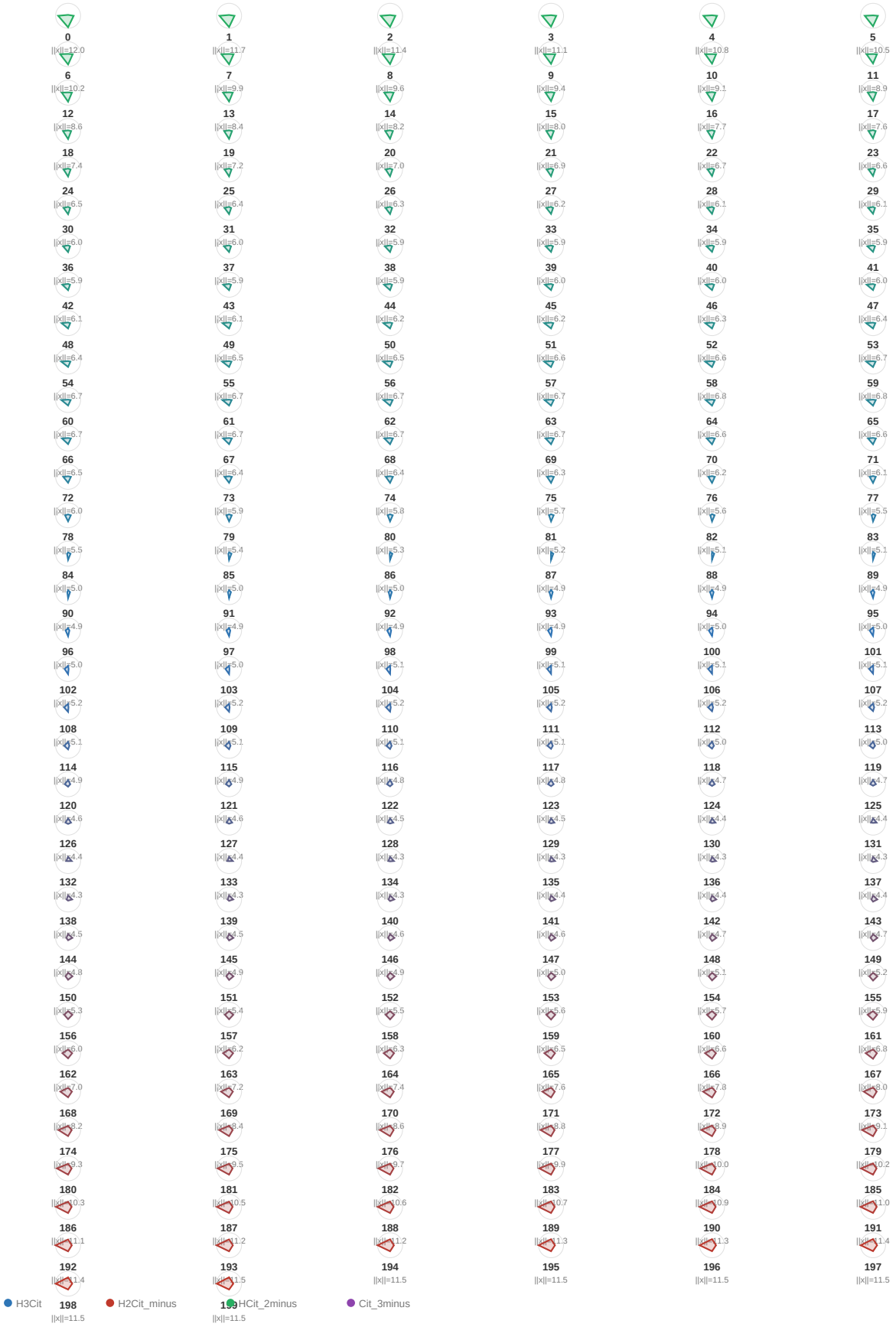
POLAR STACK SUMMARY — ALL INTERVALS

CITRIC_ACID_TITRATIO | Citric Acid Titration | 200 intervals compared



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